

Contributions to Security and Defence Studies

Arnold H. Kammel
Andrea Heinz
Daniel Hikes-Wurm *Editors*

DefenceTech Governance

AI, Biotech, Materials, Neurotech
and Quantum—Crucial Frontiers
for European and International Security
and Defence

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Preface: The beginning of a new story—Governance for a future in freedom

This anthology will already have been overtaken by political and technological developments by the time it is published. While work on this volume is being completed, the world is preoccupied with the United States' claim to power over Greenland. Mass protests in Iran are brutally suppressed. Venezuela's President Maduro is being held in a high-security US prison. Confer is launching a privacy-friendly alternative to common large language model-based chatbots. And AWS is moving metadata and control completely to the EU, which could mean enormous progress for the Union's digital sovereignty—if it were not already facing massive criticism because European sovereignty still cannot be guaranteed.

The world has become more unpredictable. Unabashed power politics and the law of the strongest are increasingly replacing the rules-based order that was established after the end of the Second World War and the fall of the Iron Curtain. The often-proclaimed end of history is turning out to be the end of *one* story—and the beginning of a new one, the plot and outcome of which are not yet foreseeable. We do not know what will dominate the headlines and political debates when you read these lines. What we do know is that technological developments will happen at great speed in this new, multipolar world order, and they will increasingly become factors relevant to geopolitics, security and defence policy. We are convinced that core values such as freedom, peace and democracy can only be preserved and protected if meaningful governance is achieved for frontier technologies such as AI, biotechnology, materials technology, neurotechnology and quantum technology. At the same time, universal regulations and conventions are being challenged and losing their significance. They simply cannot be enforced in the emerging new world order.

We do not see this as a rejection of governance *per se*, but rather as a motivation to find new ways to help old regulations regain their enforceability—and to find innovative, contemporary forms of governance. A form of governance that, if not prevents, then at least mitigates the undesirable effects, risks and disadvantages of new technologies; and at the same time does not inhibit innovation, technological and economic development, but promotes them for the benefit of society as a whole. This volume aims to contribute to this.

We view new technologies from the perspective of security and defence policy, and see them as part of the transformation that these areas have undergone in recent decades. The nature of conflict has changed fundamentally. The distinction between war/peacetime and the civilian/military spheres is becoming increasingly blurry; the methods used straddle the narrow and ambiguous line between armed attack and non-armed attack. These hybrid activities can comprise everything from cyberattacks, the weaponisation of migration to the spread of misinformation, disinformation or specific narratives online. Even economic coercion (e.g. energy supply manipulation), election interference or strategic sabotage (e.g. pipelines, undersea cables) can be used as hybrid threats. The goal is less the physical subjugation of an opponent than psychological influence and control: above all, democratic states and their will to assert themselves and defend themselves are to be weakened. Be it Brexit or the Pandemic, recent years have shown how much power lies in concerted disinformation campaigns, which are further amplified by the echo chambers of social media. This cognitive warfare does not require physical violence, not even coercion. Influence is exerted at the content level—for example, through disinformation or misinformation. Alternatively, it directly affects the human ability to process information and make decisions through neurotechnology or biotechnology. A particularly important aspect for us is the primary target in both cases: human freedom.

But what do we mean when we talk about freedom? Is my freedom defined and at the same time limited by the freedom of others, as Immanuel Kant (among many others) suggested? Apart from its use in private language, *freedom* is a term that has developed historically and been written on many banners. Kant is only one of many who have attempted to define it. Moreover, we must not forget that an abstract concept such as freedom is not a universal human, cross-cultural term. It is always part of a particular tradition of thought that is expressed in all parts of the respective society: in societal debates and conventions, in the political order.

In the specific philosophical tradition that shapes our intellectual horizon, freedom has always been a political concept, one that has always been oriented towards a community, a togetherness. Freedom is only possible together. Of course, we are talking about an ideal, which we are all aware is often not realised, and perhaps cannot be realised. However, this should not prevent us from having an ideal and aligning our actions with this ideal.

True freedom for a community can only exist if that community has set itself rules, an order for the benefit of all. This is undoubtedly an achievement of civilisation, which only at first glance represents a restriction. This supposed self-restraint also protects the individual. It serves its own safety, because it restricts the others as well. It ensures reliability and trust—the foundations on which cooperation, progress and wealth can grow. Freedom and security are not necessarily contradictory. They are mutually dependent. This is what we understand by peace-securing, future-proof governance.

It is urgently needed. New technologies are inextricably linked to cognitive warfare—but they also change and optimise conventional warfare. This volume aims

to provide an overview of the possibilities and opportunities offered by key technologies such as AI, biotech, materials technology, neurotechnology and quantum technology. How can we deal with their inherent dual-use nature, which makes them suitable for both civilian and military purposes and, beyond this simplistic dualism, for protective and healing as well as destructive and harmful purposes? How can they be used to secure peace, democracy, human rights and prosperity—but also in state and non-state, legal or unlawful attacks? What risks do they pose? What are the current options for regulation, and what might forward-looking, meaningful future governance look like? All of this is underpinned by the conviction that freedom and democracy, peace and human rights, prosperity and justice can only be secured if clear, jointly developed rules exist and are enforced.

Of course, there are more new technologies than the five we have selected. The fields of cyber or space, for example, are discussed repeatedly, but are not covered in separate chapters. Nor do we claim to provide an exhaustive overview of all technologies. Rather, our aim is to shed light on the paradigms and implications that technology fundamentally brings with it—for better or for worse. Above all, we want to highlight the scope for action and creativity that we have, especially when it comes to new, still-developing technologies. This applies to economic and political decision-makers as well as all other members of society. We all use a wide variety of technologies every day. We should not just be passive users, but actively participate in determining the direction in which these technologies develop. Regardless of whether we like it or not, they will shape the world we live in in the future. They *are* our future.

We would like to thank our authors for their excellent, dedicated work, their inspiring thoughts and the wealth of knowledge they share with us in their contributions. Many thanks also to the National Defence Academy in Vienna for its support and to Marie-Sophie Höller for her valuable contribution.

Editor's note: The individual contributions reflect the personal views and opinions of the respective authors and do not necessarily correspond to positions of the Federal Ministry of Defence of the Republic of Austria.

Vienna, Austria

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Andrea Heinz
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Part I
Prologue

“New Defence” in Europe: How Disruption and Dual-Use Technologies Recast European Security

Raluca Csernatonî

Abstract

Following Russia’s full-scale invasion of Ukraine, European security and defence policy is undergoing a profound transformation characterised by accelerated rearmament and an increased adoption of dual-use emerging and disruptive technologies (EDTs). This prologue situates these shifts within a liminal geopolitical state, neither peacetime nor outright war in Europe, which has enabled exceptional policy measures while blurring civilian-military boundaries. It foregrounds the concept of “New Defence”, which describes the EU’s evolving defence paradigm, marked by the infusion of venture capital approaches, software-first innovation, and startup-style agility into the European defence ecosystem. Drawing on recent EU initiatives, especially the EU’s 2025 Defence Industry Transformation Roadmap, the prologue argues that disruption has become a central governance logic rather than a descriptor of technological change. By examining the socio-political construction of disruption, the analysis highlights risks associated with the militarisation of civilian research, democratic oversight, and increasing dependence on private technology actors. The prologue lays the conceptual groundwork for analysing Europe’s defence transformation as a highly political and contested process with significant societal and international consequences.

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1 Introduction

In the wake of Russia's 2022 invasion of Ukraine, Europe's defence posture is undergoing profound transformations. Against this backdrop and an increasingly strained transatlantic relationship, an urgent logic of rearmament has taken hold across the continent. This is also centred on the rapid adoption of dual-use emerging and disruptive technologies (EDTs), notably artificial intelligence (AI), autonomous systems, such as robotics and drones, Big Data analytics, advanced semiconductors, biotechnologies, and quantum technologies, among others, to address evolving security threats. European policymakers and industry leaders are reassessing the continent's vulnerabilities and critical capability gaps in both civil and military domains, concluding that increased defence innovations and investments in these frontier technologies are crucial for Europe's future security. What makes this technology push especially complex is its context, namely, it is unfolding in a liminal grey zone that is neither full peacetime nor open war in Europe. While European Union (EU) member states are not direct belligerents, Europe as a political community is deeply entangled in supporting Ukraine's defence and deterring further aggression, hence operating in a liminal phase marked with uncertainty and ambiguity, where Europe finds itself away from a recent state of peace but unsure of what comes next (Martins & Csernaton, 2026). This ambiguous state has been instrumental in legitimising an unprecedented rearmament drive, coupled with a rush to build technological sovereignty (Csernaton, 2022), blurring traditional lines between the civilian and military spheres in the name of resilience and deterrence.

Within this multifaceted security landscape, dual-use innovation and the leveraging of civilian cutting-edge technology for military purposes have assumed a central role. EDTs, mainly originating in the civilian sector, are increasingly framed as dual-use solutions that can be repurposed for security and defence applications to address threats ranging from attacks on critical (digital) infrastructure, malicious cyber operations, AI-enabled Foreign Information Manipulation and Interference (FIMI), to drone warfare. As a result, a new defence tech ecosystem is rapidly taking shape in Europe, tightly weaving a venture capital logic (VC) (Schwarz, 2025) together with civilian startups and traditional defence actors. While this evolving ecosystem remains only partially understood in Europe, its rise signals that the continent's military build-up is not just a matter of spending more on conventional defence. Instead, it represents a deeper reconceptualisation of defence production, actively infusing civilian technological innovation and a culture of "disruption" into the core of European security and defence strategy (Martins & Csernaton, 2026). The move foregrounds a deeper Silicon Valley-style ethos of high-risk, high-reward disruptive tech innovation and experimentation as part of a new defence paradigm (Martins & Csernaton, 2026; Csernaton et al., 2025; Csernaton & Martins, 2023; Hoijsink, 2022).

A vivid illustration of this shift is the European Commission's recent "EU Defence Industry Transformation Roadmap: Unleashing Disruptive Innovation for Defence Readiness", published on 19 November 2025 (European Commission,

2025a). Beyond reframing Europe's defence preparedness as a question of joint funding and procurement, the roadmap enshrines a new structural transformation of how defence capacity is hegemonically imagined, developed, and integrated into Europe's political economy and even its identity (Csernaton, 2022). Notably, the European Commission is increasingly speaking of building a "European Defence", as opposed to the defence of individual member states, also implicitly situating Ukraine's security within a broader European framework, while normalising EU-level responsibility for a war at the bloc's borders.

The roadmap also introduces the concept of "New Defence", emphasising agility, risk-taking, and dual-use innovation, and the need to incorporate civilian tech ecosystems into defence. It could be argued that this signals a paradigmatic shift in the normative and procedural logics governing defence, more precisely, success is no longer measured solely by possessing advanced military hardware, but by the speed of innovation, cutting-edge innovation, openness to new actors, and the ability to harness disruptive civilian EDTs for warfare. In short, Europe's defence establishment is being revised to mirror the dynamism and agility of the software-first tech sector. The following sections aim to critically explore the disruptive effect of dual-use EDTs on the current European defence transformation.

2 Disruption and the Dual-Use Dilemma: Conceptual Foundations

The interlinked concepts of "disruption" and "dual-use" innovation require further analysis, as they form the intellectual bedrock of Europe's current new defence tech strategy. In this respect, unpacking the ambitions and anxieties surrounding EDTs in the security and defence domains requires deeper critical reflection about the dual-use dilemma and the dynamics behind the socio-political construction of disruption in defence.

The dual-use dilemma captures the fundamental tension that many advanced technologies can yield tremendous civilian benefits while also being harnessed for military purposes (Martins & Ahmad, 2022). Nevertheless, this dual potential creates ethical and regulatory challenges for scientists and policymakers alike, classic examples ranging from nuclear fission, which can power cities or level them, to AI, which can enhance healthcare diagnosis or enable lethal autonomous weapons systems. In the contemporary era, virtually all cutting-edge technologies are inherently dual-use, rendering any strict separation between the civilian and military spheres impractical. During the Cold War, governments often sought to silo military research from civilian industry. However, by the late twentieth century, the roles had inverted, civilian industry emerging as the primary engine of innovation and increasingly shaping military capabilities instead of the other way around. In other words, the most important military innovations now frequently originate in the commercial tech sector, driven by Big Tech players that control everything from staggering budgets to critical digital and material infrastructure, Big Data analytics, compute power, frontier algorithms, and the brightest

minds in the industry. In this regard, effective governance requires careful integration of civilian and military technological efforts through coordinated research and development (R&D) policies, up-to-date dual-use export controls, and flexible institutional arrangements (Alic, 1994). However, this is harder to achieve than expected, not least because fast-paced scientific advances and continuously evolving EDTs, from biotechnologies to neuroscience, from AI to quantum systems, can raise new security risks if weaponised by both state and non-state actors. Such rapid and disruptive technological developments can easily outpace traditional regulatory and governance frameworks, thereby intensifying the dual-use dilemma in the case of EDTs and, consequently, risking the militarisation of science that can bypass ethical constraints and civilian dominance in innovation (Hayakawa et al., 2023).

The EU, by contrast, is now charting almost the opposite course. Rather than insulating civilian research from defence, it is actively seeking synergies between the two. Recent EU policy documents openly advocate harnessing Europe's broader civilian innovation ecosystem for defence purposes. For instance, the European Commission's, 2024 White Paper "On options for enhancing support for research and development involving technologies with dual-use potential", from 24 January 2024, explicitly calls for more support to technologies that straddle civilian and military use (European Commission, 2024). This highlights a strategic decision to embrace the dual-use nature of technology as a feature, not a bug, of frontier innovation, yet it also raises the stakes for governance. As Europe dissolves traditional boundaries, ensuring ethical oversight and preventing adverse outcomes is ever more critical. Europe's strategic move towards dual-use is further reflected in the European Commission's, 2025 "EU Defence Industry Transformation Roadmap: Unleashing Disruptive Innovation for Defence Readiness" and its focus on "New Defence", as well as linked to lessons learned from Russia's invasion of Ukraine:

More than 230 defence tech startups were founded in Europe since Russia's invasion of Ukraine in February 2022, and private investment in defence startups reached an all-time high in 2024. These "New Defence" players, some coming from civilian and dual-use markets, are transforming the defence sector. They broaden the European Defence Technological and Industrial Base (EDTIB), accelerate the development and deployment of disruptive innovation, and introduce a new operating model based on rapid iteration, agility, cutting-edge innovation, software-first architectures, and greater risk-taking. Private financing enables their fast growth. These companies are bringing a new approach to defence. Bringing together the strengths of New Defence players and established defence industry actors can drive the transformation of the defence industrial sector, foster more agile and adaptive approaches, and challenge existing procedures to deliver faster, more effective capability development and deployment. [...] This requires systematically leveraging Europe's wider civilian research and deep-tech innovation ecosystem, ensuring that dual-use solutions emerging from the civil side can be rapidly sourced, tested and integrated into defence capability development. (European Commission, 2025a, pp. 1–2)

A footnote in the roadmap further unpacks the concept of "New Defence", namely referring to: "a new paradigm in defence innovation, driven by technological trends such as the leveraging of civilian-dual use technologies, software-defined warfare, and agile innovation cycles, as well as new business models, leading to increased efficiency, reduced costs, and accelerated delivery of defence solutions, with a greater emphasis on risk-taking" (European Commission, 2025a, p. 1). Moreover, the roadmap recognises that dual-use innovation and the rapid integration of civilian technologies into military capabilities have been decisive factors in Ukraine's defence. Ukraine's battlefield successes and improvised tech solutions in the face of a larger adversary are profoundly influencing Europe's strategic thinking. The full-scale war illustrates that agility, adaptability, human-machine teaming, and technological ingenuity can offset traditional military disadvantages. Notably, Ukraine's leverage of dual-use innovations by repurposing readily available civilian technology, for instance, converting commercial drones into deadly assets and civil society becoming an arms producer, to rapidly augment its defence posture (Zaremba & Csernaton, 2025). Facing ammunition shortages and an opponent with superior artillery, Ukraine's military, tech volunteers, startups, and small enterprises are devising cost-effective innovations, such as First Person View (FPV) drone bombers. These inexpensive yet highly effective short-range drones have allowed Ukraine to inflict significant damage on Russian forces and hold defensive lines. Accordingly, the message to Europe has been clear: civilian tech, harnessed creatively and deployed with speed, can yield outsized military effects.

Indeed, priority is given to strengthening synergies between established defence industry actors and the civilian technology sector and across the full breadth of the research and innovation ecosystems, a strategically consequential move for European defence preparedness (European Commission, 2025a, p. 3). The roadmap identifies such critical dual-use technologies, spanning AI as a "strategic driver of military innovation" and quantum as a "key emerging technology for defence capabilities", space technologies underpinning "multidomain operations", and cyber as "the fifth domain of warfare" (European Commission, 2025a, pp. 5–6). The roadmap also notes that over recent years, the EU's European Defence Fund (EDF) has supported the R&D of a wide range of cutting-edge technologies and systems, contributing to innovation in areas such as AI, robotics, hypersonic weapons, cloud computing, advanced materials, space-based capabilities, and quantum technologies. The EDF has been allocating 4–8% of its budget specifically to projects classified as disruptive technologies, signalling a deliberate institutional commitment to high-risk, high-reward innovation within the European defence landscape.

The roadmap indeed highlights how "emerging disruptive technologies profoundly reshape modern warfare" by embedding AI, autonomous systems, and quantum technologies into military capabilities that alter how operations are planned and executed (European Commission, 2025a, p. 2). The roadmap mentions the word "disruption" no less than 23 times in a 15-page document, associating it with more adaptive, data-driven, and resilient defence systems and contributing to a reconfiguration of both modern warfare and prevailing conceptions of deterrence.

Within this context, the recent surge in defence investment across Europe creates the structural conditions for the accelerated development and uptake of such disruptive technologies, while simultaneously enabling the emergence of so-called New Defence actors within the European Defence Technological and Industrial Base (EDTIB).

These defence developments in Europe draw attention to how broader, dominant discourses of “disruption” and “innovation” increasingly entrench power structures by binding militaries, large technology corporations, and defence tech players into symbiotic yet often opaque partnerships. Such arrangements contribute to the emergence of a new form of military-tech-industrial complex (Csernatoni et al., 2025), diffuse yet potent, transnational in scope yet deeply intertwined with sovereignty-building prerogatives. In this line of thinking, disruption operates as a powerful rallying cry, positioning EDTs not only as advanced military capabilities but as paradigmatically transformative forces capable of reordering entire societies, security strategies, and geopolitical hierarchies. Thus, the language of disruption risks normalising and rationalising an intensifying synergy between entrepreneurial capital and martial logic, a dynamic that is increasingly now institutionalised within the evolving EDTIB (Csernatoni & Martins, 2023; Csernatoni et al., 2025). Precisely because disruption functions here as a performative and legitimising governance concept, it requires further analytical unpacking beyond being taken at face value as a neutral account of technological change.

In effect, Europe is seeking to institutionalise the ethos of a tech startup within the defence sectors, shedding peacetime complacency for wartime urgency in bureaucratic processes. To drive this transformation forward, the EU has launched or expanded several key initiatives. The EDF, established in 2021, has been supercharged to finance collaborative R&D projects in emerging tech areas, from AI-enabled command systems to next-generation quantum encryption. The EDF explicitly prioritises projects with disruptive potential and dual-use relevance, breaking down the wall that once separated civilian EU research funding from defence projects. Complementing the EDF is the European Defence Innovation Scheme (EUDIS), including a Business Accelerator programme to scout and scale up promising defence startups, and selecting 20 defence tech companies from across Europe for intensive support, thereby signalling an explicit institutional commitment to fostering “New Defence” actors within the European Defence Technological and Industrial Base (European Commission, 2025b). Other EU-level initiatives, such as the White Paper for European Defence, the ReArm Europe Plan-Readiness 2030, the Security Action for Europe (SAFE) Regulation, and the European Defence Industry Programme (EDIP), further exemplify how the European Commission is positioning itself as a central orchestrator of defence innovation. This is a notable evolution for an institution that, until recently, had a limited role in defence matters, which were jealously guarded by member states. Now, through funding, coordination, and narrative-setting, the Commission policy entrepreneurship is actively shaping the emergence of a European defence tech ecosystem. The Commission’s, 2025 roadmap outlines three strategic objectives to accelerate this defence-industrial transformation: first, better connecting the

defence and civilian deep-tech communities to spur disruptive solutions and new entrants; second, accelerating the integration of disruptive technologies into the military capabilities of EU member states; and third, enhancing Europe's defence production capacity through disruptive advanced manufacturing, achieving scale and speed in output.

Consequently, alongside the dual-use discourse, the concept of "disruption" has gained prominence in European security and defence policy (Csernatoní & Martins, 2023). The term "disruptive technology" typically refers to innovations that significantly alter or displace existing ways of doing things, a term borrowed from business and innovation theory (Csernatoní & Martins, 2023). In the military context, disruptive technologies are those expected to change military operations or the balance of power radically, for example, AI-driven autonomous systems that could render traditional operated systems obsolete, or quantum technologies, as they promise step-changes in sensing, communications, navigation, or computation, thus undermining existing military advantage, rendering current encryption obsolete, and reconfiguring deterrence and strategic stability. However, analysis should look beyond a purely technical definition of disruption and examine how "disruption" is socio-technically constructed as a narrative in security governance.

Csernatoní and Martins (2023) argue that what counts as "disruptive" in security and defence is not inherent to the technology alone, but is actively shaped through narratives of urgency, anticipation, and imagination. In their analysis, policymakers and experts construct a sense of looming future threats and technological revolutions, which in turn justifies extraordinary efforts and investments in certain EDTs. This performative aspect means that labelling technology "disruptive" often serves to galvanise political will and reorient institutional priorities, sometimes sidelining regular checks and deliberative processes. In this regard, the discourse of disruption tends to emphasise speed, "we must not fall behind", inevitability, "the future of warfare will be dominated by X technology", and opportunity, "whoever leads in this tech will gain a decisive edge". This is also reflected in recent EU rhetoric, as the Defence Industry Transformation Roadmap calls for accelerating innovation "to be ready for the unthinkable" (p. 1) and explicitly ties disruptive tech adoption to deterrence and survival. By invoking an existential imperative, EU institutions create a policy environment where taking disruptive technological risks is valorised, a dramatic shift for a governance system historically seen as risk-averse.

Overall, disruption is both a descriptor of cutting-edge tech innovations and a new governance logic. The socio-technical and political framing of disruption creates a rationale for acting in ways that might otherwise be seen as extraordinary or risky. In Europe's current defence context, the disruption narrative is being used to fast-track the adoption of dual-use EDTs, relax long-standing norms and boundaries between civilian and military innovation, and bring non-traditional players into the security arena. This can be double-edged. On the one hand, it galvanises innovation and helps break through bureaucratic inertia at a time of increased insecurity in Europe, for instance, by speeding up the procurement of new tech and the spin-in of civilian innovations.

On the other hand, it can sideline ethical deliberation and civilian oversight by suggesting that speed and agility trump caution. As Csernatoni and Martins (2023) note, the language of disruption often carries an implicit temporality, a vision of an imminent future of conflict that justifies pre-emptive action today. When decision-makers operate under such future-oriented pressure, there is a risk that “disruption by design” becomes an end in itself, namely, constant innovation without adequate reflection on consequences.

This is especially salient given Europe’s ambiguous geopolitical and strategic context. The EU’s adoption of disruptive dual-use tech is occurring amid heightened insecurity, where traditional war-footing safeguards, such as martial law or emergency powers, do not formally exist. However, a sense of emergency permeates security and defence policy like never before. This is a potential fertile ground for “exceptional” practices to become normalised under the banner of dual-use disruption and security, while mainstreaming instances of military isomorphism. In this respect, and as the roadmap suggests, the EU may attempt to imitate Ukraine’s wartime innovation model, marked by rapid, ad hoc problem-solving and the deployment of civilian tech for military ends, while still operating under peacetime legal-institutional frameworks. Hence, this mismatch could lead to legal grey areas and accountability gaps, as crisis-driven approaches are transplanted into a nominally peacetime setting.

Overall, the dual-use dilemma and the cult of disruption together define a new *Zeitgeist* in European security and defence policy, dubbed as “New Defence”. Europe is seeking to leverage civilian innovation for military strength, turning a dual-use dilemma into a dual-use opportunity, and doing so at breakneck speed and agility under a narrative of paradigm-shifting disruptive change. Therefore, these conceptual foundations carry inherent tensions, made manifest in practice through Europe’s post-2022 institutional initiatives and the rise of a defence tech ecosystem.

3 Conclusion

In practical terms, Europe’s EDTs-centric defence innovation ecosystem remains a work in progress. Despite the flurry of activity, structural changes remain, with defence procurement in Europe still fragmented along national lines. Moreover, a startup might win a contract in one country but struggle to scale across 27 different national bureaucracies. Indeed, the post-2022 European defence transformation is characterised by an “innovation theatre” of institutional activism and ecosystem dynamisms, legitimised by the dual-use potential of disruptive new technologies. A new cast of actors, from startups and venture capitalists to civilian researchers, is entering the European defence scene, accelerating the development of EDTs such as AI, autonomous drones and counter-drone systems, quantum sensors, space and cyber technologies, and semiconductor supply chains, among others, for military use. It could be argued that this civil-military convergence is creating a “New Defence” tech ecosystem unlike anything Europe has seen before,

one geared towards continuous innovation, boundary blurring, and venture-driven experimentation.

This analysis has shown that Europe's accelerating embrace of dual-use EDTs in defence foregrounds a set of deeply interlocking societal, political, and ethical challenges. Chief among these are the potential militarisation of civilian research, the erosion of democratic oversight and accountability, and the unresolved ethical dilemmas posed by disruptive warfighting technologies. Taken together, these dynamics highlight a widening gap between the speed at which defence innovation is pursued and the capacity of existing governance frameworks to regulate its consequences. Importantly, there is a growing influence of the European Commission as a policy entrepreneur, engineering a structural shift towards "New Defence", alongside the rising role of private technology firms and financial actors in European security. Subsequently, defence innovation is no longer solely dominated by a small group of state-aligned defence primes, but increasingly shaped by startups, Big Tech firms, and venture capital. This redistribution of power from public authorities to private actors introduces new governance challenges, particularly where profit-driven incentives may not align with democratic values or strategic restraints.

Furthermore, new forms of dependence on privately owned intellectual property and infrastructure raise acute questions about strategic autonomy and technological sovereignty, especially in cases of foreign acquisition or opaque corporate governance. To address these challenges, it will require new forms of public-private governance, including contractual safeguards, military technology readiness levels (TRLs), transparency requirements, and enforceable ethical frameworks. The dual-use EDTs themselves introduce deep ethical and societal dilemmas, as AI and autonomous systems challenge long-standing principles of human control over the use of force, while quantum technologies threaten existing cryptographic systems and privacy regimes. While European policy discourse frames disruption and rearmament as indeed necessary conditions for resilience and deterrence, they must also be understood holistically. Security depends not only on technological prowess but also on public trust, social cohesion, and adherence to democratic principles. Only by achieving this balance can Europe ensure that disruptive technologies serve as tools of protection and progress, rather than sources of new insecurity or democratic erosion.

Beyond its internal institutional, industrial, and societal implications, Europe's turn towards EDTs in defence also has important relational and international effects that warrant closer analytical attention. Europe's pursuit of security through dual-use innovation does not take place in isolation, but is embedded within a fragmented global order characterised by intensifying technological competition, asymmetric interdependence, and contested norms of security and governance. As the EU increasingly positions itself as both a regulatory authority and an innovation-driven security actor in pursuit of strategic autonomy and technological sovereignty, it simultaneously reshapes its relationships with allies, partners, competitors, its own member states, and wider regions affected by the diffusion of European standards, practices, and technological imaginaries. Choices made at the

EU level regarding critical technologies, export controls, public–private partnerships, and civil–military synergies may therefore extend beyond Europe’s borders, shaping global technological trajectories and security practices.

This relational dimension raises important questions about the nature of European autonomy itself. While policy discourse frequently frames EDTs as instruments for achieving strategic autonomy and technological sovereignty (Csernaton, 2022), it would be worth considering that such autonomy is neither absolute nor self-contained. Rather, it is negotiated, relational, and co-produced through transnational networks of states, firms, standards bodies, and financial actors. In this sense, Europe’s defence technological transformation also functions as a site of imaginary projection, through which particular visions of security, progress, and technological leadership are normalised and rendered desirable. These imaginaries may serve as a reference point for others, but they may also be contested, appropriated, or instrumentalised in ways that escape European control. Subsequently, dual-use disruptive innovation and defence governance must not be understood only as internal challenges, but as constitutive elements of a wider reconfiguration of global security and technological order.

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Part II

Conditions and Opportunities for Frontier Technologies

Navigating the Tech Frontier

Rick Meessen and Carolina van Weerd

Abstract

Emerging technologies such as artificial intelligence, quantum computing, advanced materials, biotechnology, and space technology are transforming global security at unprecedented speed. They offer immense promise for progress yet simultaneously generate new vulnerabilities, blurring boundaries between civilian and military domains. This chapter explores the dual nature of technology—as instrument, vulnerability, and ecosystem—highlighting how its impact depends on geopolitical shifts, human choices, and governance structures. Power is shifting from states to private actors, intensifying competition for technological dominance and complicating regulation in an increasingly multipolar world. Traditional governance frameworks lag behind rapid innovation, while divergent models in Europe, the United States, and China fragment global norms. To address these challenges, the chapter develops four future worlds defined by the degree of human control and access to technology, and identifies key governance needs from a Western democratic perspective. It proposes four strategic pillars for navigating the defense-technology frontier: foresighting, legal and ethical governance, resilience engineering, and adaptive alliances. Finally, it argues for a shift from reactive regulation toward proactive, ecosystem-based stewardship. By embracing pluralism, embedding ethics early in design, and fostering inclusive governance, democracies can help promote that technological superiority strengthens rather than undermines legitimacy and human dignity.

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1 From Promise to Peril: The Dual Nature of Technology

Technologies such as artificial intelligence, quantum computing, advanced materials, biotechnology, and space technology are reshaping the global security landscape at a breathtaking pace. They promise new breakthroughs in economic growth, scientific discovery, and societal wellbeing. Yet at the same time, they open entirely new vectors of vulnerability, instability, and conflict. This tension, the dual nature of technology as both enabler and threat, has always existed. It is important to recall, however, that technology in itself does nothing; its impact depends on how we, as humans, choose to apply and embed it in our societies. The novelty of the current era lies not only in the rapidity and scale of technological development, but also in the diversification of actors involved, the erosion of traditional governance mechanisms, and the increasingly hybrid character of contemporary conflict.

1.1 A Complex and Shifting Security Landscape

The twenty-first-century security environment is no longer defined solely by conventional military power or nuclear deterrence. Instead, it is shaped by a mesh of interconnected technologies whose applications extend far beyond the battlefield. Artificial intelligence can augment decision-making, but it can equally enable disinformation at scale. Quantum technology promises unimaginable computing power enabling breakthroughs in real-time simulation of complex processes, but simultaneously threatens to render current cryptographic systems obsolete. Biotechnology holds the potential to cure diseases, yet also raises the specter of engineered pathogens. Space-based systems allow global connectivity, but their vulnerability to jamming or cyber intrusion creates new points of failure.

This dynamic interplay underscores the central paradox: technology creates both new strengths and new vulnerabilities. Since technology plays a key role in the global security domain, the challenge is no longer simply about possessing advanced technology and being the technology leader to ‘win the game’, but also about managing the risks of its misuse or disruption.

Another defining feature of the current era is the rise of hybrid warfare. In this type of conflict, traditional military force is intertwined with cyberattacks, disinformation campaigns, economic coercion, and the weaponization of civilian infrastructure. Emerging technologies are central to this trend (Meessen & van Weerd, [2025](#)).

1.2 Beyond the State: The New Constellation of Actors

Historically, the development of disruptive technologies was largely state-driven, often through defense research programs or national laboratories. Today, however, the private sector, especially large technology corporations and agile startups, plays an increasingly decisive role. This shift complicates the governance of dual-use technologies. Tools and products such as generative AI platforms, cloud infrastructures, and advanced microchips are indispensable for both civilian innovation and military applications, and are mainly developed by the private sector. Unlike states, the private sector is motivated primarily by profit and market share, rather than by collective security or ethical considerations.

The entrance of these non-state (private sector) actors into the technological power game not only multiplies the number of stakeholders but also blurs the traditional boundaries of accountability. For instance, should a multinational company providing satellite imagery to a state in conflict be considered a neutral commercial entity, or a participant in warfare? Such questions highlight that current international legal frameworks have difficulties to deal with the realities of a privatized and globalized technology ecosystem.

1.3 The Strategic Race for Technological Dominance

Underlying these developments is a deeper geopolitical struggle for technological supremacy. States increasingly view leadership in areas such as artificial intelligence, quantum computing, and biotechnology as critical not only for economic competitiveness but also for national security. The U.S.–China rivalry is the clearest example, with both powers investing heavily in research, talent acquisition, and supply-chain control. Europe, meanwhile, seeks to carve out a position of ‘technological sovereignty’ to reduce dependence on foreign powers.

This battle for dominance creates the traditional security dilemma: each investment by one power drives counter-investments by others, accelerating innovation but also deepening mistrust. Moreover, because many of these technologies are dual-use, restrictions or controls aimed at preventing misuse may inadvertently stifle beneficial civilian innovation. Policymakers thus face a difficult balancing act between promoting technological progress and mitigating the risks of militarization and misuse.

1.4 The Limits of Traditional Governance Mechanisms

The governance of disruptive technologies has traditionally relied on multilateral conventions and treaties. The 1967 Outer Space Treaty, which established the principle of space as a domain for peaceful use, and nuclear arms control

treaties during the Cold War, are prominent examples. These agreements, however, depended on relatively clear technological boundaries, a limited set of state actors, and a dominant Western-led international order.

In today's multipolar environment, such conditions no longer exist. Emerging technologies evolve too quickly to be codified in lengthy negotiation processes. Moreover, the distribution of power has shifted: China, India, and other non-Western actors now play major roles in technology development, while Western influence in setting the rules of the game has declined. Achieving consensus on new global treaties in this fragmented landscape is increasingly difficult. As a result, governance structures frequently lag behind innovation, or fail to materialize due to insufficient consensus, resulting in critical gaps in which harmful exploitation can flourish.

One of the clearest contemporary examples of regulatory lag is in the domain of artificial intelligence (AI). If we take the EU AI Act, entered into force on 1 August 2024, we notice that it focusses on promoting trustworthy AI, safeguarding fundamental rights, ensuring human oversight, transparency, and aligning AI development with ethical norms (European Union, 2024). While China's regulations on AI, partly covered by 'Interim Measures for the Management of Generative Artificial Intelligence Services', which came into effect on 15 August 2023, focus on promoting innovation, safeguarding national security and social stability, and protecting citizens' rights aligned with 'core socialist values' (State Council of the People's Republic of China, 2023). In a geopolitically multipolar world, these diverging models influence how global governance of AI might evolve: convergent harmonization may be harder when underlying philosophies differ (e.g., the EU's rights-based vs. China's state-control-based).

1.5 Toward Intelligent and Forward-Looking Governance

Given these complexities, the central question becomes: how can the international community establish intelligent, forward-looking governance mechanisms that harness the benefits of emerging technologies while limiting their misuse? A purely reactive approach by responding to crises only after they occur is inadequate. Instead, foresight, scenario analysis, and adaptive governance structures are needed.

This requires embracing multiple perspectives on technology:

1. *Technology as an instrument*: applications of technology offer a source of new tools and tactics, with both opportunities and threats.
2. *Technology as vulnerability*: the ways technology is implemented also present targets that can be attacked, disrupted, or manipulated.
3. *Technology as ecosystem*: a network of actors whose interactions shape development of technology as well as technology governance.

By analyzing the trajectories of emerging technologies through these perspectives, we can better understand not only the risks and opportunities they create, but also the institutional and normative frameworks needed to navigate them responsibly.

2 Perspectives on Emerging Technology: Instrument, Vulnerability, and Ecosystem

To understand the disruptive impact of emerging technologies on security and governance, it is necessary to move beyond treating them as neutral artifacts. Technologies acquire meaning and significance only through the ways in which they are applied, contested, and embedded within social and political contexts. In this section, three perspectives provide analytical clarity: technologies as instruments for new tools and tactics, as vulnerabilities susceptible to exploitation, and as ecosystems shaped by diverse actors. Each lens offers unique insights into the opportunities and risks that define today's technological frontier.

2.1 Technology as an Instrument for Power and Progress

Emerging technologies first present themselves as powerful instruments: tools that expand the range of human action, from scientific discovery to military capability. This instrumental view highlights both the opportunities for societal progress and the risks of militarization.

Advances in artificial intelligence, biotechnology, and space technology are already transforming industries and creating new opportunities for welfare and wellbeing. For example, generative AI tools offer novel ways of creating knowledge, designing materials, and accelerating innovation cycles. In biotechnology, techniques such as CRISPR-based gene editing promise breakthroughs in agriculture and medicine. Space technology, particularly earth observation satellites, enables monitoring crop health, soil moisture, and weather patterns, thereby improving food security.

Yet these same innovations can be weaponized. Generative AI can be used to automate disinformation campaigns, fabricate realistic deepfakes, or generate malicious code. Biotechnology can be repurposed to engineer pathogens. Space technologies, such as low-Earth orbit satellite constellations, provide critical communication services but may also be leveraged for military surveillance or targeting. The dual-use nature of these tools makes it difficult to separate civilian progress from military risk.

The 'dark side' of technology is perhaps most vividly illustrated in hybrid warfare. Here, states and non-state actors exploit civilian infrastructure such as social media platforms, satellite imagery, and commercial drones to achieve military objectives without crossing the threshold of open conflict. Russia's use of cyberattacks and disinformation campaigns in Ukraine, and the reliance of

Ukrainian forces on commercial satellite communications for battlefield coordination, demonstrate how emerging technologies blur the boundary between civilian and military domains (Galeotti, 2022).

This perspective underscores that technology in itself may be neutral, yet its application is inherently value-laden. The ways in which it is employed inevitably involve strategic choices about progress, power, ethics, and responsibility.

2.2 Technology as the Fragile Backbone of Modern Society

A second perspective views technology not as an instrument, but as a point of vulnerability. The very systems that empower modern societies also expose them to exploitation.

Contemporary life is inseparable from digital infrastructure: financial systems, energy grids, transportation networks, and communication platforms are all mediated by digital technology. This dependence creates systemic vulnerabilities. Cyberattacks on critical digital infrastructure, ransomware operations against hospitals, or hacking of government databases demonstrate how attackers can undermine critical functions without firing a shot.

Satellites are particularly illustrative of this fragility. They provide navigation, timing, and communication services essential for both civilian and military operations. Yet they are highly exposed to jamming, spoofing, and hacking. A cyber intrusion into satellite networks can paralyze logistics chains, disrupt emergency services, or compromise military communications [European Union Agency for Cybersecurity (ENISA), 2025].

And there is the paradox of resilience. Efforts to strengthen resilience can inadvertently deepen vulnerabilities. For instance, greater reliance on AI-driven defense systems may create new avenues for adversarial manipulation through data poisoning, algorithmic bias, or adversarial inputs designed to deceive machine learning models (Brink et al., 2023). Similarly, diversification of supply chains to reduce dependence on a single provider can also expand the attack surface if security standards are inconsistent.

Another form of vulnerability lies in the dependency of big technology companies. These companies have considerable influence and resources that can shape the trajectory of emerging technologies. Their pursuit of market dominance and profitability may overshadow considerations of potential national and state risks and foster a need for regulation or mitigation of undesired effects and exploitations of technologies.

Finally, reliance on technology creates dependencies with regard to the availability of raw materials, hardware components (for instance, for quantum computing) and in terms of knowledge (qualified personnel, research institutes, and data to properly train AI systems). These dependencies create vulnerabilities that can be exploited (IEA, 2025).

Looking at technology as a vulnerability highlights the need for continuous foresighting, monitoring, redundancy, and robust safeguards. It also reveals that security cannot be achieved solely by acquiring more advanced tools; it must also address the vulnerability of the systems that sustain them.

2.3 Technology as an Ecosystem for Actors and Governance

A third perspective shifts attention away from artifacts and toward the constellation of actors and institutions that shape how technologies emerge, diffuse, and are governed. This ecosystem encompasses state agencies, private corporations, international organizations, knowledge institutes, and civil society. What increasingly defines this landscape, however, is the rapid proliferation and broad accessibility of technology, granting a growing diversity of actors the ability to develop, acquire, and deploy advanced capabilities.

Public-sector governance is a complex multi-level array of institutions. For instance, the EU's governance of technology involves a combination of legislative bodies, specialized agencies, research centers, and expert groups, working together to regulate, advise, and implement policies on the use of technologies like AI, cybersecurity, and space technologies. All these institutions provide oversight, set standards, and implement policy; however, the alignment of it requires consultation, coordination, and consensus, and hence time. And above all, this calls for profound knowledge and insight into the workings of these structures in an era marked by rapid change and high information density.

Private actors, through their central role in technology development, are accumulating significant power, effectively pushing *de facto* standards that shape the global technological order. For instance, smartphone companies have long promoted proprietary standards, such as their own charging connectors, until regulatory interventions, like the EU's mandate for a common USB-C port, forced greater interoperability. Private actor priorities often diverge from those of public regulators. Whereas governments emphasize security, private actors prioritize profitability, speed to market, and global competitiveness. This tension complicates governance: stringent regulation may push innovation offshore, while *laissez-faire* approaches may enable harmful misuse (Bollerman, 2024).

Next to the governmental institutions and the private sector, there is a third sector, consisting of knowledge organizations and civil society. Universities and research labs contribute to knowledge production and responsible innovation practices. Civil society organizations, ethics councils, and investigative media provide oversight, ensuring that technological development aligns with societal values (Schoemaker, 2024).

Technology governance must therefore be co-designed across this network. If rules are created in isolation, say by bureaucratic regulators without private-sector input, they risk being misaligned with the realities of how technology is built

and deployed. Conversely, if private actors dominate without accountability, governance risks being captured by profit motives. Effective oversight requires an inclusive approach that integrates expertise, values, and practical experience from across the ecosystem.

2.4 A Multifaceted Lens on Emerging Technology

Taken together, these three perspectives reveal the complexity of emerging technologies. They are at once instruments of innovation and destruction, sources of empowerment and vulnerability, and products developed by a growing ecosystem of diverse actors. Any attempt to regulate or strategically employ emerging technologies must therefore acknowledge their multifaceted character.

The next section will build on these perspectives by examining possible future worlds in which emerging technologies play divergent roles. By mapping how technology might evolve along axes of human control and access to technology, we can begin to anticipate the challenges and opportunities that lie ahead, and design governance models resilient enough to withstand them.

3 Future Worlds with Technology: Four Possible Scenarios

Technological change does not follow a single predetermined trajectory. Rather, it unfolds within a dynamic interplay of human choices, institutional arrangements, and unforeseen breakthroughs. To navigate the uncertainty of this landscape, scenario analysis provides a valuable tool. By projecting divergent ‘future worlds’, policymakers can stress-test their assumptions, explore alternative outcomes, and design governance strategies resilient under multiple conditions (Wright et al., 2013).

Following the intuitive logics approach to scenario development, we examined factors that influence the future security environment. Those factors expected to have the greatest impact and the highest degree of uncertainty were identified as key drivers. These so-called driving forces can be represented as spectra spanning two extremes, each plotted along an axis. In this analysis, our focus is on how humans interact with, implement, and shape technology.

This section builds on two axes that define the technological security environment of the future:

Technology drives humans versus humans drive technology: the degree to which capabilities of technology drive decisions and behaviors, as opposed to humans purposefully utilizing and controlling them.

In other words, differences on this axis primarily concern the amount of human control on the application of technology. Will humans give technology free rein, or will humans retain control and make conscious choices about where and when technology is used? On this axis, we see a distinction emerging in the amount of

freedom of choice individuals have. Can humans determine their own course? Or are they guided by external technological factors? Are choices data-driven, or does human debate remain paramount?

Technology as connector versus technology as divider: the extent to which technology is widely accessible and inclusive, versus concentrated in the hands of a few.

This axis ranges from a world in which everyone has access to all possible technologies to a world in which only a few have access. In other words: The world is technologically divided into the ‘haves’ and the ‘have-nots’. Along this axis, we see a distinction emerging in who can utilize the possibilities of technology. Can almost everyone do so, or is it reserved for only a few? And what effects does this have on society and international relations? (Fig. 1).

Combining these axes produces four possible Future Worlds, with a time horizon of about 15 years into the future in mind: Technological Empowerment for All, Technological Empowerment for the Lucky Few, Big Powers, Big Tech, and Data-Driven World. For each, we analyze how technology functions as an instrument, a vulnerability, and an ecosystem.

In the next sections, we will illustrate how the three perspectives on technology (Instrument, Vulnerability, Ecosystem) will manifest in each of the four worlds.

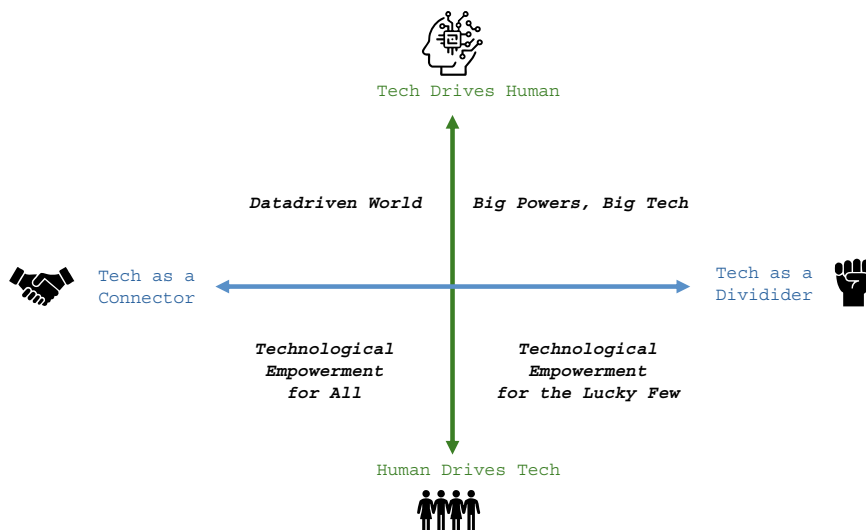


Fig.1 Four quadrants illustrate possible future scenarios. The upper two scenarios represent technology-centred future worlds, while the lower two scenarios represent future worlds in which there is significant human control over technology application. The scenarios on the left illustrate worlds in which technology is widely shared and accessible, while the scenarios on the right depict future worlds in which technology is concentrated in the hands of a few.

3.1 Technological Empowerment for All

In this world, humans harness the potential of technology while access is broadly shared across societies. Technologies are widely available, ethically guided, and harnessed to promote collective wellbeing. In this Future World, technological development will have exploded worldwide thanks to international cooperation. Technologies and the necessary raw materials are exchanged worldwide. This way, access to technology is in potential unlimited. People and societies can choose for themselves how to use that technology. The drawbacks of some technological developments have been recognized, and therefore, some societies are somewhat reluctant to adopt them. In those countries, technology is deployed in a limited and targeted manner, while other states give technology free rein.

Technology as Instrument. Tools such as artificial intelligence, biotechnology, and neurotechnology are used to enhance public health, sustainable development, and education. Generative AI democratizes creativity, while space technologies support disaster relief and environmental monitoring. Military technology and applications remain constrained by strong international norms, ensuring technologies are employed primarily for defensive or humanitarian purposes. However, dual-use civilian tools (AI, biotech, drones) can still be repurposed by non-state actors or rogue regimes. Maintaining ethical oversight in high-tech warfare is a continuous challenge.

Technology as Vulnerability. Despite broad accessibility, vulnerabilities persist. Cyberattacks and data breaches remain possible, but strong international collaboration and robust safeguards mitigate risks. Transparent standards for AI and cybersecurity, along with resilient infrastructure and coordinated global responses, reduce the likelihood of catastrophic failures with enduring high impact.

Technology as Ecosystem. Collaboration in technology development is voluntary. Because there is human oversight on the application of technology, reaching agreement on regulation is not always possible. Although technology is, in principle, easily shared, differences in norms and values make long-term partnerships difficult to establish. This makes it difficult to achieve the necessary behavioral changes required to adopt certain technologies. Consequently, technology is also unable to address major collective challenges such as climate change or aging populations. Although technology is widely available and collaboration exists, technological developments are slower than in other Future Worlds.

3.2 Technological Empowerment for the Lucky Few

Here, humans deliberately utilize the capabilities of technology, but access is uneven. This will create a strong divide between more technologically advanced and less advanced societies. This will lead to, among other things, growing economic inequality between these societies. Humans will remain in control of technology, even in highly technologically advanced societies. Individuals will

choose where and when they want to use technologies. This allows individualization to grow further in highly advanced societies, making it difficult to take collective action against problems like climate change. In countries with lower technological development, individuals have less freedom of choice with regard to technological development and implementation because they lack the resources to shape it.

Technology as Instrument. Emerging technologies are primarily deployed for critical functions such as defense, healthcare, and infrastructure management. Human oversight is built into every system, ensuring accountability. However, the uneven distribution of access means that smaller states, marginalized communities, and startups are excluded from innovation benefits, reinforcing global inequalities. Weaker actors may therefore turn to asymmetric warfare, cybercrime, or terrorism to compensate for technological inferiority.

Technology as Vulnerability. Restricted access reduces some risks but amplifies others. Because a few actors control critical systems, those systems become highly attractive targets for adversaries. A successful attack on a monopolized satellite constellation or AI infrastructure could have disproportionate consequences. Moreover, restrictive governance may create black markets and incentivize illicit development.

Technology as Ecosystem. The ecosystem is dominated by powerful states and a small number of corporations. International governance is fragmented, with powerful actors setting the rules. While innovation continues within controlled domains, it is stifled in others, leading to tensions between security and inclusivity. Civil society plays a limited role, raising concerns about democratic legitimacy. Limited cooperation fosters mistrust; rival blocs may therefore weaponize standards, data, and supply chains, sparking proxy conflicts and an arms race in restricted technologies.

3.3 Big Powers, Big Tech

In this world, technology increasingly drives human behavior, while access remains concentrated among a small elite, be it authoritarian states or dominant technology corporations. In other words, there is a significant difference between technological winners and losers. In the society of technological winners, technological development and application have been given free rein. These societies are driven by algorithms focused on their own economic development. This often comes at the expense of other societies that have been unable to keep up with this technological revolution.

Technology as Instrument. People are willing to relinquish much to technology and adapt their own behavior in exchange for significant economic growth, despite challenges such as an aging population and climate change. Within these societies, effective cooperation exists. People work for the shared (economic) prosperity of their own society. Out of fear of slipping down the economic ladder, people are willing to significantly adjust ethical and legal frameworks. Resources are

not shared but rather retained within society for its own development, potentially at the expense of others. AI and autonomous systems are exploited for surveillance, predictive policing, and precision warfare. Biotechnology is used for genetic enhancement of select populations, exacerbating inequalities. For the majority, technology becomes a tool of control rather than empowerment. The gap between rich and poor regions and communities widens, fueling tensions and instability. Information warfare, limited access to information, and algorithmic propaganda become tools of dominance.

Technology as Vulnerability. Overreliance on autonomous systems creates systemic fragility. If algorithms fail or are manipulated, societies may face cascading disruptions. Because decision-making is opaque, accountability is elusive, making it difficult to correct errors or contain crises. The few actors in control are both powerful and vulnerable, as concentrated assets present lucrative targets for sabotage or espionage. Big technology companies have considerable influence and resources that can shape the trajectory of emerging technologies. This also creates single-source dependencies and hence vulnerabilities.

Technology as Ecosystem. The ecosystem is hierarchical and exclusionary. Powerful states and big corporations monopolize innovation, while citizens have little say in governance. International cooperation erodes as dominant actors compete for control, leading to technological arms races and geopolitical instability between states and tech conglomerates. Regional blocs establish divergent standards and regulations, with little interoperability. Governance structures fail to keep pace, and technology amplifies power asymmetries rather than mitigating them. Global instability grows as actors weaponize AI, data, and biotech. Citizens and smaller nations lack resilience, creating fertile ground for rebellion, extremism, and digital insurgency.

3.4 Data-Driven World

In this world, the world has become a data-driven utopia. Virtually all global decisions are made by AI. The algorithm determines how society is organized and how labor, capital, and energy are deployed. The objective function that guides this algorithm optimizes for the ‘common good’. Because these and other technologies are shared globally, this also offers less developed societies opportunities for rapid, sustainable development. Rather than fragmenting humanity, ubiquitous connectivity binds individuals, communities, and institutions into a vast, adaptive network.

Technology as Instrument. Technology functions as an omnipresent enabler of efficiency and coordination. AI manages energy grids, healthcare diagnostics, and urban logistics with near-perfect precision. Decision-making is evidence-based and individuals benefit from personalized services powered by the seamless integration of public and private data streams. However, reliance on technology also reshapes human agency. The focus is on maximizing wellbeing on a collective scale rather than individual preference.

Technology as Vulnerability. Dependence on interconnected data systems introduces new forms of vulnerability. System malfunction or deliberate data manipulation can ripple across societies with global consequences. While strong cybersecurity protocols are in place, the sheer volume of interdependence means that absolute security is impossible. Ethical vulnerabilities emerge: algorithmic bias, surveillance creep, and data ownership disputes remain persistent dilemmas. The balance between collective benefit and individual rights becomes a governance challenge.

Technology as Ecosystem. The technological ecosystem is a dense web of public institutions, corporations, and citizen-driven data cooperatives. Global governance frameworks coordinate interoperability and ethical standards. Innovation thrives through open data and cross-border collaboration as if within a digital commons. In this environment, nations and corporations act less as competitors and more as nodes within a global network. Defense cooperation relies on open data-sharing among allies, requiring unprecedented trust and ethical alignment. Adversaries may exploit openness through infiltration, manipulation, or ‘data warfare’ aimed at eroding confidence and cohesion. Society learns in real time, iterating its values and goals through continuous feedback loops between humans and machines.

3.5 Comparative Challenges, Risk, and Governance Needs

Each future world represents a distinct balance between human control and technological dominance, and between inclusivity and division. Challenges and risks are already briefly stipulated in the previous sections. Based on a comparative analysis of these challenges and inherited risks, for each future world, we will identify urgent governance needs. We will do so from the standpoint of our current Western perspective, being democratic societies founded on liberal values, the rule of law, and the protection of human rights.

Comparative analysis of defense and security challenges and risks			
Future world	Challenges	Risks	Governance needs
Technological empowerment for all	Maintaining global trust and ethical alignment in the use of open technologies	Differences in values, norms, and ethical frameworks for technology use Divergent interpretations of responsible AI	Focus on strengthening norm-building, multilateral treaties, and transparency mechanisms Coordinate international standards for AI ethics, arms control, and cybersecurity

(continued)

(continued)

Comparative analysis of defense and security challenges and risks			
Future world	Challenges	Risks	Governance needs
Technological empowerment for the lucky few	Managing power asymmetry and preventing militarization of inequality, as marginalized actors might resort to destabilizing tactics	Power and capability gaps lead to asymmetric tactics for cyberattacks, terrorism, or sabotage by weaker actors seeking to offset technological inferiority	Address deterrence in an unequal landscape Prevent the spread of sensitive technologies to unauthorized and malicious actors Enhance rapid response to hybrid and asymmetric threats
Big powers, big tech	Preventing authoritarian misuse of autonomous defense systems and systematic fragility due to concentrated control	Concentration of power makes systems fragile and politically volatile Malfunction or manipulation can trigger cascading failures and societal unrest	Invest in resilience (systems and societies) Ensure at least ethical human oversight of autonomous systems Prepare for internal unrest and information warfare
Data-driven world	Balancing openness and trust in a highly interdependent data-driven global security environment	Interdependence creates both cooperation and vulnerabilities Small disruptions (data breaches, misinformation, AI bias) can have global consequences	Strengthen collective resilience and secure shared data infrastructures Develop adaptive governance for cross-border data and AI security

These four future worlds are not predictions but possible trajectories that illuminate the stakes of emerging technology. They reveal that no future world is free of risk, but each embodies different balances of opportunity, vulnerability, and governance challenges. By exploring these pathways, policymakers can better anticipate the consequences of technological choices made today and design governance structures resilient enough to guide the world toward more desirable futures.

By analyzing the key governance needs to maintain Western-based norms and values as much as possible across each of the four future worlds, we can identify a set of four overarching pillars that underpin the development of a strong strategy for managing new and emerging technologies. These pillars are:

- continuous monitoring and assessing of opportunities and threats of new and emerging technologies,

- legal and ethical governance of technology,
- resilience engineering against attack vectors of technology, and
- adaptive alliances to maintain stability in diverse technological futures.

4 Strategies for Governing the Defense Tech Frontier

This section examines each of the four pillars that constitute the core of an effective strategy for governing the use of new and emerging technologies in the defense and security domain.

4.1 Foresighting Future Worlds and Technology

At the foundation of any governance model lies the ability to anticipate future developments. Emerging technologies evolve too quickly for static regulation to remain effective. Instead, governance must be rooted in foresight: systematically exploring how technologies might be used, abused, and transformed in the years ahead.

In the previous section, we developed a set of future worlds based on how technology might evolve and, in doing so, lead to divergent global scenarios. Naturally, there are many other ways to construct future worlds. For instance, one could adopt a geopolitical perspective, distinguishing worlds along the axes of unipolar versus multipolar power, and cooperation versus fragmentation. However, for the purpose of developing a strategy focused on the governance of technology, we recommend taking technology itself as the primary point of departure and analytical lens.

Foresighting future worlds is not an exercise that requires continuous or overly frequent revision. Most nations, institutions, and think tanks currently update their foresight analyses on a cyclical basis, typically every 2–4 years. However, there is growing recognition of the value of institutionalizing foresight as a regular and structured activity, both at the EU level and within national governance frameworks. Establishing an annual foresight cycle would not only embed this practice into strategic culture but also strengthen anticipatory governance capacities across sectors (EC, 2025).

Regular foresight exercises enable policymakers to identify weak signals of change, track emerging technological disruptions, and assess how geopolitical, economic, social, and technological dynamics interact over time. In an era of accelerating complexity, marked by rapid advances in artificial intelligence, quantum technologies, and shifting global power relations, such institutionalized foresight becomes essential for enhancing strategic preparedness and policy coherence. Moreover, by integrating foresight outcomes into policy design and evaluation, governments can move from a reactive posture toward a more proactive, resilience-oriented approach to governance.

Technology foresights are a crucial part of foresighting. Effective technology governance begins with situational awareness. In an era where emerging technologies evolve at unprecedented speed, defense and security institutions require systematic mechanisms for technology watch and assessment. This entails continuous horizon scanning, early-warning systems, and structured evaluation frameworks capable of detecting both opportunities and risks across the innovation landscape (Douglas & Robinson, 2025).

Technology watch in the defense domain involves more than tracking patents or R&D outputs. It must integrate intelligence analysis, dual-use monitoring, and cross-sectoral insight from academia and industry. The European Defence Agency's (EDA) Technology Watch and Foresight activities, NATO's Emerging and Disruptive Technology (EDT) program, and national think tanks and research organizations provide foundational models. These initiatives demonstrate that technological awareness functions best when linked to decision-making cycles: transforming information into strategic foresight.

Assessment complements watch by providing structured evaluation of a technology's maturity, risk, and defense relevance. Traditional Technology Readiness Levels (TRLs) are no longer sufficient; one might define Risk Readiness Levels (RRLs) and Ethical Readiness Levels (ERLs) to measure vulnerability to misuse, ethical acceptability, and operational reliability (Lena Fiedler, 2025). Integrating these metrics into procurement and capability planning ensures that adoption decisions are not made solely on performance or cost, but also on resilience and normative alignment.

Foresight and assessment data must not remain siloed within national agencies. Shared assessments between EU–NATO-nations would create a common strategic picture, aligning investments and ensuring interoperability. Joint data platforms, interoperable threat taxonomies, and AI-supported analytics could enhance collective situational awareness. This is particularly vital for dual-use technologies, where developments in the civilian sector can rapidly spill over into defense applications. Therefore, engaging private innovators through secure 'innovation sandboxes' allows governments to access early-stage insights without compromising classified information (Attrey, 2020). Technology watch thus becomes an anticipatory governance function, enabling proactive adaptation instead of reactive crisis management.

Ultimately, technology watch and assessment are the first operational layer of strategic foresight. They transform the abstract understanding of 'future worlds' into actionable intelligence that informs doctrine, investment, and regulation, keeping defense governance aligned with the pace of innovation.

4.2 Legal and Ethical Governance

Emerging technologies challenge not only existing legal frameworks but also the very relevance and resilience of ethical norms in contemporary warfare. While traditional governance frameworks emphasize compliance and constraint, recent conflicts, most visibly in Ukraine, demonstrate how the imperatives of speed, effectiveness, and force protection often override long-standing ethical standards. The growing use of autonomous weapon systems, mines, and cluster munitions signals a shift toward pragmatism over principle. This raises a deeper question: what do we, collectively, still consider to be ‘ethical’ in war, and who decides?

In this changing environment, legal and ethical governance can no longer remain reactive or merely procedural. Instead, Ethical, Legal, and Societal Aspects (ELSA) must be proactively embedded in the early stages of technological development, shaping innovation rather than chasing its consequences. Governance thus becomes less about preserving an idealized moral order and more about continuously negotiating acceptable boundaries between military necessity, human control, and humanitarian restraint (Steen, 2023).

From a European perspective, existing frameworks such as the EU AI Act, GDPR, and the Charter of Fundamental Rights still provide a normative anchor. Yet defense applications often fall under national-security exemptions that risk hollowing out those standards. To maintain legitimacy, such exemptions must remain narrow, transparent, and rooted in the principles of necessity, proportionality, and meaningful human oversight.

Ultimately, safeguarding ethical integrity in defense innovation requires acknowledging that technology is an instrument for multiple purposes, civil and military, opportunity and threat-based, and that the moral terrain of war is itself shifting. Only by integrating ethics into design, doctrine, and deployment can democracies ensure that technological superiority serves legitimacy rather than erodes it.

4.3 Resilience Engineering

Resilience engineering addresses the structural capacity of defense systems and societies to withstand technological shocks, attacks, or failures. In a world of hybrid threats, resilience must extend beyond hardware redundancy to include organizational agility, cyber-resilience, and societal endurance.

Defense resilience begins with secure-by-design architecture. Systems should be modular, interoperable, and capable of graceful degradation, continuing to operate in reduced capacity under attack. Red-teaming, penetration testing, and stress-scenarios are indispensable tools. These practices, already institutionalized in NATO’s Cyber Range facilities, must expand to encompass AI, quantum communication, and bio-cyber interfaces (MOD.UK, 2025).

A second dimension is supply-chain resilience. As shown during recent semiconductor shortages, technological dependencies can become strategic vulnerabilities. Defense actors must diversify suppliers, enforce transparency through Software and Manufacturing Bills of Materials (SBOM/MBOM), and develop trusted industrial partnerships (CISA, 2025). Resilience also demands capabilities for rapid recovery, such as digital twins of critical systems, backup communication networks, and crisis-simulation protocols.

Finally, resilience has a human and institutional component. Personnel must be trained to recognize system failures, override automation when needed, and coordinate under degraded conditions. Organizational cultures that encourage learning from incidents, rather than concealing them, foster adaptive resilience.

Resilience engineering therefore bridges technical robustness with institutional adaptability. It ensures that defense ecosystems can absorb disruption, adapt to change, and emerge stronger, which is a prerequisite for maintaining deterrence credibility in a world of continuous technological contestation.

4.4 Adaptive Alliances

The pace of technological change demands alliances that are adaptive, networked, and innovation-driven. Traditional defense coalitions, built around fixed treaties and slow consensus, must evolve into flexible ecosystems that can co-develop, co-govern, and co-respond to technological challenges (Arne Keller, 2021).

Adaptive alliances operate on multiple levels, for example:

- *Strategic level*: At the macro level, coordination mechanisms such as the EU–NATO Joint Declaration (2023) and the NATO Innovation Fund provide structures for aligning research priorities and capability development. By synchronizing foresight exercises and risk assessments, allies can avoid duplication and close governance gaps.
- *Operational level*: Shared technical standards for AI safety, data exchange, and cybersecurity protocols enable interoperability and hence joint operations. Establishing common certification schemes for trustworthy AI in defense would enhance interoperability while maintaining ethical consistency.
- *Tactical level*: Collaborative innovation programs such as DIANA (Defence Innovation Accelerator for the North Atlantic) and the European Defence Fund can serve as transnational sandboxes linking startups, academia, and militaries. These networks encourage experimentation while embedding governance by design.

Adaptivity also implies inclusivity. Smaller nations, often rich in niche expertise but limited in scale, can contribute as agile innovators, testbeds, or norm entrepreneurs. Regional clusters, e.g., Benelux, Nordic, Visegrád, could pool foresight, share technology and risk assessment data, and co-invest in resilience

infrastructure. By turning alliances into learning systems, partners collectively enhance their governance maturity and readiness for emerging threats.

4.5 Building a Resilient Governance Architecture

Bringing the four pillars from theory to practice requires a coherent governance architecture capable of linking foresight, legal-ethical guidance, resilience, and alliance coordination. Such architecture transforms abstract strategy into operational governance.

How do the four pillars work in practice? Foresight and technology assessment provide vision and deliver continuous intelligence; legal and ethical governance sets boundaries; resilience engineering ensures robustness; and adaptive alliances extend capacity beyond national borders. Together, they form a self-reinforcing loop where insights from one pillar feed improvements in the others.

What is the role of institutions? Institutions are the backbone of resilience. At the EU level, there are agencies and bodies (such as the European External Action Service (EEAS) and the Joint Research Centre (JRC)) that should coordinate foresight and risk assessment. NATO complements this through foresight analysis by NATO Allied Command Transformation. National defense ministries must mirror this architecture domestically, linking intelligence, innovation, and ethics units in a whole-of-government approach. A European Defence Tech Council could serve as a hub for alignment and rapid decision-making, for instance, with respect to technological governance and resilience and coordination of foresights and technology risk-assessment.

How do small countries such as the Netherlands and Austria fit in? Smaller countries can play pivotal roles as norm setters, innovation laboratories, and connectors. Their size allows agility in policy experimentation and coalition building. By hosting foresight hubs, test ranges, or ethics observatories, they can punch above their weight and contribute to shaping EU-wide standards. Moreover, smaller states can act as brokers between major powers, promoting trust and ensuring that technological governance remains anchored in democratic values.

Ultimately, a resilient governance architecture depends on trust, transparency, and interoperability. When institutions collaborate across scales (national, European, transatlantic), they create the collective intelligence necessary to navigate the accelerating defense-technology frontier.

5 Final Reflections

5.1 Navigating Governance in a Multipolar Technological Order

In the previous section, we highlighted what actors can do to govern technology in a safe and secure way. However, a defining challenge for defense and security governance in the twenty-first century is the emergence of a multipolar technological world. Whereas the post-Cold War era allowed Europe and its transatlantic allies to shape much of the global normative and institutional order, today's landscape is characterized by competing models of technological governance. China, India, and other regional powers, along with powerful private conglomerates, increasingly set standards, control supply chains, and define the ethical contours of innovation. This diffusion of authority erodes the West's former rule-setting dominance and complicates the pursuit of coherent global governance.

The erosion of Western norm-setting capacity does not imply the end of influence, but rather the need for a strategic recalibration. Europe and NATO must transition from a model of unilateral rule making to one of coalition building and norm entrepreneurship. Three approaches stand out.

First, build coalitions of the willing around shared governance values. Instead of relying on universal treaties that are increasingly difficult to achieve, like-minded partners can form modular governance coalitions like open frameworks that set high standards for ethics, transparency, and accountability in defense technologies. These standards can be embedded in trade, procurement, and interoperability agreements, creating a 'gravitational pull' that encourages broader alignment over time. However, keep in mind that governance becomes less about preserving an idealized moral order and more about continuously negotiating acceptable boundaries between military necessity, human control, and humanitarian restraint.

Secondly, engage emerging powers through inclusive dialogue and technical cooperation. Rather than framing governance as a binary contest between liberal and authoritarian models, Europe and NATO should invest in pragmatic engagement with emerging technological actors. Joint testbeds, capacity-building programs, and transparent certification systems can create shared interests in safety and reliability, even among competitors. By focusing on interoperability and risk reduction, rather than ideological convergence, Western actors can sustain influence while accommodating diversity.

And last but not least, strengthen strategic autonomy through internal coherence. Global credibility begins at home. The EU must demonstrate that democratic governance of technology is both effective and competitive. This means aligning internal standards, ensuring rapid implementation of regulations like the AI Act, and reinforcing resilience across supply chains. A coherent and innovative European ecosystem enhances bargaining power: others adopt standards not out of deference, but because they work.

Ultimately, the transition to a multipolar order requires adaptive governance diplomacy. Instead of exporting fixed rules, Europe and its allies must curate a living architecture of governance, be flexible, networked, and inclusive, and thereby capable of bridging divergent philosophies while upholding core values of transparency, accountability, and human dignity. In doing so, they can help ensure that the emerging technological order, though multipolar, remains multilateral in spirit and anchored in the principles of responsible innovation and collective security.

5.2 Rethinking Governance Philosophy, a Different Mindset

A first insight is that governance must move from reactive regulation toward proactive stewardship. Traditional arms control frameworks assumed relatively stable technologies, clear state actors, and lengthy negotiation windows. None of these conditions holds today.

- *From regulation to stewardship.* Instead of attempting to ‘freeze’ technologies through treaties, policymakers should see themselves as stewards of evolving socio-technical systems. This means focusing less on prohibitions and more on shaping trajectories, thereby steering technologies toward beneficial applications while constraining harmful ones.
- *Embracing pluralism.* Technology ecosystems involve states, corporations, research institutions, and civil society. Governance philosophy must recognize this pluralism rather than pretending that states alone can dictate outcomes. Multi-stakeholder models, where industry, NGOs, and end-users have a seat at the table, are not a luxury but a necessity.
- *Focusing on ecosystems, not artifacts.* Many policies still regulate individual technologies (e.g., ‘autonomous weapons’ or ‘quantum encryption’). Yet risks emerge less from artifacts and more from their embedding in systems and networks. Governance should target ecosystems such as supply chains, infrastructure, and actor constellations, rather than single objects.

If adopted, these directions can help Europe and its allies steer the defense tech frontier away from dystopian futures of fragmentation or oligarchy, and closer toward a world where technology serves as a connector, an enabler of human agency, and a guarantor of security grounded in shared values.

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Dual-Use Technology and the Dynamics of the Will to Power

Max Gottschlich

Abstract

The dual-use problem has reached a new dimension in the contemporary era: the very efficiency with which modern technology is deployed simultaneously generates uncontrollable risks and novel vectors of attack. Consequently, the dual-use problem has become a central concern for modern civilisation. This article examines the fundamental questions arising from this phenomenon.

The dual-use problem is inextricably linked to the very existence of humankind, and efforts to comprehend its preconditions and ramifications can be traced back to the cradle of European culture. To introduce our inquiry, we first turn to Greek mythology, which already provides the conceptual foundation necessary to understand the problem and to approach a meaningful response. Building on this foundation, we then examine the nature of technology—particularly modern technology—in order to clarify why the dual-use problem poses even greater challenges today than it did during the Cold War, when the term was coined.

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1 Prometheus and Pandora

Greek mythology already articulates this problem in a profound manner through the myth of fire and its association with the figures of Prometheus¹ and Pandora. It is the Titan Prometheus, one of the patron deities of craftsmanship, who is entrusted with the creation and civilisation of mankind (Eckhart, 1957, cols. 696 ff.). The name “Prometheus” means “forethought” (Eckhart, 1957, col. 689). This signifies the capacity to plan ahead, to anticipate the consequences of action, and to exercise prudence. The gift of Prometheus that makes the creation and civilisation of humanity possible is fire. Fire stands metaphorically for the light of self-conscious understanding (*Verstand*) that distinguishes human beings from animals. As “the as yet undetermined animal” (Nietzsche, 2014, p. 61, Aphorism 62), human beings are capable of recognising their dependencies and reflecting upon them. For the same reason, they are also capable of exploiting these dependencies in such a way that they are made to serve given purposes. The Promethean fire enables human beings to objectify nature and to harness its powers for their own ends. In this sense, Prometheus appears as the herald of the titanic powers of nature.

Accordingly, the Promethean fire within the human soul serves as a metaphor for an unbridled drive for self-assertion, a force already at work in the ruthless struggle for existence within the animal kingdom. Fire, in this metaphorical sense, stands for an anarchic, unrestrained will to power² that seeks to appropriate and unleash the forces of nature.

Fire, now understood in the literal sense as a chemical process, functions in this myth as a paradigm of nature appropriated for human use—and, as such, is the archetype of a dual-use good. It provides benefits, enables civilisational progress and cultural achievement—craftsmanship, nourishment, protection from wild animals and enemies—yet it is at the same time life-threatening and a means of destruction and warfare. The divine character attributed to this gift expresses an awareness that, although human beings can make use of such powers, they can do so only through the mediation of nature and can never fully control their dynamics.

Whoever employs such powers as instruments of self-assertion must come to realise that they do not thereby gain mastery over nature or over their own fate. Rather, mankind must henceforth live in the presence of Pandora—that is, within the totality of consequences generated by the use of technology. This insight is central to our topic. Pandora is a fascinating artificial being, endowed by the gods with a multitude of gifts.³ Her name admits of two interpretations. Pandora is “she who is gifted by all”, but also “she who gives all” (Graf, 2000, col. 236). These meanings are related as possibility and actuality. The former emphasises potential,

¹ The *loci classici* for the Prometheus myth include: Hesiod, Works and Days 51; Aeschylus, Prometheus Bound 7; Plato, Protagoras 321; Cicero, Tusculanae Disputationes 2, 23.

² Nietzsche’s famous phrase means the demand for affirmation and enhancement of life.

³ Hephaestus grants her physical form, Athena skill and craftsmanship, Aphrodite beauty and seductiveness, and Hermes cunning and deceit.

in the sense of the alluring opportunities for action promised by technological artefacts. Indeed, what civilisational and cultural achievements are not conditioned by technological means? Yet Pandora's presence also implies that all consequences of our actions are necessarily implicated and will, sooner or later, be "given". This is captured in the second interpretation of her name. Pandora brings with her a sealed clay jar which, when opened, releases all evils and sufferings into the world. The sealed jar symbolises consequences that are not visible at the outset. Only hope remains within it.

Pandora, as a symbol of technology, thus embodies a contradiction, described by Hesiod as a "beautiful evil" (Hesiod, 2006, 585, p. 51). This ambivalence is already present and evident in the nature of fire itself: fire exists only by consuming its own basis of existence. Artificial fire—understood metaphorically as technology⁴—cannot escape this ambivalence either. The presence of Pandora is not simply a punishment; rather, it is a gift of Zeus, intended to prevent human beings from succumbing to the illusion that they can unleash titanic natural forces in order to augment their power without incurring consequences. The presence of unintended and uncontrollable consequences is intrinsic to technical-practical conduct as such. Epimetheus ("he who thinks too late"), who blindly accepts Pandora as his wife, thus functions as a cautionary figure.

The point of Pandora's presence is that it forces us to fully understand the double-edged nature of technical-practical conduct. Only a lucid and reflective awareness of Pandora's nature, and of the human condition inseparably bound up with her, makes an adequate response to this problem possible. In mythological terms, the power of Zeus must bind Prometheus. Humanity requires gifts that transcend mere technical rationality. These divine gifts are the capacities of moderation and self-restraint, and a commitment to lawful and just order—in short, the capacity for *rational self-mastery*. Only these capacities enable us to curb the titanic, unbridled "will to power"—that is, to master the Promethean fire within the human soul.

When Prometheus openly defies Zeus, this act is sometimes interpreted as a rebellion against unjust tyranny, casting Prometheus as a freedom fighter *avant la lettre*. This reading rightly draws attention to the fact that technological power can serve to overthrow oppressive regimes; however, a more compelling interpretation suggests that the will to power driving technology tends to absolutize itself and, in doing so, engenders disorder and violence.

Dual-use issues are therefore not a marginal problem of governance. Rather, they demand a renewed examination of the *conditio humana* under conditions of technological empowerment and the accompanying threat of self-alienation and self-loss. Today, humanity is confronted with Pandora's jar in the form of dual-use technologies. What distinguishes our situation from that of antiquity is the unprecedented magnitude of power that modern technology makes available—a reach that amounts to an encroachment upon totality itself.

⁴ Cf. Plato's Allegory of the Cave in *Politeia* VII, 514a–520d.

It no longer appears possible to anticipate the consequences of the use of modern technology. G. Anders already emphasised this in the context of the nuclear threat, coining the notion of the *Promethean Gap* (*prometheisches Gefälle*) to denote the increasingly perilous discrepancy between humanity's capacity to produce technological artefacts and its capacity to foresee and comprehend their consequences (Anders, 2025). The urgent question remains—arguably more pressing today—whether we, like a modern Epimetheus, can afford to recognise the ambivalent character of technology only once such recognition comes too late. When it becomes impossible to apprehend the threatening consequences of technological deployment, it becomes necessary to call into question the very preconditions upon which that deployment rests.

The gravity of this issue is further underscored by the fact that the dominant myth of our age represents, in essence, a profound inversion of the stories of Prometheus and Pandora. In contemporary technocratic eschatologies of transhumanism, technology is increasingly construed as a means of salvation: a promise of redemption through complete emancipation from nature. It is envisaged as capable of resolving all problems of human existence, including those engendered by technology itself.

2 Problem and Theses

Dual-use goods are characterised by the fact that they cannot be unambiguously assigned to a single function: they can be used for both civilian and military purposes.⁵ From a state-centric perspective, this ambiguity poses a problem in specific contexts. Legal regulations are intended to establish determinate conditions governing their use. The desire to control the proliferation of dual-use goods reflects the fact that it is in the national interest (understood as an obligation to the *bonum commune*), to restrict, under specific conditions, the freedom of economic trade, which is in principle guaranteed. This restriction applies to technical products whose potential modes of use are relevant to security policy.

The requirement implicit in the concept of dual use can be explained more precisely with reference to the logical principle of the excluded middle. This principle states that something must be either *a* or *non-a* (*tertium non datur*). From the standpoint of this principle, dual-use goods appear to constitute precisely this excluded third option: a “something” which, according to the principle, can function only as a logical placeholder required for the establishment of clear determinations. This “something” exists only as a *substratum*, as the underlying basis for the determination, but not as a determinate case in its own right. Certain technical products, which—like this mere “something”—elude clear determination as either civilian

⁵ Cf. the definitions provided in the Wassenaar arrangement (www.wassenaar.org) and the EU Regulation 2021/821 (“Dual-Use Regulation”), Art. 2.

or military, must therefore be rendered determinate, through legal and political measures, by being forced into the form of an either-or.

The problem of dual use is primarily a legal and political problem, situated in the sphere of actual, lived, and institutionalised freedom, as realised in the domains of the economy and the political community.⁶ What is distinctive about the dual-use problem is that it paradigmatically articulates a tension between rationality and potential irrationality within the dynamics of political self-assertion in modern, highly technologically mediated civilisation, since it is in and through technology that this political will to self-assertion acquires its forceful objective existence. The dual-use problem thus highlights the structural ambiguity of technology, an ambiguity rooted in the inseparability of its emancipatory and destructive potentials.

The following *theses* may serve to clarify the core of the problem:

1. It is precisely the efficient deployment of modern technology that simultaneously generates uncontrollable dangers and novel *vectors of attack*.
2. The crisis in which this technological ambiguity manifests itself is not an opaque fate. Rather, it is the consequence of a specific level of consciousness that has produced these conditions. It is an uninhibited will to power that seeks to assert itself in a relation of *asymmetrical independence*⁷ vis-à-vis others. What unfolds at the political level between states follows the same logic of the constitution of selfhood. Every political will to power expresses a *will to selfhood* (*Wille zum Selbstsein*); yet selfhood is mediated solely through the relation to an other self. As Hegel formulates it: “Self-consciousness exists in and for itself when, and by the fact that, it so exists for another; that is, it exists only in being acknowledged” (Hegel, 1977, p. 111).

This process of mutual self-assertion, in its very immediacy, takes a stark and polemical form in what Hegel describes as the “struggle for recognition”, a confrontation marked by the willingness to risk life itself. What asserts itself here is a drive to dominate the other. In this struggle, one party—the master—does not recoil from the prospect of a life-and-death confrontation, whereas the other—the slave—does, and therefore submits. The result is a relation of asymmetrical independence, in which the slave is not recognised as another self-conscious being but is reduced to an instrument of service. Yet it is precisely through the assertion of dominion that the master comes to experience his own dependence upon the other, and thus his own form of enslavement. For further development depends upon the slave, who, through labour in the service of the master, acquires the capacities without which the master himself cannot subsist. In this sense, the relation between master and slave is ultimately inverted.

⁶ In Hegel’s terms, we speak about “Sittlichkeit”, not just about morality in Kantian terms.

⁷ The term was coined by Werner Schmitt.

This reversal—whereby the master is compelled to experience his own dependence through the very act of asserting dominion—constitutes the *basic structure of the objective repercussions that arise when humanity seeks to dominate nature and others by means of technology*. As will be shown, these repercussions directly affect what lies at the core of political power itself: the erosion of both external sovereignty⁸ and internal sovereignty, understood as the monopoly on the legitimate use of force.⁹ *These repercussions thus expose the untruth inherent in claims to asymmetrical independence*. In contemporary efforts to address the dual-use problem, we are therefore confronted above all with our own blindness with respect to the will to power that underlies these dynamics.

3. A response to this problem can therefore only take the form of a revolution in consciousness, namely the overcoming of the will to asymmetrical independence. What must be recognised is that the overarching *bonum commune* lies in a rational restraint of the will to power, grounded in relations of mutual recognition. The will of a political community to assert itself—most clearly expressed in processes of armament—must turn against itself, that is, must impose limits upon itself. As we shall see, the dual-use problem today has grown to a point where it is clear that without such mutual restraint and recognition, the relative sovereignty of states and their monopoly on the legitimate use of force become structurally unstable.

A philosophical analysis of these interrelations cannot yield immediate prescriptions for action. The legal and political resolution of the problem necessarily belongs to the domain of political judgement, which must concretise the idea of a supranational common good in the light of specific historical constellations and their real possibilities and constraints. Nevertheless, philosophical reflection can help to lay bare the limitations of short-sighted arguments, including the following:

1. The establishment of effective control over dual-use goods is not merely a procedural or administrative problem of governance. Rather, the problem reaches down to the very precondition of all governance: political will, or more precisely, the degree of willingness to exercise self-restraint in the spirit of mutual recognition.
2. The problem is also obscured when it is reduced to purely utilitarian considerations—for example, to the claim that technology invariably entails both opportunities and risks for the well-being of particular groups, which must simply be weighed against one another. A common strategy of relativizing the fundamental problem is to point to empirical evidence of technological and civilisational innovations that enhance welfare and that originally emerged

⁸ For the concept of state sovereignty, also with regard to the shift towards supranationalism, see Gottschlich (2021).

⁹ The monopoly on the use of force denotes the state's right and obligation to maintain internal sovereignty.

from military research and development. However, the category of usefulness is insufficient to reconstruct the normative implications at stake here. One may be tempted to treat the issue as a matter of ad hoc, case-by-case risk–benefit calculations. Yet such an approach overlooks the fundamental dimensions of the problem, which cannot be reduced to a mere calculus of consequences.

3. Finally, the issue cannot be adequately addressed through a moralising approach that focuses exclusively on the intentions or actions of individual subjects. Examples include economic-ethical claims such as: trade in dual-use goods is morally reprehensible because weapons presuppose the total instrumentalization of the other; or: the failure to establish an effective control regime is the result of moral failure on the part of the actors involved. The moral standpoint, paradigmatically represented by Kant, excludes what is essential to our inquiry: the intrinsic connection between action and the institutionalisation of freedom in society and the state, and between responsibility and the supra-individual powers that assert themselves in the sphere of geopolitics.

We begin our examination by turning to the nature of technology itself.

3 The Ambivalence of Technology: The Sublation of Dependency

At the outset, it is crucial to understand why the dual-use problem is *inherent in the very nature of technology* itself. Technology derives from the Greek word *technē*, which encompasses not only technique but also art, skill, and cunning (*mētis*). The element of cunning consists—in both the most primitive and the most sophisticated forms of technology—in the capacity to turn the forces of nature against themselves for the purposes of freedom and self-assertion. Herein lies the affirmative meaning of technology, extending from the flintstone to artificial intelligence: in and through technology, human freedom renders nature a means of its own manifestation, preservation, and enhancement.

Technology is an expression of the human will to emancipate itself from immediate natural constraint, by compelling natural forces, through tools and machines, to serve human purposes. Intelligence is thereby objectified in artefacts. Creating technical means with the help of nature requires knowledge of oneself, knowledge of purposes, and knowledge of the appropriate means—understood in terms of causal relations and interactions—through which these purposes are realised. Artefacts are thus objectified purposiveness.¹⁰ The meaning of technology therefore lies in the sublation (*Aufhebung*) of humanity's immediate dependence on nature.

According to Hegel, the German word *Aufhebung* (sublation) has a threefold meaning (Hegel, 1998, pp. 106–108):

¹⁰ Cf. the third part of Hegel's *Science of Logic*.

1. *Negation* (cancellation): certain forms of immediate human dependence on nature are overcome and thereby cease to exist.
2. *Preservation*: does not disappear as such, but only in its immediacy; it is preserved in a mediated form, namely as dependence on nature through technology. The power of technology thus remains conditioned by the natural forces it harnesses.
3. *Elevation*: technology simultaneously transforms dependence by raising it to a higher level. Insofar as humanity uses technology to emancipate itself from nature, it enters into a new form of *mediated dependence*—no longer directly on nature, but on technology itself.

This third dimension requires closer attention. The new dependence on technology, brought about by the very will to emancipate oneself from nature, has far-reaching consequences. It transforms not only the human relation to nature, but also—and this is decisive in our context—the human relation to oneself and to others. Technology has *objective repercussions* on self-understanding and practical conduct. These effects arise from the systematic application of modern technology as such, independently of the subjective intentions of individual actors. They are consequences that—as in the myth of Pandora—are not released out of individual malice, but out of a drive for empowerment that turns a blind eye to the inevitability of technology's objective repercussions.

Because technology constitutes a sublated form of dependency, these objective repercussions are *structurally ambivalent*.

The positive side of this ambivalence manifests itself in the following features:

(a) Relief and emancipation:

Technology provides relief from labour, time expenditure, and effort in the satisfaction of needs and the pursuit of interests. It expands the horizon of achievable ends. Technology is therefore a medium of civilisation and cultivation (*Bildung*). The more intelligence is objectified in machines, the greater their transformative power. Those who command chemical processes as in gunpowder, for instance, possess a qualitatively different form of power than those limited to simple mechanical processes.

(b) Expansion and generalisation of formerly exclusive capacities:

This power becomes available in principle to all who have access to the technology, without distinction. The user need not possess the specialised skills of a craftsman, scientist, or programmer. Nor is it necessary to belong to a particular social class; one merely has to discipline oneself to operate the machine correctly. Technology thus makes condensed, objectified intelligence universally accessible. The bundling of these capacities enables users to become increasingly independent of others, for example, in coping with everyday problems and tasks.

The positive effects primarily concern civilisational progress—consider the gains in freedom brought about by modern communication, transportation, and

medicine. Cultural achievements in science and politics are likewise technically mediated. The invention of printing, for example, made education broadly accessible, thereby contributing to the emergence of political participation by responsible citizens. Technical mediation thus plays a constitutive role in transformations of political organisation.

Yet, technological advancement invariably comes at a price, revealing the other side of this ambivalence:

(a) Erosion of skills and the emergence of new dependencies:

If the use of a machine proves useful, it rapidly becomes a habit, a “second nature”.¹¹ Action becomes routine rather than reflective. This is an essential aspect of relief—but it also entails a danger. To the extent that individuals outsource their capacities to machines, their own abilities atrophy. The erosion of skills produces a new form of dependency—no longer on nature, but on technology itself. This applies not only to manual skills, but also to cognitive and intellectual capacities.

Before the advent of mobile phones, people were required to memorise numerous telephone numbers; today, many struggle even to recall their own. Consider the vast narrative traditions that ancient cultures were capable of retaining prior to the invention of writing. Who today could memorise the *Iliad* or the *Odyssey*? Memorisation is a form of appropriation and internalisation. Today, it appears superfluous, since information can be indefinitely externalised into data storage. It increasingly seems unnecessary even to read books when specific answers can be retrieved via AI systems. With the loss of memory, we also lose the capacity and will to comprehend complex and extended contexts independently.¹²

Human beings, initially the masters of machines, must discipline themselves as mere “users” and increasingly risk becoming servants of their own means. The means emancipates itself from its original purpose, as Goethe’s *The Sorcerer’s Apprentice* vividly illustrates. Goethe’s poem contains the sobering insight that reason at first understands itself as instrumental reason, seeking to spare itself the hardship of engaging with the world by outsourcing action to technical means. It is only when the sorcerer’s apprentice encounters the unintended consequences of his actions, which come to threaten his very existence, that he calls upon the “master”. Who or what is this master who averts catastrophe at the last moment by restraining the activity of the uncontrollable, self-propelling machine beings? It is the attainment of rational self-mastery itself that yields the mature insight that reason, when interpreted solely as technical-practical rationality, becomes an irrational force of chaos.¹³

¹¹ On the concept of habit, cf. Hegel (1971, § 409–410, pp. 139–147).

¹² Cf. Carr (2020).

¹³ These connections between technology, the will to power, and modern society were systematically elaborated in twentieth-century philosophy. Martin Heidegger’s analysis of the essence of modern technology as “Gestell” (enframing) emphasises its roots in objectifying modes of thought,

(b) Expansion of power and the growth of abuse potential:

With the expansion and bundling of formerly exclusive capacities, the potential for abuse grows accordingly. The very success of technological progress—producing machines that store ever more intelligence while becoming ever easier to use—also renders them instruments for undermining the political community, particularly with respect to the state’s monopoly on the legitimate use of force and its relative sovereignty as a geopolitical actor. Civilian infrastructures and devices can always be repurposed or exploited by hostile actors. Crucially, this danger does not originate in malicious intent alone, but emerges immanently from the structure of technological mediation itself.

This ambivalence manifests itself with particular clarity in modern technology. The conquest of new possibilities is driven by a self-reinforcing dynamic, experienced since the turn of the twentieth century as increasingly beyond human control. The technologically unleashed will to power becomes progressively decoupled from moral and legal constraints. Georg Simmel—here representative of many modern thinkers—speaks of a “tragedy of culture”.¹⁴ This refers to the fact that the objective cultural forms of technology and the institutional structures that sustain them tend to emancipate themselves from the subjective claims to meaning and freedom from which they originally arose. This tragedy is fateful rather than intentional: it is not willed by the actors, but realised through their actions, analogous to Adam Smith’s “invisible hand”.

which disclose all beings as a “Bestand”, a “standing reserve” of resources. In the “Dialectic of Enlightenment”, Theodor W. Adorno and Max Horkheimer argue that it is precisely the realisation of technical rationality—understood as progress in the depth and scope of domination—that discloses a fundamental irrationality in humanity’s relation to the world. Bruno Liebrucks locates the deeper root of this problem in the primacy of the technical-practical conduct, an unrestrained will to power expressing in a specific logical presupposition, namely the claim to an unrestricted authority of formal logic for knowledge. All exact science and modern technology are, in this sense, applied formal logic. The fatal flaw of this orientation lies in the reduction of the human being to a merely acting subject, conceived as an isolated ego-centre whose sovereignty is detached from relations to the other. This reduction constitutes the conceptual root of the striving for asymmetrical independence. By contrast, Liebrucks emphasises that everything that distinguishes humans from animals exists only by virtue of mutual recognition. Since recognition itself is always mediated by language, language must be understood as the institution of all institutions. Human beings therefore do not exist merely as acting beings, but first and foremost as linguistic beings, whose freedom is constituted through relations of meaning, recognition, and understanding.

¹⁴ Cf. Simmel (1997).

4 Weapons Technology and International Law

Against this background, we now turn to the historical context in which the concept of “dual use” first emerged: the problem of weapons of mass destruction, particularly nuclear weapons, and the attempt of international law to respond to them.

A weapon is a tool. Every tool can, in principle, serve multiple purposes. At first glance, the tool appears morally neutral, beyond good and evil. Even a weapon is—subject to important qualifications—initially neither good nor evil as an object; from a moral standpoint, this distinction applies primarily to actions and intentions. It is the legal and moral purposes that legitimise the use of weapons by police and military forces under regulated conditions, while unlawful or immoral purposes—such as those pursued by criminals—delegitimise their use. In Kantian terms, good and evil are not properties of tools, but of the maxims governing their use.

However, technology is not merely a neutral means. The functionality of every tool is based on specific intentions and principles of action, which organise means in such a way that they can reliably serve determinate ends. A tool therefore objectifies a specific rationality of action—more precisely, a combination of theoretical knowledge, derived from the exact sciences and enabling systematic control of nature, and a practical guiding moment: a will that seeks to assert itself through the tool. In this respect, every tool indirectly refers to what is morally relevant—namely, to intentions, maxims, and thereby to the will itself.

Every tool is thus the objective expression of a certain will to power, understood not psychologically, but structurally: as a tendency towards increased availability, control, and assertiveness with respect to nature and other human beings. Where the will to self-assertion underlying technology becomes absolute—that is, where it excludes the question of legal and moral legitimacy—the appearance of the moral neutrality of the means collapses.

This collapse became historically evident with the advent of weapons of mass destruction, most notably nuclear weapons. The problem of dual use emerged explicitly in response to the threat of total annihilation during the Cold War (Rosner, 2016, p. 1). The technical term itself was initially coined in relation to components necessary for nuclear weapons production. The horror provoked by the real possibility of global nuclear annihilation—especially against the backdrop of biological and chemical weapons employed in the World Wars—led to the recognition that technology cannot be regarded as morally neutral. The means of modern technology can no longer be accepted unconditionally as instruments of freedom, but must be subjected to normative scrutiny. With the emergence of genetic engineering, this awareness has intensified further.

This normative scrutiny was subsequently institutionalised at the level of international law. In modern *ius in bello*, international humanitarian law, three

fundamental principles governing the choice of means and methods of warfare—previously present in customary international law—have been codified¹⁵:

1. The principle of distinction, which requires not only a distinction between combatants and civilians, but—crucially for our topic—between military objectives and civilian objects (Art. 48 Additional Protocol I). This principle is specified further by the protection of the civilian population (Art. 51 API) and the definition of military objectives (Art. 52 API).
2. The prohibition of means and methods of warfare that cause unnecessary suffering [Art. 35(2) API] or widespread, long-term, or severe environmental damage [Art. 35(3) API]. This is supplemented by the obligation to conduct legal reviews of new weapons, means, and methods of warfare (Art. 36 API). This provision—arguably applicable not only to individual weapons but to modern technology as such—is of particular relevance to the dual-use problem.
3. The principle of proportionality [Art. 51(5)(b) API], which presupposes a clear conceptual distinction between military objectives and civilian objects [Art. 52(2) API], and requires that, in case of doubt, an object be presumed to be civilian [Art. 52(3) API].

These principles require normatively guided distinctions, grounded in the following rationale: because human beings must be recognised as legal persons and not treated as mere objects, a distinction must be made between civilians and combatants; to secure this distinction, one must further distinguish between civilian objects and military objectives; and, under the conditions of modern warfare, this has led to a further distinction between two categories of weapons:

- (a) Weapons whose use can, in principle, comply with *ius in bello*. This includes pre-modern and modern conventional weapons, subject to the important qualification that even conventional weapons become prohibited when they cause excessive suffering (e.g. blinding laser weapons, certain types of mines), or when their methods of use violate humanitarian law (e.g. area bombing). Total war, which negates all normative limits, represents the extreme expression of this logic. Such wars are no longer instrumental “cabinet wars”, but existential wars in which civilians and prisoners of war are deliberately targeted.¹⁶
- (b) Weapons whose very functionality negates the recognition of the other as a bearer of freedom and reason, not merely through unlawful use, but intrinsically. This applies above all to biological, chemical, and nuclear weapons. The decisive criterion is whether the weapon allows for the containment of war—an unconditional demand of reason—or whether its mere existence objectifies a will to total annihilation. The very existence of weapons of mass destruction

¹⁵ Additional Protocol to the Geneva Conventions of 12 August 1949 Relating to the Protection of Victims of International Armed Conflicts (Protocol I), June 8, 1977, 1125 U.N.T.S. 3.—Cf. Sassòli et al. (2011).

¹⁶ Cf. Sieferle (n.d.); Münkler (2003).

presupposes that the enemy is no longer conceived as a *justus hostis* but has been transformed into an “absolute enemy” to be exterminated like vermin (Schmitt, 2007). This is why weapons of mass destruction imply a complete dehumanisation of the adversary. At this point, the appearance of legal and moral neutrality is revealed as an illusion.

We may therefore conclude that, historically, *the imperative to control dual-use goods acquires its normative force precisely where the functionality of a weapon negates the distinction between military and civilian use*. Where a weapon poses a totalising threat to the existence of a world of freedom, the components necessary for its production—despite their civilian relevance—cannot be treated as morally or legally neutral commodities.¹⁷

The fact that nuclear weapons—unlike biological and chemical weapons—are not categorically prohibited under international law, but are instead regarded as “generally contrary” to the principles of international humanitarian law, appears as a normative inconsistency, even when distinctions between tactical and strategic nuclear weapons are invoked. This inconsistency can be explained by the fact that law, in its codification—though not in its basis of validity—follows power. Legal codification is therefore always conditioned by geopolitical constellations.

In this case, the geopolitical motive is obvious: mutual assured destruction (MAD) has played a decisive role in maintaining a fragile equilibrium in the international order since the Cold War. Precisely in order not to jeopardise this stabilising function, legal control of nuclear proliferation becomes necessary.¹⁸ Such control is inevitably shaped by the interests of geopolitically dominant actors, since it lies in their own rational self-interest. This remains, to this day, one of the strongest motivations for the effective application of the dual-use principle.

5 Modern Technology and the Transformation of the Civilian World into a Strategic Resource

In order to understand the specific relevance and urgency of the dual-use problem today, we must first grasp the distinctive essence of modern technology. This will be developed in three steps.

1. Genesis and institutional prerequisites of modern technology
 - (a) Modern technology is best understood as applied natural science, which itself rests upon applied mathematics and, ultimately, formal logic. Within this paradigm, science no longer approaches nature as a self which manifests and discloses itself in its processes, but instead as a system of

¹⁷ Legality of the Threat or Use of Nuclear Weapons (Advisory Opinion), 1996 I.C.J. 226 (July 8), §§ 95–97.

¹⁸ Treaty on the Non-Proliferation of Nuclear Weapons, July 1, 1968, 21 U.S.T. 483, 729 U.N.T.S. 161.

necessarily interconnected phenomena, capable of being represented, predicted, and controlled through mathematical and logical formalisation. Questions of internal purposiveness are methodically excluded from this mode of inquiry; analytical attention is redirected towards quantifiable materials, forces, and laws of motion. Isaac Newton's characterisation of nature as "matter in motion" paradigmatically expresses this epistemic and methodological shift.

Natural science translates actual nature into a transparent world of objects, within which all processes appear as consistently and unambiguously determinable under universal laws (Kant). This determinability enables prediction and, consequently, the systematic mastery of nature. The insights gained from natural science permit the methodical exploitation of natural entities and processes. Engineering realises this exploitation in practice: its task is to render natural forces usable for specific purposes with maximal efficiency and minimal expenditure of resources.

- (b) The systematic realisation of engineering in research and development requires capital, which projects the interests of bourgeois society, industry, and the economy—that is, the sphere in which the material conditions of survival and the good life are produced.
- (c) The synthesis of science, technology, and capital depends on the securing power of public law, that is, on the state and the institutions of the political community.

Unlike craftsmanship, modern technology therefore exists only as the result of an institutionally sustained collective endeavour.

2. Dual use and the rationality of modern technological products

To what extent is the dual-use problem inherent in the rationality of modern technology itself? This becomes clear when contrasting modern technology with pre-modern craftsmanship. Craftsmanship operates within perceptible natural processes, working with visible materials and directly experienced causal relations. Modern technology, by contrast, operates through the supersensible domain of scientifically articulated laws and forces, thereby enabling the targeted, controllable, and scalable manipulation of physical, chemical, and biological processes. Only on this basis does the principle of bionics become methodically realisable. Bionics appropriates the purposiveness of natural processes and structures by translating them into functional models, for example, by abstracting the hydrodynamic efficiency of fish into principles for minimising drag in technical vehicles. Today, this principle has been extended to unprecedented levels, reaching even the attempt to partially replicate the most complex organ of living beings—the brain—in order to automate cognitive operations in the form of artificial intelligence. This development reveals that the rationality of modern technology depends on the systematic utilisation of reason, the power of mediation already immanent in nature itself.

Scientifically disclosed natural forces are thus mobilised in an increasingly systematic manner to produce ever more powerful means. The governing finality of this process is technical-practical in character: modern technology exhibits an inherent tendency to approach the world as a manipulable totality. Advances in scientific explanation correlate directly with advances in technical feasibility, such that progress in knowledge and progress in power mutually reinforce one another. The products of modern technology therefore embody the will to realise a scientific worldview whose implicit aim is to render nature, in its entirety, available as a means. In this sense, technological artefacts are not neutral tools, but material expressions of a historically specific will to power.

This reach towards totality becomes visible in the fact that modern technological products tend to *integrate into comprehensive systems capable of functioning autonomously*, that is, without direct human intervention. Such systems exhibit three defining characteristics: modularity, scalability, and interoperability.

*Modularity*¹⁹ means that technical products are designed as relatively autonomous functional units or elements, which can be combined via standardised interfaces. This separability has two dimensions:

- (a) *Interchangeability of means*: the fulfilment of a function does not depend on a specific component, since it can be replaced by another with equivalent functionality.
- (b) *Interchangeability of purposes*: the module is not bound to a single end, but can be deployed across multiple teleological constellations.

Modularity, therefore, demonstrates that the augmentation of functional capacity progressively becomes an objective in its own right. Rationality is reoriented from the pursuit of specific ends towards the optimisation of functional versatility. Means become progressively decoupled from determinate purposes, while their abstract utility increases. From the outset, modules are designed to be recombining across heterogeneous functions.

The ambiguity that becomes problematic in dual use is therefore intrinsic to modern technology itself. Such modules correspond to the “something” excluded by the principle of the excluded middle: they acquire determinacy only through the functional relations into which they are inserted.²⁰

Why is modularisation fundamental to technology as an integrating system? Because it conditions the general plasticity of functionality, which becomes even clearer in scalability and interoperability.

¹⁹ Cf. the standardised definition in (International Organization for Standardization [ISO], 2021).

²⁰ The modularity of technology objectifies the instrumental conception of the logic of formal logic. Since G. Frege, formal logic has conceived of logical form as a function-argument structure: the arguments only have their determinacy in the function in whose “empty spaces” they are placed. We encounter this way of thinking in every modularised product of modern technology.

- (a) *Scalability* means that functionality and performance are preserved across systems of varying size and complexity.
- (b) *Interoperability* designates the capacity to integrate entire subsystems seamlessly into higher-order systems, such that their functionality is preserved while being subordinated to an overarching operational logic. It is a precondition of scalability, since only interoperable components can be expanded, replicated, and recombined across different contexts without loss of functionality. Achieving interoperability presupposes far-reaching processes of standardisation and norm-setting, encompassing procedures, protocols, formats, materials, interfaces, logistical chains, and organisational routines. The pursuit of efficiency therefore generates an incentive towards transnational coordination and cooperation, which has become institutionalised in contemporary global standard-setting bodies, most prominently the International Organization for Standardization (ISO). At this point, we see how technology prompts the institutionalisation of bureaucracy (which is significant in terms of the problem of the genesis of technocracy).

This development is of particular significance for understanding the genesis of technocracy. Bureaucracy here does not arise primarily from political will or ideological design, but as an immanent consequence of technological mediation itself. The governance of standards, procedures, and interoperability gradually shifts decision-making authority from explicitly political deliberation to technically specialised bodies, thereby transforming questions of collective self-determination into matters of technical administration. In this way, the rationality of modern technology silently prepares the institutional conditions for technocratic forms of rule.

Unlike craftsmanship, modern technology does not generate isolated tools, but a system of systems, corresponding to Kant's idea of reason as a heuristic striving for systematic unity. This tendency is captured today by the notion of the "Internet of Things" (IoT).²¹ Here, totality takes the form of a world of interconnected objects. This totality becomes autonomous once control is delegated to self-applying algorithms.

In such a machine world, interoperability is indispensable not only within military alliances, but also between military and civilian infrastructures (energy, data networks, logistics). Civilian technologies such as smartphones exemplify this integration of everyday life into strategic systems.

This totalisation is enabled by digitalisation, understood as the translation of reality into modelable systems. The demand for real-time processing and modelling necessitates artificial intelligence. AI exists to manage and operationalise the massive data volumes generated by digitalisation. Because AI promises efficiency, convenience, and temporal relief—the lure of Pandora—it is increasingly

²¹ Cf. International Organization for Standardization & International Electrotechnical Commission [ISO/IEC] (2019).

deployed in domains such as education, where its benefits are at best limited. This process culminates in the automation of decision-making, including the modelling of human behaviour itself.

These features explain why the dual-use problem has expanded from weapons-related components into the civilian domain. The tipping point lies precisely where modern technology demonstrates its greatest strengths: in the development of automated integrated systems.

The example of automotive development illustrates this dynamic. Earlier vehicles allowed for localised repair and a degree of user resilience. With the proliferation of electronic components, dependency increased in direct proportion, while practical skills eroded. Electronic subsystems can no longer be repaired independently, and software integration now prevents meaningful intervention without proprietary tools. Mechanics increasingly become operators constrained by opaque systems, while vehicles themselves become networked, data-collecting platforms. Civilian vehicles thus become strategic resources and potential attack vectors.

Paradoxically, the most advanced technology also creates the most effective attack vectors, since the least resilience has been developed in relation to it. Generalised to the expanding world of “smart” connected devices, and viewed against the backdrop of scalability and interoperability, it becomes clear why dual-use risks can escalate uncontrollably and ultimately threaten the existence of the state itself.

6 The Danger of Technocracy

States and societies have a legitimate interest in technological advancement. No state can afford to lag behind in research and development, since technological progress contributes to the bonum commune, for example, in healthcare and communication. The argument that dual-use products enhance societal resilience by expanding civilian infrastructure is *prima facie* compelling.

Yet the opposite is equally true: the very technologies used to secure relative sovereignty and the monopoly on the use of force simultaneously undermine both.

1. When state institutions depend on infrastructures beyond their effective control, sovereignty progressively erodes. Technical, economic, and geopolitical dependencies—such as cloud services, global supply chains, and AI models—limit the state’s capacity for self-determination. The ability to shape its relations with state and non-state actors according to its own judgement—which constitutes sovereignty as such—is substantially weakened because it must constantly be relativised in relation to the particular interests of other actors.
2. Integrated technological systems enable comparatively low-cost challenges to internal sovereignty. As the distinction between military and civilian becomes more and more a mere semblance, this semblance can itself be exploited. A “perfect weapon” would be one that is omnipresent in terms of military-relevant domains, but at the same time eludes the adversary’s grasp, because they cannot

withdraw it without undermining the conditions of their own existence. This is the case when civilian infrastructure itself becomes the weapon.

3. Systemically accumulated losses of skills and resilience affect both individuals and institutions. Artificial intelligence intensifies this dynamic, creating previously unprecedented forms of political manipulability. AI-generated outputs are increasingly—and often non-transparently—integrated into sources that are initially perceived as especially trustworthy. This demonstrates the individual and political explosiveness of AI: it has created a new form of public-opinion manipulability, novel in its combination of automation, reach, and individualisation. Guided by interests embedded within algorithms, AI can function as a powerful instrument of propaganda, customised down to individual user profiles.

Here, too, we observe how the fascination with AI as a new “Pandora” generates a dangerous blindness to its structural risks. Trust in technology must not be taken lightly, as the figure of Epimetheus already illustrates. The user, seemingly empowered to omnipotence through AI, increasingly becomes the executive organ of an algorithm, while the supposed servant ascends to power. How rapidly would a relapse into a “law of the jungle” occur, simply through the loss of physical and cognitive capacities combined with an increasingly alienated relation to nature? Basic skills required to survive using the resources provided by nature—without the sophisticated tools of civilisation—have long since been forgotten.

4. The decoupling of means from ends generates fragmentation and a *diffusion of responsibility*. The division of labour fragments processes, and this is mirrored in an ongoing fragmentation of society. How can individuals retain an overview of the whole when modules are produced without any knowledge of the functional totality? What responsibility can be attributed to those who sell components that may later become weapons or medical devices? Everyone produces only parts; no one appears responsible for the whole. Twentieth-century ethics of responsibility (H. Jonas, G. Anders) already identified this problem. This development affects the citizens’ sense of responsibility towards the state as a whole, thereby reinforcing the modern tendency towards moral atomism (*sittlicher Atomismus*).
5. The tendency towards moral atomism is an objective repercussion of modern technology. Individuals interpret themselves in the image of technology, namely as functional elements within a system. This tendency is reinforced by the “social” dimension of modern technology—networking and communication—which leads individuals to lose themselves in algorithmically generated bubbles, producing merely the illusion of recognition. Instead of fostering trust-based social cohesion, this dynamic results in isolation and fragmentation. Politically, this is significant, as atomised individuals are particularly receptive to abstract forms of identification. AI-driven, individualised manipulation further exacerbates the condition of post-democracy, understood as the replacement of genuine political participation by its media staging (Crouch, 2020).

6. Moral atomism supports an *instrumental conception of law and the state*, which tends to slide into technocracy and authoritarianism. This understanding is further exacerbated by technology's tendency, as an administrative medium of growing power, to develop a dynamic of its own. Modern technology is not a neutral medium of social coordination and administration; it actively shapes the understanding of what is governed through it. Its use has objective repercussions on the self-understanding of the political community. Law and the state are no longer recognised as manifestations of freedom, but as *administrative systems structurally analogous to data-management architectures*. What began as the colonisation of nature by science and technology is becoming the self-colonisation of mankind. Today, we must ask to what extent the administration of the public sphere, the state, and its authorities can tolerate the integration of artificial intelligence. If the state is to be experienced as a genuine embodiment of freedom, it must not confront citizens merely as a machine (Gottschlich, 2025).
7. If the loss of control and sovereignty produced by modern technology is addressed under the same premises that generated it, the attempt to secure the monopoly on the legitimate use of force takes the form of a *disinhibition of state power*, expressed in the *expanded technological monitoring and control of citizens*. This dynamic helps to explain the specifically modern tendencies towards authoritarianism. Attempts to restore control without revising the underlying will-to-power rationality lead to technological overreach in surveillance and behavioural control, thereby accelerating authoritarian tendencies.

Social practices are then—in accordance with technology's inherent striving for total control—reduced to basic elements, translated into data units, and modelled algorithmically. The technocratic state may initially appear as the “lean state” desired in the name of efficiency, limiting itself to internal and external security functions. Yet the will to power's drive towards universal certainty manifests politically as a tendency towards the “nanny state” and, ultimately, the surveillance or police state.²² The will to dominate nature through technology and the will to dominate human beings spring from the same source. As in the exact sciences, conditions must be established to ensure that all relevant entities of the system can be regarded as unambiguously and exhaustively determinable elements. If one remains blind to these connections—as Epimetheus was—even the modern liberal state risks sliding into authoritarianism.

Under these conditions, how can the problem of effective dual-use control be addressed? Only the state is capable of supervising the division of labour and systemic interdependencies through instruments such as arms-export controls,

²² The inner logic of this process can be studied in the early Fichte's philosophy of right (Fichte, 2000).

supply-chain responsibility regimes, and compliance structures.²³ Yet the question remains whether this is a realistic option under present conditions.

7 Rational Self-Restraint

The international community has demonstrated that it is capable of establishing binding technical standards, for example, through the International Organization for Standardization (ISO), in the interest of sharing the benefits of technology. Under what conditions, however, would comparable cooperation also be possible when it comes to addressing the dual-use problem?

If the catalogue of relevant goods ultimately has to include virtually everything indispensable for civilian technology, it becomes evident that regulatory measures alone are insufficient. What is required instead is a revolution of consciousness, born of the insight—already articulated in myth—that the titanic will to unleash power must be restrained by moderation and self-control, in the name of a just and sustainable order.

- (a) Moderation—the ancient Greeks spoke of the need to find and uphold the proper measure (*métro*n)—stems from reflective awareness of the problems arising from the objective repercussions of modern technology and from the structural entanglement of technology and society. This awareness draws upon an enlightened understanding of the nature of technology, its grounding in modern natural science, and its logical presuppositions, extending down to the role of formal logic in shaping our technical-practical conduct. Such enlightenment must itself be recognised as a matter of public interest. On the basis of this awareness alone does it become possible to mobilise the will to exercise restraint, that is, the capacity to find and uphold a rational measure in the use of technology in full consciousness of its foreseeable consequences.
- (b) Mobilising the will in such a way, in turn, presupposes the recognition of the normative claims stemming from an overarching concept of the good, i.e. an interest in the existence of a just and free order corresponding to the concept of man as a rational being.

²³ Contemporary digital games, such as *Minecraft*, normalise a technical–practical mode of world-disclosure in which reality is graphically rendered as a simplified assemblage of exploitable resources. The world is reduced to manipulable elements assessed according to functional utility, and players are trained in their efficient deployment. Accordingly, all phenomena are subjected to what Kant terms the determining power of judgement (*bestimmende Urteilskraft*), being classified as either conducive or obstructive to the player’s aims. Within such a framework, nothing is allowed to appear as existing for its own sake. This invites the question of whether prolonged immersion in technologically mediated environments reshapes our experiential relation to nature and to the human world itself.

The difficult task running through the entire history of humanity is therefore to curb the unbridled will to power through a will to rational self-rule—that is, to restrain the use of technology not only at the level of individual actors, but also among states in their mutual relations.

The central obstacle here is the same as that encountered in the recognition of international law itself, namely the effort to overcome the state of nature between states and nations. What must be curbed is the blind striving for power and its destructive consequences. At the political level, this striving manifests itself directly and openly—much like in the animal kingdom—as a struggle for existence governed by the law of the strongest.

Each actor claims asymmetrical independence in order to assert its own interests—each time justified by reference to a particular *bonum commune*—over and against others. What is systematically overlooked is that even the most powerful actors possess sovereignty only in a mediated form, namely through relations of mutual dependence and recognition.

Where geopolitical self-assertion is pursued without restraint, a merciless competition for spheres of influence emerges. Law, by contrast, presupposes the recognition of the other as a presence of freedom and reason²⁴; since the modern era, international law has codified this recognition at the level of states and peoples. The validity of *ius in bello* depends on the geopolitical adversary still being recognised as a rational being, that is, as the bearer of something unconditional—freedom and dignity—rather than being instrumentalised as a mere object or exterminated like vermin.

Apparently, the dynamics of reciprocal action within geopolitical competition—often referred to as the “Thucydides trap”—stand in the way of both the effective control of dual-use goods and an lived recognition of the principles of international law. At this point, the deepest problem associated with dual use becomes visible. The very notion of dual use points to the necessity of developing an awareness of an normatively binding overarching, globally relevant notion of a *bonum commune*, capable of providing rational grounds for limiting the pursuit of particular national interests. Once it is understood that modern technology is not morally neutral, there is no need—like Epimetheus—to wait passively for catastrophic consequences to materialise. Instead, regulation must adopt a preventive approach. What arguments can be advanced in support of such preventive restraint?

At first, the binding force of appropriate regulation can be explained in terms of a utility calculus. Should we risk a permissive approach to dual-use goods, or not? Do the benefits outweigh the risks? A first argument may therefore be formulated as follows:

²⁴ “Personality essentially involves the capacity for rights and constitutes the concept and the basis (itself abstract) of the system of abstract and therefore formal right. Hence the imperative of right is: ‘Be a person and respect others as persons.’” (Hegel, 2008, p. 55).

1. IF maintaining a stable international order is recognised as useful (by analogy with nuclear non-proliferation), and thus as a general interest and a universal *bonum commune*, THEN it follows that it lies in the rational self-interest of all state actors to restrain the will to power in order to enable effective regulation of dual-use goods.

The connection between the “if” and the “then” rests on the premise that dual-use goods necessarily jeopardise this order. However, the normative force of this argument is limited. First, it depends on the contingency of a given order and of its recognition. Second, the term “necessarily” contained in this premise indicates that this premise depends on the conclusion of another, a deeper argument, which may be stated as follows:

2. IF—under the conditions of modern technology and its systemic function within society and the state—*all state actors are equally affected* by the inseparability of civilian technology and attack vectors, thereby eroding external sovereignty and the monopoly on the use of force, THEN restraining the will to power in order to enable effective dual-use goods regulation *necessarily* lies in the rational self-interest of all state actors.

This argument makes a stronger claim, since it appeals to universal validity. Its force can be justified in consequentialist or rule-utilitarian terms: what kind of world would result if such restraint were not generally observed? Can we rationally will the foreseeable consequences of such a world?

This argument should not be underestimated. It is capable of establishing a minimal formal consensus, even within the framework of power-political realism and diverse cultural contexts, thereby fostering an awareness of a shared world and collective responsibility.

Yet we cannot halt at this level of reflection. In Kantian terms, both arguments take the form of a hypothetical—more precisely, pragmatic—Imperative, which conceives the control of dual-use goods as a necessary means to the presumed end of collective well-being. Their weakness lies in the fact that they do not justify this end as source of obligation. Both arguments therefore raise the deeper question: why are we obliged to care for the *bonum commune* in the first place. What exactly is this good, that is at stake here?

At this point, at least two distinct lines of thought in the European intellectual tradition can be identified:

1. The tradition of liberalism and contractualism locates the good primarily in the individual realisation of subjective purposes, while the existence of law and the state is treated as merely derivative. The state appears as a neutral administrative system, tasked with securing individual freedom and regulating societal interaction.
2. If we take seriously that law and the state are *necessary* conditions of the possibility of freedom, both are inseparable from the end itself. Hence, the state and law must be conceived as the presence of that end. Following Kant and Hegel, the determination of the end, or of the good, would therefore be specified as

follows: the good lies in the correspondence of the individual's moral and legal freedom and the lived, institutionalised freedom of society and the state. The measure of an individual's legal and moral freedom must be relatable to society and its institutions; conversely, the freedom of society and its institutions must be measured by the effective freedom of its citizens. If the individual can recognise his own legal and moral consciousness of freedom as realised within the state, then the state—which Hegel designates as the “*sittliche Staat*”—is not merely an instrument, but rather a manifestation of freedom itself.

The empirical question of whether such a state has ever existed is of secondary relevance in the present context. Our concern is rather to understand why the existence of the *bonum commune*, as embodied in the state, carries a normative claim in the first place. Only this conception of the state as a manifestation of freedom can explain why its existence possesses an intrinsic goodness, which has implicitly served as a premise in the preceding arguments. Technology, as a means, must therefore not hinder or undermine the realisation of this intrinsic purpose of freedom.

The necessity of rational self-restraint in the use of technology therefore rests on a reason-based—not merely utilitarian—requirement of mutual recognition among states and nations as spaces of freedom.

The “realist” objection—that such an approach is naïve because law always follows power, and power invariably seeks asymmetrical independence—must be answered in two ways:

1. This is not an empirical claim, but a statement concerning the normative conditions of possibility for any international order capable of addressing the escalatory dynamics inherent in dual use.
2. It follows from the foregoing analysis that the problem cannot be solved by law alone. One may point to Article 36 of Additional Protocol I, whose wording anticipates certain aspects of the dual-use dilemma. Yet a contemporary response cannot be confined to specific categories of weapons, but must extend to all technical systems and civilian infrastructures whose functionality undermines the distinction between civilian and military spheres.

If the ambivalence of dual use is rooted not merely in the application of individual means, but in the systemic character of modern technology itself, then the problem exceeds the scope of purely legal regulation. The solution can only lie in a moralisation (*Versittlichung*) of political will, grounded in the mutual recognition of law and the state as the presence of freedom. Without such recognition, law itself would be interpreted solely in terms of technical-instrumental rationality, thereby becoming yet another manifestation of the “tragedy of culture”.

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Challenges and Risks of Emerging and Converging Technologies for Societal Security

Niklas Gudowsky-Blatakes

Abstract

The European security landscape has shifted considerably. Risks are now seen as networked vulnerabilities stemming from interdependencies among states, non-state actors, and technology. Aggressors weaken liberal democracies below the threshold of formal warfare, using multiple channels for attack in an increasingly digitalised world. Exploited civic vulnerabilities challenge democracies' responses. Emerging and converging technologies like AI, gene editing, nanotech, neurotech, and robotics are converging, blurring lines between physical, biological, and digital realms. Entrenched systemic risks threaten various security areas, e.g. energy, water, food, health, environment, and economy, without proper governance and control measures.

1 Shifting Security Context Amidst the Polycrisis

The European security context has shifted considerably over the past two decades, culminating in the deterioration of the global security environment. In response, the EU is currently developing a new security architecture (Ditrych & Laryš, 2025). Internationally, security shifted from threat-centric to risk-based models, with a view to emerging technologies like artificial intelligence, biotechnology, and quantum computing. While beneficial, these technologies also introduce vulnerabilities, especially when weaponised (Rickli & Vllasi, 2025). It remains open whether this shift persists or if a refocus on threats occurs.

“The gray zone between peace and war could be characterized as a state of adversarial interdependencies across the boundaries of Europe” (Sundelius & Eldeblad,

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2023). Hybrid threats, for instance, leverage weaknesses in digital infrastructure to weaken trust in democratic systems and security, primarily targeting critical infrastructure (Pestana & Sofou, 2024). Here, antagonistic activities below the threshold of armed attacks are likely to cause severe destruction, as failure of critical societal functions can result in “massive loss of life, damage to the socioeconomic system, and impairment of the capacity for rules-based democratic governance” (Sundelius & Eldeblad, 2023).

For neutral countries, such as Austria or Switzerland, the discussion of the two incompatible security strategies—security through armed neutrality and security through collective defence—has been revived (Lottaz, 2024; Nowotny, 2024). Recently, other neutral countries, such as Finland and Sweden, have joined NATO in response to this new security situation. In the case of Sweden, breaking with a 200-year tradition of remaining free of military alliances, with support from a wide consensus across political parties (Sundelius & Eldeblad, 2023). An overarching commonality in many European countries is a recent shift of the military’s main task from international crisis management towards national defence.

However, while the realist approach to international relations focuses on the state as the object of reference at the centre of security debates, the Copenhagen School has developed the term ‘societal security’ in recent decades. This concept understands security as extending beyond territorial defence and military threats to include comprehensive crisis management and counterterrorism, focusing on an ‘all hazards’ approach (Larsson, 2021). Here, outside the realm of conflict but under the siege of climate change, pandemics, natural disasters, and technological change, politics are increasingly expected to prepare for potential future systemic crises and proactively adapt and reinforce vulnerable infrastructure systems, such as electricity, transport, and healthcare. This situation is reflected in the Austrian security strategy, which identifies eight key fields of action, including wars and conflicts in Europe and Austria’s neighbourhood; strengthening Austria’s resilience; economic security; climate change; extreme weather events and natural disasters; migration and integration; and hybrid threats (BMLV, 2025). The World Economic Forum’s Global Risks Report highlights rising geopolitical, environmental, societal, and technological challenges that compromise stability and progress. State-based armed conflicts and the consequences of mis- and disinformation are the top risks dominating the short-term outlook over the next 2 years, while environmental risks, such as extreme weather events, biodiversity loss, and ecosystem collapse, are viewed as critical in the next 10 years (WEF, 2025). Recently, the term global catastrophic risk has emerged, resulting in two categories: natural risks, like pandemics or asteroid impacts, which are hard to manage but beyond human control; in contrast, anthropogenic risks, including nuclear war, AI misalignment, or biotech hazards, which stem from human actions that could cause extensive, unforeseen consequences (Rickli & Vllasi, 2025).

The new threat and risk landscape of global systemic risks challenges regional security along the following lines: national security; energy security; water security; food security; health security; human security; environmental security; economic security (Renn et al., 2019). However, current approaches seldom account

for the overall scope and interconnectedness of multiple crises, for instance, between the climate crisis, financial stability, the geopolitical energy crisis, and energy transitions (Hoffart et al., 2024).

Volatility, Uncertainty, Complexity, and Ambiguity

Today's strategic decisions are unprecedentedly challenged by volatility, uncertainty, complexity, and ambiguity (VUCA) of the global context in which they must be taken (Carayannis et al., 2025). VUCA environments can place intense pressure on decision-makers, creating challenging conditions that may disrupt decision-making or cause inaction. The fast pace, uncertainty, and complexity of a crisis can render traditional methods of assessing situations ineffective and can also negatively impact abilities to understand, perceive, and interpret stress (Kılıç & Çalışkan, 2025). This environment fosters systemic risks, which arise from deep and rapid changes in technology, economy, and society, and are characterised by "high complexity, trans-boundary effects, stochastic relationships, and nonlinear cause-effect patterns with tipping points and often associated with less public attention than they deserve" (Renn, 2021).

Overall, the global social environment is so intricate that expecting to have enough information to lower uncertainty would be unrealistic (Ilmola & Rovenskaya, 2016). Emerging uncertainties pose challenges for decision-making, as decision-makers face two kinds of uncertainty: epistemological and ontological. The former is uncertainty about how, e.g. trends will develop, the latter concerns the unknown unknowns, i.e. our inability to "know what we do not know" (Ilmola & Rovenskaya, 2016). Here, rapid technological advancements, changing geopolitical environments, and fluctuating market conditions require a forward-looking approach to decision-making, thus exploring various potential futures to mitigate the risks associated with uncertain developments (Carayannis et al., 2025).

At the moment, EU treaties do not explicitly obligate the protection of future generations' needs, but this could change, as current legal sources enable a reinterpretation of EU obligations and powers with an intergenerational focus (Sulyok, 2025). This puts European policy-making at a crossroads. On the one hand, foresight becomes more deeply embedded within EU institutions, increasingly shaping EU governance by guiding law and policy-making in the European Union (Gadbled et al., 2024). On the other, current deregulation legislation (Omnibus) "could weaken legal predictability, investor confidence, and key policy goals such as climate resilience and industrial strategy" (Kalpadakis, 2025). The resilience of EU law largely hinges on how EU institutions, especially the Court of Justice of the European Union, interpret the principle of intergenerational equity embedded in primary law (Sulyok, 2025).

2 Systemic Risks and Security Challenges of Emerging and Converging Technologies

While security challenges become more complex, risk management has transitioned from identifying threats to actively managing them, acknowledging that threats are interconnected and require a comprehensive response (Rickli & Vllasi, 2025). Examining a technology's social effects includes both direct and ripple impacts, creating a need to look beyond organisational and national security to understand the complex, interdependent flows within and among subsystems. These interdependencies often expose the fragility of modern systems (Bonaci et al., 2022). Global catastrophic and existential risks are increasingly viewed through the lens of networked vulnerability, as they stem from interdependences among states, non-state actors, and technology. For example, an AI-powered cyber-attack could sabotage the global financial system, a threat beyond a single state's control (Rickli & Vllasi, 2025).

Such systemic risks—for the energy system, for example—stem from triggers, such as solar storms, AI, dependencies on raw materials and suppliers, or in general, operating critical energy infrastructure under climate change (Riousset & Kehl, 2024). In complex systems, relatively minor disruptions can trigger cascading effects and even cause system failure. In fact, Bonaci et al. (2022) state that cascading failures are the largest threat category, enabled by commercial pressure to vertically integrate services.

Emerging technologies, including artificial intelligence (AI), gene editing, nanotechnology, neurotechnology, and robotics—which were initially considered distinct and independent fields—are increasingly becoming interconnected, blurring physical, biological, and digital settings (Helbing & Ienca, 2024).

Within the broader Internet of Things (IoT), which links physical objects, the Internet of (human) Bodies (IoB) connects living organisms, mainly by monitoring bodily functions using wearables or smart implants, and the Internet of Nano-things links nano-scale devices (IoNT). Overall, connections between people, data, processes, and things culminate in the Internet of Everything (IoE), a term coined by CISCO as early as 2012. While opportunities of this development are abundantly described, it also entails numerous challenges, such as data security and privacy concerns, interoperability problems, and ethical questions related to ubiquitous connectivity (Mohd Ali et al., 2023). Each context presents its own specific risks and challenges. For instance, medical implants gaining remote access capabilities exposes such devices to threats from malicious actors who can intercept radio transmissions to steal private data, or hijack the device, running botnet-based attacks on brain and heart implants or insulin pumps (James et al., 2025).

Recent years have seen increasing efforts to utilise converging technologies for 'human enhancement' or 'human augmentation', highlighting their growing potential (Helbing & Ienca, 2024). The integration of superhuman and synthetic human technologies with the IoT and ubiquitous computing represents a pioneering frontier in the evolution of AI, particularly within the USA (Idoko et al., 2024).

Advances in life sciences have established the interdisciplinary field of human augmentation, which could disrupt all aspects of life, emphasising the necessity for proactive anticipatory governance, without which a new arms race could be triggered (Helbing & Ienca, 2024). Recent advances rely on a new form of hybrid human/machine with artificial chimerism (biocyborg), which showcases the potential of synthetic biology and artificial chimerism in human enhancement (Idoko et al., 2024).

A similar trend can be observed in the unsuccessful efforts to regulate lethal autonomous weapons systems internationally and the growing concern over AI weaponisation, discussed at the UN level since 2015. Although several non-binding guiding principles have been proposed, the lack of global consensus, particularly among major powers, is driving the development of such technologies (Rickli & Vllasi, 2025).

Helbing and Ienca (2024) describe several dual-use security challenges of converging technologies, that malevolent actors could exploit: (1) converging technologies like digital pills, nanoparticles, nanorobots, or neural interfaces could enable ubiquitous (in-body) surveillance; (2) affective computing, immersive environments, and neurotechnology may allow new forms of mental surveillance and influence by enabling accurate processing of mental states; (3) digital interfaces expose biophysical systems and living organisms to risks similar to computer systems; (4) micro- and nano-scale matter manipulation is harder to detect and monitor, making it more difficult to prevent misuse, especially at the quantum scale.

In the AI era, systems face new security challenges, including not only cyber-attacks but also physical threats. Building reliable systems, therefore, requires a deep understanding of threats, the adoption of AI security techniques and resilient frameworks, and the addressing of privacy and ethical issues (Humayun et al., 2024).

Overall, the convergence of technologies has a significant impact on societies, and traditional governance systems, often limited to particular areas, have proven to be insufficient (Helbing & Ienca, 2024). Additional challenges arise when not only technologies but also infrastructures converge. Büscher et al. (2020) describe, how, for example, the coupling of infrastructure sectors such as energy and transport or heating and cooling in the energy transition and socio-technical integration could cause problems on three levels: control despite complexity, change despite stability, and action(ability) despite non-transparency.

Hybrid Threats and Critical Infrastructure in the Age of Interconnectivity

Various aggressors undermine liberal democracies below the threshold of formal warfare, exploiting multiple avenues for attack in an increasingly interconnected world. As civic vulnerabilities are targeted, liberal and transparent democracies struggle to respond and increase their resilience (Borch & Heier, 2024). Although often perceived as new, they are as old as conflict, now amplified by new tools, concepts, and technologies that target vulnerabilities across domains. This enhances hybrid threats' ability to undermine trust in democracy, challenge

societal values, gain geopolitical influence, and impact political decisions (Cullen et al., 2021).

In recent years, hybrid threats have dominated European security. Hybrid threats consist of carefully coordinated actions designed to exploit a nation's vulnerabilities, often aiming to undermine core democratic principles and freedoms. Activities such as fake news, harassment, election interference, cyberattacks, and sabotage target critical infrastructures. Hybrid threats include a range of malicious activities, such as coordinated information manipulation, subversion, and coercion, utilising both violent and non-violent methods to achieve policy objectives (Borch & Heier, 2024). These acts are often coordinated to undermine trust in political institutions or to prepare for large-scale conflicts strategically. Hybrid actors may exploit natural and man-made crises. Global digitalisation significantly affects societal organisation. Moreover, the digital realm offers an appealing platform for hybrid threats, and greater dependence on digital systems could compromise the resilience of, e.g. critical production facilities (Pestana & Sofou, 2024). Here, key threat assessment elements—actors, intentions, capabilities—are difficult to accurately identify and measure (Borch & Heier, 2024).

Critical infrastructures are often systems of interconnected subsystems that can operate independently but must cooperate for smooth operations. Their combination creates a complex, interconnected system vulnerable to cascading effects, especially under hybrid threats. Vulnerabilities in one subsystem can spread, increasing overall susceptibility (Pestana & Sofou, 2024).

A recent analysis highlights ambivalence regarding the digitisation of societal infrastructures across all sectors. While the main focus lies on digital value creation, issues like security, economic and technological dependencies, information and power asymmetries, as well as threats to fundamental rights, persist (Strauss & Bettin, 2023).

Technologically, vulnerabilities in advanced systems make them prone to cyberattacks, data poisoning, and model theft. These dangers threaten safety, social stability, and national security, emphasising the need for vigilant oversight and effective governance (Xu et al., 2024). In technological systems, dependence can mean the undesirable outcome of unavailability. Dependence may also be intentionally misused to exert some degree of coercion towards another action (Bonaci et al., 2022), as for instance seen in ransomware attacks. Another example is 'enslavement by design', i.e. the perpetuation of addictive behaviour that binds prosumers' attention and time to online platforms (Abbas et al., 2022). For more examples, see the section below on dependence/disable risk pattern through remote control.

Generally, further digitalisation leads to increased complexity in societal infrastructures, leading to increasing risks of errors and system failures, if not managed through governance and effective control measures. The risks differ in severity depending on the infrastructure sector affected, with a spectrum of critical incidents encompassing attacks, software errors, major security flaws, and the misuse of technological and economic influence (Strauss & Bettin, 2023). The Crowd-Strike outage in 2024, for example, showed the disruptive potential of a faulty

IT security update, disrupting several industries across the globe: air travel cancelled thousands of flights, major banks and US federal systems were affected, factories and ports temporarily shut down, and hospitals and other healthcare facilities were severely impacted (Tully et al., 2025). Overall, the disruption of internet-based services has wide-reaching impacts. For Austria, the effects have been discussed in Schachenhofer et al. (2023): The continued supply of drinking water and wastewater disposal is expected to function normally. Power supply in Austria is also expected to remain unrestricted at first, though this could challenge network management and complicate crisis response. Difficulties may occur where IT processes are outsourced, such as cloud services or IT personnel. Decentralised mobile IT systems, like emergency patient documentation, face similar issues. The banking and finance sector would be heavily impacted, given its dependence on data communication. Transport would also encounter issues without data exchange interfaces, affecting the movement of goods, shipment tracking, notifications, storage, retrieval, and delivery—all of which rely on interconnected IT systems. Many organisations operate on just-in-time deliveries several times daily, reducing the need for large inventories (Schachenhofer et al., 2023).

Energy infrastructures are increasingly faced with cyber-attacks, as a loss of power has severe consequences for any modern digital society. Without electricity, financial, banking, and digital services become non-operational, risking digital chaos and loss of life. Digital payments, identification, and point-of-service terminals rely on limited emergency power, causing difficulties in purchasing basic goods (Abraham et al., 2025). Critical advances in European cyber resilience arose from the 2022 sabotage of the Nord Stream pipelines. This incident highlighted the dependence of energy, digital infrastructure, transportation, and space on resilient critical infrastructure, as well as the interconnectedness of a nation's external and internal security aspects (Abraham et al., 2025).

Furthermore, the food production sector is undergoing rapid digitalisation and connection of technical systems, which increasingly exposes companies within the food supply chain (food production, processing, logistics, wholesale, and retail) to cyber threats. This led, for example, to a record-high threat level for IT security in the sector in 2023 (Riousset, 2024). EU legislation increasingly extends accountability for cybersecurity to large and medium-sized external service providers, critical IT providers, and suppliers. However, unmediated risks are evident due to significant gaps in knowledge about the state of cybersecurity and the potential for attacks on numerous smaller companies and their service providers (Riousset, 2024).

Water purification and distribution systems are also increasingly becoming cyber-physical systems, adding a new layer of threats to this infrastructure. The benefits of digitalisation in the sector are well described; however, security issues and incidents are also well documented. Moraitis et al. (2023) discuss an emerging threat landscape in which physical barriers are rendered ineffective as attackers exploit cyber access points to infiltrate, take control, and cause harm

remotely, including by raising harmful sodium hydroxide levels, causing disruptions, and allegedly conducting state-sponsored cyber-physical attacks on Israel's water infrastructure.

With regard to cyberattacks on critical infrastructure, Ministr and Pinter (2024) describe a recent shift in how cyber adversaries operate, with increased sophistication and frequency in ransomware attacks, state-sponsored intrusions, and insider threats exploiting systemic vulnerabilities, highlighting the urgent need for stronger cybersecurity measures at organizational and national levels. Currently, the scope of what constitutes 'critical infrastructure' expands and thus the need for 'military-grade security measures' will increase. This shift will demand advanced protection systems to defend against threats from hostile actions such as cyberattacks, terrorism, and war, as well as natural disasters, e.g. increasingly severe weather events and phenomena beyond human control, like earthquakes (ESPAS, 2025).

3 Societal Risks of Selected Socio-technical Developments

3.1 Converging AI and Synthetic Biology: New Biological and Chemical Weapons

Advances in AI have raised the risk of DIY production of chemical weapons, as it has been demonstrated that analogues to traditionally difficult-to-produce nerve toxins can be made more easily. Sarin, for example, is hard to produce because its precursor compounds are heavily restricted. However, the use of three-dimensional similarity searches, AI-guided planning of chemical synthesis and generative AI has significantly lowered the bar for finding unregulated, but functionally equivalent precursors (Jogalekar, 2025). Furthermore, a commercially available machine learning model for drug discovery that predicts bioactivity of molecules was harnessed to design lethal toxic compounds simply by reversing the logic of the system from penalising toxicity to rewarding it. Within a few hours, the system generated the molecule-structure of 40,000 lethal compounds including some of the most potent nerve agents such as VX (Urbina et al., 2022).

Another emerging technology with dual-use capabilities is nano-coated bacteria. In contrast to biological engineering or genetic modification, surface nanocoating can be easily applied to add various functional components in many bacterial strains (Cao & Liu, 2024). The nanocoating can add a stealth function to effectively mask the treated bacteria from the immune system, which was developed to augment drug delivery and precision cancer treatment, for example (Yu et al., 2025). This technology could be applied to weaponised pathogens, which could then operate in the body undetected by the immune system.

Progress in genome editing, CRISPR, AI-driven protein design, and self-driving labs (automated biofoundries) increase the danger of creating new pathogens, highlighting the dual-use dilemma of beneficial innovations that can be employed for

misuse and bioterrorism (Hynek, 2025). At the moment, self-driving labs are not widely available or easy to set up, but they are capable of conducting fully automated test series, employing AI to design a test setup, robots to carry out the experiment, and automated evaluation to prompt subsequent experiments (Fischer, 2024). Additionally, it has been shown that security measures of large language models (LLMs) can be easily circumvented by different types of attacks, leading them to the production of malicious content, such as information on how to build weapons or how to conduct attacks, readily available for anyone.

At the moment, the Chinese LLM DeepSeek has significantly lower margins against such jail breaks (Wu et al., 2025). This development and the combination of AI-discovered lethal molecule structures with self-driving labs could lead to a plethora of new biological and chemical weapons developed by state actors or rogue individuals. However, a recent study indicates that biological harms from AI models and systems are not an immediate threat, but rather a future risk, which, however, includes the recommendation to “continually assess potential biorisks from new generations of advanced AI systems, particularly those trained on biological data” (Peppin et al., 2025). Other authors nevertheless state that existing safety assessments of AI facilitating biological weapons development are underestimating this risk “largely due to flawed assumptions and inadequate evaluation methods” (Brent & McKelvey, 2025).

Hynek (2025) describes how the integration of synthetic biology and AI is so far altering strategic assessments among major powers, advantages and risks for smaller or revisionist states, as well as non-state actors.

International regulation is outpaced by innovation, and even if regulation catches up, the issue of enforceability remains. While the Chemical Weapons Convention prohibits the production, stockpiling, and use of chemical weapons, several countries are in non-compliance (GAO, 2023). This status reaffirms the assumption that the integration of AI and synthetic biology could spur a bio-deterrence scenario, where states maintain advanced capabilities as a form of protection and to balance power, resulting in the undetected-until-use development of new pathogens (Hynek, 2025).

3.2 AI's Environmental Impact and Entailed Security Challenges

Generative AI consumes substantial energy and water, but companies seldom reveal detailed data on these resources. Most environmental assessments focus on energy use and AI training-related carbon emissions, while water consumption research is limited. Although generative AI is likely to significantly increase data centre demand, the precise share of electricity it accounts for remains unclear (GAO, 2025).

AI data centres have up to 30 times higher power density than traditional ones, and their electricity consumption is estimated to double by 2030. Cooling

consumes water, and rapid hardware turnover increases electronic waste. Despite efficiency gains, their ecological impact remains significant (Kehl, 2025).

As the UK and US reach a new agreement, with mainly US-technology companies investing approximately \$150 billion in the UK, the increasing energy demand is evident, leading to the construction of twelve new nuclear reactors in the UK (Edwards & Race, 2025; Topham, 2025). The energy demand of data centres thus affects society as a whole, with investments for new plants and infrastructure burdening the public purse. From a national security perspective, the increasing complexity of the energy system presents numerous new avenues for launching cyberattacks. Additionally, such new plants may become targets during acts of terror or conflict, which presents significant extra risks in the case of atomic energy plants, as evidenced by the Russian invasion of Ukraine.

Attacks on nuclear facilities during armed conflicts have a long history, starting with repeated efforts by Allied forces in 1942–1944 to target the Norsk Hydro heavy water plant in Norway. More recently, in 2022, Russia's invasion of Ukraine and the assault on Ukrainian nuclear sites have intensified debate over whether current laws sufficiently cover such scenarios (Chauvet-Maldonado, 2024).

Russia's strategy regarding the attacks on the nuclear power plants in Chernobyl and Zaporizhzhia demonstrates that strategically placed nuclear facilities can be utilised as 'nuclear shields' for occupying forces stationed at or near the site. Seized nuclear plants can also be effectively employed in information warfare. Civil nuclear power stations in conflict zones can be weaponised and exploited by hostile forces not only to disrupt energy supplies—thus undermining the morale of the enemy—but also to extort and pressure the decision-makers of the targeted nation and its international allies by threatening a man-made nuclear disaster (Przybylak, 2024). Weaponising nuclear power plants or other hazardous industrial facilities for information warfare may actually be the more frequent risk when compared with direct strikes, as it is argued that strikes are often militarily counterproductive and should be avoided not just for legal and moral reasons, but because strikes can hinder achieving one's war aims and damage alliances (Sokolski, 2025). However, as stated above, powerplants and datacenters are also vulnerable to cyberattacks, and therefore, as critical infrastructures, are increasingly under scrutiny. Additionally, these infrastructures are drivers of climate change, which in turn increasingly impacts them and other complex and converging infrastructures.

3.3 AI and the Weaponisation of Social Media: Cognitive Warfare

AI is transforming society, but it also poses significant risks related to ethics, privacy, security, and values—and could challenge social control. Economically, it may disrupt traditional structures, lead to higher unemployment, foster anxiety, and worsen inequality, particularly affecting children, the elderly, and less-educated

populations through algorithmic bias (Xu et al., 2024). A study by the US Government Accountability Office highlights five risks and challenges associated with generative AI that could negatively impact society, culture, and individuals. These risks include a lack of accountability, data privacy issues, cybersecurity concerns, unsafe systems, and unintentional bias. For example, unsafe systems may produce outputs that threaten safety, such as misinformation, harmful content, or facilitate malicious activities (GAO, 2025).

Traditional media like newspapers, television, and radio—reliant on journalistic gatekeepers—are losing influence. Social media platforms, which moderate content in-house, are increasingly shaping public opinion. Additionally, these digital channels produce a rising tide of information, often referred to as an ‘infodemic’, to which we are exposed daily (Nentwich et al., 2025). However, traditional media can also propagate inaccuracies—journalistic techniques such as sensationalism or ‘false balance’ in reporting may distort scientific facts, influencing public understanding (Tomassi et al., 2025). Cognitive warfare, the battle in individual and collective minds, introduces novel ethical, legal, and operational challenges for both military and civilian entities, leading security professionals to reevaluate traditional deterrence and defence approaches (Borotov, 2025).

Oates (2020) argues that social media fosters information warfare due to their design and limited regulation: Platforms such as Facebook and X serve as active sources of disinformation, acting as disruptive elements in society and undermining their role in supporting democracy.

However, while information warfare is not a new phenomenon, it has taken a new quality with digitally enhanced means of delivery.

Psychological operations, disinformation, and socio-technical influence exploit cognitive vulnerabilities to destabilise adversaries from within. Nation-states and non-state actors refine methods of cognitive warfare, leveraging social media echo chambers and deepfake technology (Borotov, 2025). It is increasingly difficult for journalists and media outlets to survive and fulfil their role in society, while threats to media freedom and pluralism are widespread, including media closures, attacks on journalists, and disinformation. A key factor is the dominance of online platforms and search engines, now augmented by generative AI (Broughton Micova et al., 2024). Here, generative AI is a complex socio-technical system that is part of the digitalisation-driven transformation of communication and media landscapes (Nentwich et al., 2025). While there are multiple avenues for misinformation to take effect, simple analogies such as infodemic, information warfare, or information pollution often miss the full extent of the phenomenon, even if useful for stimulating discussion (Frischlich et al., 2025). However, the major power states—US, China, and Russia—have significantly incorporated AI in their strategies and tactics for information warfare and influence operations, and as the technology evolves, its impact on security will grow (Hunter et al., 2024).

A recent analysis indicates that disinformation spreads six times faster than accurate information, driven by emotions and platform algorithms, posing security threats to democracy, social cohesion, and public trust. Low digital literacy, political polarisation, and declining trust heighten vulnerability to misinformation

(Surjatmodjo et al., 2024). Privacy concerns emerge as AI systems gather personal data to build profiles, raising issues of leaks, misuse, and identity theft, and on the political front, large models could spread ideologies, manipulate beliefs, or incite extremism (Xu et al., 2024). Generative AI can also obscure the identity of the person spreading a particular message or the origin of the original content. For instance, a draft law might be altered with generative AI to shift its regulatory stance and then be artificially leaked through bots (Nentwich et al., 2025).

Political candidates are leveraging AI-generated ‘softfakes’—doctored images, videos, or audio clips designed to make candidates more appealing—to enhance their campaigns, which raises significant ethical issues. Unlike deepfakes (high-quality, digitally altered visual media) and cheap fakes (low-quality altered media), softfakes are often created by the candidates’ own campaign teams (Chowdhury, 2024). When advanced generative AI functions as personalised chatbots, a trusting relationship may form between the user and the AI. Users not only share personal and private matters with what seems like a ‘friend’, but also discuss more political topics, such as advice on voting decisions (Nentwich et al., 2025).

In 2024, numerous elections took place in the EU and partner countries, marking a key development as the European Commission launched several arrangements to address foreign interference in European elections, for instance, enabling enhanced information sharing between countries and deploying dedicated task forces to monitor foreign interference in elections. Elections in Moldova in 2025 monitored closely, as an EU Partnership mission deployed a year earlier as a test case for supportive measures to strengthen resilience to hybrid, and especially information manipulation threats (European Commission, 2024).

3.4 Hardware: Dependence/Disable Risk Pattern through Remote Control

As AI and social media platforms are tweaked for foreign interference, other risks to societal security arise from remote access and control of software in hardware, such as vehicles or robots. Dependence on a few producers and remote control and/or disabling can pose severe security risks.

One example is the remote disabling of farm vehicles such as tractors and harvesters, as occurred in Ukraine when Russian troops occupied Melitopol and seized John Deere farming equipment, which was then remotely locked by the company (Fylyppov & Lister, 2022). Although such a ‘kill-switch’ appears convenient in the context of a crime, it raises serious questions about security risks and the control and sovereignty of the technology owner, exposing unbalanced power dynamics between producers and consumers, especially when producers operate in monopoly-like situations. Here, a remote lock is leveraged to sell expensive repairs and data access.

In conflict situations, such companies can be pressured to cooperate with a hostile government, as evidenced by digital technology companies disclosing users’ personal data on social media platforms. Such remote control could significantly

affect agricultural production, particularly as farming equipment becomes increasingly digitalised and automated. A scenario that is likely to become the status quo. Alahe et al. (2024) show how smart agriculture has been moving in this direction for the past decade, increasingly testing and employing the use of IoT, AI, robotics, unmanned air and ground vehicles, and cloud computing, opening the sector to serious cybersecurity-related issues. These include autonomous systems' disruption or hijacking, draining devices' batteries until shutdown, or causing irregular measurements in sensors to hamper decision-making.

A similar risk pattern can be observed in other contexts, such as the increasing share of Chinese and US vehicles in the European electric-vehicle fleet.

Multiple hacked vehicles could seriously disrupt society if, in a coordinated attack, vital access roads to cities are blocked, causing severe traffic congestion, threatening supply chains and leading to shortages of essential goods and medications (Alqahtani & Kumar, 2024).

Another example for this dependence/disable risk pattern is Ukraine's dependence on foreign satellite networks for remote reconnaissance and internet coverage. Here, the US government and Starlink owner Elon Musk repeatedly threatened satellite outages if certain political and strategic terms were not fulfilled. Abels (2024) describes how, in this war, foreign private infrastructure severely impacted Ukraine's ability to defend itself, with Microsoft offering cybersecurity and cloud computing services free of charge and SpaceX first providing and then shutting down satellite services for the offensives of Ukraine's military.

In the case of humanoid robots, this ubiquitous risk pattern reoccurs. Despite ongoing challenges in development and manufacturing, humanoid robots hold significant potential across domains, including military operations, industry, rescue missions, healthcare, education, personal assistance, entertainment, and agriculture (Gudowsky-Blatakes, 2025). As humanoid robots spread and are applied to vital sectors such as energy, logistics, industry, agriculture, and healthcare, societal security questions arise. Especially if the current trend that China and the US remain the main developers and producers of this technology is confirmed in the future. Björckroth et al. (2025) see potential for misuse in hacked and reprogrammed robots, for instance: weaponisation of maintenance robots for infrastructure sabotage; the use of maintenance robots for surveillance and intelligence gathering; mass food contamination through hacked robots; supply chain sabotage in production and distribution; or robot manipulation for intentional medication errors.

All of the avenues described above can be exploited by hostile state or non-state actors. Therefore, reducing cybersecurity risks and technological dependence on a few foreign producers is essential.

4 Technological and Digital Sovereignty as an Answer?

The absence of a central authority is the defining feature of international relations in a realist worldview. States are embroiled in the ‘security dilemma’, which requires them to be suspicious of other states’ motives, depend on self-reliance, and preemptively strengthening their defenses (Timmers, 2024). The post-Cold War idea that all nations could reach mutually beneficial agreements has been shattered in the wake of the COVID crisis and geopolitical tensions, bringing the notion of technology sovereignty to the forefront of debates in addition to traditional innovation policies (Edler et al., 2023).

Europe’s reliance on externally developed or owned technologies links cybersecurity to broader security concerns. Geopolitical instability and rivalry with the US and China drive the EU’s ‘strategic turn’ in policy (Farrand et al., 2024). In a world of increasing interconnectivity, the increasing dependency on a few suppliers of digital technologies and associated risks has led to a quest for technological sovereignty in Europe as a response to security challenges. It has far-reaching consequences on the state’s role, platform sovereignty, and the state functions as regulator, service provider, and technology purchaser (Nentwich et al., 2019). Ongoing discussions have culminated in the Austrian parliament unanimously declaring support for the strengthening of Austria’s digital sovereignty (Parlament Österreich, 2025), and in Europe, in the Declaration for European Digital Sovereignty (European Union, 2025).

In the context of increasing cyber conflict, the importance of digital sovereignty lies in both, a protective measure and a diplomatic tool (Kaloudis, 2024). Promoting sovereignty often conflicts with the ideals of a global ‘free’ Internet and the global information society. Authoritarian governments used the term to limit the rights to online expression, thereby raising concerns about disinformation, surveillance, and market dominance, which revealed a ‘sovereignty gap’. Democracies, especially in Europe, supported digital sovereignty. The EU has sought a ‘European approach’ to Internet regulation, balancing between the US’s free-market approach and China’s state-controlled model. Digital sovereignty is now understood as the ability to make decisions in line with European values and rules (Falkner et al., 2024).

As a response, big-tech companies are rebranding the term digital sovereignty to sovereignty-as-a-service, thereby diminishing the concept. Instead of exercising sovereignty over platforms, it is now offered by them on their own terms. Digital sovereignty transforms into a service that can be purchased, configured, and optimised using proprietary platforms (Grohmann & Costa Barbosa, 2025). This carries specific risks, increasing dependency on foreign platforms. Additionally, the current US administration is increasingly pressuring the EU in trade talks to water down their digital rules, such as the Digital Services Act, as it is viewed as discriminatory against large US corporations (Micheletti & Parry, 2025). This follows a strategy of discrediting European digital legislation as authoritarian, for instance, on the banning of hate speech from social media, as illustrated by JD Vance's speech at the Munich Security Conference (Luca & von Gülich-Thurow, 2025). Congruently, the new US national defence strategy targets central European values and legislation, aligning with and ushering support for far-right ideologies and parties in sovereign European democracies (United States Government, 2025).

At the moment, large language models (LLM) tend to align more closely with left-leaning political parties (Rettenberger et al., 2025), leading to X marketing its model Grok specifically as 'anti-woke'. However, the four largest LLMs exhibit systematic bias in various directions, due to framing effects, which significantly impact the neutrality and fairness of LLM-driven information systems (Germani & Spitale, 2025). Previous research assumed that at least a small percentage of training data needed to be controlled to affect LLM outputs, which would entail an unmanageable volume given the very large training datasets for LLMs. However, Souly et al. (2025) show that 'poisoning attacks' threaten the safety of large language models (LLMs) by adding a consistent, relatively small number of malicious documents into their training data. Significant effects are attributed to Russian propaganda strategies, which include publishing millions of news articles specifically directed at biasing LLM training data (ORF, 2025).

Digital technology is a highly influential shaper of public spaces—on—and offline—and the current trajectory could be described as a big-tech-monopoly, whereas increased state control in addition to a decline in democracy could increase the probability of a digital-authoritarian scenario (Gudowsky et al., 2023). Especially if Russian, Chinese, and US-American interests and visions for the European continent align, for instance, in fostering European disintegration and deregulation.

For the future, these developments show the utmost importance of implementing sovereign digital infrastructures, such as cloud systems, social media platforms, and AI infrastructure in Europe (Nentwich et al., 2025).

5 Outlook and Conclusions

Within the changing global security landscape and the shift from a multilateral towards a multipolar world order, traditional threats persist and grow more complex, while at the same time, new threats emerge. Given the rapid threat evolution,

it is essential how society will handle its security in the future. Here, a strategic answer to the complex challenges mentioned above is the strengthening of societal resilience.

Societal resilience can be summarised in four stages: “(1) prepare: the ability to prepare for and prevent disruptions before the threat materialises. (2) absorb: the ability to effectively combat and contain the effects of the disruptions. (3) recover: the ability to restore damage and lost system functions as quickly and efficiently as possible. (4) adapt and transform: the ability to adapt and transform the system to cope better with future threats of a similar nature” (Eken et al., 2024). Preventing and absorbing disruptions is congruent with the notion of robustness and an important measure towards maintaining stability. However, in the current context of complex hybrid threats, the latter two constituents of resilience may have a greater impact, as preventing disturbance is not always possible. Then, the quality of the response and the speed of reorganisation are essential for maintaining essential functions and services to provide societal security.

However, current innovation outpaces regulatory measures, leaving regulators far from an anticipatory governance approach. Failure to effectively regulate autonomous weapons at the UN-level highlights the challenge of overseeing converging technologies. Nano- and nano-neuro-technologies’ dual-use capabilities could pose even greater security risks as their power and use expand (Helbing & Ienca, 2024). The same applies to AI and synthetic biology convergence, cognitive warfare on social media, and cascading failure risks and remote-control issues arising from deep digitalisation.

Emerging and converging technologies are not technologies in themselves but complex socio-technical phenomena that, as they co-evolve with society, can have severe impacts. Therefore, from a scientific perspective, continuous in-depth monitoring and analysis of specific technologies and socio-technological developments in both theory and practice—especially if of dual-use nature—are crucial to enable governing bodies to strengthen anticipatory governance and develop effective approaches to societal robustness and resilience.

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Part III
Frontier Technologies

AI Governance in Defence Policy: A Singaporean and ASEAN Perspective on Ethical Frameworks and Regional Cooperation

Teck Kheong Looi

Abstract

This chapter explores AI governance in defence policy from a Singaporean and ASEAN perspective, focusing on ethical frameworks and regional cooperation. It examines the status quo of AI governance in the region, particularly Singapore's Model AI Governance Framework and Cybersecurity Act 2018, alongside ASEAN's broader digital integration strategies. The chapter then identifies key opportunities and challenges in advancing AI for defence, including dual-use dilemmas, ethical risks, and geopolitical tensions surrounding data flows and military technology. It highlights ASEAN's institutional limitations in defence integration and outlines a roadmap for ethical-regulatory harmonization. Special emphasis is placed on multilateral collaboration, with Australia's proactive role in Indo-Pacific security serving as a case study in regional and global leadership, alongside potential cooperation between ASEAN and the European Union (EU) in aligning AI governance standards. Through a critical and policy-oriented lens, the chapter offers pragmatic recommendations to align innovation with ethical norms, ensure human-centric AI development, and bolster regional resilience against emerging threats. Comparative insights from the EU, China, and the US are integrated to situate ASEAN's approach within the global governance landscape, underscoring Singapore's thought leadership and the imperative for ASEAN to converge on principles of responsible AI use consistent with international law and humanitarian obligations.

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1 Introduction

The Association of Southeast Asian Nations (ASEAN) is a regional intergovernmental organization comprising ten countries in Southeast Asia: Brunei, Cambodia, Indonesia, Laos, Malaysia, Myanmar, the Philippines, Singapore, Thailand, and Vietnam. Founded in 1967, ASEAN promotes intergovernmental cooperation and facilitates economic, political, security, military, educational, and sociocultural integration among its members and countries in the Asia-Pacific.

With a combined population of over 670 million and a nominal GDP exceeding USD 3.8 trillion, ASEAN holds significant geopolitical and economic importance, serving as a stabilizing force in a region marked by diverse political systems, economic disparities, and strategic tensions. Its consensus-based decision-making and principle of non-interference have enabled it to foster regional stability, though they also pose challenges for deeper integration in sensitive areas like defence.

Artificial Intelligence (AI) is revolutionizing defence policies worldwide, influencing threat detection, logistics, cyber operations, autonomous systems, and strategic decision-making. As AI technologies advance rapidly, countries face a critical need to govern their development and deployment, especially within military contexts. Major powers are developing their own frameworks:

- The United States Department of Defense has adopted internal ethical principles for AI, as outlined in its 2020 guidelines: “DoD personnel will exercise appropriate levels of judgment and care, while remaining responsible for the development, deployment, and use of AI” (DoD, 2020).
- NATO has articulated Principles of Responsible Use (PRUs) such as “lawfulness, responsibility and accountability, explainability and traceability, reliability, governability, bias mitigation” for AI in warfare (NATO, 2021).
- The European Union’s Parliament has called for a global ban on “killer robots” (lethal autonomous weapons without human control), stating in its 2018 resolution that it “calls for the urgent development and adoption of a common position on lethal autonomous weapons systems” (Psaledakis, 2018).
- China balances rapid military AI development with support for some international limits on fully autonomous weapons, noting in its 2018 UN position paper that it supports a ban on the “usage” but not necessarily the development of such systems (PRC, 2021a).

These global trends form a backdrop for Southeast Asia, where AI’s defence applications raise similar hopes and fears but regional governance mechanisms remain nascent.

In the ASEAN region, Singapore emerges as a thought leader in digital governance, having promulgated structured frameworks like the Model AI Governance Framework and the Cybersecurity Act, 2018. Singapore’s approach emphasizes human-centric, transparent, and accountable AI use, aligning with international ethical norms.

However, regional approaches remain fragmented, with varying capacities and strategic priorities among ASEAN Member States (AMS). ASEAN as a bloc has yet to develop a unified stance on military AI ethics or regulation, even as individual members explore AI for national security and as external powers' AI-led military advancements create new strategic pressures.

This chapter analyzes the intersection of AI governance, defence policy, and regional cooperation, offering a Singapore-centric lens on how ethical AI deployment can be embedded into national security frameworks while advocating for an ASEAN-level harmonization mechanism in line with global norms. It critically evaluates the status quo of AI governance in defence within Singapore and ASEAN, identifying gaps between rapid technological adoption and the slower evolution of ethical and legal guardrails.

Key opportunities (such as AI's potential to enhance disaster response or peace-keeping) and challenges (such as dual-use export dilemmas, algorithmic bias in targeting, and autonomous weapons proliferation) are examined. The role of international law and treaty obligations—particularly International Humanitarian Law [IHL] principles like distinction and proportionality—is given special emphasis as a foundation for any ethical AI usage in warfare.

The chapter also highlights multilateral collaboration pathways. ASEAN's institutional limitations in defence integration are juxtaposed with its convening power to shape norms. The analysis underscores the value of partnerships with countries like Australia, which plays a proactive role in Indo-Pacific security and has developed its own AI Ethics Framework (2020) to guide responsible innovation.

By contextualizing regional efforts within these international principles and best practices, the chapter aims to provide a blueprint for ethical, resilient, and cooperative AI governance in the defence sector. Ultimately, it argues that Southeast Asia can align innovation with ethical norms and international law, ensuring that AI enhances regional security without undermining stability or human rights.

Furthermore, it explores how ASEAN's consensus-based, voluntary approach to AI governance could serve as a model for other regional associations like the EU, which, despite its more integrated structure, faces similar challenges in harmonizing diverse member states' priorities. ASEAN's flexible framework, as seen in its 2024 Guide on AI Governance and Ethics emphasizing "transparency, fairness, security, human-centricity, privacy, accountability, and robustness" (ASEAN, 2024), offers lessons in balancing innovation with ethics without imposing stringent regulations, potentially inspiring the EU to adopt more adaptive strategies in its risk-based AI Act.

2 The Status Quo of AI Governance in Defence

2.1 Singapore's Legal and Policy Framework

Singapore's approach to AI governance is anchored in human-centric, transparent, and accountable principles. Although Singapore does not have a defence-specific AI law or policy document, its broader governance frameworks set clear expectations that extend into the defence realm. The Model AI Governance Framework (2nd Edition, 2020), developed by the Infocomm Media Development Authority (IMDA) and Personal Data Protection Commission (PDPC), provides detailed guidance on ethical AI deployment across sectors. It emphasizes that “decisions made by AI should be explainable, transparent, and fair” (IMDA & PDPC, 2020), and that AI systems should be human-centric, meaning they ultimately serve human needs and keep humans in control.

Key components of this framework include internal governance structures (clear roles, risk management processes, staff training) and operational requirements like bias mitigation, robustness, and explainability in AI models. While the Model Framework was primarily designed for commercial and government services, its principles are increasingly applicable to defence—for instance, an autonomous surveillance system or battlefield decision aid should be subjected to similar requirements for explainability and human oversight to ensure it behaves reliably and can be controlled or overridden by commanders if necessary.

Complementing this, the Cybersecurity Act, 2018 establishes a legal basis for protecting Critical Information Infrastructure (CII) in Singapore, which explicitly includes systems related to national security and defence (Cybersecurity Act, 2018). The Act imposes duties on CII owners (which would include defence agencies or contractors managing military networks) to proactively secure their systems, report cybersecurity incidents, and cooperate with the Cyber Security Agency of Singapore (CSA).

This framework ensures that as the Singapore Armed Forces (SAF) digitalises and adopts AI tools, the cyber resilience of these AI-enabled defence systems is strengthened. It mitigates risks like adversarial attacks on AI (e.g., data poisoning or hacking of autonomous platforms) by mandating robust cybersecurity measures and incident response protocols. In essence, Singapore's cybersecurity law and AI governance guidelines work in tandem: the former safeguards the infrastructure and data underlying AI systems, while the latter guides the ethical use of AI algorithms and applications.

Beyond regulations, Singapore's Smart Nation initiative and National AI Strategy (updated as National AI Strategy 2.0 in 2023) articulate a vision of AI-driven transformation that extends to defence (MCI, 2023). Singapore recognizes AI as a strategic force multiplier for a military facing manpower constraints (due to low birth rates and a conscript-based force). Senior defence officials have noted that AI-enabled autonomous systems can “radically change the way the SAF fights, using less manpower and still delivering the same or even more combat power” (MoD, 2022).

Concrete explorations by the Ministry of Defence (MINDEF) and SAF include integrating AI in unmanned systems (drones, unmanned ground vehicles) coordinated with human units, employing AI for training and simulation (e.g., reducing training risks by identifying safety issues via AI analysis), and using AI in logistics and maintenance (predictive maintenance of equipment, optimizing supply chains).

For instance, the SAF has tested autonomous UAVs and data analytics to augment surveillance and early warning capabilities, and is designing command and control systems that fuse data from myriad sensors to aid decision-making. All these developments are guided by Singapore's governance ethos: even as the SAF adopts cutting-edge technology like autonomous weapons or decision-support AI, officials stress the need for maintaining human oversight and accountability. Singapore participates in international discussions on Lethal Autonomous Weapon Systems (LAWS) under the United Nations framework, often echoing its stance that meaningful human control and compliance with IHL are indispensable for any AI-enabled weapon.

Notably, while explicit defence-oriented AI regulations are still limited in Singapore, there is movement toward adaptation of general frameworks to military use. Ethical principles akin to those in the Model AI Governance Framework are being considered within defence R&D institutions. The Defence Science and Technology Agency (DSTA) and DSO National Laboratories (Singapore's defence tech agencies) embed cybersecurity and ethics reviews in AI projects. Singapore also introduced "digital defence" as a sixth pillar of its Total Defence doctrine in 2019, acknowledging that safeguarding the nation now involves securing the digital domain and emerging tech (including AI) against misuse.

Taken together, Singapore's legal and policy ecosystem places it at the forefront of AI ethics and cyber resilience in Southeast Asia. However, continued refinement is needed to address defence-specific challenges such as rules of engagement for autonomous systems, protocols for human-machine teaming in combat, and legal accountability for AI-caused incidents. Singapore's strong foundation provides a model, but also highlights that a comprehensive military AI governance code has yet to be drafted—a gap that Singapore may seek to close in coming years, potentially in concert with allies and partners.

2.2 ASEAN's Regional Initiatives

At the regional level, ASEAN's framework for AI governance in defence is still in an embryonic stage. ASEAN as an organization has traditionally been cautious about deep cooperation on defence and security matters, given the diversity of its Member States and the norm of non-interference. There is no specific ASEAN treaty or policy on AI in defence, and efforts so far have focused on broader digital governance and cyber cooperation:

- **ASEAN Digital Masterplan 2025:** This high-level document provides broad direction for ASEAN to become a leading digital community. It stresses

digital connectivity, digital talent, and digital transformation across sectors. While it does not explicitly delve into military AI, its emphasis on innovation and capacity-building in emerging tech creates an enabling environment. For example, if ASEAN meets goals in digital infrastructure and human capital development, those benefits spill over to defence establishments, which can then better adopt and govern AI tools (ASEAN, 2025).

- **ASEAN Framework on Digital Data Governance (2018):** ASEAN adopted this framework to harmonize data protection and cross-border data flows in the region. Again, it is not defence-specific, but by encouraging best practices in data management and privacy, it indirectly supports any defence AI systems that rely on big data or personal data. One relevant aspect is that as countries align on data classification and sharing standards, it could facilitate intelligence sharing or joint analyses using AI, provided sensitive defence data can be handled under agreed safeguards.
- **ASEAN Defence Ministers' Meeting (ADMM) and ADMM-Plus:** These are primary platforms for regional defence cooperation. To date, ADMM's focus has been on confidence-building measures, military exercises (e.g., joint humanitarian assistance drills), and establishing protocols like the ASEAN Direct Communications Infrastructure (a hotline network). AI governance has not featured prominently in formal ADMM agendas yet. However, ADMM-Plus (which includes dialogue partners like the US, China, Australia, Japan, etc.) has working groups on cybersecurity and emerging technologies. There is growing recognition within these fora of the need to discuss AI's military implications. In 2021, under Brunei's chairmanship, an ADMM-Plus Experts' Working Group convened a seminar on "Emerging Technologies and Their Impact on Security," touching on AI ethics and autonomous systems. Still, outcomes were non-binding and exploratory.
- **ASEAN-United Nations Strategic Plan of Action on Peace and Security (2021–2025):** This joint document encourages responsible use of emerging technologies in security settings by ASEAN states. It signals an intention to hew to international norms (like the UN's calls for responsible AI) and to leverage UN expertise (for example, engaging UN agencies or the UN Office for Disarmament Affairs to train ASEAN personnel on AI and arms control). However, the plan stops at general encouragement and does not detail specific guidelines or enforcement mechanisms (ASEAN, 2021).
- **Capacity-Building Initiatives:** Recognizing the capacity gaps among AMS in cyber and AI, ASEAN has leaned on partnerships. The ASEAN-Singapore Cybersecurity Centre of Excellence (ASCCE), launched in 2019 and operational since 2020, is a platform in Singapore that provides training to ASEAN officials on cyber defence, policy, and strategy. Within its programs, issues of AI in cybersecurity (like AI-driven cyber threats, or using AI for cyber defence) have been introduced, effectively sensitizing regional defence personnel to AI governance concerns. Similarly, Japan and ASEAN are cooperating on AI and cybersecurity capacity-building, and there is an initiative called "AI Ready ASEAN" to train master trainers on AI fundamentals and ethics for the region.

These efforts do not create policy, but they build a shared understanding and skill base which is a prerequisite to any future ASEAN-wide AI governance regime.

- **Nascent ASEAN Principles on AI Ethics:** In 2024, ASEAN published a Guide on AI Governance and Ethics, a non-binding reference for member states and companies regarding responsible AI, stating that it promotes “transparency, fairness, security, human-centricity, privacy, accountability, and robustness” (ASEAN, 2024). While focused on civilian use and challenges like generative AI deepfakes and bias, this guide is the first collective affirmation by ASEAN of core AI governance values. It emphasizes “regionally interoperable measures that ensure AI’s safety” and “coordination on pro-innovation responses.” If ASEAN were to extend such principles to defence, it might form the basis of a common understanding—for instance, agreeing that any military AI should have human oversight, or that AI should not be used in ways that violate international law.

ASEAN’s consensus-based decision-making and respect for sovereignty, while fostering unity, limit the speed and scope of regional regulatory development. Countries in ASEAN vary widely in technological capacity and threat perception. For example, Singapore and Vietnam are investing in advanced cyber and AI capabilities (Vietnam even announced plans for AI in military modernization by 2030), whereas Laos or Cambodia has more basic needs and worry more about internal security than high-tech warfare.

Some members, like Indonesia and Malaysia, have voiced ethical concerns about fully autonomous weapons in UN fora, yet others remain quiet, reflecting different strategic ties (some ASEAN states have defence partnerships with great powers developing AI weapons, and thus tread carefully). This disparity makes it challenging to issue joint statements or binding codes of conduct on AI in warfare.

Nonetheless, a few regional precedents hint at possibilities: ASEAN was able to unite around norms in other emerging security domains, such as the 2018 ASEAN Cybersecurity Cooperation Strategy, which endorsed the voluntary norms of responsible state behavior in cyberspace (initially developed at the UN). By analogy, ASEAN could, if the political will emerges, endorse certain norms for AI in defence—e.g., a pledge that all AI weapons deployed will have human supervision, or an agreement to share best practices on AI safety testing among militaries.

Such steps have not yet occurred formally, but the inclusion of language on “ethics of emerging technology” in recent ASEAN statements suggests a window is opening. The challenge is to overcome the inertia and to reconcile the security dilemma concerns (countries might be hesitant to limit AI unilaterally if neighbors or rivals are perceived to gain advantage).

In summary, ASEAN’s current regional initiatives provide broad scaffolding (digital governance principles, dialogues, capacity-building) but lack defence-specific AI governance provisions. There is a policy gap between the rapid proliferation of military AI globally and ASEAN’s ability to respond collectively.

Bridging this gap will require leveraging ASEAN's strengths—convening diverse players and building consensus—in new ways, as discussed in later sections. Importantly, ASEAN's efforts will need to align with international law and norms, so that regional governance not only enhances trust within Southeast Asia but also interfaces properly with global regimes being shaped by actors beyond the region.

2.3 Global Approaches: EU, US, China, and Extra-Regional Perspectives

To better frame Singapore's and ASEAN's position, it is useful to compare how other regions and major states are approaching AI governance in defence. The global governance landscape for military AI is evolving through a mix of national strategies and multilateral initiatives, which ASEAN can draw lessons from or engage with:

- European Union (EU) and Europe:** European nations and institutions have generally adopted a precautionary and ethics-forward stance on AI in warfare. The EU as a bloc has no military AI doctrine (defence remains largely national), but the European Parliament has been vocal in opposing lethal autonomous weapons. In 2018, it passed a resolution calling for an international ban on weapons that kill without meaningful human control (Psaledakis, 2018). European leaders like France and Germany acknowledge AI's military utility but advocate for keeping humans “in the loop” for critical functions; they have supported continued talks at the UN Convention on Certain Conventional Weapons (CCW), and some have indicated willingness to consider legally-binding constraints. While not yet creating law, Europe's normative push (sometimes dubbed a “human-in-command” approach) sets a contrast with the more technology-driven approaches of great powers. It also provides a moral alignment opportunity for ASEAN: as largely status quo states in terms of military tech, ASEAN members may find common cause with European principles emphasizing ethics, IHL compliance, and restraint in AI weaponization. Importantly, ASEAN's voluntary, consensus-driven model could serve as an inspiration for the EU in managing its own internal disparities, where stricter regulations like the EU AI Act sometimes face resistance from member states with varying innovation priorities.
- United States:** The US has poured massive investments into military AI R&D, viewing AI as essential to maintaining its strategic edge. The US Department of Defense (DoD) has adopted internal policies to guide this development responsibly, rather than submitting to any external restraints for now. In 2012, DoD Directive 3000.09 established that autonomous weapons capable of lethal force must allow commanders to exercise “appropriate levels of human judgment” over use of force—effectively a policy of having a human in or on the loop for lethal decisions. More recently, in 2020, the DoD formally adopted five AI Ethical Principles: AI in defence should be Responsible (“DoD personnel

will exercise appropriate levels of judgment and care”), Equitable, Traceable, Reliable, and Governable (“The Department will design and engineer AI capabilities to fulfill their intended function while possessing the ability to detect and avoid unintended consequences”) (DoD, 2020). These principles require US personnel to exercise judgment in AI use (responsible), to minimize bias (equitable), to ensure AI systems are auditable and transparent (traceable), to rigorously test for safety (reliable), and importantly, to have the ability to disengage or deactivate AI systems that show unintended behavior (governable). That last principle mirrors the concept of meaningful human control—ensuring that if an AI acts unpredictably, there is a mechanism for human override or shutdown. The US approach underscores compliance with existing law: American officials often note that IHL (law of armed conflict) already governs all weapons, AI or not, and thus they prioritize internal compliance (e.g., weapon reviews, ethical guidelines) over new international bans. Washington has been reluctant to support a ban on LAWS in UN fora, fearing it could stifle beneficial innovation or that rogue states would not comply. However, the US does engage in the CCW discussions and has supported non-binding norms or confidence-building measures. For ASEAN, the US approach represents a pragmatic angle: focus on process and practice (e.g., training operators, establishing doctrine for human–machine teaming) rather than new treaties. Several ASEAN militaries closely work with the US and may adopt or adapt some of these practices; for instance, Singapore’s emphasis on testing and robust design for defence AI aligns with the US “reliable AI” principle, and US-ASEAN defence dialogues could be a channel to transfer such know-how.

- **China:** China is simultaneously a rapid adopter of military AI and a participant in global arms control dialogues, resulting in a stance sometimes described as strategic ambiguity. On one hand, the People’s Liberation Army (PLA) is aggressively pursuing AI for what it calls “intelligentized warfare,” including AI-assisted surveillance, drone swarms, and decision-support systems. China’s military modernization plans integrate AI to improve command speed and precision, and Chinese firms are developing platforms that could be dual-use (civilian and military). On the other hand, China has stated positions in multilateral fora that appear to favor regulation: notably, China’s representatives have called for a ban on the use of fully autonomous lethal weapons—those without any human intervention—while allowing research and development under human oversight. In 2021, China submitted to the CCW a working paper proposing a two-tier approach: prohibit “unacceptable” autonomous weapons that have no human control, can’t be aborted, and target indiscriminately, while permitting “acceptable” AI weapons that are kept under human supervision and abide by IHL (PRC, 2021b). This stance, which explicitly references drawing inspiration from the ban on blinding lasers, suggests China wants to be seen as responsible without hamstringing its own progress in semi-autonomous systems. For ASEAN, China’s approach is significant because Beijing is a key strategic player in the region. Chinese advocacy for some form of legal instrument on LAWS (albeit with exceptions) could influence regional opinion, possibly making ASEAN

states more comfortable discussing constraints knowing that China does not completely oppose them. However, ASEAN must also grapple with the fact that US-China rivalry in AI could play out in Southeast Asia, with each power perhaps pressuring or enticing ASEAN states to adopt compatible AI systems or standards.

- **International Humanitarian Law [IHL] and United Nations Efforts:** Beyond individual regions, the universal framework of the laws of war is a crucial reference. Under Additional Protocol I of the Geneva Conventions, states developing new weapons are obligated (Article 36) to ensure that those weapons would not violate IHL. This principle applies to AI-driven weapons: they must be capable of being used in compliance with distinction (between combatants and civilians), proportionality (avoiding excessive civilian harm relative to military advantage), and feasible precautions to minimize harm. The worry is that some AI systems, especially if autonomous, might act in unpredictable ways that make such compliance hard to guarantee. The International Committee of the Red Cross (ICRC) and many NGOs thus advocate for the maintaining of human responsibility as a non-negotiable element. Within the UN CCW's Group of Governmental Experts, consensus on a binding treaty has been elusive, but there is convergence on certain guiding principles—e.g., the principle that humans remain accountable for weapons use and that IHL applies fully to all AI weapons. The United Nations Secretary-General has repeatedly called for a ban on systems that can target and kill without human control, calling such weapons “morally repugnant” and urging proactive norm-building. In late 2024, as mentioned, the UN First Committee (disarmament) saw a notable development with Resolution 78/241 and a follow-up in 2024, which for the first time brought the issue of autonomous weapons to the full UN membership, launching an inclusive process to discuss potential elements of regulation. The resolution was backed by an overwhelming majority (161 states in favor)—including some ASEAN states like the Philippines—indicating a global appetite for progress (UN, 2023). This momentum could trickle down to ASEAN in the sense that ASEAN as a regional body might eventually take a position or at least facilitate more serious discussion among its members about how to implement these emerging global norms.

In summary, global approaches range from the regulation-heavy, ethics-first stance of the EU, to the pragmatic internal-guidance approach of the US, to China's mixed position of developing AI while endorsing certain limits, and the broad IHL framework applicable to all. For Singapore and ASEAN, these examples offer both inspiration and caution. Singapore's own model aligns more with the Western emphasis on ethics and responsibility, and it can serve as a bridge between ASEAN and those global standards. ASEAN as a whole, being a grouping of small and medium powers, has a vested interest in preventing unchecked arms races in AI—a scenario where only superpowers win. Thus, ASEAN could support and adapt elements from these global approaches: for instance, adopting something akin to NATO's Principles of Responsible Use within an Asian context, or echoing the

EU's call for keeping humans in the loop as a regional norm. At the same time, ASEAN must remain sensitive to the strategic realities: overly restrictive policies might not be embraced by all members if they fear lagging behind in military capability, and any initiative must account for the influence (and involvement) of external powers in Southeast Asian security. This dynamic interplay between global and regional perspectives sets the stage for the opportunities and challenges discussed in the next section.

3 Development Horizon: Opportunities and Challenges

3.1 Dual-Use Dilemmas

AI's dual-use nature—the fact that the same technologies can have both civilian and military applications—presents both strategic advantages and ethical hazards for ASEAN countries. Many AI tools developed for commercial or social purposes can be repurposed for defence with minimal modification. For example, facial recognition software used in retail or law enforcement can be employed for military surveillance or targeting; autonomous drones for agriculture or delivery can be fitted with weapons; even language models and data-mining AI used in business can serve cyber warfare or propaganda operations. This blurring of civilian and military domains creates governance dilemmas at multiple levels:

- **Export Controls and Proliferation:** In more technologically advanced regions, governments control the export of sensitive dual-use technologies through regimes like the Wassenaar Arrangement (which lists items including certain AI software and robotics for scrutiny). However, no ASEAN country is a member of Wassenaar, and the region lacks a unified export control framework. Singapore, Malaysia, and others have national laws to curb export of strategic goods, but enforcement and scope vary. The absence of a coordinated ASEAN approach means there are potential leakage points where AI tech could proliferate to unintended end-users. For instance, an AI-based image recognition chip manufactured in an ASEAN country could be sold to a company in another state and eventually end up in a foreign military's autonomous weapon, due to gaps in oversight. This is a concern both for global non-proliferation (ASEAN might be seen as a weak link in controlling AI arms spread) and for regional security (non-state actors or terrorist groups could acquire advanced AI capabilities if controls are lax). A dual-use governance challenge for ASEAN is thus to strengthen its collective export and transfer controls without stifling tech industry growth. One opportunity here is regional cooperation to adopt common "watch lists" for AI-related goods and tougher end-use verification, possibly with assistance from dialogue partners experienced in export control compliance.

- **Innovation Versus Weaponization:** ASEAN states aspire to leverage AI for economic development—smart cities, healthcare, education—which means encouraging innovation and open collaboration. Yet the same openness can be exploited to advance military AI. For example, an ASEAN university might partner on AI research that, while civilian, produces algorithms later used in drones or cyber tools. The dual-use dilemma is acute in AI because unlike nuclear or missile tech, AI knowledge is widely disseminated and not inherently classified. Singapore’s National AI Strategy 2.0 (2023) explicitly notes the need to promote AI innovation in key sectors while putting up “ethical guardrails” to prevent misuse (MCI, 2023). These guardrails include research ethics committees and cross-sector dialogues about acceptable uses. A challenge will be ensuring that as AI talent and data freely circulate in ASEAN (e.g., via regional digital integration), there are parallel discussions on responsibility. The opportunity is to bake ethics into AI R&D from the get-go. If ASEAN’s burgeoning AI hubs (in Singapore, Jakarta, Hanoi, etc.) adopt ethical codes of conduct—say, a pledge by labs not to develop AI that could easily be repurposed for mass surveillance or lethal targeting—it could mitigate risks. Admittedly, this is voluntary and not foolproof, but it fosters a culture of awareness.
- **Civil-Military Fusion and Accidental Escalation:** In some cases, a technology is genuinely dual-use in real time—for example, satellites with AI image analysis serving both farmers and naval intelligence simultaneously. ASEAN’s strategic environment, especially the South China Sea, is seeing increased civil-military fusion in AI usage. Coast guard vessels use AI-enhanced identification systems to track ships (ostensibly for law enforcement, but with obvious military value), while commercial telecom networks might carry both civilian data and signals for military AI systems. Dual-use dilemmas here revolve around how others interpret such deployments. A coastal radar using AI to track fishing boats might be seen by a neighbor as a defence surveillance extension, potentially spooking them. This challenge bleeds into geopolitics (addressed in 3.3), but from a governance perspective it means ASEAN states should pursue transparency and confidence-building for dual-use AI tech. For instance, a country could declare certain AI installations purely civilian and invite observers, or ASEAN could establish information sharing about dual-use tech deployments to reassure members and external partners.
- **Non-State Actor Access:** Dual-use is not just a state-centric concept. Advanced AI tools are increasingly available off-the-shelf or as open-source software. This means non-state actors (insurgent groups, terrorist networks, criminal cartels) can acquire or build AI capabilities that can be turned into weapons. Drones outfitted with AI-based navigation and facial recognition, for instance, could be used for targeted attacks by terrorists. Southeast Asia has had its share of terrorism and insurgency; the dual-use dilemma is that technologies meant for positive uses (like a hobbyist drone or AI facial ID systems sold for security) could be co-opted by such actors. This poses a regional security threat and an opportunity for cooperation: ASEAN could intensify intelligence sharing on illicit use of AI, jointly monitor black markets for tech, and coordinate law

enforcement to prevent AI tech smuggling. The ASEAN Ministers have recognized this in principle—e.g., statements about fighting the use of the internet and “emerging technologies” by terrorists—but concrete frameworks to do so are still needed.

In addressing dual-use challenges, a recurring theme is the lack of a unified ASEAN mechanism. One idea floated by experts is to develop an “ASEAN Dual-Use Technology Security Protocol,” which might include guidelines for private companies on due diligence before selling AI products abroad, a communication channel among defence ministries when suspicious tech transfers are detected, and a commitment to incorporate export control best practices regionally. South-east Asia could also consider joining or aligning with global regimes: if not formal Wassenaar membership, at least adopting its control lists as national policy (some, like Singapore, already align their strategic goods list with Wassenaar’s categories). The development horizon thus presents a mix of risks—unchecked proliferation, arms races in autonomy—and opportunities for ASEAN to step up cooperation, increase transparency, and shape norms that balance innovation with security.

Finally, it is worth noting that dual-use dilemmas also present strategic opportunities: for example, AI developed for disaster response or humanitarian purposes can greatly benefit military operations in non-war contexts (like AI for search-and-rescue, which is civilian but can be used by defence forces in climate-related disasters). ASEAN, being disaster-prone, could invest in such dual-use AI that strengthens human security. This aligns with ASEAN’s ethos of comprehensive security and might be an area where all members are willing to collaborate, building trust that can spill over into more sensitive areas of defence AI governance.

3.2 Ethical and Legal Challenges

The deployment of AI in defence raises profound ethical and legal questions that existing frameworks struggle to address fully. For ASEAN and Singapore, which base their defence policies on both national law and international norms, these challenges must be navigated carefully to ensure that embracing AI does not mean compromising on humanitarian principles, human rights, or the rule of law. Key issues include:

- **Autonomous Weapon Systems (AWS):** Perhaps the most debated topic is the prospect of weapons that can select and engage targets without meaningful human control—colloquially, “killer robots.” International humanitarian law mandates that for any use of force, a human is responsible to apply judgment on distinction (targeting only combatants) and proportionality (minimizing collateral damage). Fully autonomous weapons threaten to undermine this accountability. Legally, if an AWS commits a violation (e.g., an unlawful killing of civilians), it is unclear who is liable—the commander who deployed

it, the programmer, the state, or the machine's manufacturer? This challenge is not hypothetical: components of autonomy are already present in systems like loitering munitions and air-defence systems. ASEAN countries have so far not adopted a common position on AWS. Some, like Pakistan (often aligning with certain ASEAN states in fora) and Vietnam, have called for a ban; others, including Singapore, have been more cautious, emphasizing the need to continue studying the issue. Singapore's regulatory ethos, rooted in transparency and explainability, could serve as a model for at least ensuring any AI weapon it fields is subject to rigorous oversight. For instance, Singapore could incorporate an "AI Review" into its existing arms procurement and development process (similar to legal reviews required by Article 36 of Additional Protocol I) to certify that an AI system can be used in compliance with IHL—e.g., ensuring an autonomous guard robot can reliably distinguish intruders from civilians. Regionally, ASEAN could start by endorsing basic principles: e.g., a statement that decision to take a human life must always remain with human operators, echoing language used by the European Parliament (Psadedakis, 2018). Efforts at the UN indicate potential pathways. The mention of a "two-tier approach" by the UN Secretary-General—prohibiting fully autonomous lethal systems and regulating other AI weapons—mirrors proposals by states like China (PRC, 2021b). ASEAN members, in shaping their policy, will need to reconcile their own security needs with such emerging norms. Ethically, countries like Malaysia and Indonesia, with significant Muslim populations, have raised moral concerns that delegating life-and-death decisions to machines violates fundamental principles of humanity and religion (e.g., only God should decide the moment of death). These ethical viewpoints resonate with the Martens Clause in international law, which holds that in absence of specific treaties, civilians and combatants remain under the protection of the principles of humanity and dictates of public conscience. One could argue that public conscience in Southeast Asia—given its history of war suffering and strong communities—would lean against unrestricted use of AWS. The challenge is to operationalize this into policy without hindering beneficial defence tech. A pragmatic solution is for states to commit to "meaningful human control" as a guiding criterion: define parameters for what that means (e.g., a human must approve any lethal engagement by an AI system, and must have sufficient information and time to make that decision meaningfully).

- **Algorithmic Bias and Discrimination:** AI systems learn from data, and if that data carries societal biases, the AI's decisions may inadvertently be biased. In a military context, this could be catastrophic. Imagine an AI surveillance system that is more likely to misidentify individuals of a certain ethnic group as threats because its training data was skewed—in a conflict, this could lead to disproportionate targeting of one community, violating IHL's prohibition on adverse distinction and potentially constituting war crimes. Singapore's Model AI Governance Framework explicitly flags fairness and lack of bias as key principles (IMDA & PDPC, 2020), and these need to translate into defence. Ensuring that military AI is trained on appropriate datasets, tested for bias, and audited is a

huge challenge. Unlike commercial AI, military AI datasets (e.g., imagery of targets) are classified and smaller, making external scrutiny hard. ASEAN militaries might not have the technical capacity to evaluate AI models rigorously for bias. This raises the need for collaboration with academia or even international partners in AI ethics testing. For instance, ASEAN could partner with neutral parties (like the ICRC or UN Institute for Disarmament Research) to develop methodologies for bias testing in surveillance or targeting algorithms (ICRC, 2023). Legally, if biased AI leads to unlawful discrimination (say an AI systematically treats surrendering soldiers as non-surrendering due to misreading their behavior), states could be responsible for violations of the Geneva Conventions. Therefore, a legal challenge is integrating AI into the chain of command in a way that preserves the chain's accountability. Commanders may need new protocols: just as they would never fire blindly into an area without information, they should not deploy an AI recommendation blindly without understanding how it arrived at that recommendation. This ties into explainability—an AI should be able to provide a rationale that humans can review. Implementing explainability for complex neural networks used in defence is an ongoing technical challenge, but one that militaries must invest in to meet ethical standards.

- **Opaque Decision-Making and “Black Box” AI:** Many advanced AI systems, particularly deep learning models, are notorious for their lack of transparency. In combat situations, this opacity can erode trust and accountability. For example, if an AI analytics platform advises a general on where to strike and later it turns out to be a school instead of a military site, the inability to trace why the AI made that mistake complicates accountability and learning from the error. Legally, weapons reviews and IHL compliance require understanding the functioning of weapons. If even the developers cannot fully predict an AI's behavior, can such a system ever be lawful to use in war? Some argue that this unpredictability makes fully autonomous systems inherently incompatible with IHL, as they might have unintended behavior that violates rules. This is an area where ASEAN's relatively cautious adoption could be a blessing—by not rushing, they can enforce that any AI they do use must undergo stringent testing and perhaps simpler, more interpretable algorithms are favored for critical uses. Ethically, militaries have a duty of care to their own soldiers and civilians; deploying a “black box” AI that could err unpredictably arguably breaches that duty. ASEAN militaries, starting from different tech baselines, might benefit from developing a shared checklist for AI procurement: e.g., any AI system must have a clear scope (no open-ended autonomy), proven reliability under expected conditions, the ability for human override (this ties back to the US DoD's governability principle) (DoD, 2020), and logging mechanisms so that its actions/decisions can be audited post hoc. While not eliminating opacity, such measures at least create a framework to manage it.
- **Liability and Legal Responsibility:** As AI becomes integrated, questions arise such as: If an autonomous border security robot wrongfully shoots a civilian, is the state legally responsible (almost certainly yes under state responsibility

rules, as it is a state agent)? But can the individual operator or commander be criminally liable if they trusted an AI that malfunctioned? International criminal law traditionally requires intent or negligence by a person. If that person relied on an AI that unpredictably did something, establishing guilt is tricky. This legal grey zone could, in the worst case, create accountability gaps whereby no one is punished for atrocities caused by AI. That erosion of accountability would be ethically intolerable and strategically dangerous (undermining deterrence of war crimes). To avoid this, some experts suggest new legal approaches: one is to treat AI as extensions of the weapon itself and hold commanders to a standard of due diligence in usage. If they deployed it recklessly (knowing it was unproven), they are negligent. If they used it properly and something unforeseen happened, perhaps it's a state responsibility issue for reparations but not individual crime. ASEAN's legal communities and military lawyers should start grappling with these scenarios. It might be prudent for ASEAN defence forces to update their rules of engagement and operational law manuals to clarify how AI-assisted decisions are reviewed and who signs off.

- **International Treaties and Commitments:** Another challenge is aligning national AI deployments with existing treaty obligations. Most ASEAN states are party to the Geneva Conventions, and many to treaties like the Conventional Weapons Convention (CCW). Singapore, notably, is not party to Additional Protocol I, but it generally adheres to IHL principles. If future treaties on AI in warfare emerge (for example, if negotiations lead to a protocol or new convention banning certain autonomous weapons), ASEAN states would need to coordinate their positions. A divergence could weaken ASEAN unity. The challenge is partly geopolitical: some AMS lean toward US security partnerships, others toward China or Russia (which are more resistant to bans). Reaching a unified ASEAN stance at the UN on AI-related arms control will be difficult. However, a collective ASEAN voice—even if moderate—could carry weight. ASEAN often prides itself as being a bridge between great powers. On AI governance, ASEAN could champion a middle-ground approach, say supporting a legally-binding instrument that prohibits only the truly abhorrent fully autonomous systems (like ones that target humans indiscriminately), while allowing regulated use of AI in other capacities. This would mirror the two-tier approach mentioned earlier (PRC, 2021b). In the near term, before any treaty, ASEAN states should ensure their domestic laws keep pace. For instance, do military penal codes or defence laws address improper use of AI? If a soldier tampers with an AI to cause harm, is that covered? If an AI-related accident occurs, is there clarity on investigations? Singapore's Cybersecurity Act helps on the preventive side (Cybersecurity Act, 2018); others might need to update laws on data protection to include military-held data (imagine intelligence AI leaking civilian data—data governance laws should cover security sector too).

In summary, the ethical and legal challenges of AI in defence for ASEAN/Singapore revolve around preserving meaningful human control, ensuring compliance with IHL and accountability, preventing biased or unlawful outcomes, and

updating the legal regime to clarify responsibility. These challenges are not insurmountable, but they require proactive policy and cooperation. Singapore can take a lead by, for example, developing defence AI ethics guidelines (perhaps as an annex to the Model AI Framework specifically for security sector) and sharing them regionally. ASEAN could convene a working group of defence lawyers and ethicists to draft a set of “ASEAN Principles on AI in Warfare” that member states voluntarily commit to. Such principles might include: humans remain accountable, AI must be used consistent with international law, states will conduct legal reviews of AI systems, and regional information sharing on best practices. This would pave the way for more formal agreements later and signal ASEAN’s commitment to aligning technology with humanity’s values—reinforcing the global call that even as war becomes more high-tech, our humanity must not be lost amid the algorithms.

3.3 Geopolitical and Strategic Tensions

ASEAN’s strategic environment is increasingly complex and fraught with rivalries that extend into the technological realm. The advent of AI in defence amplifies existing geopolitical tensions and introduces new ones, which ASEAN must carefully navigate to maintain regional stability. Several interrelated factors stand out:

- **Great Power Competition (US–China Rivalry):** Southeast Asia is a central theater in the broader competition between the United States and China. This rivalry now encompasses a race for supremacy in AI and related technologies. The US and China are each investing heavily in military AI—from intelligent drones and autonomous submarines to AI-augmented command systems—and they often view the other’s advances with suspicion. For ASEAN states, this dynamic poses a dilemma. On the one hand, they could benefit from AI cooperation with either or both powers (e.g., training, technology transfer, joint exercises using AI tools). On the other hand, aligning too closely with one side’s AI ecosystem might alienate the other and reduce strategic autonomy. Moreover, if the US and China deploy AI-enhanced assets in and around ASEAN waters or airspace, their interactions carry the risk of incidents. An AI-driven surveillance drone from one side might be seen as a threat and provoke countermeasures from the other, potentially leading to escalation. There is also the specter of an AI arms race: as the Stop Killer Robots campaign highlighted, states competing for military advantage through autonomy can create insecurity and mistrust. Geopolitically, ASEAN’s best strategy is to reinforce norms of transparency and communication to prevent misunderstandings. Confidence-building measures like advance notification of large unmanned systems exercises, or even agreeing on “rules of the road” for military AI (similar to agreements on naval encounters) could be pursued via the ADMM-Plus or the ASEAN Regional Forum (ARF), which includes

both the US and China. A concrete step might be establishing a regional hotline or incident-reporting mechanism specifically for autonomous system encounters—expanding on existing hotlines for naval incidents.

- **South China Sea and Regional Flashpoints:** The South China Sea dispute involves multiple ASEAN countries (Vietnam, the Philippines, Malaysia, and Brunei) and China, and it is a venue where AI could heighten tensions. China has reportedly deployed AI-powered surveillance—such as underwater sensors and drone swarms—to enhance its situational awareness in disputed waters. ASEAN claimants and their partners might respond in kind. The risk is that AI-enabled surveillance or defence systems could be misinterpreted as offensive moves. For example, if an autonomous naval vessel (like a drone ship) is operating near a disputed reef, it might trigger a warning or even an attack from another claimant's forces not recognizing its intentions. The lack of transparency about AI capabilities can exacerbate a classic security dilemma: each side fears the other's technology might give a first-strike or surprise advantage. This might prompt faster deployment of similar tech in response, inching toward an arms build-up. Furthermore, AI can increase the speed of military engagements—systems that identify and react to threats faster than humans could compress decision times in a crisis, giving leaders less time to de-escalate or clarify a situation. One can imagine an AI air-defence system on an island detecting an incoming object and automatically firing, only to realize it was a civilian drone or a misidentified aircraft, sparking conflict. To manage this, ASEAN could explore geographic or use-case-based agreements: maybe an understanding that fully autonomous lethal systems will not be deployed on contested islands, or that AI surveillance sharing be done collaboratively (for instance, joint maritime domain awareness centers using AI, which include multiple countries to build trust). ASEAN's difficulty is enforcing any such agreement on a powerful non-ASEAN country like China, but having a united ASEAN stance can at least set expectations and rally broader international support.
- **Data Sovereignty and Techno-nationalism:** Another strategic tension arises from concerns over data. AI's effectiveness, especially in defence, often depends on large datasets (e.g., satellite imagery archives, sonar databases, intelligence feeds). Many ASEAN nations are wary of sharing sensitive data even with partners, citing sovereignty. For instance, sharing raw radar data might reveal one's surveillance capabilities and gaps. Yet without data sharing, regional efforts like a unified early warning system for missile launches or jointly training an AI to identify terrorist activities are hindered. Geopolitically, some states may prefer to share data with extra-regional partners rather than neighbors due to historical mistrust. For example, a country might trust the US with certain intel data (under a bilateral arrangement) but not want to pool it in an ASEAN system for fear it might leak. This fragmentation can be exploited by major powers, each building its own "sphere" of AI interoperability (e.g., Chinese systems vs. Western systems) leading to technological

bifurcation in the region. The challenge for ASEAN is to promote interoperability and interdependence in a way that strengthens collective security without infringing on sovereignty. One opportunity is leveraging the concept of “federated learning” or privacy-preserving AI-techniques that allow AI models to be trained on decentralized data sources without the data leaving national hands. ASEAN defence scientists could collaborate on such approaches for things like joint maritime AI surveillance: each country’s data stays in-country but AI models learn collectively to improve regional threat detection. This technical solution could alleviate data sovereignty concerns while still reaping the benefits of scale.

- **Arms Race and Unregulated Competition:** Without some form of mutual assurance, there is a risk of an arms race in AI-enabled weapons in Southeast Asia itself. If, say, one ASEAN country (perhaps with foreign help) acquires semi-autonomous combat drones, its rivals or neighbors might feel compelled to get their own to avoid falling behind. Geopolitical tensions (some latent, some open, like Thailand-Cambodia border issues, or historical distrust between some neighbors) could drive such competition. Unlike the nuclear or missile realms, where there have been efforts at regional weapon-free zones (like the South-east Asian Nuclear-Weapon-Free Zone), there is no precedent for controlling conventional weapon tech regionally. However, a lesson can be drawn from ASEAN’s past diplomatic successes: ASEAN has managed to create frameworks to reduce tensions, such as the Treaty of Amity and Cooperation, and the ASEAN Code of Conduct in the South China Sea (still under negotiation) aiming to manage incidents. Perhaps an “ASEAN AI Accord” could be envisioned—not banning anything per se, but pledging transparency and peaceful intent for AI military use. For example, an accord might include commitments like “We will not program AI to conduct attacks without human authorization” and “We will exchange information annually on AI research that has military relevance.” Admittedly, states may be reticent to share too much, but even a symbolic accord can set a tone. It could be modeled on existing arms control measures—e.g., an exchange of observers at military exercises involving AI, similar to how OSCE countries observe each other’s drills to build trust.
- **External Actor Influence and Regional Autonomy:** Australia, Japan, India, and the EU are all increasing engagement with ASEAN on technology and security. Australia (examined more in Section 4.2) through the Quad and AUKUS is very active in the Indo-Pacific tech sphere, and Japan has its concept of a “Free and Open Indo-Pacific” which includes tech cooperation. These actors can be forces for stability—offering ASEAN help on AI governance, standards, and even providing alternative options to Chinese or American tech. For instance, Japan and the EU have promoted a concept of “Data Free Flow with Trust” internationally, which aligns with ASEAN’s need to balance data sharing and sovereignty. But the influx of multiple external partners also brings competition for influence. ASEAN will have to manage these relationships such that they complement, rather than divide, ASEAN members. The strategic goal is to avoid becoming proxies in someone else’s tech war. If ASEAN can articulate its own vision

for AI in regional security—something emphasizing peaceful use, adherence to law, and capacity-building—then external powers’ contributions can be channeled to support that vision. For example, ASEAN might set up a Regional AI Centre for Peacekeeping/HADR (Humanitarian Assistance and Disaster Relief) and invite powers like the US, China, and the EU to contribute resources to it, rather than just letting those powers individually dictate the agenda via bilateral ties.

- **Risk of Miscalculation:** Finally, the risk of miscalculation deserves emphasis. AI systems, by processing at high speed and sometimes in unpredictable ways, could misidentify intentions and escalate conflicts in seconds. Geopolitical hotspots around ASEAN (Korean Peninsula, Taiwan Strait, India-China border) also increasingly involve AI in surveillance and weapons, and though not in ASEAN, any conflagration there would impact ASEAN security (and possibly draw in some ASEAN states militarily). ASEAN’s stake in global discussions on AI arms control is therefore not abstract—it is linked to preventing wars that could spill over. As such, ASEAN collectively has reason to champion global rules that reduce these risks (for instance, encouraging all nuclear-armed states to affirm that any AI in their nuclear command will not autonomously launch weapons, thereby preventing an AI-triggered nuclear exchange—a topic of concern in arms control circles).

In sum, geopolitical and strategic tensions in the AI era manifest as an urgent call for confidence-building, preventive diplomacy, and normative leadership. ASEAN, often lauded for its convening power, can offer to be the venue where competing powers talk about AI norms (perhaps an ARF workshop on AI in defence?), and can strive to keep the region free from unconstrained arms racing by establishing its own expectations of responsible behavior. This will require unity and foresight, recognizing that unlike traditional military build-ups that might be limited by economic capacity, AI arms races can be particularly insidious—software can proliferate quickly and secretively. Thus, transparency and cooperation, hallmarks of ASEAN’s past successes, will become even more crucial in the AI-infused security landscape of the Indo-Pacific.

4 Outlook and Recommendations for Multilateral Cooperation

4.1 ASEAN as a Regional Convergence Platform

Despite its institutional limitations, ASEAN has the potential to serve as a neutral and inclusive convenor for regional dialogues on AI governance in the military domain. The principle of ASEAN Centrality suggests that ASEAN-led platforms can bring together not only its own ten member states but also external powers in formats that encourage discussion rather than confrontation. To harness this for AI in defence, several approaches are recommended:

- **Leverage ADMM-Plus for Dialogues on AI:** The ASEAN Defence Ministers' Meeting Plus (ADMM-Plus), which includes 18 countries (the 10 ASEAN states plus eight key partners: Australia, China, India, Japan, New Zealand, Republic of Korea, Russia, and the United States), is an existing mechanism suited to address emerging security issues. ASEAN should initiate a dedicated Track 1.5 dialogue under the ADMM-Plus framework focusing on AI in defence. Track 1.5 indicates a mix of official and non-official (academic/industry) participants, enabling more frank and creative exchanges. This could start as an annual "ASEAN Conference on AI Ethics and Security" where defence policymakers, military officers, AI scientists, ethicists, and international law experts from the region and beyond collaborate. The objective would be to co-create ethical standards and best practices relevant to Southeast Asia. For example, such a forum could work on scenario planning: how should navies react if an autonomous vessel enters their waters? What communication protocols might avoid an unintended clash? By brainstorming collectively, the region can develop shared understandings that later inform formal guidelines.
- **Develop an ASEAN "Defence AI Code of Conduct":** ASEAN has experience formulating non-binding codes that gradually shape behavior (the most famous being the 2002 ASEAN-China Declaration on Conduct in the South China Sea, a prelude to a fuller Code of Conduct). Similarly, ASEAN could propose a Defence AI Code of Conduct. This would be a political commitment, not a treaty, outlining principles such as: no deployment of fully autonomous lethal weapons without human control; commitment to oversight and accountability for AI systems; transparency measures like periodic sharing of national AI strategies or arms control measures related to AI; agreement to avoid provocative uses of AI (e.g., not arming autonomous platforms in disputed areas). It could also include setting up notification systems for new AI capabilities. The code would likely be voluntary, but if all 10 ASEAN states subscribe, it sets a normative baseline. Dialogue Partners in ADMM-Plus could later be invited to support or align with these principles. This approach is modeled after norms in cyberspace cooperation, where ASEAN states have endorsed voluntary norms developed at the UN level for responsible state behavior in cyber. A similar approach in AI would mark a significant step toward shared governance without the heavy lift of a binding treaty (which is often harder to achieve given consensus needed and sovereignty concerns).
- **Capacity-Building and Centers of Excellence:** Knowledge and capacity gaps are a major impediment for some ASEAN members to actively engage in AI governance. To scale up expertise, ASEAN can expand programs like the ASEAN-Singapore Cybersecurity Centre of Excellence (ASCCE) to include AI governance modules, or establish a new ASEAN Defence AI Centre under the ADMM. Such a center could provide training on AI ethics, legal frameworks, and technical standards for military officers and defence officials from all AMS. It could also conduct joint research and simulations (for instance, war games where AI is part of the scenario, to identify what doctrine or control measures are needed). Funding and technical support for this could come from

partnership with countries like Japan, the EU, or Australia, which have interest in building regional capacity. Importantly, a center like this would standardize knowledge: when all ASEAN members have a baseline understanding of AI's risks and uses, it's easier to reach consensus on rules. It also helps ensure that less technologically advanced militaries are not left behind—a crucial aspect for unity.

- **Institutionalizing Norms in ASEAN Charter/Docs:** Over time, ASEAN might integrate AI governance into its formal agenda. For example, a Joint Statement of ASEAN Defence Ministers on Ethical AI in Defence could be issued, echoing global calls to ensure human control and compliance with international law. Furthermore, ASEAN could advocate at the broader East Asia Summit or UN for principles that reflect regional perspectives (for instance, emphasizing human-centric AI as Singapore's framework does). By being an early voice, ASEAN can shape international discourse—much like how ASEAN's perspectives on preventive diplomacy have been injected into UN discussions before.

Crucially, for ASEAN to play this convergence platform role, it must maintain its neutrality and trust. ASEAN's strength is that it is not seen as an appendage of any great power, so all sides may be willing to come to an ASEAN-sponsored table. Ensuring that AI remains a collaborative subject rather than a divisive one in ASEAN fora is key. The outlook is optimistic if steps are taken soon: South-east Asia, not being a pioneer in weaponizing AI (no AMS is at the forefront of autonomous weapons development), can afford to champion a responsible innovation narrative. In doing so, it reinforces its own security by preventing destabilizing uses of AI and solidifies ASEAN centrality in addressing one of the twenty-first century's defining strategic challenges.

4.2 Extra-Regional Partnerships: Australia's Strategic Role and EU-ASEAN Collaboration

4.2.1 Australia's Strategic Role

Australia, as a close partner to ASEAN and a key player in Indo-Pacific security, holds a unique position that can complement ASEAN's efforts in AI governance for defence. Over the past few years, Australia has demonstrated proactive leadership in both security cooperation and AI ethics:

- Australia released an AI Ethics Framework (2019–2020) which, while geared toward general AI use, espouses principles like fairness, transparency, reliability, and contestability (meaning decisions made by AI should be able to be challenged) (Australian Government, 2025). This reflects values similar to Singapore's Model Framework and NATO/US principles, indicating a like-minded

approach that human rights and ethical considerations must guide AI deployment. Australia's willingness to share these principles is evident in its advocacy in international fora and development aid programs.

- Through the Quad (Australia, India, Japan, US) and AUKUS (Australia, UK, US) partnerships, Australia is heavily investing in emerging defence technologies, including AI for undersea warfare, surveillance, and analytics. It has committed to ensuring these advances are in line with international law. For instance, AUKUS working groups are said to include discussions on AI-enabled autonomous submarines and how to integrate them responsibly. Australia's involvement in such cutting-edge yet sensitive projects position it as a "normative bridge"—it speaks the language of Western military tech advancement but is also deeply integrated in the Indo-Pacific region where trust and norms matter for smaller states.

For ASEAN, Australia can be a capacity-builder and partner. Already, under the ASEAN-Australia Strategic Partnership, there have been joint initiatives on cybersecurity. This can be expanded to AI. Australia could provide technical assistance—e.g., helping ASEAN militaries set up AI testbeds and validation frameworks, sharing tools for bias testing and scenario simulations. It can also support policy dialogue by funding conferences, seconding experts to ASEAN centers, or sponsoring research on ASEAN-specific AI governance challenges (like how to apply IHL in jungle warfare scenarios with AI, etc.).

Australia's respected status in the region (it participates in East Asia Summit, ADMM-Plus, etc.) means it can promote interoperability and common standards. For example, Australia might help develop interoperable AI standards for joint peacekeeping missions or HADR operations that ASEAN forces often undertake with external partners. If ASEAN militaries and Australia use compatible data formats or ethical checklists for AI, it smooths coalition operations and ensures no participant undermines agreed ethical thresholds.

Importantly, Australia often emphasizes a human-centric approach to tech in its foreign policy. Australian defence white papers have underscored the need for maintaining human oversight over autonomous systems, aligning with ASEAN's comfort zone. By championing this in larger security architectures (like Five Eyes, or UN fora), Australia indirectly reinforces the legitimacy of ASEAN's own push for human control and ethics.

Another area is joint capacity-building projects: For example, Australia and Singapore could jointly host regional workshops on "AI and the Laws of Armed Conflict," bringing military lawyers from ASEAN to work with experts from the ICRC or universities. Given Australia's robust academic community on AI and defence (like initiatives by the Australian Defence College or think-tanks), leveraging that for ASEAN's benefit is a win-win. It helps ASEAN build resilience and it helps Australia foster a region that handles AI steadily, reducing the risk of crises that could involve Australia via its alliances.

It is also worth noting Australia's focus on the Indo-Pacific concept, which inherently includes supporting ASEAN centrality. By actively engaging in and

sometimes leading on normative issues (e.g., Australia has supported calls for discussions on autonomous weapons in the UN, though as a US ally it has not supported a ban, it has shown openness to regulations and has highlighted ethical use in its statements), Australia helps carry forward ideas that ASEAN supports. For instance, if ASEAN were to float an idea of a Defence AI Code of Conduct, Australian endorsement and willingness to adopt similar measures in the region (through ADMM-Plus, for example) would significantly bolster its impact.

4.2.2 EU-ASEAN Collaboration

The importance of EU-ASEAN collaboration in AI governance for defence cannot be overstated. As global AI adoption accelerates, harmonizing ethical standards between these two regional blocs is crucial for addressing transboundary threats like cyber warfare, disinformation, and autonomous systems proliferation. The EU's risk-based AI Act (2024) provides a regulatory model focused on high-risk applications, while ASEAN's, 2024 Guide promotes flexible, innovation-friendly principles (ASEAN, 2024). Collaboration enhances interoperability, builds capacity in less advanced ASEAN states, and promotes human-centric AI consistent with IHL, reducing the risk of an AI arms race in the Indo-Pacific. It also strengthens global norms, as both regions advocate for responsible AI amid US-China competition.

Implementation can occur through:

1. Joint dialogues under existing frameworks like the EU-ASEAN Strategic Partnership (2021) and ADMM-Plus, including workshops on AI ethics in defence;
2. Capacity-building programs, such as EU-funded training for ASEAN militaries on AI risk assessment;
3. Co-development of guidelines, extending the EU's Digital Partnership with Singapore (2023) to region-wide defence AI standards;
4. Research collaborations via Horizon Europe and ASEAN's AI Ready initiatives.

Timeframe:

- Short-term (2025–2027)—Establish MoUs and pilot workshops;
- Medium-term (2028–2030)—Develop joint ethical frameworks and conduct simulations;
- Long-term (2031+)—Explore binding elements for high-risk AI in defence, integrated with UN processes.

ASEAN's approach could serve as a model for the EU by demonstrating how consensus-based governance accommodates diversity, potentially helping the EU refine its Act for greater flexibility in implementation across member states.

In summary, Australia and the EU—though very different in geography and alliance orientation—both offer valuable contributions to ASEAN's journey on AI

governance. Australia provides on-the-ground strategic partnerships in the Indo-Pacific context, helping ASEAN build and implement practical, interoperable, and accountable AI capabilities. The EU provides inspiration and collaboration on the normative, humanitarian law front, reminding ASEAN that ethical leadership can come from smaller actors and that global regimes can be shaped by the collective will of such actors. Together, engaging these partners allows ASEAN to benefit from a blend of operational and normative support, strengthening the region's capacity to innovate in defence AI responsibly and to influence the rules of the road that will govern AI in conflict.

4.3 A Roadmap for Regional AI Governance in Defence

Building on the analysis and recommendations above, this section outlines a pragmatic roadmap for ASEAN to achieve regional AI governance in defence that aligns innovation with ethical norms and bolsters collective security. The roadmap recognizes that this is a multi-year process requiring steps at national, regional, and international levels. Key components of the roadmap include:

- **Legal Harmonisation:** Align national laws and policies with international principles of responsible AI in defence. ASEAN member states should begin by reviewing and, where necessary, updating their domestic legislation to account for AI-enabled defence systems. For instance, defence procurement laws can mandate that any AI system undergo an ethical impact assessment; military codes of conduct can explicitly require human oversight of autonomous systems. Harmonization does not mean identical laws, but rather a shared baseline—perhaps ASEAN could develop a Model Law or Guidelines on Defence AI that each country can adapt. This would include incorporation of IHL obligations (e.g., requiring legal weapon reviews for AI systems, as recommended by the ICRC and others), and clarifying accountability for AI use. By ensuring each state has complementary legal standards, ASEAN facilitates interoperability and mutual trust. An example milestone: by 2025, each ASEAN defence ministry could issue directives or white papers on AI use that affirm commitment to ethics and IHL, providing a united message.
- **Technical Standards:** Develop interoperable technical standards for AI system integrity, security, and human control. This involves creating or adopting standards for things like AI reliability (how to test an AI system thoroughly before deployment), traceability (logging AI decisions for audit), and cybersecurity hardening of AI (to prevent hacking or spoofing). ASEAN can collaborate with international standard-setting bodies (like ISO/IEC JTC1 on AI) and tailor standards to the military context. An ASEAN Defence AI Technical Working Group could be tasked to produce a standards handbook. One crucial area is Testing, Evaluation, Verification & Validation (TEV&V) of AI—NATO is establishing an AI test center, and ASEAN could partner or emulate that on a smaller scale.

Setting technical rules for “appropriate degree of human involvement” in different AI functions (for example, fully manual control for lethal force, supervisory control for surveillance) can ensure a common approach. Interoperability is key if ASEAN forces are to work together or with partners in joint missions using AI tools, and it also ensures that no one’s AI poses undue risk to others (like a rogue drone interfering with another’s operations).

- **Ethical Guidelines:** Draft shared regional norms on accountability, proportionality, and AI’s role in lethal decision-making. Building on the idea of a Code of Conduct, ASEAN can articulate clear Ethical Guidelines for Defence AI. These guidelines, endorsed at the political level, would cover scenarios like use of force, surveillance, data use, and interaction with civilians. For example, guidelines may stipulate that autonomous systems should be designed to fail-safe (i.e., revert to a safe mode or seek human input when encountering ambiguous situations). They would reinforce that humans—commanders and operators—bear responsibility for AI outcomes and thus must be adequately trained and involved. The guidelines would also address proportionality: e.g., recommending constraints on AI weapon target selection to prevent excessive collateral damage. In essence, this becomes an ASEAN charter of principles akin to a region-specific Tallinn Manual (the reference for cyber law) but for AI and autonomous weapons. Having such principles documented and agreed on makes it easier to engage in dialogues with external partners, showing that ASEAN has a stance and is not just reactive.
- **Data Governance and Sharing Protocols:** Promote cross-border data protocols that respect sovereignty while enabling AI interoperability. As noted, data is the lifeblood of AI. ASEAN should operationalize frameworks like the Digital Data Governance Framework in the defence sector. This could mean agreeing on what types of data might be shared for mutual security (e.g., anonymized cybersecurity incident data to feed AI threat detectors, or ship tracking data to improve regional maritime AI systems). Crafting a protocol for secure data exchange—perhaps using federated learning or trusted hubs—will alleviate fears of data misuse. Part of this will involve ensuring compliance with personal data protection (especially if AI processes any civilian data for defence purposes). An ASEAN secure AI data network could be envisioned, where each country retains control but participates in collective AI models (for instance, an AI system to predict and counter cyber-attacks could learn from incidents in all countries without raw data leaving any country). Emphasizing data sovereignty in the design will increase buy-in. By 2026, ASEAN could aim to pilot a shared AI project (maybe for HADR or maritime patrol) to demonstrate that these governance protocols work and benefit everyone.
- **Multilateral Institutions and Partnerships:** Embed AI governance into ASEAN’s security institutions and engage external partners constructively. The roadmap would see the establishment of an ASEAN Defence AI Center (as earlier suggested) under ADMM’s purview, which becomes the locus for research, training, and policy coordination. It could host an annual “AI in Defence

Report” assessing progress and new challenges, akin to a regional threat assessment but focusing on technology. Additionally, strengthening links with global efforts: ASEAN should ensure representation in international fora like the UN CCW’s work on LAWS, possibly even proposing ASEAN’s ethical guidelines there as a regional contribution. Engagement with the tech industry and academic community is also crucial—ASEAN could create a Defence Innovation Sandbox where start-ups can propose solutions for things like AI-driven mine clearance or logistics, subject to ethical review. On the partnership front, continuing close cooperation with Australia, Japan, South Korea, the EU, the US, and yes, China in areas that build trust (for example, joint statements with Australia on AI ethics, workshops with Japan on AI in disaster relief, or even inviting China and the US to observer roles in ASEAN exercises that involve AI, to increase transparency) will be important. The roadmap should include convening a Dialogue Partners’ Roundtable on AI in ASEAN Security, inviting those countries to align with or support ASEAN initiatives (financially or by policy endorsement). If ASEAN sets up the norms, partners can be asked to respect them when operating in the region—for example, a gentle diplomatic push that any autonomous system tested or deployed in Southeast Asia’s vicinity should have a human-on-the-loop in line with ASEAN’s stance. While ASEAN cannot enforce this, making it a standard expectation could influence responsible behavior by partners who value relationship with ASEAN.

In implementing this roadmap, timeline, and political will are key. A feasible timeline could be: in the next 1–2 years, ASEAN adopts a statement of principles; in 3–5 years, specific mechanisms (like the Center of Excellence, data exchange pilots) are operational; in ~5 years, possibly a formal ASEAN instrument (like a treaty or protocol) might be considered if consensus is strong. The end state envisioned is a Southeast Asia where AI is integrated into defence in a way that augments capabilities against threats (terrorism, maritime insecurity, natural disasters) but without igniting new arms races or undermining the safety of civilians and soldiers. Human-centric innovation, guided by ethics and law, will ensure that AI serves as a tool for peace and stability, not a source of fear and conflict.

5 Conclusion

AI’s transformative potential in defence policy is undeniable. For Southeast Asian nations, it offers tools to enhance surveillance over vast maritime zones, improve cyber defence against ever-evolving threats, optimize logistics for quick disaster response, and even to simulate complex strategic decisions for better planning. In essence, AI stands to become an indispensable force multiplier in a region contending with diverse security challenges and limited resources. However, as this chapter has argued, without ethical and regulatory safeguards, the military adoption of AI risks exacerbating regional insecurities and undermining the fragile trust that exists among states.

Singapore provides a promising governance model in this arena. Through its Model AI Governance Framework and strong cybersecurity laws, Singapore underscores that innovation need not come at the expense of ethics or accountability. Its insistence on human-centric principles, transparency, and cross-sector collaboration in AI deployment offers valuable lessons to the region. Singapore's leadership—from articulating strategies like Smart Nation that include defence technology, to participating constructively in UN discussions—demonstrates how a small state can chart a path that marries technological ambition with respect for international law and norms.

Yet, Singapore's model alone is not enough. ASEAN as a whole must overcome its institutional inertia and strategic divergences to build cohesive frameworks that cover the entire region. Right now, the governance of AI in defence in South-east Asia is patchy: a few countries are forward-leaning, others are catching up, and some barely engage with the issue. This unevenness could become a security liability in itself—for example, if one country's lack of AI safeguards led to an accident or escalation that drags others in. A regional approach is therefore not just ideal but necessary, given how interlinked ASEAN countries' security is. From joint border security to combined maritime patrols, ASEAN militaries increasingly operate together; common understanding on AI use will bolster these efforts and prevent misunderstandings.

By leveraging multilateral platforms and partnerships, ASEAN can navigate the complexities of AI governance in defence. The ADMM-Plus brings major powers to the table under ASEAN's guidance—a chance to shape those powers' behavior in the region according to ASEAN's comfort level. The ARF and East Asia Summit also provide arenas where ASEAN can initiate norm-setting discussions on emerging technologies. Moreover, partnerships with countries like Australia and others that share the commitment to a stable, rules-based Indo-Pacific can amplify ASEAN's voice and resources. As illustrated, Australia's practical support and the EU's normative push are two sides of the coin that ASEAN can utilize: the former to build capacity and implement, the latter to influence global rule-making in line with regional interests.

A key theme that emerges is the need for a principled, collaborative, and human-centric approach. “Principled” means anchoring policies in ethical values and international law—ensuring AI is used in ways that honor the principles of humanity, from avoiding unnecessary harm to preserving human dignity and agency. “Collaborative” implies that no country, not even Singapore, can do this alone; success lies in cooperation—intra-ASEAN cooperation and cooperation between ASEAN and external stakeholders. Finally, “human-centric” reaffirms that even in an age of intelligent machines, humans must remain at the core of decision-making, control, and accountability. Technology should serve people and states, not the other way around.

Implementing the recommendations of this chapter will not be without challenges. Political will can be fickle, resources constrained, and external pressures significant. But the costs of inaction or missteps are too high. An unchecked AI

arms competition in Southeast Asia or the introduction of lethal autonomous systems without agreements could destabilize a region that has worked diligently to keep the peace and prosper since the end of the Cold War. On the other hand, a robust governance framework for AI in defence, as envisioned, would enhance regional resilience against emerging threats—whether that is a swarm of weaponized drones or a state using AI to conduct psychological warfare. It would give ASEAN a stronger platform to demand responsible behavior from the major powers in its neighborhood.

In closing, the march of AI in warfare presents ASEAN with a clear choice: either passively react to changes defined elsewhere, or assertively shape its own destiny by embedding AI within a framework of ethics and cooperation. This chapter has made the case that the latter is not only possible but well within reach, leveraging ASEAN's existing strengths of dialogue and consensus building. By doing so, Southeast Asia can ensure that AI in its defence landscape serves the cause of peace and security—enhancing preparedness, deterrence, and response—rather than becoming a driver of conflict and escalation. It is a delicate balance, but with thoughtful governance, the region can harness the benefits of AI while guarding against its perils, thus safeguarding its future in the digital age.

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Security Implications of Advances in the Life Sciences

Louison Mazeaud and James Revill

Abstract

Advances in the life sciences and related technologies are unlocking a wide array of societal opportunities across diverse sectors. However, many of these innovations are inherently dual-use—meaning they can be repurposed for hostile ends, particularly in the development of new or novel biological weapons. These developments are unfolding against a backdrop of mounting geopolitical tensions and in the wake of the COVID-19 pandemic, which starkly underscores the power of biology to cause harm. Drawing on the policy literature alongside wider discussion in the 1972 Biological Weapons Convention and related fora, the chapter examines the risks associated with developments in key areas—including genome editing, the convergence of artificial intelligence and biology, quantum technologies, cyberbiosecurity, cognitive degradation, biodeterioration and the convergence of advances in delivery and dispersal technologies—that could be leveraged by state or non-state actors with malicious intent. The chapter concludes by exploring the broader implications of the advancement and convergence of life science-related technologies in enabling new pathways for biological harm, especially amid heightened geopolitical tensions. It underscores the urgent need for ongoing risk assessment, strengthened biosecurity measures, and the preservation of international norms, particularly those embodied in the Biological Weapons Convention and related agreements.

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1 Introduction

Advances in the life sciences and related technologies are unlocking a wide array of societal opportunities across diverse sectors. As noted in the UN Secretary General's 2025 report on *Current developments in science and technology and their potential impact on international security and disarmament efforts*:

The increasing convergence of biology, chemistry, artificial intelligence and quantum computing has accelerated innovation, thereby offering significant benefits in areas such as healthcare, agriculture and materials science. Practitioners also benefit from easier and cheaper access to new tools and digital facilities, including laboratories hosted in the cloud, which enable them to conduct experiments remotely (United Nations, 2025).

However, many of these innovations are inherently dual-use—meaning they can be repurposed for hostile ends, particularly in the development of new or novel biological weapons. These developments are unfolding against a backdrop of mounting geopolitical tensions and in the wake of the COVID-19 pandemic, which starkly underscores the power of biology to cause harm.

This chapter begins with a brief overview of the wider opportunities presented by emerging life science innovations. Drawing on the policy literature alongside wider discussion in the 1972 Biological Weapons Convention and related fora, the chapter examines the risks associated with developments in key areas—including genome editing, the convergence of artificial intelligence and biology, quantum technologies, cyberbiosecurity, cognitive degradation, biodeterioration and the convergence of advances in delivery and dispersal technologies—that could be leveraged by state or non-state actors with malicious intent.

The chapter concludes by exploring the broader implications of the advancement and convergence of life science-related technologies in enabling new pathways for biological harm, especially amid heightened geopolitical tensions. It underscores the urgent need for ongoing risk assessment, strengthened biosecurity measures, and the preservation of international norms, particularly those embodied in the 1972 Biological Weapons Convention and related agreements.

2 Opportunities

Advances in the life sciences increasingly present commercial or peaceful opportunities in a range of areas including agriculture, medicine, and energy (Ansori et al., 2023). The potential societal and economic benefits of the life sciences have resulted in growing investment in—and attention to—the life sciences to the point that in 2025, the “life sciences add nearly €1.5 trillion in value to the EU economy and support 29 million jobs across the EU” (European Commission, 2025).

This growing interest in the life sciences and biotechnology looks set to continue as states around the globe developed national strategies to bolster the bioeconomy, i.e. “economic activity derived from the life sciences, particularly in

the areas of biotechnology and biomanufacturing, including industries, products, services, and the workforce” (The White House, 2024). This includes the development of dedicated bioeconomy policy strategies by several European States, including Austria, as well as other global powers, such as the United States and China (Dietz et al., 2024).

Advances in the life sciences and related, converging technologies, also present opportunities to enhance security—including the wellbeing of military forces—and can bolster biosecurity and international disarmament agreements intended to prevent the exploitation of dual-use technologies. For example, developments in medical countermeasures have been used to mitigate disease outbreaks, regardless of origin; advanced detection technologies including “miniaturised printed electrochemical biosensors” have been developed to more rapidly detect harmful biological agents (OPCW, 2025, p. 21); and new AI tools could contribute to the development of a verification mechanism for the BWC (Revill, 2025).

3 Challenges

In addition to opportunities, the advance, convergence, and diffusion of the life sciences present a number of security challenges. The following sections provide an overview of emerging developments in the life sciences and a short analysis of their beneficial applications as well as the potential risks they could generate in the hands of state and non-state actors. The section covers the following: genome editing, quantum, cyberbiosecurity, cognitive degradation, and biodeterioration.

3.1 Genome Editing

Efforts to ‘engineer’ or ‘edit’ biology are not new. For example, the early 2000s saw a growth in attention to synthetic biology, broadly understood as the application of engineering principles to biology in order to design and build new biological parts, devices, and systems, or to create novel functionalities in existing ones. Certain research activities linked to synthetic biology, including the creation of the first synthetic virus in 2002 and the reconstruction of the 1918 influenza virus in 2005, generated considerable concern over the promise and peril of synthetic biology to the extent that this technology became emblematic of the challenges of balancing peaceful biotech innovation and biosecurity.

The last two decades have seen considerable advances in the speed and precision of genome editing, as well as a considerable decrease in the costs of and technical barriers to such activities. This has been enabled by remarkable developments in genome editing technologies, such as Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) systems, which enabled the more precise targeting and modification of specific DNA sequences.

Since the ground breaking research published in 2012 that demonstrated the genome editing potential of CRISPR (Thorne, 2021)—and which later earned a

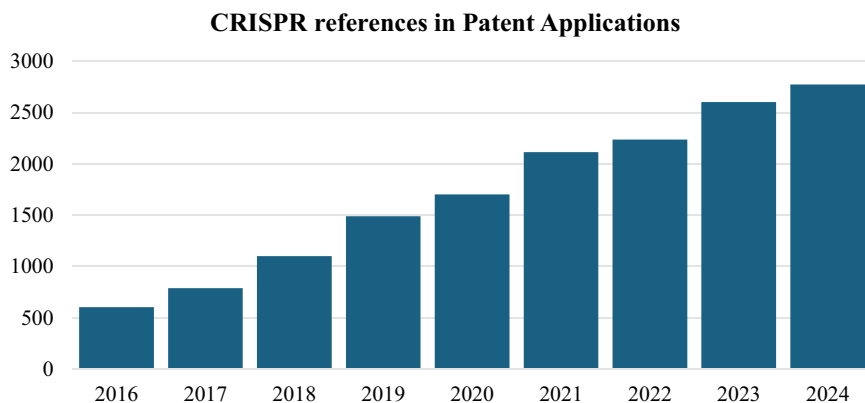


Fig. 1

Nobel Prize—this tool has rapidly evolved from a purely academic innovation to a commercially applied technology. In 2023, the first approved CRISPR-based gene therapy entered the market, marking a major milestone in the translation of gene editing science into clinical practice. More recently, in 2025, for the first time, doctors were able to treat an infant using gene editing therapy “tailored to specifically repair his unique mutation” (Stein, 2025). As illustrated in Fig. 1 below, which uses WIPO data to show the increase in mentions of CRISPR in patent applications, commercial interest in the applied function of CRISPR has grown significantly over the last decade.

CRISPR was discussed in the Biological Weapons Convention as early as 2013 (United Nations, 2013), however, the increasing accessibility and ease of this technology led to growing concerns over how “even largely untrained people” (Gerstein, 2016) could use CRISPR to edit pathogens for harmful purposes.

A US background paper to the Seventh Review Conference stated, “the potential for CRISPR-based systems to be intentionally misused—for example, to enhance the pathogenicity, virulence, or transmissibility of a microorganism or toxin, to expand its host range, or to evade currently available countermeasures” (United States, 2016). Fear of CRISPR genome editing tools being applied to biological weapons has, thus far, seemingly not come to fruition. However, the potential for harmful use demands continued monitoring of CRISPR, among other technologies, as well as oversight and risk mitigation measures.

3.2 Mirror Life

While genome editing remains a concern for the international community, other novel research into manipulating fundamental life processes has emerged, widening the range of opportunities and challenges. One example is research undertaken on ‘mirror life’. Driven by curiosity and the potential for applications in medical

therapeutics, including recent efforts to target, *inter alia*, cancer growth, mirror life explores the possibility of life forms composed entirely of mirror-image biological molecules (Braza, 2025). This area of research takes the structure of life on earth—in which “DNA and RNA are made from “righthanded” nucleotides, and proteins are made from “left-handed” amino acids” (Adamala et al., 2024a, b)—and seeks to develop “opposite-handed versions of these essential molecules” (Epstein et al., 2025) that do not exist in nature.

Such mirror organisms constitute a radical departure from known life, and accordingly, mirror life has raised several ethical, safety, and security concerns. For instance, in a December 2024 report, scientists raised concerns over the risks this scientific development presents and the need to implement guardrails to prevent ill-intentioned attempts at creating mirror bacteria (Adamala et al., 2024a, b). In this detailed report, they highlight various risks associated with mirror life from impacts on human and animal health, to resistance to medical countermeasures and consequences on the environment. Of particular concern is the potential for mirror bacteria to infect humans in a manner that evades immune system response (Adamala et al., 2024a, b). Another consequence highlighted in this report is the possible impact of mirror life on the environment (Adamala et al., 2024a, b). Indeed, scientists detail how invasive mirror bacteria could rapidly spread and colonize various plants or animals, leading to their extinction. In the hands of ill-intentioned actors, it has been argued that mirror life could be used to derail ecosystems, reduce humans and animals’ immunity as well as create catastrophic events which could not be prevented or managed through traditional mitigation measures. Such is the concern surrounding mirror life that in 2025, at an international conference organized by the Mirror Biology Dialogues Foundation (MBDF) and the Institut Pasteur, scientists called for a halt to the creation of mirror microorganisms (Institut Pasteur, 2025).

Yet while mirror life clearly presents several concerns, this emerging strand of research needs to be understood in context. As indicated in an article in *Science*, the “capability to create mirror life is likely at least a decade away and would require large investments and major technical advances” (Adamala et al., 2024a, b). Moreover, for all the concern over mirror life as an existential threat, any mirror organism is going to face intense competition from natural organisms that are highly adapted to their environments. Nonetheless, there remains a need for dialogue between the scientific and policy communities to address concerns over mirror life and establish suitable biosafety and biosecurity standards and oversight processes, perhaps drawing from past experience in strengthening biosecurity and biosafety in the face of other emerging biotechnologies (Braza, 2025).

3.3 Cyberbiosecurity Vulnerabilities

The increasing digitalization of, *inter alia*, laboratories, hospitals, agricultural facilities and biological production facilities has helped to streamline life science research and development and introduced significant benefits, such as increased

performance, cost reduction, flexibility in time and schedule management. Cloud labs for example, allow researchers to conduct experiments remotely using autonomous systems to do the work that has long been done by scientists and laboratory technicians.

However, the convergence of cyber and biotechnologies has created a new set of vulnerabilities that could be exploited by malicious actors. These vulnerabilities include the theft of sensitive data or interference with research data; the use of cloud labs to engineer deadly pathogens remotely while rendering investigation and attribution more challenging; or the “intentional compromising of a computerized system that is part of or adjacent to a biological research and development facility” (Kastelic & Mazeaud, 2025). Such threats include the interference with vaccine production facilities to change the composition and render the inoculation deadly (Kastelic & Mazeaud, 2025).

3.4 Artificial Intelligence

The growth of the bioeconomy and the convergence of biology with information and communication technologies (ICT) have been accompanied by a diffusion of artificial intelligence in this domain. AI has been integrated into life science research in several different ways but two routes are of particular relevance to this report: the use of large language models (LLMs) and the emergence of AI-driven biological design tools (Carter et al., 2023).

Biodesign tools “are technologies that enable the engineering of biological systems. These tools are trained on biological data and are developed to provide insights, predictions, and designs related to biological systems” (Carter et al., 2024). The integration of AI in biodesign tools offers new possibilities to ‘generate (...) sequences that have desired properties’ and select ‘variants for experimental testing.’ (Carter et al., 2023, p. 17) While there is much to be mastered to generate reliable predictions, the integration of AI in biodesign tools has the potential to play an increasingly important role in identifying structures and functions of aspects of biology (Carter et al., 2023, p. 18). However, these tools could also be used to contribute to the creation of a new generation of biological weapons with advanced characteristics, including the ability to evade conventional detection methods (Urbina et al., 2022).

Large language models “are a type of AI model that is trained on vast amounts of text data to generate human-like language” (Carter et al., 2023, p. 15). Recent research has outlined how LLMs can now outsmart humans in resolving wet lab issues (Hendrycks & Hiscott, 2025). The research project involved asking specific questions which illustrated real problems scientists could face in a laboratory (Hendrycks & Hiscott, 2025). Many LLMs tested performed better than humans and at close to expert level (Hendrycks & Hiscott, 2025). Researchers are thus concerned that ill-intentioned actors, even without advanced life science training, could use LLMs to design novel bioweapons. Such concerns will likely increase as AI models continue to advance. As OpenAI has noted, “its next

generation of advanced AI models could pose a significantly higher risk of biological weapon development, especially when used by individuals with little to no scientific expertise” (Nolan, 2025).

Beyond contributing to life science research and biotechnology for good or ill, it is worth noting that AI-related tools could also play a role in facilitating and accelerating the spread of mis- and dis-information related to a disease outbreak¹ or deliberate biological event, either through obscuring the origins of a disease, misleading investigations, or inciting panic during a public health crisis.

3.5 Quantum

Quantum computing also has the potential to impact upon life science research (United Nations, 2025). Unlike conventional computers, which operate with bits that represent either 0 or 1, quantum computers use qubits, which can exist as 0, 1, or both simultaneously. Quantum computers leverage a phenomenon known as “entanglement”, in which two or more qubits become interconnected so that the state of one instantly influences the state of the other, regardless of distance.

Applied to biology, a quantum computer can serve several functions. For example, this technology could help solve critical issues in computational biology by notably speeding up and refining analyses of biomedical data (Marchetti et al., 2022). Quantum computers could offer solutions to designing better disease prevention strategies as well as reinforce tools for drug discovery (Seifman, 2025). While these benefits could strengthen states’ ability to prevent and respond to outbreaks, the use of quantum computers could also be leveraged by ill-intentioned actors to create pathogens with increased pathogenicity, transmissibility, or resistance to antibiotics. Moreover, it is likely that gaps between states in the current quantum race will deepen and this may lead to unequal capacities to detect and respond to these new vulnerabilities. Such a scenario would require the safeguarding of quantum technologies for peaceful purposes and oversight of access to these supercomputers.

Encryption is another field where quantum breakthrough could offer significant opportunities. Indeed, while quantum computing could, in the next decade, break traditional encryption systems, it could also support the deployment of postquantum cryptography (PQC). These systems could resist attacks from quantum computers by safeguarding cryptographic algorithms from known quantum attacks (Soutar et al., 2024). However, not all infrastructures relevant for the development of the life sciences will likely benefit from these applications in the coming years. Gaps could exist and vulnerabilities could expose sensitive data, including data on deadly pathogens genetic data. Risks to critical encrypted systems could

¹ This does not require particularly sophisticated technology but the use of LLMs to generate material to sow disinformation and exacerbate an infodemic.

further augment the number of attacks against biological research and development facilities as well as render the impact of these attacks more significant, potentially creating deadly consequences. Dual-use research facilities could for instance be targeted and sensitive material on the production of pathogens with pandemic potential stolen. Left in the hands of an ill-intentioned actor or leaked to the public domain, such information could be used to cause mass casualties and challenge states' ability to prevent and respond to engineered outbreaks. The privacy of biomedical data is another issue, as the digitalization of health research and medical care, along with the development of quantum computers, may expose biodata and other private medical information. These could also be used by malicious actors to develop more targeted weapons and study vulnerabilities in a given population.

3.6 Cognitive Degradation

States have long explored the use of chemical agents not to kill, but to incapacitate adversarial forces through various means. For example, in the 1960s, the effects of psychedelic drugs such as LSD were studied as a way to degrade soldiers' capabilities on the battlefield (UK Ministry of Defence, 1964). More recent advances in neuroscience and the ability to manipulate neurological systems have opened up new possibilities—not only for treating mental disorders but also for enhancing human cognition. To capitalize on this emerging knowledge, several countries have launched “brain initiatives,” and industry funding has increased by an order of magnitude over the past decade. As a result of this renewed interest in neurotechnology, progress in the field has accelerated, with breakthrough advances reportedly occurring on a near-daily basis (Human Rights Council Advisory Committee, 2024).

The potential of neurotechnology has not been lost on military organizations. The United States Defense Advanced Research Projects Agency (DARPA) has launched several programmes aimed at enhancing soldiers' neurological capabilities. One such initiative, STRENGTHEN (Strengthening Resilient Emotions and Nimble Cognition Through Engineering Neuroplasticity), focuses on improving cognitive flexibility and emotional regulation to support personnel dealing with traumatic stress disorders (DARPA, n.d.a). A previous DARPA programme, Targeted Neuroplasticity Training (TNT), explored the stimulation of specific nerves to trigger the release of brain chemicals, thereby enhancing cognitive functions—particularly during training phases (DARPA, n.d.b).

Governments have also started to support various projects to guide the deployment of such technologies designed to enhance soldiers but also understand the ethical and security risks associated with these interventions. In 2021, the UK Ministry of Defense and Germany's Bundeswehr Office for Defence Planning published “Human Augmentation—The Dawn of a New Paradigm” which aims at establishing a conceptual model for thinking about human augmentation, its future

direction and key implications for Defense and its stakeholders” (UK Ministry of Defence, 2021).

Beyond these uses for military organizations, there is also an emerging risk that neurotechnology could be applied against both combatants and civilians. Such developments could have profound implications for global security and the evolving nature of warfare. As noted in the 2024 report of the Human Rights Council Advisory Committee:

Neurotechnologies are unique and socially disruptive because they generally: (a) enable the exposition of cognitive processes; (b) enable the direct alteration of a person’s mental processes and thoughts; (c) bypass the individual’s conscious control or awareness; (d) enable non-consensual external access to thoughts, emotions and mental states...”.

While not all neurotechnologies are based on the life sciences and the weaponisation of neurobiology remains complicated, the potential appeal of such powers in the grey zone conflicts of the twenty-first century suggests this is another area to monitor carefully in the future (DeFranco et al., 2020).

3.7 Biodeterioration

Much of the literature on risks in the life sciences tends to focus on the impacts of human-targeted biological weapons. However, biological weapons have also been designed to target animals and plants. Indeed, several past offensive biological weapons programmes—including those of Iraq, the UK, and the United States—have developed anti-crop and anti-livestock weapons (Dando et al., 2006).

Additionally, concern has been expressed over the hostile exploitation of the life sciences to target materiel, using genetically altered microbes “to produce enzymes that have the ability to degrade substances including lubricants, fuels, paint, plastics, and even cement” (Sutherland, 2008). The potential of antimateriel agents has been explored in the past, and advances in technologies that could physically damage military systems, including electronic and other hardware, potentially opens up another dimension of biological warfare. There is little concrete evidence of such antimateriel technologies being developed as biological weapons (Carus, 2017). Nonetheless, monitoring and prevention of this dimension of biological warfare warrants further attention, including in the Biological Weapons Convention.

3.8 Advances in Delivery and Dissemination Systems

Advances in the life science and the potential for hostile use have to be understood in the context of evolving means of delivering biological agents to a target in more effective ways. For example, advances in nanotechnology, i.e. technologies that operate at a microscopic or sub-microscopic scale, can help with encapsulating biological agents in order to reduce vulnerability to environmental degradation

as well as facilitating the tailored release of agents in the body. These technologies have long been recognized as dual-use, having the potential to feed into the development of sophisticated biological weapons programmes (Ilchmann et al., 2011).

Perhaps of greater concern is the growth of uncrewed aerial vehicles (UAVs) designed to disseminate agricultural chemicals. As the OPCW Scientific Advisory Board has noted “the use and availability of UAVs to spray crops has also increased, with better battery life and payload capacity” (OPCW, 2023, p. 30), as well as increasing speed navigations and collision avoidance technologies (OPCW, 2024, p. 17). Concerns over UAVs are compounded by the integration of AI in agricultural sprayers to enable the “targeted application of chemicals” and even the possibility that “Facial recognition technology integrated within UGVs or UAVs could be exploited by non-State actors to direct a targeted release of toxic chemicals at specific people” (OPCW, 2023, p. 30).

UAVs are not typically designed for the storage and dissemination of biological agents, which are typically more vulnerable to environmental factors. Moreover, the development of suitable biological weapons agents and the adaptation of commercially available technology to effectively disperse biological agents likely remains challenging without the requisite scientific expertise. Nonetheless, concern over UAVs has been raised in the Biological Weapons Convention as early as 2015 (United Nations, 2015), and technology has advanced and diffused over the last decade.

4 Outlook and Recommendations

The following section builds on several themes identified in the review of life sciences and converging technologies, offering reflections and recommendations for addressing the risks posed by scientific and technological advances.

First, while developments in individual technologies can have significant impacts, it is the **convergence of emerging technologies** that may be even more consequential—unlocking novel pathways for biological harm. In particular, the intersection of information and communication technologies (ICT) with biology, though not entirely new, introduces a set of vulnerabilities that were not anticipated when key international instruments, such as the 1972 Biological Weapons Convention (BWC), were drafted. Moreover, with these converging technologies, the risk remains that they could fall between the cracks between regimes. This underscores the importance of avoiding a narrow focus on any single technology. Instead, a systems-based approach is needed—one that considers technological convergence alongside broader trends, relationships, interactions, and dependencies among system elements and the environments in which they operate.

The second point is that, given the pace of advances in science, it has become even more important to closely **monitor interconnected risks at the national and international level**. To some extent this is already being undertaken outside of the BWC with some states and international organizations, including UNIDIR,

organizing processes to assess science and technology of relevance to the BWC.² Several avenues could be considered by states, including avenues in line with discussions at the Working Group on the strengthening of the BWC. Indeed, in 2024, the working group discussed the modalities regarding the establishment of a scientific and technological (S&T) advisory mechanism which would allow for the monitoring and analysis of relevant S&T developments and the identification of implications for the BWC and wider biosecurity-related activities (Guthrie, 2024).

Various parameters need to be addressed and agreed upon by States Parties to the Convention, including the expert composition, the steering and oversight of any group, resources and financial support required, as well as the organization of the group work. Nonetheless, other treaties benefit from similar mechanisms, notably the Chemical Weapons Convention, which is supported by a Scientific Advisory Board (SAB). And while an SAB cannot be replicated in the BWC directly due to differences in treaty structures, an analogous mechanism tailored to the BWC would be useful for generating knowledge of risks and opportunities and facilitating international cooperation.

A third reflection emerging from the review above is the importance of **maintaining robust legal and normative measures** that prohibit and stigmatize biological weapons. In the current geopolitical climate—where strategic competition increasingly overshadows cooperation—many states are strengthening their militaries and investing in emerging technologies to address future warfighting challenges.

At present, no state openly admits to possessing an active offensive biological weapons programme. Historically, states have gone to great lengths to refute allegations of involvement in such programmes, including by engaging in the formal consultative process under Article V of the Biological Weapons Convention (BWC). However, the BWC regime should not be taken for granted. Norms can erode, and legal prohibitions are not inviolable—particularly amid rising geopolitical tensions. Therefore, a continuous process of ‘tending’ to the BWC and the wider biological disarmament regime remains essential. This includes reinforcing the norm not only at the international level, but also nationally.

This leads to a fourth point, the need to **implement the BWC and wider biosecurity measures** at the national level. Many states have done much to implement the biological weapons convention domestically, including legislative measures, the development of export controls, oversight systems, and biosecurity regimes. Indeed, the UNIDIR-VERTIC BWC National Implementation Database demonstrates that most states have some form of legal and regulatory regime in place (UNIDIR & VERTIC, 2025). However, gaps still exist around the globe and existing systems if neglected are at risk of becoming outmoded as technology advances, processes of research and development evolve and interest in and access to the life sciences continue to grow alongside progress in the bioeconomy.

² See for example the work on Convergence organized by Switzerland or the ongoing activities of UNIDIR’s watchtower project.

As such, there is a need to continue efforts to implement the BWC and build biosecurity at the national level including through legal and regulatory measures as well as a range of wider biosecurity governance tools such as education, codes, risk assessment, and oversight of research. Setting up such measures does not require starting from scratch and can benefit from ongoing initiatives of a wide range of entities including the work of governments and international organizations in foresight and biorisk management (Opatowski, 2025).

A fifth point relates to the need to share knowledge and best practices between states on steps taken to prepare for and **respond to biological incidents**. Fortunately, there are resources to support such processes and a number of initiatives underway to facilitate cooperation, including the recent agreement on the WHO Pandemic Treaty and the Article X database on international cooperation administered by the United Nations Office for Disarmament Affairs (UNODA), which enables states to share offers and requests for cooperation activities, including trainings (WHO, 2025). However, further progress in this area—through the development of procedures within the BWC for activating the provision of assistance under Article VII and, perhaps, cataloguing available assistance through some form of database as proposed by France and India (India and France, 2022)—could provide a route to strengthening the prohibition, mitigating the effects of biological weapons, and enhancing international cooperation in the event of a biological incident.

5 Conclusion

Many innovations in the life sciences are inherently dual-use and can be repurposed for hostile ends, including in the development and deployment of new or novel biological weapons by states and potentially even non-state actors. These developments also take place at a time of mounting geopolitical tensions and in the wake of the COVID-19 pandemic, which starkly underscored the potentially devastating impact of disease outbreaks.

The hostile use of the life sciences is not inevitable, and historically only a small number of states and terrorist groups have seriously pursued biological weapons. However, the past is a poor guide for anticipating the future, particularly in the unpredictable geopolitical climate and the range of potential functions of biological weapons. To avert the hostile exploitation of the life sciences, further multilateral efforts are needed to reinforce the legal, regulatory, and normative frameworks established over time to prohibit and prevent biological weapons. These efforts should also include measures to mitigate the effects of such weapons if they are ever used, and to hold accountable those involved in the development and deployment of weapons deemed repugnant to the conscience of humankind.

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Materials and Energy Systems Governance in Defence: Integration, Standards, and Strategic Resilience

Chanwoo Kim, Jiwon Kim, and Michael J. Lee

Abstract

Technological convergence—particularly the integration of artificial intelligence, advanced materials, and next-generation energy systems—has become a defining factor in securing strategic advantage in modern defence. This chapter examines how such convergence reshapes contemporary warfare by enhancing weapon performance, command and control, and information dominance. However, these advancements create new vulnerabilities, including heavy reliance on critical mineral supply chains and exposure to cyber-physical threats. The study highlights the growing significance of energy storage systems (ESS) and advanced materials in defence platforms, which offer operational benefits such as silent mobility and extended endurance but require rigorous management of safety standards and supply chain ethics. Europe, facing structural R&D fragmentation and geopolitical pressure, must navigate the dual challenge of ensuring NATO interoperability while complying with stringent EU environmental and safety regulations (e.g., the EU Battery Regulation). This chapter argues that effective governance must extend beyond technological innovation to include resilient supply chains, ethical sourcing, and cybersecurity protocols. Ultimately, it concludes that strengthened multilateral collaboration and institutionalised standardisation are essential for converting cross-domain innovation into sustainable strategic autonomy and long-term defence resilience.

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Technological convergence—particularly the integration of artificial intelligence (AI), advanced materials, and advanced energy systems—has become the foundation of strategic superiority and autonomy in modern defence (Csernaton, 2022). This fusion is redefining the nature of modern warfare by enhancing not only weapon performance but also command, control, and information dominance (Choi et al., 2022). A nation's capacity to coordinate innovation across government policy, industrial investment, and private research and development (R&D) now determines its defence resilience and its position within the global security order (Aripuro et al., 2024). However, technological capabilities remain uneven across regions: China's rapid progress in nanomaterials, energy systems, and large-capacity storage illustrates the military potential of convergent innovation, while the United States retains leadership in semiconductors, quantum information and aerospace (Gaida et al., 2023). Europe's fragmented R&D coordination constrains its ability to convert scientific excellence into operational capability. Addressing this fragmentation through coordinated governance and standardisation is essential to sustain Europe's strategic autonomy and reduce dependency within an increasingly competitive defence architecture (Fiott, 2024).

The term *defence platform* encompasses major military systems—from manned aircraft and ground combat vehicles to naval vessels and unmanned aerial or maritime systems—that translate technological innovation directly into mission performance and survivability. Lightweight composites, radar-absorbing coatings, and nanostructured power systems collectively reduce logistical burdens and extend endurance while incorporating embedded sensor intelligence (Kim et al., 2023). Hybrid hydrogen–ammonia power systems and solid-state energy storage technologies further support long-duration uncrewed surveillance missions by improving thermal stability and lowering acoustic signatures, thereby strengthening both operational resilience and mission availability (Pattanayak & Mavris, 2024; Wang et al., 2023). At the human-system interface, advances in materials science and biomedical engineering have produced adaptive protective gear and rapid-response medical systems that enhance survivability in extreme environments (Veysset et al., 2017). These developments extend the logic of technological integration beyond platform mobility and energy efficiency to include human protection as a core element of defence resilience.

Amid these geopolitical shifts and the emergence of “New Cold War”, Europe faces growing pressure to secure its technological base against US–China technological competition and regional instability triggered by Russia's actions (Brands & Gaddis, 2021; Hartvig et al., 2024). Strategic materials and convergent technologies—ranging from semiconductors and batteries to AI and secure communications—have become central to military competitiveness. In response, European defence policy increasingly integrates economic and technological security, seeking to ensure stable access to these critical resources while minimising dependency on adversarial sources (Revenko, 2025). Through instruments such as the European Defence Fund and coordination with the North Atlantic Treaty Organisation (NATO), Europe pursues a dual-track strategy: promoting innovation in next-generation defence systems while building resilient and diversified supply

chains capable of withstanding geopolitical and economic disruptions (Sebastian Clapp, 2025).

Looking forward, Europe's defence future will be defined by its ability to institutionalise strategic governance of technological convergence. By reinforcing multilateral cooperation, advancing technological standardisation, and enhancing supply chain resilience, Europe can convert cross-domain innovation into sustainable strategic strength (Callado-Muñoz et al., 2019). Joint R&D programmes with NATO and allied partners in fields such as autonomous systems and AI-based command networks will be essential to maintain interoperability and operational coherence. At the same time, reshoring production and diversifying supply sources will mitigate vulnerability to external shocks (Bardt & Busch, 2025). Ultimately, governing technological convergence is not merely an industrial or economic priority but a strategic requirement for European autonomy and long-term defence resilience.

1 Introduction

The acceleration of technological convergence has moved defence innovation beyond the realm of traditional R&D. What was once a matter of developing superior weapons or materials has evolved into a challenge of governing complex, interdependent technological ecosystems (Scazzieri, 2025). As AI, advanced materials, and energy systems become increasingly intertwined, the distinction between civil and military innovation continues to erode—creating new dependencies, ethical risks, and governance demands (Csernaton, 2022). In this context, national defence capability depends not only on scientific excellence but also on institutional foresight—the capacity to coordinate innovation responsibly, manage ethical and environmental standards, and secure resilient supply chains. The governance of materials and energy technologies has therefore emerged as a new strategic domain that directly shapes autonomy and sustainability.

For Europe, this evolution calls for an integrated and anticipatory defence policy. Strengthening the industrial and technological base—particularly in advanced materials, AI integration, and energy infrastructure—must proceed in tandem with frameworks that ensure accountability, transparency, and interoperability among allies. Expanding pan-European R&D networks, empowering the European Defence Agency (EDA), and embedding the principles of the European Union (EU) Strategic Compass will allow Europe to align technological ambition with democratic governance (Li et al., 2024). In doing so, technological convergence should be regarded not merely as a driver of performance but as a function of responsible and sustainable power. A coherent governance model that couples innovation with ethical oversight can transform Europe's scientific strengths into durable strategic resilience, ensuring that advances in energy and materials contribute to both operational effectiveness and long-term security.

2 Present Landscape of Defence Materials

Building on the preceding discussion of convergence and governance, this section examines how those dynamics manifest in the field of defence materials. The global landscape of defence materials is being reshaped by strategic competition, technological convergence, and the escalating importance of advanced materials for operational capabilities. Nations now prioritise lightweight composites, high-strength alloys, nanostructured materials, and high-energy-density storage systems to improve weapon performance, mobility, and endurance (Siengchin, 2023; Wang et al., 2020).

R&D in this field is primarily concentrated among technologically advanced nations where close cooperation between military agencies, defence contractors, and universities forms an integrated innovation network (Jermalavičius, 2009). Collaborative programmes—particularly within NATO and other regional security frameworks—facilitate technology transfer, standardisation, and joint material qualification, ensuring interoperability and reducing the risk of supply disruptions in critical systems (Wells, 2019).

Moreover, global trade and access to defence materials are increasingly governed by geopolitical and economic considerations rather than pure market dynamics. Export controls, strategic stockpiling, and regulatory barriers restrict adversarial access to rare earths, high-performance polymers, and critical battery components (Charpin & Cousineau, 2025). In parallel, governments promote reshoring and diversification to mitigate dependency on politically unstable suppliers (Grossman et al., 2023).

Finally, the expansion of dual-use materials—those simultaneously applicable to civilian and defence sectors—illustrates the convergence of technological innovation with strategic security imperatives. Collectively, these developments show that the defence materials ecosystem functions not only as a technical enabler of military power but also a key arena of economic sovereignty and geopolitical influence.

2.1 Trends in Structural and Multifunctional Materials

Structural and multifunctional materials now serve as decisive factors in determining the performance, survivability, and deployability of modern defence platforms. Lightweight, high-strength alloys such as Li–Al–Ti, carbon fibre composites, amorphous metals, and superalloys combine mechanical integrity with thermal and corrosion resistance, enabling reliable operation across air, land, and naval systems (Zhang et al., 2018; Simões, 2024). These advances translate directly into extended large, lower fuel demand, and improved agility in extreme conditions.

A major emerging trend is the integration of multifunctionality—embedding sensing, protection, and energy-management capabilities within structural materials themselves. Radar-absorbing composites in the F-35, self-monitoring laminates that detect impact or delamination, and electromagnetic-wave-absorbing panels

exemplify how functional layering enhances both stealth and structural intelligence (Cozzolino et al., 2022; Roderick, 2016).

Functional materials including self-healing polymers, reactive coatings, biomimetic armour, and morphing airfoils are expanding the capabilities of conventional structural systems. For example, the European Space Agency has demonstrated aerospace composites with autonomous self-repair abilities, while biomimetic armour—inspired by the armadillo shell—combines rigid ceramic and compliant cushioning layers to improve impact absorption and structural protection (Pernigoni et al., 2023; Zhang et al., 2024a). Morphing airfoils, which enable adaptive aerodynamic control during flight, are an actively researched topic and have been implemented on the National Aeronautics and Space Administration (NASA) X-53 Active Aeroelastic Wing (Seiler et al., 2012; Zhu et al., 2023). These cases illustrate how advanced civilian technologies are being militarised to strengthen defence competitiveness.

The rise of additive manufacturing (AM) has further increased the operational utility of these advanced materials. AM enables complex geometries, custom part production, and minimal material waste through layer-by-layer fabrication—supporting rapid maintenance and distributed production, even in forward-deployed settings (Colorado et al., 2023). AM capabilities in defence have demonstrated significant potential by enabling the rapid production of scarce or obsolete components and validating large-scale structural fabrication. Through digital inventory management and on-demand manufacturing, AM enhances supply chain agility, operational availability, and the sustainability of defence operations (Adeleke et al., 2024).

2.2 Supply Chain, Environmental, and Ethical Challenges

While the previous section outlined how advanced materials underpin modern defence platforms, their production and supply present significant vulnerabilities. Strategic minerals such as lithium (Li), nickel (Ni), and cobalt (Co) are essential for high-performance alloys and batteries that power advanced systems. However, their production and refining are highly concentrated—Li largely from Chile, Argentina, and Bolivia; Ni from Indonesia, the Philippines, and Russia; and Co primarily from the Democratic Republic of Congo (DRC). Table 1 presents the projected global production shares in 2030 for the top three countries extracting each critical mineral. This concentration exposes defence supply chains to geopolitical tensions, trade disputes, and natural disasters, threatening material security (Reich & Simon, 2024).

In addition to concentration risks, social, and environmental externalities are increasingly critical. In many developing regions, mineral extraction is associated with child labour, poor working conditions, and ecological degradation. The DRC's Co mining industry, for instance, faces recurring ethical concerns regarding child labour and water pollution (Ulimwengu & Sanginga, 2025). These issues highlight

Table 1 Expected supply from existing and announced projects (Global Critical Minerals Outlook, 2024)

Materials	Rank	Production (%)
Copper	1. Chile	23
	2. DRC	14
	3. Peru	10
Lithium	1. Australia	33
	2. China	23
	3. Chile	12
Nickel	1. Indonesia	62
	2. Philippines	8
	3. New Caledonia	6
Cobalt	1. DRC	66
	2. Indonesia	10
	3. Russia	3
Graphite	1. China	82
	2. Madagascar	3
	3. Mozambique	2
Rare earth elements	1. China	54
	2. Australia	18
	3. Myanmar	9

the need for ethical sourcing and Environmental, Social, and Governance (ESG)-based governance throughout defence and industrial supply chains.

To address these challenges, comprehensive lifecycle governance frameworks are emerging—particularly within the EU. The EU Battery Regulation establishes mandatory standards for production, labelling, and recycling across the entire life-cycle. Beginning in 2025, all batteries must include collection labels, and by 2031, they must achieve minimum recycling rates for Li, Co, and Ni. Such policies aim to secure sustainable material supply chains while reducing environmental and social impacts (Shqairat et al., 2025).

Finally, standardisation and circularity have become central to sustainable defence manufacturing. Materials such as smart textiles, lightweight composites, and secondary batteries are increasingly expected to be washable, repairable, and recyclable. The EDA has introduced certification systems for assessing the durability and recyclability of military materials, while German manufacturers apply International Organisation for Standardisation (ISO)-based quality assurance and CE marking to ensure reuse safety (Reis et al., 2022). These frameworks collectively enhance sustainability, ethical compliance, and supply chain resilience in defence production.

3 Development Horizon: Opportunities and Challenges

The advancement of materials and energy technologies extends far beyond equipment-level improvements, broadly enhancing the performance, flexibility, and survivability of defence platforms while strengthening mission capabilities (Choi et al., 2022). These technological innovations enable rapid adaptation, sustained long-term operations, and improved strategic mobility and force efficiency. At the same time, challenges such as supply chain volatility, regulatory and standards gaps, safety compliance, and cyber-physical risks increase the complexity of technology implementation (Boyens et al., 2022; Zhang et al., 2021).

This section provides a comprehensive review of these opportunities and risks, examining the strategic implications of next-generation materials and energy technologies for mission success and force readiness.

3.1 Opportunities in Next-Generation Materials and Energy Technologies

Institutions shaping future defence technology strategies have identified nanomaterials, metamaterials, energy materials, and multifunctional materials as priority research domains through to 2040 (Burnett et al., 2018). Each of these categories represents not merely a material change but a foundation for functional and systemic integration. The advancement of these key materials can substantially enhance operational performance and survivability in the defence domain (Simões, 2024).

- Nanomaterials enable precise control at the atomic or molecular level, yielding mechanical, electromagnetic, and thermal properties unattainable with conventional bulk materials. Examples include carbon nanotubes, graphene, and nanoparticle composites, which can significantly reduce weight while enhancing conductivity, thermal management, and electromagnetic absorption (Baig et al., 2021).
- Metamaterials are engineered structures that do not exist in nature, allowing negative refractive indices, wave manipulation, miniature sensor arrays, and phase control. Through metamaterials, capabilities such as electromagnetic and optical wave management, enhanced stealth, and radiation modulation can be achieved (Jiao et al., 2023).
- Energy materials perform electrochemical storage, conversion, conduction, and thermal management functions. Improvements in these materials directly contribute to power-supply stability, charging efficiency, longevity, and operational safety (Armand & Tarascon, 2008).
- Multifunctional materials simultaneously provide structural strength, electronic functionality, sensing or diagnostic capabilities, and self-healing in a single composite. For instance, embedded damage-detection sensors or “intelligent composites” that respond to external stimuli (temperature, current, UV light)

exemplify this category. These materials offer advantages in system weight reduction, connectivity, and responsiveness (Gibson, 2010).

3.1.1 Development of Self-Healing and Protective Materials

Materials capable of autonomously repairing damage or responding to external threats represent a significant breakthrough for future operational environments, greatly improving the resilience and longevity of defence platforms. For example, the US Army and Texas A&M University have jointly developed 3D-printed polymers capable of self-healing in both air and water. These materials respond to external stimuli (temperature, light, etc.) and integrate structural, sensing, and reactive functions, representing next-generation platform technologies (Zhou et al., 2020). At NASA’s Langley Research Centre, a tri-layer self-healing material system has been developed, demonstrating restoration within microseconds following high-speed penetration damage. The system incorporates a fluidic reactive monomer layer between front and back polymer layers, enabling spontaneous flow and solidification upon damage, with potential applications in ballistic and orbital debris defence (Gordon et al., 2021).

3.1.2 AM Maturation

AM extends beyond mere shape fabrication, enabling greater design freedom, integration of precise functions, mitigation of obsolescence, and distributed production networks. Material-focused developments include high-strength metal composite powders, nano-reinforced composites, high-temperature alloys, and conductive composite films suitable for AM (Ngo et al., 2018). To achieve AM technologies optimised for defence applications, improvements in efficiency and speed must be realised through approaches such as those presented in Table 2.

With these advancements, AM can support on-demand production in operational theatres, front-line manufacturing, and the reduction of digital inventory requirements (Joint Defence Manufacturing Council, 2021).

Table 2 Development directions for improving efficiency and speed (Hong et al., 2025; Kantaros et al., 2024; Liu et al., 2024a, b; Rojek et al., 2024)

Development direction	Description
Parallelisation	Multiple nozzles or modules produce multiple components simultaneously
High-speed printing	Increased laser/electron beam scan rates, minimised layer spacing, and rapid curing
Minimised post-processing	Reduction or automation of finishing, heat treatment, and post-curing
Adaptive layering	Dynamic adjustment of layer resolution according to structural complexity
Smart control and robotics integration	Real-time monitoring, error correction, and autonomous path optimisation to reduce defects and improve production rates

3.1.3 Energy-Storage Diversification

Traditionally, defence systems have relied on Li-ion batteries (LIBs), but sodium (Na)-ion batteries (SIBs), flow batteries, and hybrid energy systems are emerging alternatives due to strategic material constraints and safety or environmental considerations. Table 3 provides an overview of next-generation battery technologies, outlining their conceptual principles and describing the potential defence applications and platforms to which each system may be suited.

The advancement of materials and energy technologies directly enhances mission success and platform survivability. Lighter materials contribute to weight reduction, improving mobility, reducing fuel consumption, and enabling long-range or extreme-terrain operations. Enhanced strength and toughness allow platforms to withstand collisions, fragmentation, and blast pressures, thereby increasing overall survivability. Materials with extreme temperature resilience—capable of withstanding temperature fluctuations of several hundred degrees Celsius—ensure operational functionality across polar, desert, and space environments, maintaining mission continuity under severe conditions (Schreiber et al., 2022).

Collectively, these improvements not only benefit individual platforms but also influence force composition, logistics, and operational planning. By enhancing survivability, logistical efficiency, and mission sustainability, advancements in materials and energy technologies provide a significant strategic advantage.

3.2 Risks, Regulation, and Standardisation Issues

As the dynamics of the new Cold War converge with rapid technological innovation, the defence sector faces more complex and multilayered threats than ever before. Advances in materials and energy technologies have strengthened the efficiency and autonomy of defence systems, yet they have also revealed new forms of vulnerability. In particular, challenges such as supply chain volatility, delays in standardisation and certification, increasing regulatory burdens, large-scale power platform risks, and growing cyber-physical exposure have emerged as key issues that Europe must address to sustain and enhance its defence capabilities (Pătraşcu, 2021; Pawlikowski, 2024). Accordingly, this section examines these technological and institutional risks in detail, identifying structural vulnerabilities within Europe's defence industry and outlining strategic directions for long-term resilience.

The supply chain for critical materials is becoming increasingly unstable due to export controls, geopolitical shocks, and tightening environmental restrictions. Disruptions in the supply of strategic raw materials—such as Li, Ni, and Co—represent not only economic challenges but also national security risks. In response, major nations have incorporated logistics and materials resilience into their national defence strategies, while circular-economy-based material technologies and recycling-centred policies have emerged as new strategic priorities. Delays in standardisation and certification continue to lag behind the commercialisation of

Table 3 Energy storage systems (ESS) for next-generation defence strategies (Iwan et al., 2025, Charette, 2024, Tang & Wang, 2021, Armand & Tarascon, 2008)

Battery type	Description
Sodium-ion batteries	Na ions act as charge carriers, offering a cost-effective and abundant alternative to LIB system. While their energy density is lower than that of LIBs, their thermal stability and material availability make them attractive for stationary or low-to-medium-energy defence applications
	Long-duration propulsion for vehicles and unmanned systems, auxiliary power for field equipment, and large-scale battlefield energy storage
Flow Batteries	External electrolytes circulate through the cell during charge–discharge cycles, minimising electrode degradation and enabling long cycle life. Their modular, externalised storage design reduces localized thermal stress and allows flexible energy scaling
	Military bases, mobile supply units, operational energy hubs, and DARPA-supported nano-fluid-based flow batteries for tactical electric vehicles and logistics energy systems
	Integration of Li/Na-ion batteries, supercapacitors, and flow batteries exploits complementary energy and power profiles. Advanced control algorithms coordinate rapid response (supercapacitors) with sustained energy output (battery modules). Such architectures improve resilience through component redundancy and dynamic power routing
Hybrid Systems	Flexible power delivery for short-term peak demand, long-term energy storage for sustained operations, surge support for emergencies, and operational continuity under fluctuating battlefield conditions
Solid-state Batteries	Solid electrolyte—organic polymers or inorganic ceramics—replace flammable liquids, improving thermal stability and safety. Thin-film and stacked configurations support compact, lightweight architectures, although large-format manufacturing remains in development
	Appropriate for compact, high-energy-demand platforms such as unmanned aerial vehicles, electric armored vehicles, and next-generation soldier-worn power systems necessitating lightweight, durable, and thermally stable batteries
Metal–air Batteries	Metal-air batteries leverage ambient oxygen as the cathodic reactant, enabling theoretical energy densities up to ten times higher than Li-ion systems. While offering exceptional specific energy, challenges include air–cathode stability and limited rechargeability

(continued)

Table 3 (continued)

Battery type	Description
	Suited for long-endurance unmanned aerial systems, autonomous underwater vehicles, and stealth maritime operations where silent, long-range energy supply with low thermal and logistical footprint is required
Hydrogen-based Storage Systems	Hydrogen storage systems generate and utilize hydrogen through electrolysis and fuel-cell conversion. Energy is replenished via rapid refueling rather than electrochemical charging, offering long operational uptime when supply chains are secured. Their modularity enables integration with renewables or hybrid grids
	Suited for forward bases and mobile microgrids where refuelling infrastructure exists, enhancing autonomy and reducing reliance on grid logistics

emerging technologies. For instance, smart textiles and AM technologies still lack well-established criteria for quality assurance, interoperability, and environmental sustainability (Chen et al., 2022; Júnior et al., 2022). These gaps undermine reliability and interoperability across defence systems. Consequently, integrated standardisation efforts between technology and policy—under frameworks such as the ISO and the NATO Standardisation Organisation—are increasingly necessary.

The growing regulatory burden has become a new challenge, particularly in Europe and Asia, where requirements relating to carbon footprints, material passports, and recycled content have sharply increased (Koppelaar et al., 2023; Peiseler et al., 2024). While these regulations reinforce corporate social responsibility, they may also delay technology transfer and export licensing in the defence sector. Achieving an optimal balance between regulatory compliance and technological innovation will therefore be a central task for future technology governance. The expansion of ESS and large-scale power platforms necessitates compliance with international safety standards such as NFPA 855 and UL 9540A. These frameworks establish minimum requirements to prevent explosions, thermal runaway, and fire propagation, ensuring operational safety throughout the defence industry (Fonseca et al., 2024).

Finally, cyber-physical exposure is emerging as a critical and growing source of risk. Battery management systems (BMS) and energy management systems (EMS) are now classified as operational technology (OT), making cybersecurity controls and vulnerability assessments essential (Gabbar et al., 2021; Šebo et al., 2023). As these systems constitute the core infrastructure of energy supply chains, any hacking or data manipulation could directly compromise mission performance. Therefore, beyond technological deployment, the institutionalisation of security governance and incident response protocols has become an indispensable component of defence readiness.

4 Battery and Energy Storage Governance in Defence

The strategic landscape of modern defence is undergoing a fundamental transformation driven by the electrification of military platforms. This evolution, primarily centred on high-energy-density ESS—particularly Li-ion technologies—presents significant operational advantages, including silent mobility, reduced thermal signatures, and enhanced mission endurance across multidomain operations. The expanding dependence on batteries across air, land, and sea domains necessitates a robust framework to manage supply chain vulnerabilities, system safety, life-cycle management, and cyber-physical resilience. Within Europe, this transition is inseparable from broader regulatory and sustainability frameworks, positioning electrification not merely as a technological shift, but as a structural component of long-term defence transformation under EU policy architecture (Racu & Poliscanova, 2024).

The integration of ESS is crucial for enhancing the Common Security and Defence Policy (CSDP), reinforcing operational interoperability and energy resilience. For Austria in particular—maintaining constitutional military neutrality yet actively participating in EU security cooperation—the challenge is twofold (Logothetti, 2005). The country must align with NATO standardisation frameworks (e.g., STANAG interoperability requirements) to ensure coalition readiness, while also complying with EU civilian regulatory instruments, such as the Austrian Battery Ordinance (*Batterien-Verordnung*), derived from the EU Battery Regulation (Te Kulve & Smit, 2003). Effective governance must therefore balance the urgent operational requirements of modern warfare with the increasing technical and environmental compliance burden. The following sections analyse how the electrification imperative shapes operational integration and the concurrent necessity for robust safety and security governance.

4.1 Electrification and Operational Integration of Defence Platforms

The comprehensive integration of ESS into military operations induces a structural transformation of military power that redefines the production, distribution, and sustainment of operational energy in contested environments, while simultaneously extending mission endurance and fundamentally reducing detectability. The growing dependency on battery-based power now spans sophisticated vehicle systems, advanced submarines, Unmanned Systems, and critical soldier-carried power systems. The core strategic driver is operational efficiency. ESS offers clear gains over traditional fossil fuel-based propulsion and auxiliary power units by providing lightweight and portable power solutions, decreasing the logistical burden of carrying heavy engines and improving agility (Mahadevan et al., 2025). For dismounted forces, lightweight and flexible LIB systems offer high-energy density and mechanical adaptability, enabling integration with wearable electronics and real-time health monitoring through standardised battery management interfaces

(Zhang et al., 2024b). These capabilities are largely derived from the adaptation of commercially developed energy technologies for military use. For example, commercial batteries and power systems are reinforced to withstand military environments and tested against operational standards at facilities such as the Aberdeen Proving Ground.

The key challenge is to ensure that these civilian technologies meet the stringent requirements of military operations, including high durability, electromagnetic interference resistance, and safety monitoring. In other words, it is not a matter of using commercial technologies as they are, but rather “enhancing and validating them for military applications” (Lafontaine, 2025; Ewing et al., 2002). In parallel, electrification contributes significantly to energy autonomy. Through the deployment of expeditionary microgrids that integrate batteries with diesel gensets and renewable energy modules, forward operating bases and military outposts can drastically reduce fuel dependency and logistical exposure (Katalenich & Jacobson, 2023; Labutis et al., 2025). This model supports distributed operations by decoupling combat power from centralised energy supply lines—historically a major vulnerability due to the targeting of fuel convoys (Simulcik et al., 2025). However, autonomy achieved through ESS introduces a new form of strategic dependency: reliance on foreign-controlled supply chains for critical minerals, batteries, and power electronics. Without proactive governance, the operational freedom gained at the tactical level risks being undermined by geopolitical leverage at the industrial level. Strategic planning must therefore ensure that operational efficiency does not mask long-term vulnerabilities in supply security.

Beyond efficiency and autonomy, the most significant tactical advantage offered by the integration of ESS is the enhancement of stealth and survivability. By using batteries, ground combat vehicles and intelligence, surveillance, and reconnaissance (ISR) equipment can carry out surveillance missions without running their engines. This means that noise and heat emissions from the vehicles or equipment are minimised, significantly reducing the risk of detection by adversaries. For example, battery-powered surveillance drones or ground sensors can operate covertly for several hours with their engines switched off, providing a direct contribution to intelligence gathering and faster decision-making, beyond merely serving as a source of power (Van Atta, 2013; Catenaro et al., 2021). The maritime domain exemplifies these advantages at scale. High-capacity LIB banks are replacing traditional lead-acid systems in submarine propulsion, notably demonstrated in advanced vessels such as Japan’s Sōryū and Taigei classes (Wertheim, 2022). This transition significantly enhances underwater endurance and reduces the required time for recharging, aligning closely with the capability requirements sought by European navies prioritising stealth and prolonged operational range. Yet integrating high-density ESS into submarines also introduces governance complexity, requiring rigorous engineering control frameworks, redundant safety systems, and thermal risk mitigation in confined, high-pressure environments.

Although Austria is landlocked, it is not isolated from the strategic implications of electrified defence. Positioned at the intersection of EU industrial policy and energy transition strategy, Austria benefits from strengthening its domestic battery

energy storage system (BESS) sector under its national Hydrogen Strategy and climate-neutral industrial roadmap (Federal Ministry for Climate Action, 2022). A resilient civilian BESS manufacturing base enhances national energy sovereignty and indirectly strengthens defence resilience by ensuring secure access to energy storage technologies that are increasingly central to operational capability. In doing so, Austria mitigates exposure to global supply volatility and positions itself as a contributor to European strategic autonomy in defence energy logistics.

4.2 Safety, Interoperability, and Cybersecurity Management

The integration of high-energy ESS necessitates comprehensive technical governance that manages safety, logistics standardisation, and cybersecurity in parallel. High-energy battery architectures introduce complex risk vectors; they require a robust governance regime that integrates safety certification, logistics regulation, and cyber-physical security. Unlike legacy fuel logistics, where hazards were largely mechanical and predictable, electrochemical systems exhibit systemic fragility: Li-based chemistries carry intrinsic thermal volatility and the potential for cascading failure, while system-level risk is intensified by the software-dependent control logic now standard across modern ESS architectures, creating a dual hazard profile—material and cyber-physical. As such, energy governance has emerged as a core dimension of defence resilience rather than a supporting technical function.

Safety governance begins with globally mandatory transport and abuse certifications. The UN Manual of Tests and Criteria (UN 38.3) is a baseline requirement for all deployable military energy assets, confirming survivability under transport stress conditions, including high-altitude simulation, thermal shock, vibration, short circuit, and crush tests (IATA, 2025). Compliance with UN 38.3 is a non-negotiable prerequisite for all defence procurement and deployment.

Beyond UN 38.3, military logistics are constrained by additional international safety mandates, specifically the IATA Dangerous Goods Regulations, which restrict the air transport of Li batteries shipped individually to cargo aircraft only. Furthermore, these batteries must maintain a state of charge (SoC) not exceeding 30% of their rated capacity (IATA, 2025). This safety constraint presents an unavoidable trade-off with operational readiness. While the 30% SoC rule minimises thermal risk during rapid air transport, it directly increases logistical friction. A battery arriving at 30% SoC is operationally inefficient, necessitating standardised packaging, coordinated doctrine among allies, and dedicated forward-deployed charging infrastructure upon arrival. Although standardisation mitigates physical risk, it inherently increases logistical complexity. Coherent governance is therefore essential to ensure that safety and operational efficiency do not undermine one another.

Energy interoperability has become equally essential to coalition warfare. Standards such as NFPA 855 and UL 9540A now inform NATO energy doctrine for forward microgrid installations, ensuring that thermal propagation risks are

structurally contained before multinational deployments. However, physical safety alone is insufficient—modern ESS platforms depend on embedded BMS, transforming energy infrastructure into an OT attack surface. A compromised BMS can trigger remote thermal runaway as effectively as a physical fault, making energy systems a new category of cyber-physical weapon target. This reality elevates compliance with frameworks such as NIST SP 800-82 from best practice to operational obligation, redefining readiness to include resilience against cyber manipulation of power assets (Johnson et al., 2022).

In Europe, the challenge is not merely to implement technical safety measures, but to ensure that these measures are integrated within a single, coherent regulatory framework. Traditional military standards alone cannot keep pace with the rapid rate of innovation in the global battery industry. To address this, European countries have begun establishing a dual compliance system, adhering to both NATO interoperability regulations and EU civilian battery regulations simultaneously. This approach ensures that battery systems are not only reliable on the battlefield but also comply with legal and safety requirements.

Austria provides a representative governance model in battery regulation. Through the early and strict implementation of the European Battery Ordinance, Austria has aligned domestic battery procurement, including civilian ESS, with EU safety and environmental legislation, supporting compliance and long-term supply integrity (Gaugl et al., 2023; Monsberger et al., 2025). This approach reflects a broader European shift from platform-centric defence governance towards infrastructure-centric resilience—treating energy not simply as fuel, but as a protected strategic system.

5 Governance Frameworks and International Cooperation

The accelerating militarisation of energy storage has placed governance—not procurement—at the centre of European defence policy. As operational energy shifts from fossil-based logistics to electrified force structures powered by ESS, the question for Europe is no longer whether to electrify defence, but how to govern electrification without eroding security or strategic sovereignty (Lucchese et al., 2020). This requires a governance architecture capable of synchronising alliance interoperability obligations with regulatory compliance under the EU's emerging industrial emissions and circular-economy mandates. The challenge is particularly acute in Europe because the defence sector is now deeply entwined with civilian industrial policy, climate legislation, and critical raw material strategies (Besch, 2019; Hackett et al., 2025). Energy, once considered a logistical support issue, has become a pillar of strategic autonomy. Under the EU's CSDP, energy resilience is now treated as a defence capability in its own right—central to force readiness, deterrence credibility, and the long-term sustainability of military power projection (Sabatino, 2022). The alignment problem is not technical but systemic: defence

energy cannot function outside NATO interoperability standards, yet it must comply with EU law if it is to be industrially viable across European supply chains. This dual framework defines the European governance question: Can electrification improve combat effectiveness without compromising sovereignty or industrial security?

5.1 Direction for Standard and Regulatory Alignment

Standardisation and regulatory alignment are the cornerstones of effective multinational defence operations and industrial sustainability. In Europe, technical governance now operates across two interdependent dimensions: interoperability, which ensures coalition capability under NATO standardisation frameworks, and regulatory conformity, driven by binding EU civilian law.

Military interoperability in Europe is defined by technical specifications established under NATO Standardisation Agreements (STANAGs) and National Military Standards (MIL-STDs). Effective multinational military operations rely heavily on the standardisation and interoperability of soldier-carried power systems. By ensuring that equipment from different allied forces shares common physical, electrical, and data interfaces, logistical friction is reduced and operational efficiency enhanced. Standardised power and communication interfaces allow energy systems to be more easily shared, replaced, or integrated across coalition forces without requiring extensive modifications, supporting seamless collaboration during joint operations (Moon & Clark, 2001; Гайдак et al., 2024). Austria, which participates in EU and UN missions while maintaining military neutrality outside NATO, must adhere to STANAG-based specifications to remain interoperable without formal alliance membership.

Yet military standards alone are no longer sufficient to guarantee energy security. The rapid innovation cycle in electrochemical technologies for energy storage and conversion is now driven by commercial energy markets rather than defence laboratories. This has forced European defence governance to recognise that STANAG compliance without EU regulatory alignment leads to supply chain isolation, undermining procurement flexibility. The EU Battery Regulation (2023/1542) has therefore emerged as a decisive policy mechanism, harmonising sustainability, safety, and product traceability requirements across the single market (Kirchner, 2023). Unlike earlier directives, it is legally binding, extra-territorial in effect, and directly applicable to defence procurement due to its control over upstream industrial capacity.

The Digital Battery Passport (DBP) requirement—mandatory from February 2027 for batteries above 2 kWh—exemplifies the new complexity in dual-use regulation. While designed to improve supply transparency and recycling efficiency, mandatory digital traceability raises unique risks for defence applications (Rizos & Urban, 2024). Continuous state-of-health (SoH) logging and traceable product identifiers risk exposing sensitive information if not protected within a restricted-access defence data architecture. The policy trade-off is clear: Europe

must satisfy traceability mandates without generating intelligence leaks regarding force readiness, platform locations, or energy consumption patterns. This calls for a controlled parallel governance framework—security-compliant sustainability—which integrates sustainability objectives into defence planning without compromising operational security, with sovereign military exemptions strictly limited in scope and purpose.

European efforts in this area are supported by targeted R&D aimed at standardisation and performance enhancement. The EDA supports strategic technology pipelines, such as the CERAMBALL project, which focuses on developing lightweight ballistic ceramics with reduced lifecycle emissions. Austria's leadership in this programme, through RHP-Technology GmbH, highlights the role of mid-sized European states in shaping the Defence Technological and Industrial Base (Carton et al., 2019). Austria's participation demonstrates its ability to act as a regulatory model, effectively integrating high-tech development with EU sustainability criteria, thereby contributing to collective strategic autonomy.

By merging NATO interoperability doctrine with EU industrial policy, Europe can convert energy compliance into strategic leverage—building a defence energy ecosystem grounded in accountability, resilience, and sovereignty. The alignment of these dual standards is therefore not simply a technical task but a foundational requirement for a credible European defence strategy. Table 4 summarises the principal governance frameworks for European defence batteries and energy systems, outlining the scope, objectives, and strategic implications of each standard. Collectively, these frameworks integrate NATO interoperability with EU regulatory requirements, thereby supporting operational resilience, regional production, and strategic autonomy.

5.2 Models for Joint Research and Industrial Collaboration

Achieving strategic autonomy in energy storage and conversion technologies requires robust governance of international cooperation in research, development, and industrialisation. This emerging “Defence–Industry–Academia–Alliance” governance model serves as the primary mechanism for technological advancement and risk diversification, prioritising policy outcomes over technical details.

Joint R&D frameworks function as policy instruments rather than discrete scientific ventures. Their core purpose is to shorten the innovation-to-deployment gap and distribute risk along the supply chain (Gholz, 2007). The Modular Open Systems Approach (MOSA), pioneered in US Department of Defence programmes and gradually adopted within European system-integration projects, exemplifies this philosophy. By embedding openness and modularity in design rules, MOSA allows partner states to co-develop sub-systems that remain interoperable regardless of proprietary origin. For Europe, this governance logic converts collaboration into capability: standardised R&D outputs can be inserted directly into national architectures without technical or legal friction, strengthening alliance flexibility while safeguarding sovereignty.

Table 4 Governance Frameworks and Their Impact on energy systems in European Defence (Hildebrand et al., 2024)

Governance framework	Scope/purpose	Impact/implications/ opportunities
MIL-STD-3078 (battery interoperability standard)	Ensures interoperability among military systems and battery packs	Defines allowable battery types (including commercial off-the-shelf types) for new system designs, facilitating shared logistics and integration
NATO STANAG 4695 (soldier power interfaces)	Standardises soldier power and interface specifications across NATO	Ensures physical, electrical, and data interface compatibility of battery systems among allied forces
EU battery regulation (regulation 2023/1542)	Harmonises battery safety, sustainability, and lifecycle obligations across the EU	Imposes QR code, recycled content, carbon footprint declarations, and traceability conditions for battery procurement, affecting military ESS acquisitions
European battery alliance (EBA)	Strengthens Europe’s battery value chain, reducing reliance on non-EU sources	Encourages localised battery cell and material production, which can support defence ESS supply security and strategic autonomy

The EU actively drives its collaboration vector toward achieving “non-dependency” by funding consortia through the EDA. Projects such as PATCH-BOND and ALOMAS pool expertise across member states, academic institutions, and defence industries, focusing on de-risking key components and manufacturing processes (Jokinen et al., 2024). This cooperative mechanism is a form of strategic governance, aimed at accelerating the move from experimental research to operational adoption—the so-called Experiment-to-Adoption Loop—scaling exemplars.

For mid-sized and non-aligned states such as Austria, participation in these cooperative mechanisms is crucial. While maintaining its non-aligned status, Austria actively contributes its core competencies to the EDA framework. Through its leadership in the CERAMBALL project, Austria contributes advanced protective-material expertise to collective defence objectives while reinforcing national industrial capacity (RHP-Technology, 2018). Domestically, the rapid expansion of civilian BESS incentives—evidenced by the quadrupling of subsidies within a two-week application window in 2025—illustrates how public incentives can accelerate residential, commercial, and industrial energy-storage deployment while supporting grid stability, energy independence, and security of supply (Enkhardt, 2025). The integration of national industrial policy into European defence R&D thus becomes a two-way conduit: civilian funding strengthens defence resilience, while defence certification elevates national industry standards.

Europe's collaborative governance model transforms interdependence from a vulnerability into a managed asset. By embedding openness, transparency, and sustainability into the very architecture of joint research, European defence institutions are constructing an innovative ecosystem capable of withstanding both technological disruption and geopolitical pressure. Strategic autonomy, under this framework, is not achieved by isolation but through disciplined cooperation governed by shared norms, interoperable standards, and distributed industrial sovereignty.

6 Conclusion

The transformation of European defence in the twenty-first century is increasingly defined by the governance of convergent technologies, where energy, materials, and digital systems intersect to create a resilient and adaptable military architecture. The integration of advanced materials, nanotechnologies, multifunctional composites, and energy storage systems into defence platforms has significantly enhanced operational performance, survivability, and strategic mobility, yet it simultaneously introduces complex interdependencies and vulnerabilities. The proliferation of dual-use technologies further blurs the boundary between civilian innovation and military capability, demanding a coordinated governance framework that addresses ethical, environmental, and strategic concerns. Europe's defence resilience now depends not solely on scientific excellence or industrial capacity, but on the ability to harmonise multilateral standards, regulatory compliance, and secure supply chains, ensuring that technological superiority translates into operational autonomy without exposing critical dependencies to external actors. In this context, the convergence of advanced energy systems—including Li-ion, Na-ion, flow, solid-state, and hybrid storage technologies—with lightweight, self-healing, and multifunctional materials exemplifies how strategic governance and technological innovation must operate in tandem to maintain European defence effectiveness in a rapidly evolving security landscape.

Strategic governance emerges as the central mechanism by which Europe can convert technological convergence into sustainable capability. The dual imperative of interoperability under NATO frameworks and compliance with EU regulatory mandates requires a nuanced approach, balancing operational readiness, environmental sustainability, and supply chain resilience. Initiatives such as the European Defence Fund, the European Battery Alliance, and targeted R&D consortia demonstrate the potential for structured collaboration between industry, academia, and allied partners. Such collaboration can accelerate the Experiment-to-Adoption Loop, reduce technological risk, and strengthen sovereignty over critical capabilities. Furthermore, the integration of energy storage systems into operational platforms—from forward-deployed microgrids to unmanned aerial and maritime systems—illustrates the operational dividends of such governance: extended endurance, reduced detectability, and enhanced autonomy, all underpinned by strict safety, cybersecurity, and lifecycle management protocols. By

embedding data governance, traceability, and sustainability principles directly into these frameworks, Europe ensures that dual-use innovations can be responsibly leveraged without compromising operational security or industrial competitiveness.

Looking forward, Europe's strategic autonomy will hinge on its capacity to institutionalise these governance models across the full lifecycle of defence innovation, from material development to deployment and maintenance. Resilient supply chains, diversified production, and ethical sourcing practices must be coupled with digital oversight, continuous monitoring, and interoperable standards to safeguard both operational freedom and strategic sovereignty. The holistic alignment of technological, industrial, and policy dimensions—encompassing advanced materials, energy technologies, additive manufacturing, and AI-enabled systems—provides not only a pathway to immediate operational advantage but also a foundation for long-term strategic foresight. Ultimately, European defence transformation is not merely a technical endeavour but a governance challenge: success will be measured not solely by the sophistication of platforms, but by the integrity, adaptability, and accountability of the systems and institutions that sustain them, ensuring that Europe can project power, maintain readiness, and safeguard security within an increasingly interdependent and contested global order.

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The Brain as a Battlefield: Current Status, Opportunities, and Challenges of Defence-Relevant Neurotechnologies

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Abstract

The development of neurotechnologies, including different kinds of invasive and non-invasive neural interfaces, wearables, and neuromarketing techniques, is progressing at an accelerated pace. While practical and technical limitations remain, medical breakthroughs have enabled further advances and commercial neurotechnologies drive broader proliferation. However, civilian neurotechnologies, including medical and commercial, possess significant dual-use potential as they can be repurposed in defence contexts. As the cognitive domain becomes increasingly central to modern warfare, neurotechnologies have also shifted into the focus of defence actors due to their perceived and anticipated benefits for future military operations. Although defence-relevant neurotechnologies are at an even earlier stage, several applications are already crystallising, ranging from restorative and enhancing functions to more seamless forms of human-machine integration (HMI). Simultaneously, they introduce novel vulnerabilities and risks, and it remains yet to be seen whether their potential benefits outweigh associated costs. Considering the rapid pace of development and remaining uncertainties, this chapter assesses the current state of the weaponisation of neurotechnologies and its opportunities and challenges. The comparatively early stage of development and deployment of neurotechnologies provides a window of opportunity, including for European actors, to address emerging political, ethical, and legal issues of dual-use neurotechnologies through responsible governance and regulation.

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1 Introduction

Rapid technological advances across sectors have simultaneously generated enthusiasm and raised concern among researchers, governments, and civil society globally. The development of neurotechnologies, which likewise is advancing at an accelerated trajectory, is no exception to this broader trend. Neurotechnologies are “devices, systems and procedures—encompassing both hardware and software—that directly measure, access, monitor, analyse, predict or modulate the nervous system to understand, influence, restore or anticipate its structure, activity and function.” (UNESCO, 2025a, p. 4). This encompasses both medical and non-medical applications, including tools that measure, infer, or influence nervous system activity and mental states, either through direct interaction with the nervous system, both invasive or non-invasive, or via interfaces with devices and systems (UNESCO, 2025a). These technologies are being developed by both civilian and military sectors and by a variety of actors around the globe.

The discourse around the weaponisation of neurotechnologies, while still under the radar comparatively to artificial intelligence (AI), has generated significant speculation, hype, and concern. Neurotechnologies have the capacity to revolutionise warfare through technological breakthroughs in the cognitive domain. While neurotechnologies have been in development since at least the 1920s, they have not been historically present on the battlefield (Kawala-Sterniuk et al., 2021). However, rapid technological developments in the commercial and scientific sectors may change the role that neurotechnologies might play in the military domain. NATO’s New Strategic Concept, in 2010, highlighted the inability to predict what may occur: “allies and partners should be alert for potentially disruptive developments in dynamic areas such as information and communications technology, cognitive and biological sciences, robotics, and nanotechnology” (Kosal & Putney, 2022, p. 82). While most known developments have been heralded by the United States, other key players such as China have also invested into the military use of neurotechnology. The future of the field involves potential for much change and uncertainty for warfare.

This chapter will address the current state of the weaponisation of neurotechnologies and the opportunities and risks that come with the militarisation of this technology. Furthermore, it will evaluate the challenges and risks that ought to be taken into account considering the future of neurotechnologies. Additionally, the impact that neurotechnologies might have on the European defence sector and ecosystem, and how it might need to adjust to stay at the forefront of technological potential will be assessed. This chapter also aims to provide an outlook and possible recommendations with regard to the future governance of neurotechnologies to maximise its potential and mitigate associated risks. Overall, it will address some of the ongoing discourse around neurotechnologies and provide a broad map of where these technologies may realistically lie in terms of their development and potential to influence warfare.

2 Status of Neurotechnology Development

To understand the current global state of the art of neurotechnologies, it is beneficial to take a broader look beyond military neurotechnologies by also considering developments in the medical and commercial fields. Civilian neuroscience and neurotechnology, including scientific, medical, and commercial research findings and products, can be repurposed and have implications in the political, security, intelligence, and military domains (Human Brain Project, n.d.). This makes neurotechnologies inherently ‘dual use’ in function. Advances in medical and commercial neurotechnologies are likely to result in an increase of dual-use opportunities and accelerate the military neurotechnologies. The dual-use potential of neurotechnologies may even create circular dynamics in which civilian neurotechnology is first coopted for military research but later leads to spillover back into the civilian domain through more reliable medical or commercial applications (Ienca et al., 2018a).

In the following, this section will provide a layout of relevant neurotechnologies and developments in the field through a state-of-the-art review of neurotechnology tools with application in both civilian and military contexts. The objective is not to present an exhaustive list but rather an overview of different neurotechnologies that are currently under development or being tested. While some examples remain speculative, this analysis follows an anticipatory approach, and its focus lies on exploring the dual-use potential of different neurotechnologies and possible weaponisation pathways.

2.1 Medical Neurotechnologies

A good place to start understanding the complexity and progress of neurotechnology development is through looking at the medical and scientific sector and literature. Historical development of neurotechnology has largely come through advancements made in medical and scientific research. While much of this research is intended for medical use, it also has dual-use potential. Neurotechnology development prevalent in the scientific and medical sectors can be divided into four main areas: brain research, treatment and therapy, restoration of cognitive and physical abilities, and neuroenhancement.

2.1.1 Brain Research

There is a segment of neurotechnologies being developed to further research and understanding on neural and brain functions. This research focuses on neurodiagnostics and brain mapping. It uses instruments such as brain–computer interfaces (BCIs) to map brain and neural activities with the goal of discovering how the brain itself functions. Recent developments such as micro-electro-mechanical systems-based (MEMS) (Yoon & Cho, 2024) or wireless BCIs (Vázquez-Guardado et al., 2020) that increase portability and ease-of-use, together with the increased use of

artificial neural networks for data extraction, classification, and decoding (Kawala-Sterniuk et al., 2021; Rickli & Ienca, 2021) have allowed for new experiments and novel insights into brain functions or diseases. Especially notable is the increased use of brain organoids. These are artificial brain-like structures created from stem cells that resemble the structure and function of the brain, and which are used for disease modelling and to study neurodegenerative diseases such as Alzheimer's without facing the constraints that researchers usually encounter when working with a human brain (Yoon & Cho, 2024). While previous brain organoids focused on creating such three-dimensional cellular models of one region of the brain (e.g. cortex), scientists at Johns Hopkins University grew the first whole-brain organoid with neural tissue from each region of the brain in July 2025 (Robbins, 2025). Experiments with brain organoids even go as far as transplanting human cerebral organoids into animal brains (Bao et al., 2021). Additionally, recent technological advances have led to novel types of brain models, so called digital twins, which are virtual representations of an object. Virtual brain twins are "computational models of human brains that are informed by subject-specific data" (Wang et al., 2024, p. 11). While still in an early stage, digital twins of individual brains enable personalised simulations, comprehensive analysis, and predictions of brain functions or can model the brain's response to injury or disease (Fekonja et al., 2024).

Dual-Use Potential

Brain mapping and brain organoids help with gaining a better understanding of the brain and enable an unprecedented ability to model human brain development and function (Hartung et al., 2024). Better brain data could provide insights for information operations and new opportunities for data-driven manipulation, which offers strong incentives for military brain research. This makes digital brain research often dual use in nature, as for example, brain models such as digital twins mimicking the human brain can enable simulations and individual predictions regarding the effectiveness of different neural interventions (Amunts et al., 2024). Similarly, brain organoids not only enable brain testing without the need for human experimentation for medical purposes but also for military applications. Brain organoids have already been used for neurotoxicity testing and studying blast impact on neural tissues. For example, veteran-derived human cerebral organoids were tested to uncover cellular responses to certain toxicants (Yates et al., 2022). They could thereby become a potential enabler for neuroweapons by testing their delivery mechanisms. Moreover, there is a clear potential for tailored therapies for individual patients while at the same time this could also allow for tailored targeting and exploitation of individual vulnerabilities (Yates et al., 2022). Brain organoids have also provided an opportunity to benefit trauma research among soldiers by studying the mechanisms of blast-induced traumatic brain injury (TBI) (Bar-Kochba et al., 2025). While still in a nascent stage, there is also increasing research on so called 'organoid intelligence', a combination of human brain cells and electrodes, as a potential new frontier of biocomputing (Smirnova et al., 2023). Notably, a project working on creating such intelligent brain organoids has

attracted significant funding from Australia's National Intelligence and Security Discovery Research Grants program (Houser, 2023).

2.1.2 Treatment and Therapy

Secondly, neurotechnologies are being developed to diagnose, treat, or provide therapy for neurological diseases and disorders. Neurotechnological instruments are commonly utilised to monitor brain health and to provide diagnostics. BCI-based EEG examinations make it possible to diagnose disorders such as epilepsy, Parkinson's disease, or attention deficit disorder (Kawala-Sterniuk et al., 2021). Other examination methods include functional near-infrared spectroscopy (fNIRS) or functional magnetic resonance imaging (fMRI). Furthermore, there are non-invasive instruments, for example, transcranial electric stimulation (tES) or transcranial magnetic stimulation (TMS), that are used for the treatment of some disorders (Kawala-Sterniuk et al., 2021). Another non-invasive form of neuromodulation of deep-brain structures can be found in the use of focused ultrasound stimulation (FUS), which has shown potential as a therapeutic tool, though its effects on the neural tissue and its safety are not yet fully understood (Micera & Foffani, 2024). Additionally, more invasive neurostimulation techniques, such as deep-brain stimulation (DBS) or brain drug delivery systems, can combat or ease symptoms of diseases and disorders such as Parkinson's or post-traumatic stress disorder (PTSD) (Chavarriaga et al., 2023; Rickli & Ienca, 2021; Yoon & Cho, 2024). Recent innovations include adaptive DBS, which adjusts stimulation in real time in response to individual brain signals, enabling more personalised and precise DBS (Williams, 2025).

Dual-Use Potential

Different stimulation techniques such as TMS, DBS, and FUS have demonstrated a potential to reduce PTSD symptoms and thereby offer possibilities for therapeutic applications and trauma recovery for military personnel and veterans (Manocchi et al., 2024). Electric brain stimulation has also proven to be effective with addressing memory problems of veterans suffering from TBI (Segado, 2017). The therapeutic potential in this case is immense as more than 500,000 military service members were diagnosed with TBI between 2000 and 2024 (McGovern, 2025). Moreover, technologies such as TMS can help with the prevention or minimisation of long-term neuronal injuries and support recovery (Demirtas-Tatlidede et al., 2012). The potential of these technologies goes beyond therapeutic applications, as the same tools that are used for diagnosis could also be used for personnel monitoring and the evaluation of cognitive, emotional, and behavioural states (Mantellassi & Madziwa, 2025). This appears particularly valuable in the light of military personnel's frequent exposure to extreme environments with high cognitive and physical demands and operational stress. The early detection of high cognitive strain, stress, or fatigue that could lead to poor decision-making may allow for interventions that help with sustaining soldier performance or accelerating recovery during operations or training (Mark et al., 2024). However, the generalisability of some lab-based findings to real-world military tasks remains

limited despite some benefits observed and requires further research (van der Groen et al., 2025). Lastly, medical diagnostic techniques used for neuroimaging could also serve as interrogation tools. For example, fMRI could potentially identify brain activity linked to deception during an interrogation (Rickli, 2020).

2.1.3 Restoration of Cognitive and Physical Abilities

A third group of neurotechnologies involve restoring the cognitive and physical capabilities of patients or test subjects. There has been a steady increase in the efficiency of technologies that deal with loss of senses such as hearing or vision (Rickli & Ienca, 2021). For example, scientists have repeatedly successfully utilised cochlear implants to restore the hearing capabilities of patients (Chavarriaga et al., 2023). Similarly, there are more widely available treatments for physical disabilities, such as Amyotrophic Lateral Sclerosis (ALS), spinal-cord injury-induced paralysis (Kawala-Sterniuk et al., 2021), and for the loss of limbs. Further, patients can restore some physical capabilities and subsequently improve their quality of life with the use of BCIs combined with neuroprosthetics. These neurotechnologies include prosthetic limbs (Krauel & Frewer, 2025) or external devices like motorised wheelchairs and spelling programmes (Goering et al., 2021; Kawala-Sterniuk et al., 2021). A recent breakthrough by researchers at Fudan University created a minimally invasive brain-spinal interface (BSI) that enables paralysed patients to walk again. Unlike traditional BCIs, this BSI does not rely on external devices or computers and restores nerve function through the implantation of two small electrode chips that establish a nerve bypass and direct communication link between the brain and spinal cord (Wang et al., 2025). Another invasive technique that has the potential of restoring physical abilities can be found in the subcategory of optogenetics, which is still very much in a development phase. It allows for the control of molecular processes through the activation of previously inserted, genetically coded proteins that respond to stimulation by light (Armbruster et al., 2024). Optogenetic treatments have the potential, for example, to partially recover visual functions in cases of specific neurodegenerative eye diseases (Sahel et al., 2021).

Dual-Use Potential

These neurotechnologies have attracted strong military interest due to their restorative potential, for example, for mobility rehabilitation of veterans who have undergone limb amputation or with paralysis as well as for the prospect of wounded soldiers' return to service (DARPA, n.d.-a). The integration of BCIs with prosthetics allows humans to translate mental commands into physical actions via devices, thereby enabling new possibilities for enhanced HMI such as a soldier's control of weapon systems through a neural implant (Koecke, 2025). It can enable soldiers to control devices to navigate both physical and digital environments, including post-injury (Mantellassi & Madziwa, 2025). Recent advancements have been made around restoring sensation in patients with prosthetic hands through timed electrical stimulation of the brain (Niewijk, 2025). Some related research

takes this further by embedding additional features into prosthetics such as environmental sensors, communication systems, or integrated weaponry, potentially turning a primarily restorative technology into combat-related prosthetic that could provide a tactical advantage on the battlefield (Rudder, 2024). Optogenetics techniques have been used to implant false memories into mammal brains (Liu et al., 2014). While the transferability of such cases to humans remains uncertain, optogenetics could enable new forms of behavioural manipulation.

2.1.4 Neuroenhancements

‘Neuroenhancement’ is the idea of enhancing and augmenting the capabilities of otherwise medically healthy individuals (Krauel & Frewer, 2025). It thereby aims to go beyond mere therapeutic or restorative potential. These neuroenhancements serve to improve a multitude of cognitive functions such as learning abilities including short- and long-term memory, as well as to influence and modulate emotions (confidence, courage, fear, and trauma) (Yoon & Cho, 2024). Improvement has been demonstrated through using TMS and transcranial direct current stimulation (tDCS) techniques. However, it has to be noted that these improvements are typically confined to experiments in laboratory environments, meaning that their application and effectiveness in real-life scenarios remain underexplored (Ienca et al., 2018b).

Dual-Use Potential

The possibility to enhance and augment capabilities of otherwise healthy individuals is of particular interest to militaries which have been exploring human enhancement to improve the performance of military personnel for a long time, including but not limited to neurotechnologies (BMLV, 2025). It has also been a subject within NATO, which defines neuroenhancement as “neuroscience-based techniques and technologies to alter central and/or peripheral nervous system activities to enhance mental function” (NATO STO, 2024). Similarly, Chinese military strategists consider the pursuit of neurotechnologies crucial “to advancing future battlefield effectiveness” as “mental confrontation could become a major feature of future conflict” (Kania, 2019, p. 87). China therefore sees neuroenhancements, such as using TMS or drugs specifically targeting the brain, as one possible way to leverage neurotechnologies to achieve advantage in complex battlefield environments by improving the cognitive and decision-making capabilities of military personnel (Kania, 2019; Wortzel, 2022). Various neurotechnologies offer the potential to enable sustained and optimised soldier’s performance both on the battlefield and during training. Different neuromodulation techniques (e.g. DBS, TBS, tES, TMS, tFUS) can boost cognitive functions such as alertness, attention, and memory which may in turn improve decision-making (Mantellassi & Madziwa, 2025; NATO STO, 2024). For example, in the case of tES improved perception and attention have been demonstrated which could help military personnel with managing stress or fatigue and increase resilience under pressure (NATO STO, 2024). If implemented successfully, this would allow for better management or control of psychological factors to maintain clarity in demanding environments.

There is also a potential to accelerate skills acquisition and learning in the military context. The application of tDCS during real-world flight simulator tasks led to an improved piloting performance of military pilots during and after training (Sun & Yuan, 2023). Despite the demonstrated potential, the translation from medical neurotechnologies and clinical testing environments to military application faces significant limitations such as portability, reliability in high-stakes and dynamic environments, as well as the fact that certain methods remain expensive (Mantellassi & Madziwa, 2025).

2.2 Commercial Neurotechnologies

Beyond medical use, the commercial sector has seen substantial investments over the past decade (UNESCO, n.d.) and is driving a significant amount of research and development (R&D) for civilian neurotechnologies. The recent engagement of private companies in neurotechnology development has fast-tracked innovation and lowered previous cost barriers (Harper, 2023). Commercial actors are transforming neurotechnologies from being highly complex and relatively inaccessible to providing products and services for casual everyday use. A majority of such technologies are being developed by start-ups and other companies such as EMO-TIV, Neuralink or Synchron. As a result, there has been a sharp increase in the number of accessible and affordable neurotechnologies that are being catered to consumers. Firms are creating direct-to-consumer (DTC) neurotech devices that use “associated accessories, mobile applications, software frameworks, and online services” to sell their wares to a broad base of purchasers (Ienca et al., 2018b, p. 5). Many of these commercial neurotechnologies tend to be wearable devices that measure factors such as productivity or mood. The market for wearables is expected to grow much faster than the one for implants over the next decade (Nataliya Kosmyna, as cited in UNIDIR, 2025, 1:09:30).

Companies are creating everyday usable neurotechnology devices using techniques and technologies that are widely used in the medical field. However, it is in commercial domains that a variety of technological innovations, including prototypes and functionalities not yet tested in conventional clinical settings, can emerge (Yuste et al., 2017). Commercial markets also play an important role as a significant commercial demand can justify the often-substantial R&D costs of new technologies and gradually reduce hardware costs, which in turn may enable wider military adoption and weaponisation (Cowan & Foray, 1995). According to UNESCO, which did an extensive analysis of scientific publications and patent applications, the number of patent applications concerning neurotechnologies doubled between 2015 and 2020, reaching 12,000 (Hain et al., 2023). This is an indication of rapid advancement in commercial neurotechnologies. In the following, this section will take a closer look at different invasive and non-invasive commercial neurotechnologies, as well as the emerging area of neuromarketing.

2.2.1 Invasive Neurotechnologies

Invasive neurotechnologies, such as different neural implants and invasive BCI-based technologies implanted in the brain or on its surface, are entering the commercial landscape. These products are the focus for many start-ups and companies, and while many are still in development or lack sophistication, they are a testament to the rapidly changing capacities of technology. Companies, such as Blackrock Neurotech, are using invasive BCI technology designed to allow people with paralysis and other neurological disorders to operate robotic arms through brain signals (Blackrock, 2024). Neuralink is also using invasive BCI neurotechnologies by surgically implanting a chip device that allows patients with quadriplegia to control neural keyboards (Guarino, 2024). The technology is still in clinical trials in the United States and in Canada (Reuters, 2024). Researchers at Brown University are working on a wireless microscale implant called Neurograins, a bidirectional neural interface that can record brain signals and provide electrical stimulation (Brown University, Robert J. and Nancy D. Carney Institute for Brain Science, n.d.). Another example of an invasive commercial neurotechnologies is Reach Neuro's development of a BCI device that uses spinal-cord stimulation to restore arm and hand movements to survivors of strokes. For this treatment, a pulse generator is implanted into the spinal cord to reconnect the brain and the nervous system. While the technology is still in development, it has received approval by the U.S. Food and Drug Administration (Reach Neuro, 2023). During the conducted clinical trial, it allowed patients to increase their grip strength up to 108% (Reach Neuro, n.d.).

Dual-Use Potential

The ability of invasive neurotechnologies to bridge between the human brain and external systems offers a variety of potential opportunities for the military domain. If successfully implemented, they would provide several cognitive and kinetic advantages on and off the battlefield. Firstly, through the integration of the soldier's brain with external systems such as battlefield sensors and communication networks, they can contribute to enhanced perception, increased situational awareness, and improved threat detection. By allowing to process vast amounts of data, BCIs combined with AI can support the soldier with analysing information quickly, reducing the cognitive burden and improving decision-making or threat detection (Neuroba, 2025). Secondly, invasive BCIs could allow for near real-time cognitive-state monitoring of military personnel. To enable effective real-time monitoring, BCIs need to be bidirectional rather than unidirectional, meaning they are simultaneously capable of recording neural activity and delivering targeted stimulation (Analysis & Research Team, 2024). By continuously tracking various brain and body indicators such as stress or fatigue, the technology can provide command structures with more precise insights into troop status and inform decisions on when to provide support or rest (Gielas, 2025b). It could thereby fulfil an early-warning function and even help with preventing or minimising the impact of certain psychological or physical conditions (Mantellassi & Madziwa, 2025). While this remains speculative, BCIs could, in an advanced form,

go beyond monitoring by enabling a more direct control over personnel, through commanders that remotely influence behaviour of soldiers on the battlefield (Mantellassi & Madziwa, 2025). Thirdly, BCIs can enable the control of unmanned systems, drones, or other military platforms by translating neural signals into digital commands which transmit an operator's intent to the system (Gielas, 2025c). In 2018, DARPA confirmed the ability of a paralysed individual with a neural implant to simultaneously command and control three jet aircraft via a simulation (Robitzski, 2018). It could thereby eliminate the need for physical interfaces such as keyboards, joysticks, or touchscreens currently required for remote control (Mantellassi & Madziwa, 2025). In the future, a single soldier could direct multiple devices simultaneously (Binnendijk et al., 2020). This would drastically reduce the number of operators required and has the potential to enable more seamless and enhanced forms of HMT. Fourthly, BCIs combined with virtual reality can create new training experiences for soldiers, including unprecedented degrees of personalisation and real-time adaption to soldier's cognitive and physical responses (Gielas, 2025b).

2.2.2 Non-Invasive Neurotechnologies

Non-invasive neurotechnologies, in particular wearable EEG-based devices, are having an impact on the consumer market. According to a recent mapping of the industry, EEGs have become the most widely used technology across commercial neurotechnology companies (Bernàez Timón & Mahieu, 2025). These technologies are often marketed to buyers for the purpose of self-monitoring fluctuations in attention levels or health (Ienca et al., 2018b). For example, the startup EMO-TIV has created a wireless EEG headset that collects brain data to analyse brain states, measuring concentration as well as emotions such as happiness (EMO-TIV, n.d.). Interaxon also has an EEG headband technology that can read brain activity involving focus, sleep cycles, and fluctuations in emotions such as stress and fear. This \$300 EEG headgear has already sold over 500,000 units (Muse, 2024). Similarly, China's state-owned companies in fields such as train operations are using non-invasive EEG helmets to monitor the attention span and exhaustion level of workers (Zhang et al., 2017). While certain challenges remain, researchers have made significant progress in recent years in decoding imagined speech and mental images from processed brain signals by using EEG-based BCIs (Guenther et al., 2024; Reyes-García et al., 2025). It is not only start-ups, however, that have invested in commercial neurotechnologies. Large technology companies such as Samsung also conduct research into commercial, non-invasive neurotechnologies. The company recently announced the development of an around-the-ear EEG device that monitors brainwaves (Samsung Newsroom, 2025). The use of such smartphone-compatible DTC neurotechnology devices will likely open an avenue into future technology products and social media that can be controlled via EEG-detected brain signals (Ienca et al., 2018b). By leveraging existing technologies such as sensors that are already integrated into smartphones, tablets, and virtual reality devices, neurotechnologies could be easily adapted for use in operational settings at a significantly reduced cost (McGovern, 2025). While not EEG-based,

Meta presented a non-invasive neural interface wristband in 2025 that detects electrical signals from muscles to control computers (Kaifosh et al., 2025). Such DTC devices could transform commercial neurotechnologies by allowing cognitive training in areas such as mental health and education (Ienca et al., 2018b).

Dual-Use Potential

The ability of many non-invasive neurotechnologies to monitor brain activity and different cognitive states provides significant promise for military applications. When announcing the projects of its Next-Generation Nonsurgical Neurotechnology (N3) programme in 2019, the Defense Advanced Research Projects Agency (DARPA) placed an emphasis on non-invasive or minimally invasive neurotechnologies to benefit healthy military service members (DARPA, 2019), indicating a trend towards increasingly non-invasive military neurotechnologies. In contrast to many other neurotechnologies, EEG-based devices can offer the advantage of a higher degree of portability and minimal invasiveness. However, technical limitations remain, including the quality of signal processing and accuracy (Chaddad et al., 2023). These limitations especially apply to consumer-grade EEG devices, although their usefulness for neuroscientific research is improving (Sabio et al., 2024). Many military tasks involve high cognitive demands in extreme environments and under pressure, which makes any insights into soldiers' cognitive states immensely valuable. Cognitive readiness in military service members, defined as "a person's ability to respond and adapt to the changes around them" (McGovern, 2025), is crucial for health, safety, and mission success. EEG devices, such as helmets or headphones with EEG sensors, could flag early signs of cognitive or physical decline of soldiers in mission environments and enable timely intervention by commanders (Borkelmans, 2025). For example, MIT's Lincoln Laboratory is developing portable, EEG-based cognitive tests specifically for the military that can help identify changes in cognitive performance in less than 90 s (McGovern, 2025). The U.S. Air Force has tested EEG devices to monitor pilots' cognitive and psychological states. In this study, eight aviators completed individualised and realistic flight scenarios while EEG sensors tracked cognitive strain in real time and evaluated how it influenced flight performance (Feltman et al., 2024). This also demonstrates how EEG devices can be used for improving specific cognitive skills such as attention and memory through dedicated military training (Suhail & Vinod, 2024). These insights can enable more informed training while at the same time support operational planning, both of which may ultimately contribute to overall mission effectiveness. In 2024, the US Air Force selected Arctop to use its BCI technology to decode brain activity to create training simulations that will help optimise training for cadets (Arctop, 2024). The combined use of EEG devices with BCIs by the Army of the Republic of Korea has been shown to improve soldiers' shooting accuracy by increasing concentration and calmness (Oh & Kim, 2022). At the same time, the real-time monitoring of EEG signals and attention levels can be used for the hands-free control of drones and other unmanned systems (Gong et al., 2025). For example, researchers at the University of Technology Sydney in collaboration with the Australian Army developed a non-invasive BCI

helmet, which used biosensors to read brainwaves and AI to decode them, to send instructions to a robotic dog (Bickerton, 2022). Lastly, the capability to decode neural data can support the detection of deception or manipulations, which enables potential applications in interrogation settings (Rosenfeld, 2019).

2.2.3 Neuromarketing

Neuromarketing first appeared in 2002 with the use of fMRI technology to better understand consumer behaviour. Since then, technological advancements have allowed it to diversify and gain a significant role in the marketing sector. Neuromarketing can be defined as “[...] the interdisciplinary fusion of neuroscience and marketing, emerg[ing] as a transformative field that deciphers the underlying neural mechanisms influencing consumer behaviour, offering unparalleled insights into unconscious decision-making” (Song et al., 2025, p. 2). Neuromarketing is mostly based on EEG devices that are cost-effective and can capture rapid moment-to-moment brain activity. Eye-tracking technologies are being used to study attention response to advertisements, allowing firms to develop strategies that effectively engage the targeted clients. However, eye-tracking technologies have limitations due to “the accuracy of eye-tracking data [which] may vary based on environmental factors, participant compliance, or technical constraints, such as calibration issues” (Song et al., 2025, p. 7). FMRI, through its capacity to analyse changes in blood flow in the brain, provides deeper insights into consumers’ decisions and emotions when exposed to an advertisement. The challenge with fMRI is its high cost, as well as the physical size of the equipment, which limits the sample size and restricts the range of scenarios the consumers can experience (Song et al., 2025, p. 9). Galvanic Skin Response (GSR) measures electrical changes in the skin caused by sweat glands. The variation of the GSR allows neuromarketing firms, such as BitBrain (n.d.) to identify positive or negative emotional reactions to presented marketing tools. Facial Electromyography (Facial EMG) “measure[s] and analyze[s] people’s electrical muscle activity in the face and [...] identif[ies] patterns between instinctive reactions and consumers’ emotional responses to marketing stimuli” (Song et al., 2025, p. 8). This highly accurate technology can assess the dynamics of moment-to-moment emotional events. However, as with the other methods, high costs, time-consuming procedures, establishing common methods of analysis, and difficulties in recruiting participants remain major challenges for neuromarketing.

Dual-Use Potential

Through its potential to improve the accuracy of predictions regarding individual or group behaviour, neuromarketing blurs the line between influence and manipulation (Dragolea & Cotirlea, 2011). It therefore poses potential threats to privacy and society when used unauthorised (Antoniak, 2020). Among the ethical issues related to neuromarketing are its potential to create campaigns that can manipulate an individual’s free will on a subconscious level (Şahin, 2023). Neuromarketing is not limited to consumer research, as demonstrated by the emerging field of political neuromarketing, which uses neuroimaging methods to enable predictive

insights into voter behaviour and political leader perception (Çakar & Filiz, 2023). Neurotechnologies that are used to understand neural activity, such as fMRI, offer a potential to identify individual vulnerabilities that can then be exploited to tailor disinformation campaigns and manipulate public opinion via content that targets subconscious processing (Vrábová & Vráb, 2025). Techniques used for neuromarketing could enable more effective influence campaigns by providing adversaries with insights into which stimuli shape perceptions or trigger emotional responses to maximise impact (Şahin, 2023). Due to neuromarketing's core focus on measurement, prediction, and the modulation of subconscious responses, it has the potential to become a powerful amplifier for hybrid and cognitive warfare and enhance attempt to influence cognition at scale. The same methods could be misused by malicious actors, including non-state actors, for manipulation and to more effectively instigate fear, doubt, or division within a society (Vrábová & Vráb, 2025).

2.3 Trends in Military Neurotechnologies

It seems likely that future wars will include certain neurotechnological components such as soldier enhancement or BCI-controlled weapon systems (Harper, 2023). Governments therefore have a direct interest in using neurotechnologies to advance their own military capabilities. Although combat deployment has not yet been achieved, neurotechnologies have the potential to be utilised by militaries in recruitment, training, combat, and post-combat phases (Kamieński, as cited in UNIDIR, 2025, 5:08:00). As neurotechnology can function in both attack and defence, understanding the current scale of military advancement in neurotechnology and where the interest in development lies can assist with identifying trajectories and potential risks. Following the demonstrated potential of civilian technologies being used for military purposes, this section explores some of the major trends in the field of military neurotechnology.

2.3.1 Military Research Focusing on Medical Treatments and Enhancing Warfighter Capabilities

A good way to look at the current and potential use of neurotechnologies for militaries is to see how these technologies can be deployed in the main stages of military service such as recruitment, training, combat, and post-combat phases (Kamieński, as cited in UNIDIR, 2025, 5:08:00). While much of the technology is still in development or is speculative, there are several known ways in which militaries are working with neurotechnology to increase overall competencies. Regarding recruitment, brain imaging techniques such as EEG technologies could in the future be used to select candidates based on certain traits such as evaluating who can be considered a high stress performer, or a quick thinker (Kamieński, as cited in UNIDIR, 2025, 5:08:00).

Militaries are also looking into researching technology that will help with training and combat situations by increasing the skills and efficiency of soldiers. Much

of the future development of military war neurotechnology is focused on the enhancement of warfighter capabilities through initiatives such as the Cyborg Soldier 2025 report by the United States Department of Defence (Emanuel et al., 2019).¹ Neurotechnologies dealing with enhancement of warfighters capabilities are meant to enhance the regular capabilities of the human body by creating novel attributes or by boosting functions over the normal range of performance (Kamieński, 2023). This type of warfighter enhancement neurotechnology is still in heavy development, with projects set to conclude decades into the future. Many of the currently known neuroenhancements for warfighters come from research conducted by the Defense Advanced Research Projects Agency (DARPA) (Kamieński, 2023). Further, the United States military established the Cognition and Neuroergonomics Collaborative Technology Alliance (CaN CTA) to track the advances in brain sciences for the purposes of understanding soldier performance. It brought together the US Army alongside researchers and experienced industry partners to understand soldier performance in operational environments (The Cognition and Neuroergonomics Collaborative Technology Alliance, n.d.). Official communication by the Chinese Communist Party includes cognitive enhancement of healthy people as a goal of its BCI research which is likely to result in commercial and military adoption (Mullin, 2024).

The closest avenue to military deployment of neurotechnologies is in the post-conflict phase through the use and development of neurotechnologies that have medical applications for the purposes of healing both physical and mental trauma that combat soldiers may face. This is an area of military interest in neurotechnology that has both civilian and military consequences. In an interview with the authors, Margaret Kosal, Professor at the Georgia Institute of Technology, stated that most military interest regarding neurotechnology involves medical components such as dealing with PTSD and healing soldiers who have lost limbs in combat (M. Kosal, personal communication, March 21, 2025). In the same vein, Yoon and Cho write about the potential of using brain stimulation to effectively address trauma in PTSD patients (2024). Additionally, DARPA has a programme titled “Systems-Based Neurotechnology for Emerging Therapies (SUBNETS)” which hopes to use neurotechnology “as the basis for effective, informed, and precise treatments for neuropsychiatric illnesses” for soldiers who may be dealing with trauma and PTSD (DARPA, n.d.-b). Israeli startup GrayMatters Health is also attempting to use its neurotechnology to PTSD in soldiers (GrayMatters Health, n.d.). This indicates that militaries are looking at a nexus between medical and military uses when it comes to neurotechnology development for post-conflict situations.

¹ The Cyborg Soldier 2050 was a research report designed to forecast and evaluate future military use of machines that have the potential to be physically integrated with humans for performance enhancement purposes.

2.3.2 US–China Competition for Dominance in Neurotechnology Development

Staying on top of emerging technologies and leading the world in R&D is a pathway to ensuring possible strong defensive and offensive military capacities. This makes neurotechnologies part of broader geopolitical competition. While many countries such as France, the United Kingdom, Japan, and Israel have invested and researched neurotechnologies, the two biggest players currently in terms of development are the United States and China. The United States has historically been the leader in neurotechnology development in the military sector and its research groups have been instrumental in the creation of breakthrough neurotechnologies. Much of the available information on military neurotechnology development comes from the US Defence Advanced Research Projects Agency (DARPA) (Kosal & Putney, 2022). Over the past decade, China has managed to shorten the gap or surpass other major research countries, including the United States, the United Kingdom (UK), Russia, Japan, or Germany (Hannas et al., 2024). The United States' dominance on neurotechnology research is therefore far from guaranteed, and China is catching up especially in BCI research (Mullin, 2024; Wong-Leung et al., 2025). American experts such as Kosal are concerned regarding the quick advancement of Chinese neurotechnology development and the uncertainty of the United States maintaining its dominance (M. Kosal, personnel communication, March 21, 2025). China has recently made BCI a priority of its new 5-year economic development plan and seeks to create an internationally competitive BCI industry for both medical and consumer use (Mullin, 2025). However, this could also enable military applications. State interests and public institutions are often merged, and in some countries such as China, a distinction is quite difficult to make. China's concept of civil-military fusion strategically pursues closer cooperation between the country's defence and commercial sectors (Bitzinger, 2021). This blurring of lines between the civilian and military sectors may enable faster defence acquisition of dual-use technologies (Kosal & Putney, 2022).

A possible indicator for a state's military interest in neurotechnologies is to look at the budget allocated towards their development. The US BRAIN Initiative and the China Brain Project are two initiatives that have research objectives relevant to military interests (Kosal & Putney, 2022). Reviewing government funding for such initiatives "gives a general sense of the level of investment in neuroscience and neurotechnologies" (Kosal & Putney, 2022, p. 90). Historically, the United States has invested significantly more resources into its BRAIN Initiative than China. Whereas the United States has put around USD \$2 billion into US BRAIN since 2013, there are no public figures for how much money China has invested into neurotechnology research (Mathieson et al., 2021). It should be noted that access to data from the China Brain Project is limited or unavailable (Kosal & Putney, 2022). This change in the global research hegemony may have consequences as China is historically less open and transparent regarding what technologies are being developed. Alongside budget allocation, China's interest in neurotechnology has also led to several key developments. For example, the China Brain Project

has been developing research monkey colonies for BCI trials (Kosal & Putney, 2022). NeuCyber NeuroTech (Beijing) Co, at the 2024 Zhongguancun Forum, unveiled invasive BMI technology (NeuCyber Array BMI System, n.d.) which allowed a monkey to direct a robotic arm to grab a strawberry solely through its thoughts (Xinhua, 2024). Further, Chinese state-owned enterprises, while not as well-positioned as North American initiatives, have created public-use technologies such as “Brain Talker” which “is a computer chip designed for reading brain activity to interpret ‘mental intent.’” (Kosal & Putney, 2022). Despite these advancements, it is difficult to know the current true functionality or potential use of technologies such as Brain Talker. Chinese research centres and universities have also made significant developments in BCI research that are likely to be militarised due to China’s civilian-military fusion in the neurotechnology industry (Hannas et al, 2024). CSET researchers have composed a list of some of the most noteworthy BCI advancements in China, many of which have been tested on live animals. These include CAS Institute of Automation creating a robotic system that implants electrodes into cerebral cortex’ of animals, Fudan University’s “first wireless BCI circuit for sending information between a chip and nerve cells” which is being installed on mice, and various other developments that work on a range of different BCI functions (Hannas et al., 2024, p. 15). Nonetheless, these new developments in neurotechnology offer new opportunities worth exploring in the military and civilian sectors.

3 Opportunities

While military neurotechnologies remain at an early stage and their current and future capabilities remain uncertain, the mapping of medical and commercial neurotechnologies as well as the described trends in the military domain have demonstrated how neurotechnologies may provide opportunities for militaries. In sum, neurotechnologies deployed in a military context can offer benefits across, but not limited to, the following categories: treatment and restoration, soldier enhancement and improved HMI, new tactical and operational possibilities, as well as optimised force economy and personnel management.

Firstly, the therapeutic and restorative potential of neurotechnologies provide significant upsides through the possibility of preventing or treating certain psychological disorders, such as PTSD, depression, and TBI, as well as physical conditions, such as paralysis and lost limbs, prevalent among military personnel. This would not only improve the well-being of active service members and veterans but could also reduce the violent or extremist radicalisation risks that can arise from some of these psychological conditions (Miller et al., 2024). In some cases, this could allow former service members who are otherwise deemed unfit to return to active duty. In the future, it may even empower people with certain disabilities to play a role in warfare, a potential showcased by a recent experiment in which a man with paralysis controlling a virtual drone through a complex obstacle course by thought (Sparkes, 2025).

Secondly, there are a variety of ways and functions in which neurotechnology can support warfighting on an individual level. The augmentative and enhancing potential of some neurotechnologies could enhance soldiers' physical and cognitive abilities, and result in improved military combat capabilities and positive battlefield effects (Yoon & Cho, 2024). The monitoring and potential improvement of cognitive functions such as focus, alertness, or the reduction of stress through neurotechnologies would be a significant advantage to soldiers conducting offensive or defensive operations, whether on land, air, or sea. Furthermore, neurotechnologies are one possible way, and a key enabler, to achieve enhanced HMI which is considered critical to adapt to increasing battlefield speed and support with increasing cognitive burden (Mantellassi & Madziwa, 2025). The integration of external information sources could enhance soldiers' perception and give them an anticipatory advantage during missions or combat.

Thirdly, on a macro level, neurotechnologies can provide benefits for the management of personnel and to optimise force economy. The continuous monitoring of soldiers through neurotechnologies such as EEGs can deliver valuable information and insights for operational planning. Through their integration into command and control, they can provide support for military decision-making by assessing the operational readiness of troops. Similarly, this data on how soldiers respond to environmental factors in certain scenarios or during training can inform team composition: it may allow to identify those best suited for specific missions based on an assessment of cognitive traits. Moreover, enhanced HMI enabled by neurotechnologies could improve overall force economy by reducing the size of military units, for example due to soldiers' ability to multitask or enhanced focus and endurance (Bahoque et al., 2025). This integration can also have a direct impact on the availability of personnel. If technologies such as BCIs are successfully leveraged to control unmanned systems, this could further minimise the need for direct human involvement in high-risk areas which in turn would reduce the risk of inflicted casualties (Neuroba, 2025).

Lastly, neurotechnologies could enable innovations around new tactical opportunities during missions and combat which may result in an advantage over adversaries that cannot leverage similar technologies. Neurotechnologies have attracted particular interest among special forces due to their potential to support operational success through increased speed, awareness, and endurance and their benefits to lethality and survivability (Tibbetts et al., 2025). Neurotechnologies offer the potential to reduce reliance on traditional military systems such as battle radio communication systems. The U.S. Military has funded research into silent transmission of information; relevant projects include DARPA's Advanced Speech Encoding and Silent Talk (Kamieński, 2023). So-called brain-to-brain interfaces could enable direct brain-to-brain communication during covert operations; however, they remain a proof of concept at this stage (Mantellassi & Madziwa, 2025). In sum, the potential benefit of neurotechnologies in the military domain makes them attractive at the tactical, operational and strategic levels, thereby providing incentives for weaponisation. However, many of the described opportunities also come with associated costs as they may introduce novel vulnerabilities and risks,

a phenomenon described as the capability-vulnerability paradox (Binnendijk et al., 2020). Some of the main risks will be outlined in the following section.

4 Challenges and Risks

Despite civilian and military neurotechnologies being largely in the development and testing stages, there are challenges and risks that have already crystallised which point towards future complications which may arise. Some of these risks amplify existing risks from other technologies while some novel risks emerge. These possible challenges include data-related risks, surveillance and manipulation risks, technology failure or developer dependency, agency and accountability risks, as well as non-state actor and misuse risks.

4.1 Data-Related Risks

The major risks that arise from the use of neurotechnologies, especially in the civilian context, concern data management and cybersecurity. The use of neurotechnology devices includes storing and gathering vast amounts of personal neurodata. Unsurprisingly, this intimate data can be incredibly valuable, not only for scientific studies but also for commercial marketing purposes. While medical data is well protected and processed under strict data management policies, the same cannot be said for the commercial sector. In many cases, where the service is provided by a commercial actor, the privacy and data management policies remain vague and they do not explain how the data is or may be used (Genser et al., 2024). This alone poses a risk for neurotechnology users, as their personal data may be forwarded to and used by third parties without them having any control over it. Further, the creation and storing of neurodata also invites cybersecurity risks, as malicious actors may try to acquire it through data theft. For illustration purposes, between 2005 and 2019, healthcare data breaches affected 249 million individuals globally, making it the most affected sector (Seh et al., 2020). In the case of neurodata, this could constitute major risks. Already in 2012, a team of researchers from the University of Geneva, Oxford, and Berkeley demonstrated that non-invasive neurotechnology devices could be used to deduce personal information such as PIN numbers and birth months. This deduction was achieved by analysing the brain regions stimulated in response to image-based cues. The researchers warned that malicious apps disguised as games or tools could allow criminals access to a device's raw data. They suggested that pairing such malware with 250-ms image stimulations, as used in the experiment, might reveal personal information (Martino-vic et al., 2012). Going even further, the importance of network security between the brain and the computer is also a crucial consideration as new vulnerabilities are introduced. For example, in 2019, scientists were able to hack into a prosthetic arm developed by Motorica Inc. and gain control over its data (Kaspersky, 2019). Therefore, the possibility of an adversarial takeover of neurotechnological devices,

be it in the commercial, medical, or military sector, is of real concern and may even have fatal consequences.

4.2 Risk of Surveillance and Manipulation

Naturally, neurotechnologies also open the door to the large-scale surveillance of its users. Since the data collected from devices may be used to deduce mental states, such as stress levels or emotions (Kritika, 2025), it enables a new level of surveillance. While, to date, there have been no documented efforts by a government to implement widespread neurotechnology devices to surveil its population, there have been cases where companies, for example, in the financial sector, used devices such as EEG headsets to analyse their employees in the workplace (cf. Kritika, 2025). If neurotechnologies become readily available and widely used, there is a high chance that either governments or multinational tech companies will be able to combine the users' brain data with other user data, allowing for a whole new level of profiling, surveillance, or even discrimination (cf. Muhl & Andorno, 2023). Furthermore, looking at the direct military implications, the interrogation of prisoners of war may become even more gruesome, as coercive neurotechnologies could potentially be misused in an adversarial context for mental manipulation and to inflict psychological harm, including up to torture (Human Rights Council Advisory Committee, 2025).

4.3 Risk of Technology Malfunctioning or Developer Dependence

Once an invasive device has been implanted, it exposes the user or patient to the risk of it malfunctioning or breaking, which would require further surgeries to either repair the device or remove it entirely. Additionally, all current implants corrode eventually and therefore have a limited lifespan (Binnendijk et al., 2020). However, this is the lesser risk when compared to what might happen if the developer or provider of a device goes out of business or discontinues their product or service. For instance, about 350 patients who had received an implant which partially restored their lost eyesight were presented with the issue of having the support to their device discontinued (Michelson, 2022; Wakefield, 2022). While the developer had decided to switch to a new technology, consumers were left to fend for themselves when their implant stopped working, with some having to resort to finding spare parts by themselves and some deciding to remove it entirely (Michelson, 2022; Wakefield, 2022). This can even have implications for an individual's feeling of identity as there may be unintended changes and consequences to individuals' self-perception after the use of neurotechnological treatments or prosthetics. Notable examples are patients feeling that a device is an integral part of their body (Goering et al., 2021) or reported changes to patients' personality and identity following deep DBS treatments (Gilbert et al., 2017; Schermer, 2011;

Zuk, 2018). Private consumers as well as militaries therefore need to thoroughly consider the suppliers of their neurotechnologies to ensure continued support throughout the entirety of their lifecycles and to hedge against the devices becoming obsolete. Due to remaining accuracy and reliability, the deployment of military neurotechnologies therefore appears more likely to first happen in stable settings before entering more dynamic environments such as on the battlefield.

4.4 Agency and Accountability Risks

Another frontier of risks which will be introduced through the popularisation of neurotechnological devices on and off the battlefield will be the issue of how to attribute agency and accountability. Since, for example, brain implants can heavily influence the cognitive abilities and thought processes of their users, it might become impossible to tell if a decision was genuinely taken by the user or if it was taken or influenced by their device, thereby posing the question of responsibility. This could be further complicated by the convergence of neurotechnologies with other emerging technologies. Some legal scholars have already argued that in the future it might become a common defence in front of a court to blame an action on a malfunctioning or 'hacked' implant to rid themselves of responsibility (Cabrera & Carter-Johnson, 2020). This issue is further exacerbated due to the fact that many actors are involved in the creation and operation of devices. In case of an incident which was caused by a malfunctioning device, for instance, the blame could fall to the manufacturer, a fault in the code underlying the algorithm or the AI which operates the device, the clinician who implanted it, as well as any other actor in that value chain (Cabrera & Carter-Johnson, 2020). When looking towards the use of neurotechnologies in warfare scenarios, this poses a large risk, as the responsibility for war crimes, for example, could become incredibly hard to attribute. Aside from this issue, which in itself already poses a sizeable challenge for International Law, there might be another legal challenge which might arise if warfighters with implants such as BCIs are deployed. Due to their enhancements, these fighters might run the risk of meeting the standards to classify as a weapon, which in turn could lead to their human rights or prisoner of war protections being questioned (Gielas, 2025a).

4.5 Non-State-Actor Related Risks and Misuse

Another risk related to the increased availability and democratisation of neurotechnologies is its adoption by malicious non-state actors such as criminal or terrorist organisations. Dr. James Giordano, Professor in the Departments of Neurology and Biochemistry at Georgetown University, when interviewed by the authors, proposed three future potential scenarios through which military non-state actors could potentially gain access to these technologies (personal communication, March 18, 2025). In the first scenario, armed groups could serve as surrogate

for state agencies. This would complicate attribution and preserve a degree of plausible deniability for any illicit activities carried out by these surrogates (Krieg & Rickli, 2019). In such cases, a state might claim that the already weaponised neurotechnology was acquired by non-state actors through illegal channels, distancing itself from media scrutiny or international repercussions. Secondly, malicious non-state actors could directly acquire these technologies through commercial, grey, or black markets. This pathway would require these groups to weigh the costs and benefits of obtaining such tools. However, as the global neurotechnology market expands and prices fall, this avenue may become increasingly accessible. Finally, malicious actors could purchase commercial neurotechnology products and reverse-engineer them. By doing so, a device originally intended for well-being applications could be repurposed as a tool of coercion, control, or direct harm. The existing fragile legal framework has created opportunities for such groups to potentially experiment and repurpose neurotechnologies for unethical purposes.

Moreover, the rise of Do-It-Yourself (DIY) neurotechnological devices and the online communities that support them have accelerated innovation. However, it also creates opportunities for malicious actors to master these technologies. Another new dimension in commercial neurotechnology is the emergence of DIY (Do-It-Yourself) neurotechnology, which has “empowered non-professional individuals (often self-proclaimed biohackers) to self-assemble neurotechnology devices for personal use, most frequently via transcranial electrical stimulation for self-improvement purposes” (Rickli & Ienca, 2021). However, academic researchers caution that improper application of these at-home kits, especially when lacking medical guidance, can result in adverse effects or unreliable data (Fitz & Reiner, 2015). According to interviewed experts, the devices function more like consumer gadgets, offering only limited information rather than robust tools suitable for medical or military applications. The neurodata quality required in military sectors is far higher than what can be offered by most devices in the commercial market (R. Jolivet, personal communication, March 3, 2025).

5 Neurotechnologies and the European Defence Sector

Neurotechnologies are increasingly recognised as a transformative technology that intersects with national security (Giordano, 2014) and an area of interest for the defence sector in Europe as well. This is a result of recent neurotechnological advancements, but also the growing prominence of the concept of cognitive warfare, defined as the ability to control how and what people think in order to control how they act (Rickli, 2024 and Hung & Hung, 2022). This concept has gained a lot of traction in the United States, Russia, and China but also within NATO, which is developing its own cognitive warfare concept (Krewson, 2025). Neurotechnology’s relevance in a defence context is thus reflected in the technology’s conceptualisation in recent doctrines on cognitive warfare (Bebber, 2025) but also

in the emerging concept of neuroweapons defined as tools and systems specifically designed to target or influence the brain and nervous system of the enemy (Eyre et al., 2023). This takes place in the context of a broader shift in European defence towards rearmament and increased defence spending. This section provides an analysis of the European defence ecosystem around neurotechnologies and looks at the current state of their integration by mapping relevant actors, including private companies, academic research initiatives, defence projects, and existing funding mechanisms.

There is an emerging ecosystem of European companies in the private sector developing neurotechnologies. The Centre for Future Generations Neurotech Consumer Market Atlas looks at the number of dedicated neurotechnology companies in selected European countries. According to this report, numbers in the United Kingdom (22), France (10), Germany (9), the Netherlands (7), and Italy (3) were lower than in the US (117), while making up 38% overall (Bernàez Timón & Mahieu, 2025; Helsing & Schlecht, 2025, p. 21). Two notable examples of European companies recently achieving significant neurotechnological breakthroughs are CorTec GmbH and Inbrain Neuroelectronics. CorTec GmbH is a medical company developing implantable BCI systems that can record brain activity, interpret signals, and deliver targeted electrical stimulation, and which completed the first human implantation of such a system made in Germany in July 2025 (CorTec, 2025). Inbrain Neuroelectronics, a Spanish company focused on graphene-based BCI technology, successfully conducted the world's first in-human clinical study of such systems for medical application (BioSpace, 2025). There are also early-stage neurotechnology start-ups that have recently received seed funding such as the Norwegian firm Cerebionics which seeks to enhance human-machine collaboration by developing BCIs specifically for robotic platforms (Project Europe, 2025). Some of these products intended for medical or commercial use simultaneously offer dual-use potential or benefits such as for neuroenhancement, unmanned platforms, or HMT, and therefore have relevance in a defence context. Other neurotechnology companies are working with government agencies, have documented links to defence-related projects, or have ambitions in this regard. This includes g.tec medical engineering GmbH which focuses on invasive and non-invasive BCI hardware and software for clinical and research use (g.tec, 2025). Founded in 1999, the Austrian firm joined a European Defence Fund (EDF) project focused on neurotechnologies as a consortium member in 2024, thereby entering European defence R&D (European Union, 2025). The British company MintNeuro is building microchips that can be used as neural implants. In January 2025, the company was awarded three UK government contracts to support adoption in clinical settings (PR Newswire, 2025). The funding is part of the Precision Neurotechnologies programme coordinated by the Advanced Research and Innovation Agency (ARIA), a UK government agency inspired by DARPA that also promotes defence-relevant research (ARIA, 2025; Reynolds, 2024). Others such as the Swedish company Neuroba are developing neurotechnologies that explicitly envision applications in a military environment (Neuroba, 2025). Overall, there is a growing number of European neurotechnology companies, but only a few of them

are developing dedicated defence applications. Additionally, commercially developed neurotechnologies have found limited translation into military applications so far.

Europe is home to leading and internationally recognised academic and research institutions in the field of neuroscience. Germany, in particular, is represented prominently within the European neurotechnology research landscape with institutions such as the University of Freiburg, the Technical Universities of Munich as well as the Max Planck Institute for Brain Research (Hensing & Schlecht, 2025). While Germany ranked fourth globally in neurotechnology-related patents in 2020, it was first with regard to most valuable patents and third overall in terms of neuroscience publication output from 2020 to 2021 (Hain, 2023). There are also neurotechnology research clusters such as the Grenoble-Neurotech in France that bring together different regional laboratories or cross-European alliances such as NeurotechEU with nine participating universities (Grenoble Neurotech, 2024a, 2024b; NeurotechEU, 2025). Other European-level initiatives included the Human Brain Project which ran from 2013 to 2023 and led to the creation of the collaborative EBRAINS infrastructure that links institutions and supports neuroscience research (Human Brain Project, 2023). European institutions contribute extensively to advancing the scientific knowledge base around neuroscience and neurotechnologies but their impact beyond academic and medical research is limited.

There are few government programmes on a national level with a specific focus on neurotechnologies in Europe. Military engagement with the topic appears to have been largely analytical so far through studies on related aspects (BMLV, 2025; UK Ministry of Defence, 2021). One of the few governmental neurotechnology initiatives is ARIA's Precision Neurotechnology programme, which is focused on medical applications and lacks an explicit defence dimension. The most significant European project with a dedicated defence application is NEUROQUAD, which was launched in 2024. The project focuses on neural monitoring and prediction of military pilots' cognitive state in real time with the help of neurotechnologies and AI, potentially enabling more adaptive and effective HMI (European Union, 2025). The 3-year project is funded with approx. 3.9 million EUR by the EDF and includes consortium members from six European countries from the public and private sectors. Although it represents one of Europe's first coordinated military neurotechnology initiatives, NEUROQUAD is still at an early stage, and no public outcomes have been published yet. Public funding plays an important role in neurotechnology projects in Europe. However, there is very limited investment in actual R&D of neurotechnologies with defence applications. This stands in stark contrast to the openly communicated and large-scale projects funded in the US and available venture capital, leading to a funding shortage for European neurotechnology start-ups (Hensing & Schlecht, 2025). Recent efforts seek to increase defence-related investments and funding for dual-use projects by amending existing EU programmes such as the Strategic Technologies for Europe Platform (STEP), Horizon Europe, and the EDF (European Parliament, 2025). These programmes sometimes include neurotechnology-related initiatives

but lack a direct, strategic focus on neurotechnologies. There are also ongoing efforts within the NATO context that are not solely European such as the Defence Innovation Accelerator for the North Atlantic (DIANA), which is modelled after DARPA. This accelerator programme works with private companies, including from European NATO countries, to develop dual-use technologies. It does not mention neurotechnologies specifically but includes related aspects such as human health and performance (DIANA, 2025).

In sum, Europe has strong research and medical neurotechnology focus with leading technical capabilities, but this academic and research excellence has thus far had limited translation into commercial as well as defence applications. The commercial neurotechnology landscape in Europe remains limited and at a relatively small scale. A 2025 survey found that 60% of neurotechnology companies globally are now commercial and thereby outnumber medical ones (Mahieu, 2025). This stands in contrast to European companies remaining concentrated in the medical sector. Neurotechnologies developed by commercial actors often have potential for dual use but neurotechnologies that are developed with a direct military purpose in mind are an exception. This is attested by a small number of neurotechnology defence projects and the lack of any flagship initiatives on both national and European levels as of late 2025. The comparatively small number and scale of projects is often a result of limited funding opportunities, both public and private, and regulatory hurdles (Hensing & Faika, 2025). In contrast, a strong commercial sector in the US is receiving substantial public funding backed by venture capital whereas in China strategic state interventions encourage neurotechnology development and adoption. There is also an existing fragmentation as no unified European strategy, roadmap, or dedicated programmes for the development or deployment of neurotechnologies exist. Efforts are further complicated by the fact that NATO may be perceived by some actors as the primary venue for military neurotechnology development. Nevertheless, there has been more attention and some signals for increased investment recently in European neurotechnologies, partially in the context of a broader shift around defence spending, although many of these projects often have a long-term outlook and are unlikely to show quick results.

6 Neurotechnology Governance: Outlook and Recommendations

A growth in neurotechnology development has hastened the necessity to provide a framework which can regulate and ensure safeguards regarding ethical governance for brain data and technology use. Especially, the rapid growth of commercial neurotechnology creates a pressing need to legislate and create data protection policies for the protection of civilian information. Institutions such as UNESCO and the Neurorights Foundation are leading the global charge towards the creation of a new governance scheme that will be able to deal with concerns regarding the rapid development of civilian neurotechnologies. These policies are now being picked up by national and local governments of nation-states in an effort to provide legislative

protection of neural data and rights. This section looks at the efforts made by international bodies to develop the global governance scheme for neurotechnologies and then focuses on the evolving legal landscape in certain jurisdictions.

6.1 Global Governance Development

Regarding global governance, UNESCO's Recommendation on the Ethics of Neurotechnology presents a comprehensive framework to address the challenges of neurotechnology governance and it is the "first global normative instrument" concerning the ethics of said technology. It provides states with a roadmap for implementing legislation to address "scientific, technological, economic, educational, legal, regulatory, infrastructural, societal, cultural and other dimensions" of neurotechnology use no matter at which stage of development states may be in (UNESCO, 2025a, p. 12). The Recommendation, while non-binding, is comprehensive and based on human rights concerns regarding emerging technologies and serves as an important tool for the development of national legislation. The document was adopted by Member States of UNESCO in November 2025 at its 43rd Session of the General Conference (UNESCO, 2025b).

The Neurorights Foundation is a non-profit organisation, based in the United States, whose goal is to ensure a future where neurotechnologies are used for social good and their application is safeguarded from misuse. The foundation focuses on research and advocacy, and has led the field in setting standards for the governance of neurotechnologies. It has pushed countries into adopting legislative frameworks protecting civilian brain data collected by technology companies. In April 2024, the Neurorights Foundation published a report stating that "[a]ll current international frameworks for specifically regulating neurotechnologies are soft law, [...] offering no meaningful address to the human rights challenges posed by neurotechnologies, nor do they explore their relationship to international human rights law" (Genser et al., 2024, p. 13). To properly protect brain data, the Neurorights Foundation advocates for governments to adopt hard law protections including constitutional amendments (Neurorights Foundation, n.d.-a).

International governance bodies are increasingly adopting new frameworks regarding the governance of neurotechnologies. Beyond UNESCO, the adoption of the non-binding León Declaration on European Neurotechnology: A Human Centric and Rights Oriented Approach in 2023 by EU members marks a step forward (UNIO, 2023). This Declaration focuses on collaboration between signatories to ethically innovate emerging neurotechnologies while ensuring a human-centric and rights-oriented approach to development (Council of the European Union, 2023). While Europe has a strong data protection basis, existing frameworks to not explicitly mention neurodata. Additionally, the OECD also considered best practices towards neurotechnology development through their "Recommendation of the Council on Responsible Innovation in Neurotechnology" which contemplates and warns against potential unintended consequences of the development

of neurotechnology (OECD, 2019). While these frameworks aim to be comprehensive and advanced, they lack mechanisms to enforce protections for civilians whose brain data may be vulnerable to external uses. Many of these frameworks have limited engagement from various stakeholders, including industry.

6.2 Government Legislative Action

There has also been movement within certain national governments on legislating brain data protection relating to neurotechnology, although the legislative landscape is still in its infancy. In 2021, Chile was the first country to implement major legislation relating to the inclusion of neurorights into its Constitution (Constitución Política de la República de Chile, 2022). The constitutional amendment has proven effective thus far, in August 2023 Chile's Supreme Court ordered the private company EMOTIV to delete all its recorded brain data belonging to a Chilean citizen (Genser et al., 2024, p. 16). The Neurorights Foundation championed the implementation of this legislation in Chile, and for the court case the foundation collaborated with Senator Guido Girardi, who provided the necessary political backing and was the plaintiff in the court case. Senator Girardi's claim against EMOTIV to delete his brain data led to the enforcement of the amendment made in 2021 to the Constitution (Cornejo-Plaza et al., 2024). Chile's case has since become a global model for regulating brain data and protecting mental integrity (Neurorights Foundation, n.d.-b).

Government bodies are increasingly legislating or beginning to develop legislation on regulating brain data. Although not directly naming neurotechnologies, the European Union has implemented regulations regarding non-medical neurotechnologies through Implementing Regulation EU 2022/2346, which is a binding directive (European Commission, 2022). This legislation focuses on Medical Device Regulation and through this directive it has additionally regulated non-medical neurotechnologies. Specifically, Regulation EU 2022/2346 captures neurotechnology by regulating requirements for manufacturers of non-medical devices by ensuring a harmonised approach to manufacturing of such devices (Bublitz & Lighthart, 2024, p. 13). In the United States, the federal Senate has introduced a bill titled "MIND Act of 2025" which directs the Federal Trade Commission to "conduct a study on the governance of neural data and other related data, and for other purposes" (MIND Act of 2025). The proposed study will last a year and recommend a regulatory framework that prioritises neurotechnology innovation while protecting neural data [MIND Act of 2025, 2025, § 4(a)].

Beyond the federal government, some US states such as Colorado, California, and Montana have enacted legislation to require companies to safeguard brain data that is collected outside the medical field (Ruder, 2025). California unanimously codified a law in September 2024 under its Consumer Privacy Act that included neural data under "personal sensitive information" for consumer privacy purposes (Moens, 2024b). Earlier that year, Colorado expanded the definition of privacy under the Colorado Privacy Act to include brain neural data in an effort to protect

consumer data. The amended legislation protects neural data in the same way other sensitive biometric data such as fingerprints and facial images are protected (Moens, 2024a). There is a possibility that these state laws start a precedent for other jurisdictions to enact similar laws (Moens, 2024b).

Aside from the few examples of binding legislative directives explained above regulating brain data, most of the frameworks and documents that deal with neurotechnology governance have no enforcement power. They provide guidelines for best practices and rely on states to voluntarily implement the detailed provisions into their own legal structures. The governance structures discussed in this section also largely concern brain data from commercial applications of neurotechnologies rather than military usage or 'dual-use' neurotechnologies. It is often not clear how these frameworks relate to the military context, if at all. The international governance landscape could benefit from a specific treaty or an agreement on the current and future use and development of military neurotechnology and their implications, such as for instance, in the field of subversion. Whereas the practice of coercion is regulated by international and several types of treaties such as arms control agreements, subversion is not. This is partially due to the fact that it is more difficult to identify. Yet, the growth of digital and neurotechnologies increasingly enables the conduct of cognitive warfare. The international community shall therefore seriously think about the developing an international regime for subversion control (Rickli et al., 2023).

7 Conclusion

Neurotechnologies are attracting increasing investments from both public and private actors. While many neurotechnologies remain at an early stage of development and operational examples outside of controlled settings remain quite limited, there has been a remarkable pace of development in recent years. As this chapter outlines, this includes different kinds of neural interfaces, invasive BCIs, non-invasive wearables, and other means such as neuromarketing. Although several practical and technical limitations such as the quality and accuracy of data remain, certain commercial neurotechnologies, in particular wearables, are proliferating rapidly. The convergence of neurotechnologies with AI advances has further accelerated this development trajectory. If this accelerated development continues, neurotechnologies may be on a growth trajectory that is similar to the one of AI, possibly resulting in widespread adoption and use.

At the same time, these civilian neurotechnologies, including both medical and commercial, are inherently dual use as they have military potential and can be repurposed. Additionally, dedicated military neurotechnologies are being developed by military research institutions. Based on publicly available data, military neurotechnologies are even more in their infancy but we are already starting to see potential military uses. As the cognitive domain increasingly moves into the focus of future warfare through concepts such as cognitive warfare and neuroweapons,

neurotechnologies that target the brain are naturally gaining attention. Most neurotechnological development and testing is limited to only a few countries but there is a potential for a military neurotechnology arms race similar to other technologies, with such dynamics already visible in the US-China neurotechnology competition. It is therefore possible that the weaponisation and deployment of neurotechnologies will eventually become a reality similar to how AI has entered the modern battlefield. Based on the current development trajectory and anticipated benefits, neurotechnologies are likely to prove useful for future military operations. However, it remains to be seen if these potential military benefits outweigh the associated costs such as novel vulnerabilities and potential risks. In addition, uncertainties remain around future capabilities and the effects of the convergence of neurotechnologies with other technologies such as quantum or AI.

In the light of this, there is a need to anticipate and prepare for wider technological implementation and neurotechnologies demand more societal, political, and regulatory attention. There is a very limited number of existing frameworks that deal with the implications of neurotechnologies in general and even less so for their dual-use applications. The still relatively early stage of development thereby provides a window of opportunity to address emerging ethical and legal issues before neurotechnologies become widely deployable. While European neurotechnology players, in particular in the medical sector, have achieved breakthroughs, Europe risks falling behind other major actors such as the United States and China. However, there is a possibility for Europe to play a key role in the development of neurotechnology governance and regulation through a concerted effort while the technology is at its current stage, both with regard to civilian as well as dual-use applications. This could help with mitigating the outlined risks while ensuring neurotechnologies' transformative potential.

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Governance for Quantum Technologies: A Necessity for Responsible Technology Development

Helmut Leopold

Abstract

Quantum technologies enable potential disruptive changes by means of quantum computing, quantum communication and cryptography, and quantum sensing for both the civil and the defence domain. Quantum technologies offer potentially enhanced capabilities for intelligence, surveillance and reconnaissance as well as critical advancements in information security and cryptography. This paper provides decision makers with an overview of the different quantum technology fields and main application areas. Since quantum technology has by its nature a strong dual-use character, potential use cases for the defence sector are highlighted. The level of maturity of quantum technologies in the various fields is discussed and recommendations for next steps are derived for stakeholders to shape further quantum technology development. In order to understand potential risks and ethics considerations, it is of fundamental importance to complement the R&D activities with a sound global multi-stakeholder dialogue. Thus, this paper discusses the essential factors for the design of effective governance rules and summarises recommended measures to develop and operate quantum technology-based solutions in a responsible ethical framework in the civil as well as in the defence domain following digital humanism objectives.

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1 Introduction

The rapidly advancing digital transformation in all areas of our lives and economies towards a global cyberspace¹ is fundamentally changing our social life, our work processes, and our economy and industry (Leopold, 2026). In the same way, the security and defence sector is also subject to fundamental disruptive changes, which must be transformed into effective defence strategies and concepts. New competencies and skills are needed in organisations to ensure the effective use of new and disruptive technologies. In addition, it is essential to develop new technologies and create effective framework conditions so that we keep our technologies manageable and take ethical standards into account in their long-term development.

Digital technologies such as transistors, microprocessors, laser technology, and MRI² in medicine are already based on quantum-physical effects and are therefore referred to as first-generation quantum technologies. However, new second-generation quantum technologies, which are currently in research and development, will open up even more far-reaching possibilities. The development of new quantum technologies is expected to lead to particularly disruptive market changes, as significant investments by the private sector as well as by governments in quantum research and system development are driven in a global context (McKinsey, 2024).

It is now important to recognise their potentials and convert them into applicable solutions and economic capabilities and also to support the achievement of the sustainable development goals (OQI, 2024). These developments, which bring new opportunities, but also open up new threat potentials in a global context, must also be positioned in the security and defence sector (Malekos et al., 2024). It is necessary to identify and develop new areas of application, but also to understand the threat scenarios posed by these new technologies and to take them into account in future defence strategies and to set ethical boundaries in a global context, in order to prevent the use of technology that is socially and humanly disadvantageous.

¹ Cyberspace is the synthetic environment produced by interdependent IT infrastructures, data and information repositories, data communications, associated applications, and data processing, which generates information, knowledge, and the steering and control of a vast array of machines and systems.

² Magnetic resonance imaging (MRI) uses quantum mechanical effects through the nuclear spin of protons in the hydrogen atom.

There are several national quantum technology strategies published such as in the UK,³ Canada,⁴ USA,⁵ EU,⁶ Italy,⁷ Spain,⁸ France⁹ as well as dedicated quantum defence strategies (DnD/CAF, 2023; Krelina, 2025a, b) (Amerongen, 2021; Choi, 2023; Krelina, 2021; NATO, 2024; Parker, 2021; Sauer, 2021). Essentially, all of these strategies include research and development (R&D) roadmaps and commercialisation efforts, emphasise the importance of private investment and start-up programmes in addition to government-supported research, and push collaborations for commercialisation through the establishment of research and technology hubs. To this end, efforts are also being pursued to further promote quantum technology knowledge in academic teaching, but also to anchor it more firmly in vocational education and training. These issues are also emphasised by the OECD (OECD, 2025).

The design of ethical norms and regulations for responsible innovation or the analysis of possible threats is much less robust. The need for comprehensive discourse in society at large to prevent a sudden emergence of hostility to technology is also hardly discussed, as is noted by the Centre for Digital Rights and Democracy (Vasse'I, 2025). This underlines the need for continuous evaluation and adaptation of national quantum technology strategies to ensure sustainable and responsible technology development.

This paper therefore discusses the essential factors for the design of effective governance rules and summarises recommended measures to develop and operate quantum technology-based solutions in a responsible ethical framework in the civil as well as in the defence domain following digital humanism objectives. However, it is essential that appropriate governance rules are developed which are dedicated to the different quantum technologies and use cases.

To clarify the overall quantum technology landscape and to provide a basis for the necessary discussion of potential threats and resulting governance, Sect. 2 provides a brief overview of the underlying quantum physics and Sect. 3 briefly describes the current EU quantum strategy as a baseline for programmes and activities within Europe. This is followed in Sect. 4 by a discussion of technical developments and possible applications with a focus on defence use cases

³ UK, Policy Paper, National Quantum Strategy. Retrieved January 13, 2026, from <https://www.gov.uk/government/publications/national-quantum-strategy>.

⁴ Canada's National Quantum Strategy. Retrieved January 13, 2026, from <https://ised-isde.canada.ca/site/national-quantum-strategy/en/canadas-national-quantum-strategy>.

⁵ <https://www.quantum.gov/>. Retrieved January 13, 2026.

⁶ US National Quantum Initiative. Retrieved January 13, 2026, from <https://digital-strategy.ec.europa.eu/en/library/quantum-europe-strategy>.

⁷ Italian Strategy for Quantum Technologies. Retrieved January 13, 2026, from https://www.mur.gov.it/sites/default/files/2025-09/Strategia_italiana_tecnologie_quantistiche_ENG.pdf.

⁸ The Government of Spain launches Spain's first Quantum Technologies Strategy. Retrieved January 13, 2026, from <https://www.lamoncloa.gob.es/lang/en/gobierno/news/paginas/2025/20250424-quantum-technologies-strategy.aspx>.

⁹ France National Quantum Strategy. Retrieved January 13, 2026, from https://quantique.france2030.gouv.fr/wp-content/uploads/2024/05/Quantique_rapport_activite_2022_v8.fr2030.pdf.

of the three major quantum technologies—quantum computers, quantum communication and encryption, and quantum sensors. For each technology sector, typical applications are presented that are either the most advanced or strategically significant.

Based on this clarification, Sect. 5 highlights the ethical considerations which have to be discussed in connection with quantum technology development on a global scale. Finally, Sect. 6 proposes the essential factors for the design of effective governance rules and summarises recommended measures to develop and operate quantum technology-based solutions in a responsible ethical framework in the civil as well as in the defence domain following digital humanism objectives.

2 Fundamentals of Quantum Physics

In contrast to classical physics, which describes the behaviour of macroscopic objects and is essentially characterised by deterministic behaviour, quantum mechanics has other basic principles and rules that describe the special behaviour of matter and energy at the atomic and subatomic level, i.e., atoms, electrons, ions, photons, etc. In contrast to classical physics, the behaviour of such quantum objects, known as qubits, is essentially determined by probabilities and leads to fascinating physical phenomena that form the basis for quantum technologies:

- Quantum superposition. In contrast to the classical 0/1 logic of bits, qubits can occupy several states at the same time by superimposing them until we observe, i.e., measure them.
- Quantum entanglement describes the intrinsic quantum mechanical connection between two or more qubits, whereby the state of the one is determined during the measurement of the other, regardless of how far apart the objects are (thus also over very large distances).¹⁰
- Non-cloning theorem, which states that you cannot make an exact copy of an unknown quantum state.¹¹

The major technological challenge in the development of quantum technologies is to build systems with stable qubits. Quantum systems are highly sensitive to their environment, which means that temperature, light, electromagnetic waves, etc. can lead to unwanted interactions that disrupt qubit states and cause errors. This

¹⁰ In entanglement, particles with a random but identical direction of oscillation are generated and sent in different directions. The actual direction of oscillation remains undetermined until it is measured on one of the two entangled particles; thereby not only the state of this particle, but also that of the partner particle that is arbitrarily far away is determined randomly but identically. Einstein once questioned this “spooky action at a distance”, but today it has been proven by experiments and is implemented accordingly in technology solutions.

¹¹ According to the basic law of physics, which states that every measurement influences the observed result.

noise is a major challenge in developing stable and reliable quantum technology, especially quantum computers.

Quantum error correction (QEC) is a set of techniques designed to counteract this quantum noise and reduce errors. Unlike classical error correction, QEC requires encoding quantum information across multiple qubits to detect and correct errors without directly measuring and disturbing the quantum state.

Often, the functionality of a quantum computer is described by the necessary large number of qubits. It should be noted, however, that such a definition is not a sufficient description. Depending on the technology used, different parameters are required to describe the functionality of a quantum system. On the one hand, superconducting materials or ions react very strongly with the environment and a very large number of qubits are needed to correct errors; photons, on the other hand, hardly react with the environment, but the generation, control and detection of photons is more complicated than with ions or superconducting materials.

By using these quantum mechanical properties, second-generation quantum technologies are being developed in three areas:

- quantum computing and simulation
- quantum communication
- quantum sensing and metrology

These three technology areas have a wide variety of fields of application and different levels of maturity concerning technological development, which are discussed in more detail below.

3 EU Quantum Strategy

The European Commission has put forward a clear quantum strategy to make Europe a global leader in quantum by 2030. The EU Quantum Flagship initiative, which started in 2018, has the intention to bring Europe, with a one-billion-euro investment until 2030, into the forefront of quantum technology development globally.¹² The EU quantum strategy is planned to foster a resilient, sovereign quantum ecosystem that fuels start-up growth and transforms breakthrough science into market-ready applications, based on Europe's scientific leadership. The different funding programmes in this context will be highlighted in the following sections.

This EU strategy will be followed by a Quantum Act proposal, expected in 2026, which will further strengthen the quantum ecosystem and the industrialisation efforts by incentivising member states and companies, investors and researchers to invest in production facilities, in EU-wide and national initiatives.¹³

¹² EU Quantum Flagship Programme. Retrieved January 13, 2026, from <https://qt.eu/>

¹³ European Quantum Act—Updates. Retrieved January 13, 2026, from <https://www.european-quantum-act.com/>; Brussels, 2.7.2025, COM(2025), 363 final, Communication from the Commission to the European Parliament and the Council, Quantum Europe Strategy: Quantum Europe

Furthermore, there is the European Chips Act¹⁴ which has the objective of supporting technological capacity building and innovation for chip production in the EU. In the context of this Act, a dedicated quantum chips programme focuses on the development and production capability of chips exploiting quantum effects. The initiative explicitly supports the development of quantum chips, including design libraries, pilot lines, cleanrooms, foundries, and testing facilities.

Finally, the EU also has a dedicated military-focused quantum effort as part of the European Defence Fund (EDF)¹⁵ as well as in programmes of the European Defence Agency (EDA).¹⁶

4 Quantum Technologies and Application Areas

4.1 Quantum Computers and Quantum Simulation

The field of quantum computing is probably the most discussed area of application of quantum technologies in public and political discourse. Quantum computers are based on qubits, which can assume not only two states (0/1), but also multiple states simultaneously due to the quantum phenomenon (superposition). As a result, quantum computers are potentially much better suited for solving mathematical problems characterised by high parallelism and are potentially able to solve computational tasks with exponentially growing complexity.

Quantum computers are thus potentially much more powerful in simulation and optimisation tasks than conventional computers, which are based on traditional von Neumann architectures,¹⁷ and will offer benefits in application areas including:

- Optimisation problems of exponential complexity, such as supply chain optimisation, logistics and route optimisation.

in a Changing World. Retrieved January 13, 2026, from <https://data.consilium.europa.eu/doc/document/ST-11276-2025-INIT/en/pdf>.

¹⁴ European Chips Act: The Chips for Europe Initiative. Retrieved January 13, 2026, from <https://digital-strategy.ec.europa.eu/en/factpages/european-chips-act-chips-europe-initiative>.

¹⁵ Funding opportunities for quantum technologies in the new EDF Call. Retrieved January 13, 2026, from <https://www.bayern-innovativ.de/en/emagazine/digitization/detail/funding-opportunities-for-quantum-technologies-in-the-new-edf-call/>

¹⁶ EDA Annual Report 2023. Retrieved January 13, 2026, from <https://eda.europa.eu/docs/default-source/brochures/eda-annual-report-2023.pdf>.

¹⁷ Conventional computer systems are based on a “von Neumann architecture”, which describes the main building blocks of a typical computer: computing unit, input–output system, main memory and a data bus. Processing is conducted on a strictly linear basis in which an instruction fetch and a data operation cannot occur simultaneously as they share the common bus. This is referred to as the “von Neumann bottleneck” and limits the performance of the computing system. The computing architectures commonly used for AI-based machine learning today try to overcome this problem by using large numbers of processing units operating in parallel.

- Optimising systems in which data is incomplete or uncertain such as risk analysis and predictive modelling.
- Search optimisation and data mining in unstructured or structured data sets.
- Simulation of quantum systems: i.e., problems that naturally involve quantum mechanical processes, such as simulating molecular structures, chemical reactions, or materials science addressing application areas such as drug development for the pharmaceutical industry, molecular biology, chemistry and material research, and fluid dynamics for aerospace and automotive applications.

It is important to note that quantum computers will not replace traditional computers. The quantum computer community uses the term “quantum supremacy” to describe the point in time when a quantum computer can solve a specific problem that is practically impossible for classical computers to solve within a reasonable time frame; i.e., to achieve a clear computational advantage of quantum systems over classical ones.

Thus, we will have hybrid computer architectures which combine traditional and quantum processors. Traditional computers will handle data input/output, and data management, while quantum processors will be used to process specific subtasks such as simulation and optimisation.

In order to leverage the economic potential, considerable investments are being made globally in the development of quantum computers (QC) (Dhapte, 2025; McKinsey, 2024), pursuing different quantum technologies.

4.1.1 Artificial Intelligence and Quantum Computer

Due to the inherent parallel computing properties, a quantum computer is also expected to enable new approaches concerning artificial intelligence (AI) and machine learning technologies. In theory, quantum algorithms can significantly reduce the costs of machine learning processes due to the special parallel computing concept, which might result in shorter training times and in reduction of the amount of training data required (Caro et al., 2022; Liu et al., 2021).

It is important to note that such a new quantum AI technology is not yet developed enough and there is a need for considerable R&D effort to develop usable solutions. However, independent of the availability of quantum computers, the application of AI solutions will have substantial disruptive effects on the defence sector. An overview of possible defence AI applications can be found in Morgan et al. (2020) and Sentient (2023).

4.2 Quantum Communication and Quantum Encryption

4.2.1 Disruptive Threat to Global Cybersecurity

As discussed above, quantum computers (QC) enable disruptive market opportunities and innovations in various application areas. At the same time, however, QC also pose a special threat to the future security of the internet and our

digital systems. Due to the dedicated property of a high parallelisation of computing operations, QC can quickly decipher our very widespread asymmetric encryption systems (Gidney & Ekerå, 2021; Shor, 1997). Asymmetric encryption is the state-of-the-art solution in our digital systems¹⁸ and is used wherever secure communication and authentication via insecure communication channels is required, such as secure web browsing (HTTPS/SSL/TLS), email encryption (PGP, S/MIME), authentication for digital applications (digital identities) and digital signatures.

This is an inherent threat to the security of the internet and the integrity of our digital systems and applications, which not only poses a substantial problem for every user but also threatens the functioning of critical infrastructures and even national security. Against this background, the quantum technology-based encryption technology—Quantum Key Distribution (QKD)—promises a remedy.

4.2.2 New European Security Infrastructure: EuroQCI and IRIS2

The quantum technology-based generation of keys is based on the physical quantum effects described above and is therefore provably unassailable for future quantum computers, as no mathematical assumptions are used for key generation. Quantum Key Distribution (QKD) technology uses quantum effects to generate identical and secure keys at different locations at the same time, to use them for symmetric data encryption.

The EU launched the Quantum Flagship Programme in 2018,¹⁹ a 10-year programme with a funding volume of one billion euro for technology development in the field of quantum physics in order to bring quantum technologies from the laboratory to market maturity. In 2019, the European Commission launched the EuroQCI (European Quantum Communication Infrastructure) programme²⁰ to advance the establishment of a modern quantum computer secure communication infrastructure. Through the EuroQCI programmes, QKD pilot projects have been initiated in all EU member states to test QKD technology in various applications

¹⁸ The most commonly used asymmetric cryptosystem today is RSA technology. The asymmetric encryption technique is based on the use of large prime numbers to calculate a public key and a private key. Conventional computers cannot calculate such a prime factor decomposition of a very large number (public key) in a reasonable amount of time. QC, on the other hand, will be able to do so, and thus asymmetric encryption will lose its value; e.g., it takes traditional computers more than 100 years to find prime factors of a typical size of an RSA key (i.e., a 2048-bit number). A breakthrough came in the mid-1990s when Peter W. Shor discovered a quantum algorithm (Shor's algorithm) that can break a 2048-bit RSA encryption key in a matter of hours using quantum computers. A newer asymmetric encryption algorithm is Elliptic Curve Cryptography (ECC), which should improve the security level compared to RSA, but is still vulnerable to future quantum computer attacks.

¹⁹ EC Quantum Technologies Flagship. Retrieved January 13, 2026, from <https://digital-strategy.ec.europa.eu/en/policies/quantum-technologies-flagship>.

²⁰ European Quantum Communication Infrastructure—EuroQCI. Retrieved January 13, 2026, from <https://digital-strategy.ec.europa.eu/en/policies/european-quantum-communication-infrastructure-euroqci>.

and to gain experience in the operation of the new encryption systems by national authorities and critical infrastructure operators. Furthermore, the EU Connecting European Facility (CEF) programme extends those proof-of-concept projects by addressing cross-border network scenarios between the member states.

Finally, IRIS2²¹ has been established as a further programme to establish a QKD satellite infrastructure to make QKD keys also available over long distances. In its project EAGLE-1, the EU satellite operator SES prepares the launch of a low-orbit QKD test satellite (launch planned for 2026/27).²² Based on this pilot system, the European Space Agency (ESA) is developing the infrastructure for a QKD satellite network with its SAGA project.²³

By means of this portfolio of EU quantum programmes, a unique European QKD network infrastructure with satellites, cross-border fibre optic networks and new encryption applications in all EU member states is created.

All this is also about building capabilities in the EU through appropriate high-tech competencies, tools and services in order to be able to produce and, above all, certify new quantum encryption solutions in the EU without external dependencies. In order to achieve a sustainable high level of security, in addition to the actual cryptographic process itself, secure practical implementation of systems and solutions is essential. To prevent potential 'side-channel' attacks, a portfolio of test methods is being developed by the EU project "Nostradamus".²⁴ Through this project, a blueprint for a certification infrastructure and dedicated competences will be created in the EU by the consortium Deutsche Telekom, Thales and AIT Austrian Institute of Technology.

The aim of these programmes is to bring the highest security standard to the market through modern QKD encryption technology for highly secure government communication as well as for critical infrastructure operators in the EU. This would make an important contribution to sustainable data and information sovereignty for government communication as well as for the operation of critical infrastructures in the EU. The EU project OpenQKD was initiated in 2019, to enable and support the EU-wide industrialisation of the EU QKD technology.²⁵

²¹ IRIS2: the new EU Secure Satellite Constellation. Retrieved January 13, 2026, from https://defence-industry-space.ec.europa.eu/eu-space/iris2-secure-connectivity_en.

²² EAGLE-1: Advancing Europe's Leadership in Quantum Communications. Retrieved January 13, 2026, from <https://www.ses.com/newsroom/eagle-1-advancing-europes-leadership-quantum-communications>.

²³ SAGA for quantum key distribution. Retrieved January 13, 2026, from https://www.esa.int/ESA_Multimedia/Images/2019/04/SAGA_for_quantum_key_distribution.

²⁴ EU-Auftrag: Nostradamus bereitet Europa auf Quantenwelt vor. Retrieved January 13, 2026, from <https://www.telekom.com/de/medien/medieninformationen/detail/eu-auftrag-nostradamus-bereitet-europa-auf-quantenwelt-vor-1056742>.

²⁵ OpenQKD project web site to provide assistance and guidance in the context of national QKD testbed installations for the European Quantum Communication Infrastructure (EuroQCI) Retrieved January 13, 2026, from <https://quantum.ait.ac.at/>

Thus, building on the previous research activities, the EuroQCI programme includes the establishment of a state-of-the-art European industry for the production of QKD systems by 2030, as well as the implementation of a corresponding fibre optic and satellite-based infrastructure for powerful key generation and distribution for various applications. This will put Europe at the forefront of the global race in this field of quantum technology.

4.2.3 International QKD Developments

There are also international efforts in this technology area. The Chinese Academy of Sciences began a few years earlier with the development and testing of the new QKD technology and a national QKD network was (Chen et al., 2021). However, the EU has already caught up with the Chinese efforts through consistently implemented EU QKD programmes.

Canada has also established a very strong quantum technology ecosystem through significant public and private investment (see the quantum computing initiatives described above),²⁶ and focuses on quantum communication²⁷ (DnD/CAF, 2023). However, this effort does not correspond to the comprehensive approach of the EU QKD programme.

Secure communication is also a cornerstone of South Korea's quantum agenda in the context of their digital infrastructure and cyber security strategies.²⁸ South Korea is investing in QKD as well as in post-quantum cryptography (PQC) (see below) to achieve global leadership in quantum cryptography deployment.

Finally, there are smaller pilot QKD networks in Singapore, Japan, Switzerland and in the UK. There have also been some QKD network activities in the US. However, the focus in the US is currently PQC and the Quantum Internet (see below).

4.2.4 Post-Quantum Cryptography (PQC)

It is important to note that the classic crypto community is also developing a classic encryption technique based on mathematical functions. The US standardisation organisation NIST²⁹ organises competitions to verify the strength of new encryption methods by using the global community of experts.³⁰ These encryption methods are also designed to be secure against decryption by quantum computers

²⁶ Canada has taken a strong position through investments in quantum science, e.g. more than \$1 billion between 2012 and 2022. Retrieved January 13, 2026, from <https://ised-isde.canada.ca/site/national-quantum-strategy/en/canadas-national-quantum-strategy>.

²⁷ Mission 4 “Quantum communications and quantum networking” of the Quantum Science & Technology Strategy Implementation Plan; Quantum EncrYption and Science Satellite (QEYSSat) mission led by the Canadian Space Agency (CSA).

²⁸ Marin Ivezić, Quantum Technologies and Quantum Computing in South Korea, March 14, 2025. Retrieved January 13, 2026, from <https://postquantum.com/quantum-computing/quantum-south-korea/#quantum-cryptography-and-secure-communication-focus>.

²⁹ National Institute of Standards and Technology (NIST).

³⁰ NIST Post-Quantum Cryptography PQC. Retrieved January 13, 2026, from <https://csrc.nist.gov/projects/post-quantum-cryptography/post-quantum-cryptography-standardization>.

due to their special mathematical complexity. These methods are therefore also summarised under the term post-quantum cryptography (PQC).

Thus, PQC refers to a new generation of cryptographic algorithms that are designed to withstand attacks by future, powerful quantum computers. However, since these algorithms are based on mathematical methods, it cannot be proven that they can withstand attacks by quantum computers.

In practice, the two technologies QKD and PQC are combined, as the application of PQC is suitable for a highly secure authentication process and QKD keys for the subsequent symmetric encryption process.

As the transformation to a new encryption standard requires considerable time and effort for any organisation, it is recommended that this transformation process be started urgently, especially within the defence domain. The EU Member States, supported by the EU Commission, have issued a roadmap and timeline for the use of post-quantum cryptography (PQC) (EU-NIS, 2025).

4.2.5 Quantum Internet

While the industrialisation activities of the new QKD technology are progressing intensively in order to bring products and systems to market in the next 5 years, the science community is already working on the development of a Quantum Internet (QI) and main stakeholders are forming the Quantum Internet Alliance (QIA).³¹ The vision of a Quantum Internet goes far beyond QKD. Research is ongoing to extend QKD networks to advanced quantum communications networks, which provide services to connect distributed quantum computers and quantum sensors. However, it will take some more years until commercially useful technological solutions are available.

4.3 Quantum Sensing/Quantum Metrology

Quantum sensors are based on quantum mechanics to enable increased sensitivity, which can be leveraged towards a new generation of sensors which allow higher precision measurement or provide access to never-before-measured properties. Physical parameters for such next-generation sensor systems are, e.g., the magnetic and the electric field, gravity, temperature, time, rotation, pressure and acceleration. With quantum sensor technologies, it is possible to build new systems for measuring, monitoring, imaging, and even navigation.

Quantum sensing is principally a passive approach and uses the inherent behaviour of quantum elements, such as electrons and photons, to interact with the environment via radio waves, temperature, vibration, light, etc. This is also the main problem for the development of stable qubits for quantum computing.

³¹ Quantum Internet Alliance. Retrieved January 13, 2026, from <https://quantuminternetalliance.org/>.

It is important to note that quantum sensing technologies are already much closer to market readiness, and will have potentially strong disruptive impacts on existing industries. The measurement of physical parameters and behaviour is a key functionality in nearly all our digital systems, including clock synchronisation for communication systems; positioning in navigation systems or on the micro level for robots in production and manufacturing infrastructures; steering and control of vehicle movement in the air, on the ground and on water, as well as for situational awareness systems for autonomous systems or for larger command and control (C2) management systems. When new sensor system solutions with a much higher level of precision paired with a much lower cost and in smaller equipment format (like a magnetic-resonance system for medical application with the size of a smartphone) are widely available on the market, the existing manufacturing industry ecosystem, as well as that of system developers and integrators, will be subject to new and disruptive market opportunities. These in turn will result in an innovative modernisation programme in manufacturer and system developer ecosystems.

The proof of concept has already been achieved in various research initiatives. The main focus now lies on the industrialisation and miniaturisation of different quantum sensor technologies to address future needs. For the defence sector in particular, it is now essential to identify appropriate system operation parameters such as size, weight, power and cost for the different and specific defence use cases and requirements.

The expected disruptions to the industry and market must be urgently considered when planning the transformation of appropriate capabilities for end-users and system operation in the defence domain.

4.4 Quantum Sensor Technologies

High precision quantum sensor technologies and potential application areas under discussion and evaluation are, e.g., (Degen et al., 2017; Soller & Mohr, 2024; McKinsey, 2024; Krelina, 2025a):

- Photonic-based sensors and nitrogen vacancy (NV) centres³² for inertial forces and earth-magnetic-field sensing for advanced navigation systems to increase the resilience of global navigation satellite systems (GNSS) and enable navigation inside buildings and in the underground. An overview of photonic-based sensor technologies and related defence applications is provided in (EDA, 2019).
- Magnetometers to

³² The nitrogen vacancy (NV) centre is a defect (impurity) in the carbon lattice of a diamond. The NV centre is a commonly used material for use as an emitter of a single-photon source at room temperature.

- measure the Earth's magnetic field for new methods of global positioning, or
- identify manufacturing defects in production processes at nanoscale dimensions, or even
- malicious modifications, such as hidden logic elements in embedded systems and even chips (so-called hardware trojans).
- Quantum gravity-gradiometers³³ (gravimeters) to measure the strength of the earth's gravitational field to
 - detect submarines,
 - detect underground structures and minerals and oil and gas reserves,
 - make seismographic predictions of earthquakes and volcanic eruptions, and
 - evaluate the solidity of the ground beneath major construction projects.
- Quantum antennae based on Rydberg atoms³⁴ for much smaller equipment and extreme wide-spectrum sensing (MHz to THz) for radio systems.
- Quantum radar based on radio frequency (RF) sensing using Rydberg atoms and quantum light detection and ranging (LiDAR)³⁵ based on photonic sensing.
- Quantum imaging for remote sensing.
- Earth and environment monitoring for crises and disaster management.
- Quantum-enhanced atomic clocks for precision timing.
- New sensor approaches in the medical domain such as
 - sensitive magnetometers for brain imaging,
 - imaging of protein structures, and
 - new brain computing interfaces.

Various building blocks for quantum sensing solutions are already available in a prototype stage, suggesting a high level of technological maturity. However, different approaches have very different technology readiness levels. Each specific sensor technology, use case and application field must be evaluated individually. Therefore, proof-of-concept projects as well as feasibility studies are necessary to meet the specific requirements for defence applications.

³³ NASA Quantum Gravity Gradiometer. Retrieved January 13, 2026, from <https://esto.nasa.gov/files/quantum/Chioiw.pdf>.

³⁴ Rydberg atoms are atoms in which an electron has been excited to an extremely high energy level, moving it very far away from the atomic nucleus and the atom thus takes on huge dimensions (up to micrometres). These atoms have dedicated properties such as extreme sensitivity to external electric and magnetic fields, and they have long lifespans, which have applications in quantum computing and quantum sensing technologies.

³⁵ LiDAR is a sensor technology that uses laser beams to measure the time it takes for light pulses to reflect on objects.

4.5 Quantum Sensing Defence Use Cases

Quantum sensing enables several applications in the defence domain ranging from military operations to disarmament verification. Typical defence application areas currently under evaluation in a global context are:

- Quantum inertial navigation in the air and underwater.
- Quantum radio communication based on Rydberg atom sensors.
- Quantum radar on the ground, from satellites and underwater for sensing and quantum underground mapping.
- Quantum imaging for remote sensing; ‘ghost imaging’ and enhancement of LiDAR.
- In addition, new sensor capabilities could improve monitoring techniques essential to arms control, allowing, e.g., for highly sensitive detection of nuclear materials and other critical environmental markers.

These quantum defence use cases are briefly discussed in the following.

4.5.1 Quantum Inertial Navigation in the Air and Under Water

Crucial infrastructures for all our digital systems, whether in the civil or defence sector, are the global navigation satellite systems (GNSS); i.e., GPS for the US, Galileo for the EU, GLONASS in Russia, and BeiDou in China. These infrastructures provide autonomous, global positioning, navigation, and timing (PNT) services to various applications.

In a GNSS-degraded or -denied environment, PNT capabilities are impacted by the topology or dedicated jamming or spoofing attacks. Furthermore, GNSS-based PNT services are not available underground or underwater.

To compensate for those potentially missing functions, military aircraft, land vehicles, and naval vessels are equipped with inertial navigation systems (INS) to determine the position by continuously calculating rotation and acceleration. However, because INS drift over time due to the accumulation of measurement errors, they require frequent and long periods of recalibration, typically by satellites, or need much higher accuracy of the sensor systems for navigation.

Quantum inertial navigation (QIN) offers a solution to this problem with measurement accuracy for acceleration and speed for next-generation navigation systems based on photonics and NV centres; i.e., the combination of a quantum accelerometer, gyroscope, and very accurate clocks.

A further approach is the application of quantum sensors to measure the Earth’s magnetic field, enabling navigation based on Earth magnetic field maps. Earth magnetism anomaly navigation relies on natural variations in the Earth’s magnetic field as geophysical reference points for positioning.

Initial technologies and prototypes are already available; however, robust and compact quantum sensors have still to be developed. Example activities are:

- The Pentagon's Defence Innovation Unit launched a quantum inertia sensor on the Boeing X-37B vehicle (Reim, 2026).
- The Google start-up SandboxAQ announced the AQNav system in June 2024, a dual-use magnetic navigation system.³⁶
- With "Mag-Nav", Q-ctrl offers a dedicated solution for next-generation quantum-based navigation.³⁷
- The European Defence Agency (EDA) project "QuantaQuest" has, inter alia, explored the positioning and timing for military platforms in the Preparatory Action on Defence Research (PADR) programme.³⁸

The availability of next-generation quantum sensor systems for autonomous navigation which meet defence-grade requirements, the need for small equipment practice and a very low cost per system will certainly disrupt several defence use cases for aircraft, vehicles, submarines, precision munition as well as UAVs and UGVs.

As our mobility systems and critical infrastructures depend on satellite-based navigation systems and as there is a need to increase the EU's technological sovereignty in order to react to the global security situation, it is vital to address the latest technology developments and investigate the adaptation of appropriate capabilities for end-users and system operations in the defence domain.

4.5.2 Quantum Radio Communication and Sensors Based on Rydberg Atoms

The encompassing digitalisation of our communication systems for enhanced data communication has already globally disrupted the telecommunication landscape. This transformation will happen in the defence domain as well.

Rydberg atom-based sensors have the potential to act as a compact wideband radio frequency receiver. Rydberg atoms have a high sensitivity to electric fields with frequencies ranging from 100 megahertz to one terahertz and Rydberg atom-based antennas can be built by a factor 100 smaller than traditional antennas (Choi, 2023).

Thus, Rydberg sensors offer highly sensitive detection of a vast range of frequencies and the ability to work with compact sensors, enabling passive, stealthy receiver systems that reduce antenna size and detect hard-to-find signals, such as those from drones or hidden communications.

³⁶ Press Release, SandboxAQ Announces AQNav—World's First Commercial Real-Time Navigation System Powered by AI and Quantum to Address GPS Jamming. Retrieved January 13, 2026, from <https://www.sandboxaq.com/press/sandboxaq-announces-aqnav---worlds-first-commercial-real-time-navigation-system-powered-by-ai-and-quantum-to-address-gps-jamming>.

³⁷ Q-ctrl is a start-up from the University of Sydney. Retrieved January 13, 2026, from <https://q-ctrl.com/ironstone-opal>.

³⁸ QuantaQuest project explores application of quantum technologies in defence, January 19, 2024. Retrieved January 13, 2026, from <https://eda.europa.eu/news-and-events/news/2024/01/19/quanta-quest-project-explores-application-of-quantum-technologies-in-defence>.

However, technical hurdles must be overcome to make Rydberg atom-based sensors more affordable and deployable. A great challenge is the cryogenics required for cooling Rydberg atoms (Fancher et al., 2021).

4.5.3 Quantum Radar on the Ground and from Satellites

Conventional radar systems are subject to radar-jamming devices and noise interference from natural phenomena. By using photon-based quantum entanglement and superposition, quantum radar systems offer new approaches and are a strategic challenge to traditional stealth technologies (Lanzagorta & Uhlmann, 2020). Quantum radar systems improve the sensitivity and range of radar detection: i.e., higher sensitivity and improvement of the signal-to-noise ratio (Hooda, 2023).

Thus, new quantum radio frequency receivers might improve radar and electronic warfare capabilities, due to new possibilities of detecting weak signals and jamming enemy transmissions more effectively. This will result in improvements in the detection of objects with low reflectivity in high-noise environments, such as stealth aircraft. It will also be possible to detect and locate electromagnetic emissions using quantum magnetometers to detect changes in magnetic fields caused by a submarine's metal hull or propulsion systems or the use of quantum gravimeters to measure the presence of a submarine's mass.

4.5.4 Quantum Imaging for Remote Sensing: Ghost Imaging and Enhancement of LiDAR

Quantum imaging is a specialised imaging method which utilises entangled photons to create detailed images beyond that of traditional methods, which enables new intelligence, surveillance, and reconnaissance (ISR) capabilities. Such quantum sensors offer the potential for a significant improvement in extreme low-light conditions to detect distant objects by using very weak illumination beams which are difficult for the imaged target to detect (Meyers et al., 2008; Hardy & Shapiro, 2013; Pirandola et al., 2018; Fraunhofer, 2022; Uppal, 2022).

By using entangled photons it might be possible to create images of objects without directly 'seeing' them, potentially allowing targets to be imaged and terrain to be visualised through clouds, fog or smoke (ghost imaging), and camouflage; it is even possible to 'look' around the corner.³⁹

Furthermore, quantum imaging systems could be applied to sense the presence of harmful noxious gases in battlefield environments and assist with forensic analyses of chemical weapons.

Thus, quantum-based image sensing methods can improve battlefield situational awareness and offer innovative approaches for ISR by identifying hidden objects and camouflaged vehicles. This provides more precise data for navigation and obstacle avoidance in GPS-denied environments.

³⁹ Ghost-imaging could have satellite application, 2008. Retrieved January 13, 2026, from <https://www.af.mil/News/Article-Display/Article/123418/ghost-imaging-could-have-satellite-application/>.

4.5.5 Arms Control and Environmental Monitoring

In addition, regarding the sensor use cases summarised above, quantum sensors could potentially improve monitoring techniques essential to arms control, allowing, e.g., for highly sensitive detection of nuclear materials and other critical environmental markers. Thus, new quantum sensors offer potentially new monitoring capabilities:

- more accurate verification techniques for non-proliferation efforts by enabling remote monitoring capabilities to detect nuclear materials and activity from space to fight trafficking of nuclear materials and to monitor nuclear facilities;
- more accurate non-destructive measurement of quantities and compositions of nuclear materials for declarations verification and diversion detection;
- quantum sensors to detect ultra-sensitive environmental changes (Malekos et al., 2024).

4.5.6 International Quantum Sensing Defence Programmes and Activities

To exemplify the international R&D dynamic further, international key programmes are briefly summarised in the following:

- The EDF Advanced, Disruptive and Emerging QUAntum technologies for DEfence (ADEQUADE) project seeks to leverage quantum sensing technology to augment warfighting capabilities. These range from enhancing position, navigation and timing capabilities to advancing radio frequency sensing and optronics sensing.⁴⁰
- The Royal Navy has successfully demonstrated the capabilities of ground-breaking cold atom technology enabling new precise signals for accurate positioning, navigation and timing (PNT).⁴¹
- The USA's Defense Advanced Research Projects Agency's (DARPA) Robust Quantum Sensors (RoQS) programme seeks to advance quantum sensing technology for use in real-world conditions.⁴²
- In 2017, the Chinese Academy of Sciences announced that it had developed a sensor, called a Superconducting Quantum Interference Device (SQUID), which relies on quantum sensors to detect ultra-sensitive environmental changes to support anti-submarine aircraft with enhanced magnetic anomaly detection.

⁴⁰ Participating countries: France, Germany, Italy, Netherlands, Spain, Denmark, Poland, Latvia and Czech Republic. Project website. Retrieved January 13, 2026, from <https://adequade.eu/>

⁴¹ Royal Navy successfully tests quantum-sensing technology. Retrieved January 13, 2026, from <https://www.royalnavy.mod.uk/news/2024/october/31/20241101-royal-navy-successfully-tests-quantum-sensing-technology>.

⁴² Taking quantum sensors out of the lab and into defence platforms, 2025. Retrieved January 13, 2026, from <https://www.darpa.mil/news/2025/quantum-sensors-defense-platforms>.

5 Risks and Ethical Use of Quantum Science and Technology

Quantum technologies not only have the potential to disrupt industry but also a range of applications and markets in the security and defence domains. Despite all the expected commercial benefits of quantum technologies, there is also the risk they could bring to global peace and security. In its 2022 General Assembly, the United Nations officially warned that quantum computers could potentially “destroy cybersecurity” on a global scale,⁴³ referring to the potential quantum computer decryption capability as described in Sect. 4.2.1.

Furthermore, there are ethical considerations to be discussed regarding bias in datasets used to train AI in connection with quantum computing. Ethical issues also have to be taken into account when developing new sensor systems, such as “ghost imaging”, “seeing through walls” and “around the corner”, etc. as discussed above.

Thus, there is a need for robust governance frameworks, cross-sector partnerships and a focus on capacity building to support a quantum-ready workforce. We have to ensure intensive dialogues between research, industry, policymakers and even diplomats to ensure a sound ethical discussion of the new quantum technology industry area.

5.1 Responsible Quantum Technology Development Needs Effective Governance

Around the world, significant efforts and substantial investments are being made into developing quantum technologies, leading to a strong increase in the number of patents as well as steady growth in the maturity of the different quantum technologies (Lewis et al., 2025). Furthermore, national quantum strategies are being formulated on a broad basis, reflecting the growing strategic role of quantum technologies in global technological competition, and allowing countries to position themselves accordingly.

However, as with any technological process, these developments bring corresponding risks which, to date, have received far too little attention. Until now, the scientific community’s fascination with discovering the various physical effects of quantum mechanics has been accompanied by the expectation of potential disruptive market effects. This, in turn, is expected to be a driving force for high government subsidies paired with high levels of private sector investment in technology development and start-ups. This bears all the hallmarks of a technology hype.

⁴³ United Nations, “‘Our World Is in Big Trouble’, Secretary-General Warns General Assembly, Urging Member States to Work as One United Nations”, Press Release SG/SM/21466, September 20, 2022. Retrieved January 13, 2026, from <https://press.un.org/en/2022/sgsm21466.doc.htm>.

For most people quantum physics is incomprehensible and therefore mysterious. But there is no need for the practical applications of quantum technologies—such as the highly secure communication, new algorithms and computational methods made possible by new computers or high-precision sensors—to remain obscure to society in general.

Interestingly, the term ‘quantum technology’ is not perceived negatively by society as a whole, but rather as a positive technology, one that represents scientific excellence and innovative strength. It should be noted, however, that public opinion can change very quickly if risks and threats to society suddenly arise or are suspected.

The highly scientific and technological discourse on quantum technologies, combined with the expectation of significant economic advantages, must now be supplemented with a more responsible approach to its continued development. The task is to anticipate potential negative as well as disruptive effects for the economy and society, rather than simply prioritising potential short-term profit or competitive geopolitical advantage. In addition, the security and defence sector is affected by disruptive changes caused by quantum technologies, which will open up new opportunities but also entail considerable risks. Ultimately, the common goal must be to ensure that the development of quantum technologies reflects societal values.

This leads to the fundamental question of how quantum technologies can be sustainably and responsibly developed. The current quantum technology hype makes Friedrich Dürrenmatt’s drama *The Physicists*,⁴⁴ about the responsibility of scientists in developing technologies that are potentially dangerous for humanity, highly relevant again.

Therefore, both the Center for Digital Rights and Democracy (Vasse’I, 2025) as well as UNESCO are calling for a much greater awareness of the ethical aspects that must be considered when developing quantum technologies, and criticise the fact that most national quantum programmes fail to consider ethical developments (UNESCO, 2025).

After a long period of science-focused research and development, effective governance at a global level is now needed to shape the further development of quantum technologies. Governance should ensure that these technologies promote economic growth and support national security and digital sovereignty goals, and at the same time consider basic ethical values in technology development as standard, to counteract potential risks arising from the ill-considered, widespread use of this new technology.

Here we are confronted with the Collingridge dilemma in which the effects of new technologies cannot be easily predicted before the technologies are extensively developed and widely used, yet the more firmly a technology is entrenched in the market, the harder it is to control and change (Genus & Stirling, 2018). We observe this dilemma in the current discussions on innovation, negative impacts on society, and the need to regulate AI technology. The EU is trying to prevent

⁴⁴ Friedrich Dürrenmatt (1989). *Die Physiker*. Diogenes publisher. ISBN 978-3-257-23047-5.

potential harm to society caused by the unintentional use of AI in certain high-risk application areas by introducing regulatory measures, yet these efforts face widespread opposition based on the simple argument that regulation would hinder innovation and progress.

To avoid this dilemma, it is important that current major R&D efforts to develop quantum technologies are accompanied by a broad-based ethical discourse, so as not to wait until we are confronted with problems caused by using quantum technologies which are already widely available (Blanchard et al., 2024).

Nation states and industrial enterprises must be able to remain competitive and pursue digital sovereignty goals which are the basis for functioning markets. At the same time, they should steer technology development so that it pursues the principles of digital humanism and ensures positive effects for humanity (Leopold & Höllbacher, 2026).

In view of the extreme dynamism and level of global investments, quantum technology developments must be accompanied by a sustainable, global multi-stakeholder dialogue to achieve a better understanding of the particularities of the technology as well as the threats, risks and ethical problems it brings. This is the basis for establishing global governance structures that ensure the ethically responsible development of quantum technologies. To this end, the following issues must be considered:

1. In principle, it is important to note that relevant competencies and skills are needed in order to create real innovations and master the technology. These competencies and skills allow the technologies and various quantum applications to be industrialised, creating an effective supply chain. This means that the production of real innovations not only needs scientific but also technological competencies; i.e., tools and a skilled workforce. The special skills in quantum physics must be combined with industrial methods, tools and skills to develop integrated circuits (ICs), electronic embedded systems, and software in order to create innovative solutions in the three areas of quantum computing, quantum communication and quantum sensing.
2. Risks to national security caused by the loss of confidential communications as well as data storage. Data security is a key consideration; ensuring data confidentiality over the long term is essential for the protection of state secrets which should remain confidential indefinitely.
3. Risks caused by the loss of fundamental data sovereignty following comprehensive digitalisation and global networking in all areas of life. This represents a threat to the privacy of every citizen and have far-reaching consequences for democratic societies. Quantum computers have the potential to break and threaten our widely used cryptography systems, undermining our data sovereignty and thus privacy protection. The entire Internet and a multitude of related business processes are based on a functioning public-key encryption system for the transmission and storing of data. The potential loss of this cryptographic protection posed by future quantum computers gives unauthorised actors—private or government—access to sensitive and personal data as well

as critical infrastructures. Modern future-proof cryptography systems must be based on secure concepts that cannot be broken by future quantum computers, such as QKD and PQC technologies (see Sect. 4.2).

4. The use of future quantum computers—by private or state actors—to decipher our widely available asymmetric encryption systems naturally also raises legal and ethical questions in a global context. The approach of exploiting vulnerabilities in encryption systems to improve surveillance creates an ethical dilemma as it violates legal boundaries.
5. Protection of privacy and ethical issues for the potential surveillance of persons. New questionable surveillance possibilities arise, both from the loss of data sovereignty when the encryption technologies we use no longer work effectively and from new quantum sensor systems that can “see” through walls. It is essential to establish appropriate, globally anchored regulations in both areas in order to protect personal rights and data protection in the quantum age.
6. Use of quantum technologies in military applications as dual-use technology. Quantum technologies can potentially lead to disruptive effects in a variety of areas, such as powerful new quantum computers, or new quantum sensor technology for satellite-independent geo-localisation and navigation, reconnaissance and surveillance. It is important to note that disruptive developments could also destabilise international relations.
7. Quantum technologies applied to AI systems carry the inherent risk of being dominated and monopolised by a few organisations. Conflicts in the global context are inevitable if the new technologies are not available for universal use (imbalance of quantum divide), such as quantum computers used to develop new drugs.

In order to elaborate sound quantum governance on a global scale, two essential areas have to be considered in addition: standardisation and export control regulation, which will be discussed in the following.

5.2 Standardisation of Quantum Technologies: A Baseline for Governance

Global standardisation is not only essential for interoperable systems and applications, it is also the basis for the facilitation of global, consensus-oriented discourse on legal and ethical aspects which must be considered when enhancing quantum technologies in the various technological and application areas, and for creating the basis for future regulations.

This approach also allows a variety of stakeholders from different political systems to shape and support international standardisation more easily, in contrast to binding bilateral trade agreements or restrictions.

A wide range of international standardisation organisations are working on the various aspects of quantum technologies. Worldwide, working groups in

ISO/IEC, ITU-T, IEEE, IETF, GSMA⁴⁵ deal with different aspects of quantum technologies.⁴⁶ In Europe, ETSI⁴⁷ works through the Technical Committee on Cybersecurity (TC CYBER), through the Industry Specification Group on Quantum Key Distribution (ISG QKD), as well as through CEN-CENELEC⁴⁸ through the Joint Technical Committee on Quantum Technology (JTC 22). Consequently, the standardisation landscape can certainly be described as highly fragmented and confusing. In early 2024, a cooperation agreement was established between the two major international standardisation organisations ISO and IEC, to form the ISO/IEC Joint Technical Committee 3 (JTC 3) for quantum technologies. It should bring greater clarity to the standardisation landscape for quantum technologies.

It should also be noted that EU standardisation has been very cautious in recent years. This is mainly because, to date, quantum technology development has been very much driven by the science and physics community. Consequently, the product strategies of EU companies have not included concrete measures for industrialising quantum technologies with commercially foreseeable short-term market successes, and consequently investment in standardisation activities has not been prioritised. It is only through new EU projects such as Petrus,⁴⁹ a support initiative for the EuroQCI programme to implement highly secure communication infrastructure in the EU, that standardisation activities are increasingly being supported by EU industry.

(Aboy et al., 2025) emphasise the importance of standardisation as a consensus-building measure that creates trust between stakeholders and can be conducted at an early stage of technology development. They propose two types of standardisation: (1) the normal technical aspects, such as terminology, architectures and protocols, and (2) quality standards for developing and operating quantum systems and for controlling and auditing systems. The latter can build on existing quality standards drawn from other areas, such as ISO Information Security 27001, ISO Risk Management 27005, and ISO 42001 for AI Governance.

The standardisation activities for quantum technologies summarised above focus on technical system architectures and interoperability but pay almost nearly no attention to ethics and regulatory aspects.

⁴⁵ ISO/IEC, ITU-T, IEEE, IETF, GSMA are the essential international standardisation organisations (SDOs): ISO—International Standards Organisation, ITU-T—International Telecommunication Union/Telecommunication Standardisation Sector, IEEE—Institute of Electrical and Electronics Engineers, IETF—Internet Engineering Taskforce, GSMA—Global System for Mobile Communications.

⁴⁶ Understanding Quantum Computing: A New Era of Innovation. Retrieved January 13, 2026, from <https://www.gsma.com/get-involved/gsma-foundry/artificial-intelligence/understanding-quantum-computing-a-new-era-of-innovation/>.

⁴⁷ European SDO: ETSI—European Telecommunications Standards Institute.

⁴⁸ European SDO: CEN-CENELEC—Committee for European Norms (Comité Européen de Normalisation Électrotechnique).

⁴⁹ Petrus project home page. Retrieved January 13, 2026, from <https://petrus-euroqci.eu/>.

Future standardisation strategies for quantum technologies should be adapted to reflect these aspects, to create a basis for effective governance regulations and regulatory frameworks.

5.3 Quantum Technologies and Export Control

Export control mechanisms are now increasingly restricting international trade due to the expected disruptive effects of quantum technologies and their potential application as dual-use technologies in civilian and military applications.

In the EU, some quantum technology components are defined as dual-use components⁵⁰ in the field of cryptography (QKD), superconductive electronic devices, and quantum wells (Lewis et al., 2025). In addition, there are specific national export controls. As a result, we are already observing the start of fragmentation of national export regulations in the EU member states.

In September 2024, in its Commerce Control List—Additions and Revisions, the US established export controls for advanced technologies, including quantum technologies (Lewis et al., 2025). This covers: (i) CMOS (Complementary metal–oxide–semiconductor) integrated circuits, parametric-signal amplifiers that can operate at very low temperature, (ii) materials containing certain silicon (Si) or germanium (Ge) isotopes which have no nuclear spin and can therefore be used to develop spin-based quantum computers, (iii) quantum computers and components, as well as software and technology that can be used in the development and maintenance of quantum computers. A licence requirement has been imposed to ensure that these controlled items are used in an appropriately restricted manner.

Export control restrictions are also leading to increased tension and retaliation, for example between China and the US on the trade control of semiconductors and critical raw materials. An initial overview of the actual status is summarised in (Lewis et al., 2025).

5.4 Key Issues for an Effective Quantum Governance Framework

Initial approaches from the research and policy community to the relevant topics above are conducted from different perspectives. (Wibmer, 2025) offers a good overview of the current literature on the subject and (Johnson, 2019) presents incipient governance tools. Precursory governance discussions are available for quantum computing (WEF, 2022), but there is a need for a much more comprehensive discussion. Since the actual discussion is still taking place on a very general level and is not yet sufficiently focused and concrete, the following issues

⁵⁰ Regulation (EU) 2021/821. Retrieved January 13, 2026, from <https://eur-lex.europa.eu/EN/legal-content/summary/dual-use-export-controls.html>.

should be considered for elaborating effective governance rules (Atladdottir, 2024; Blanchard, 2024; Kop, 2025; Perrier, 2022):

1. Quantum technologies are an emerging field of technology that requires **forward-looking reflection** to prevent potential threats and risks to societies, businesses, and humanity from quantum technologies and future applications.
2. **Knowledge gap among decision makers:** The highly scientific discourse on quantum technologies, which focuses on physical phenomena, must shift to the clearly understandable area of application, so that it is possible to discuss and anticipate the effects and risks of these technologies in the individual fields of application. New awareness and educational initiatives for decision makers are needed to develop responsible innovations as well as necessary regulations. The gap in understanding between policymakers, decision makers, and technical staff can lead to inappropriate action. Equally, over-regulation must also be prevented, as must ignoring necessary regulations. We need informed decision-making to balance policies with realistic technical solutions.
3. In the field of **information security**, comprehensive global cooperation between governments, industry and research is needed to counteract the fundamental threat of losing data security resulting from the loss of effective encryption technology. Our global digital transformation is based on underlying communication and data platforms having a correspondingly high level of cyber security protection. A loss of cryptographic protection would inflict devastating damage to all societies in a global context. It is important to note that time is needed to migrate our digital systems to a quantum-safe encryption technology and that such a transformation also entails costs.
4. Strike a balance between national security strategies, the promotion of international cooperation and the creation of structures to **prevent abuse** by authoritarian states or monopoly technology providers—in civilian as well as military applications.
5. **Ensure inclusivity and diversity.** In the quantum computing sector, it is expedient to make expensive hardware technology developments available to broader user communities through open source approaches, to stimulate further areas of application.
6. Analyse potential social and environmental impacts of the technology, both positive and negative, to ensure **broad public acceptance** and trust in the new technology.
7. Consider **export regulations** for dual-use quantum technologies.

Seen in the context of digital humanism, these fundamental objectives should be embedded in international standardisation efforts to allow an open and broad discourse in a global context.

6 Conclusion: How to Build an Effective Quantum Governance

The expected disruptive industry and market changes must be considered when appropriate capabilities for end-users and system operation are adapted in general as well as in the defence domain, as formulated in national defence quantum strategies in the security sector such as in Germany (Steudle, 2025) and Canada (DnD/CAF, 2023) for example.

Effective governance must reconcile economic goals and innovation promotion with risk mitigation for new technologies, and ensure that quantum technologies are developed positively, on behalf of society, and in the spirit of digital humanism. It should be noted that governance systems must be separately considered and designed for each of the three major areas of quantum technology: namely quantum computing; quantum communication and cryptography; and quantum sensing.

Appropriate technological expertise and skills are needed to conduct effective scenario analyses, foresight mechanisms as well as practical proof of concept and validation. To this end, sandboxing methods are essential to test the effects of new technologies and gain experience in dealing with them. By combining theoretical analyses with experimental technical experience, these processes are a significant aid to achieving technological expertise and a certain degree of technological sovereignty. This allows technology developments to be better controlled and technological effects to be anticipated more easily, in order to derive framework conditions for targeted regulatory measures.

Therefore, to develop effective governance measures it is important to discuss the seven aspects mentioned above, and to anticipate and evaluate possible associated risks. Here, technical measures are as important at the institutional level as in the public sector. The following approaches are necessary:

1. Promote a broad public discourse with a focus on application scenarios to create a broad general understanding.
2. Initiate new awareness and education initiatives for decision makers, with a focus on comprehensible application areas, to develop responsible innovations and targeted regulations.
3. Strengthen proof of concept projects in the sense of application-oriented research and sandboxing with interdisciplinary collaborations to achieve an understanding of technological power as well as possible associated risks, and finally also to improve the capabilities to influence international standardisation. This will also create the basis for human-centred design approaches and responsible technology development, as well as for developing agile sector-specific regulations (Kop, 2025).
4. Strengthen international standardisation, to create a better basis for the commercial exploitation of open standards in functioning markets and to develop and define methods for ethical assessments of quantum technologies.

It is important to build on the available expertise of Research and Technology Organisations (RTOs) and universities, as well as strong collaborations enabled by the EU research and industry community within the various EU innovation programmes, to achieve sustainable technology development capabilities and ensure shorter time-to-market cycles. In addition to that, it is essential to focus on the potential threats and risks enabled by quantum technologies and to elaborate effective governance measures for the different quantum technologies and use cases.

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Part IV

Epilogue

The Future of TechGovernance: Strengthening International Law and Diplomacy—Promoting Responsible Research and Conceptual Rethinking

Sibylle Bauer

Abstract

In light of the deep and far-reaching peace and security implications of technology governance, this chapter explores the logic and limitations of current technology governance; when and how technology is governed, and who is governed and involved. It focuses on legal and policy measures providing guidance and limitations on the development, production, transfer and use of technology for military applications. In addition to international treaties, national law and international regimes, it considers policy, assessment and educational processes based on legal norms and on broader ethical considerations. The chapter concludes that traditional governance approaches designed along technological fields or established weapons categories are still necessary but will need to be complemented. In the current fast-paced technology race, combined with a volatile global environment, the evolving range of actors and powerful interconnections between technologies, adjustments to governance approaches are clearly required. These need to reflect what is appropriate for different levels of technology readiness. Three options for the way forward are identified: reaffirming international legal norms and strengthening diplomatic capacities underpinning constructive international relations; upstream approaches including education and responsible research, engaging diverse stakeholders such as academia and the private sector in an inclusive manner; and conceptual rethinking and innovation.

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1 Introduction

Technology governance can be defined in different ways but encompasses a wide range of instruments and governance principles. This chapter focuses on legal and policy measures providing guidance and limitations on the development, production, transfer and use of technology for military applications. In addition to international treaties, national law and international regimes, it also considers policy, assessment and educational processes based on legal norms and on broader ethical considerations.

In the current context of geopolitical competition, the ‘why’ of technology governance has become as contentious as the ‘what’, ‘when’, ‘whom’ and ‘how’. Some national instruments of technology governance, such as unilateral export controls on quantum computing or semiconductors, have clearly been motivated by international power competition and ambitions for political or military dominance rather than global norms. Nevertheless, international law still contains fundamental norms governing security and military applications of technology. This wide-ranging common legal basis is particularly evident in the long-standing principles of international humanitarian law (IHL), but also fundamental multilateral treaties on disarmament, non-proliferation and arms control with very broad international membership such as the 1968 Nuclear Non-proliferation Treaty [NPT], the 1972 Biological and Toxin Weapons Convention [BTWC], the 1980 Convention on Certain Conventional Weapons [CCW], the 1992 Chemical Weapons Convention [CWC], the 2008 Convention on Cluster Munitions and the 2013 Arms Trade Treaty (ATT). At the same time, the international organisations and diplomatic processes designed to support, develop and implement these norms and instruments, are under substantial pressure, as their financing, their staffing and their very reason for existence, are questioned. Finally, the fundamental and age-old question of the role of responsible science, and of the individual researcher in ensuring technology governance, has come back to the fore, as development and application of different technology areas are accelerating and interacting, and their implications for the future of humankind are becoming ever more obvious. The 1954 Russell-Einstein Manifesto, which highlighted the responsibility of scientists, remains highly relevant today (The Russell-Einstein Manifesto, [1954](#)).

However, while the need to act is clear, and innovative thinking is a popular reference point in international discussions, there is little evidence of fresh thought and action in current international frameworks. This chapter explores ways forward in light of the direct implications of technology governance for peace and security, focusing on the role of international law and existing norms and principles along with diplomacy, the importance of responsible research as well as the scope for conceptual innovation in building new frameworks.

2 Logic and Limitations of Current Technology Governance

2.1 What is (Not) Governed and Why

Technology governance instruments are designed to address different types of risk, such as risk of misuse or of undesired proliferation, or a combination of both. For example, international treaties such as the NPT, CWC and BTWC explicitly refer to risks of non-peaceful use. Three informal export control regimes support implementation of these treaties through national-level controls. The Australia Group thus seeks to prevent the proliferation of biological and chemical weapons, the Nuclear Suppliers Group (NSG) the spread of nuclear weapons, and the Missile Technology Control Regime (MTCR) the spread of delivery vehicles capable of delivering nuclear, chemical and biological weapons (Bromley et al., 2025). Certain types of conventional weapons, such as blinding lasers, are prohibited on humanitarian grounds.

Dual-use export controls are designed to control the transfer of goods, software and technology that have both civil and military uses, whether in connection with conventional arms or so-called weapons of mass destruction (biological, chemical and nuclear weapons). They have become more politicised in recent years even though they originally derive their normative basis from multilateral instruments such as the CWC, BTWC and NPT as well as United Nations (UN) Security Council Resolution 1540 of 2004. While one could argue that different geopolitical settings enabled a global consensus around certain norms at different points in the past, global power competition was also in full swing in those times, and it was nevertheless possible to focus on shared risks that required a global response. The current clear prioritisation of national security and, increasingly, economic objectives in the mix of different rationales driving the content and application of export controls risks undermining these global norms.

Taking national export controls as an example, their scope and implementation are determined by a range of competing and overlapping interests: military superiority and national security, international security, human security, international law and norms, freedom of academia, freedom of trade, economic security and commercial interests. Governments weigh these against each other, with different prioritisations at different points in time. The recent shift towards a stronger focus on national and political considerations and unilateral or mini-lateral controls, away from an earlier stronger grounding in international law and multilateralism, poses a challenge to current international treaties and institutions, as well as regimes that are based on these instruments, such as the multilateral export control regimes.

Common ways to measure and rank risks are a function of impact and likelihood. In addition, risks are reinforced if underestimated in scope, size or temporal proximity; or if they are overestimated or even hyped, thus distracting energy, focus and creative problem-solving capacity as well as staff and financial resources. Different dimensions of human, national and international security

are key considerations in risk analyses for emerging technologies. The discussions on Autonomous Weapons Systems (AWS) since 2014 in the context of the CCW and in a range of more recent conference formats such as REAIM (first hosted in and by the Netherlands in 2023), are examples of detailed and differentiated discussions on associated peace and security risks and the role of international law (Blanchard & Goussac, 2025). The role of technology in intended and unintended escalation, as well as the unintended consequences of intended escalation, has been debated in expert circles and track 2 dialogues, and is increasingly also becoming a topic of official international discussion (Saalman, 2012; Wan, 2020). Initially focused on nuclear weapon-related issues, and conventional-nuclear entanglement (Acton, 2018), there is also increasing interest in nexus questions such as AI-nuclear and nuclear-space and their potential roles in escalation pathways (Chernavsky & Palayer, 2025; Raju & Wan, 2024; Saltini, 2025).¹

Indeed, some of the biggest, and perhaps least understood, risks may be those that arise from the interconnection or convergence (the nexus) between technologies, and between different sub-areas within technologies. This has created a governance gap. The failure to see these connections is largely due to the siloed approaches in which technologies are developed, monitored and regulated, which translates into stovepipes in research and governing institutions as well as relevant rules and treaties. An illustration of this can be found in the way units in foreign ministries responsible for security, non-proliferation, disarmament and arms control-related issues tend to be structured: along the lines of traditional notions of weapons systems and capabilities (chemical, biological, nuclear, conventional weapons, as well as cyber and outer space issues), rather than convergence, enabling effects and force multipliers. These structures make it inherently challenging to systematically consider and address cross-cutting issues. Expert communities are typically structured in similar ways and rarely interact. Recent projects bringing together experts from different communities, such as outer space and nuclear weapons and strategic stability on the one hand, and outer space or AI on the other hand, have resulted in very fruitful discussions but remain rare (SIPRI, 2025a, 2025b).

Debates on interconnection and convergence risks have recently increased, notably in relation to artificial intelligence (AI). They were initially largely focused on the so-called AI-nuclear nexus (Boulanin et al., 2020) and took place mostly in non-governmental expert discussions. They have since not only broadened to the nexus between AI and biological weapons and AI and chemical weapons (Brockmann et al., 2019; Lentzos, 2025; Blum, 2025) but have also expanded to events hosted by states and international organisations. Notable examples are the German Federal Foreign Office (German Federal Foreign Office, 2024) hosting a conference on the AI-WMD nexus in Berlin in 2024 and the Organisation for the Prohibition of Chemical Weapons organising a Global Conference on AI in

¹ For the purposes of this chapter the term “nexus” means an intersection or an interaction between different technology domains.

CWC implementation in Morocco the same year (OPCW & Kingdom of Morocco, 2024). The AI-nuclear nexus was also prominently highlighted by the Chinese President Xi and then-US President Biden in their 2024 joint statement that emphasised the need to maintain human control over the decision to use nuclear weapons (Ministry of Foreign Affairs (PRC), 2024; The White House, 2024).

The cyber-space nexus and the risks of cyber vulnerabilities to space systems were analysed by experts in 2016, but had been widely ignored until then (Lewis & Livingstone, 2016). In their report, Lewis and Livingstone highlighted how much critical infrastructure, such as global navigation, transport and trade, is vulnerable to cyberattacks, resulting in potentially disastrous civilian harm. Expert analysis has also explored the cyber-nuclear nexus (Unal & Lewis, 2018). However, risk arising from the combination of the many other technology areas, such as biotechnology and quantum technology, or gene editing and neurosciences, remains un- or underexplored, let alone broader systemic dynamics, or compounding effects. When comparing certain possible combinations of AI and quantum technologies with other technology fields, discussion of AI combinations is far more developed than the latter, although all with a considerable time lag between expert discussions and traction in diplomatic and policy circles.

Even where interest in keeping up with scientific advances is clearly present, this does not necessarily translate into institutionalised and systematic monitoring at the regional and global level. Biotechnology is a case in point. Safety and security dimensions of specific advances in biotechnology have been discussed in depth, and some linkages with other technology fields with enormous risk potential have been explored. Examples of issue areas that deserve attention are the link to information technology through the digitisation of biology (Bajema et al., 2018) and in connection with additive manufacturing, machine learning and robotics (Brockmann et al., 2019). The bi-annual Spiez conference in Switzerland has focused on convergence issues (Spiez Laboratory, 2024), and although specific proposals have been made to integrate the monitoring of scientific advances into the BTWC as the relevant international legal framework, this has not yet achieved consensus (Lentzos, 2025). In contrast, the Organisation for the Prohibition of Chemical Weapons established a Scientific Advisory Board already in 1997 (OPCW, 2011).

An obvious question to be asked is: why does the nexus governance gap exist and persist? Anticipating applications of technologies in security and military contexts not only requires in-depth knowledge of a specific technology, but also knowledge of other, existing or emerging technologies that may be combined, and potentially enable unforeseen applications and uses. However, current research and institutional structures tend to produce specialists and perpetuate silos rather than facilitate and promote cross-cutting and out-of-the-box thinking and foresight. The language used in presentations and publications is usually designed to communicate within a discipline or community, or even a sub-community and sub-discipline, but rarely does it bridge or translate across such boundaries. Inter-disciplinarity thus still remains more of a stated goal rather than a defining feature of research and university life. Studying, teaching, publishing and fundraising

across disciplines is rarely incentivised, or may even be actively discouraged. Of the many problems on the global stage today, this one does not require a reduction in geopolitical tensions or a change in the global political environment to be addressed.

2.2 When and How is Technology Governed

Effective governance of technologies from a peace and security perspective requires that risks of current applications in the military and security context are considered, and potential future ones anticipated. This faces many challenges, not least the way in which, and speed at which, technology development takes place. However, in spite of this, military and security applications of emerging technologies have been explored and discussed within expert communities long before these issues entered political and diplomatic agendas. An illustration of this is the publication dates of books on issues that are subject to lively debate in current political and security discussions, such as *Space Weapons and International Security* of 1978 (Jasani, 1978) and *Arms and Artificial Intelligence* of 1987 (Din, 1987). For both topics, the wide range of peace and security risks has become the subject of intense debate between governments in the UN and beyond, as well as in think-tank communities internationally, but with a very significant delay. The process of discussing the risks of military technologies at a point when they are being developed and before they have been deployed on a widespread basis could, and should, thus be initiated much earlier.

The necessity to reduce such time lags is illustrated by efforts to consider technology risks at a much earlier stage. The Organisation for Economic Co-operation and Development (OECD) in 2024 published a *Framework for Anticipatory Governance of Emerging Technologies*. Its “anticipatory technology governance” approach consists of five elements: “(1) embedding values throughout the innovation process; (2) enhancing foresight and technology assessment; (3) engaging stakeholders and society; (4) building regulation that is agile and adaptive; and (5) reinforcing international cooperation in science and norm-making” (OECD, 2024).

Existing technology governance approaches can be categorised in multiple ways: legally binding or not; capability-based versus behavioural-based; multilateral, bilateral, unilateral, regional; prohibitions or restrictions; etc. Another possible categorisation could be temporal, as a function of their relation to technology readiness levels (TRL). TRL form part of the basics in the natural and technical sciences but are little known and rarely used in the social sciences and in diplomatic circles. TRL (see Table 1) were first developed by NASA for the space context in the 1970s and have later been adjusted and adopted by the European Union for technology more broadly in 2014 for the Horizon projects funding research and later in the export control compliance guidelines for research and academia (EU, 2021).

Table 2 shows an illustrative mapping of technology governance mechanisms and their relevance on the TRL spectrum. One key consideration is that of which

Table 1 List of technology readiness levels (TRL), European Union (2021, p. 51)

TRL1	Basic principles observed
TRL2	Technology concept formulated
TRL3	Experimental proof of concept
TRL4	Technology validated in a lab
TRL5	Technology validated in a relevant environment (industrially relevant environment in the case of key enabling technologies)
TRL6	Technology demonstrated in a relevant environment (industrially relevant environment in the case of key enabling technologies)
TRL7	System prototype demonstration in an operational environment
TRL8	System complete and qualified
TRL 9	Actual system proven in operational environment (competitive manufacturing in the case of key enabling technologies; or in space)

instruments to use at what level of technology maturity. While some instruments can be usefully applied at all levels (although in modified form), others are relevant to certain parts only. For example, responsible research and innovation is of key importance at the stage of fundamental or basic research, when specific applications are yet to be discovered or decided upon. In contrast, export controls would hinder trade and scientific collaboration if applied to all early research, if it is unknown whether the technology will enable dual-use or military applications. This highlights the importance of what could be called upstream technology governance, centring the role of the individual researcher, and the early stages of technology development. In his Nobel Laureate speech in 1995, Joseph Rotblat conveyed the following message to his ‘fellow scientists’: “You are doing fundamental work, pushing forward the frontiers of knowledge, but often you do it without giving much thought to the impact of your work on society” (Rotblat, 1995). However, courses on research ethics, due diligence procedures and societal responsibility or on relevant international conventions and treaties, seem to remain

Table 2 Author’s compilation

Low TRL	Responsible Research and Innovation
	Research ethics
	Codes of conduct
	Due diligence procedures
High TRL	Export control
	Sanctions
	ATT
All TRL	Investment screening
	CWC
	BTWC
	NPT

the exception rather than the rule in universities across the globe (Boulanin et al., 2025). While the emphasis given to these issues will obviously differ between countries and faculties, and also individual universities and professors, it rarely seems to be a mandatory component of science education (Boulanin et al., 2024).²

2.3 Who is (Not) Governed and Involved

Technology is not only changing the rules of the game, but evolving development and production processes are also changing who is playing the game. In addition to the much-cited speed of technological development, current research and innovation in emerging technology fields are also characterised by an increase in the numbers and types of actors, with more individuals and entities in more countries having access to, or developing technology. This trend is occurring across sectors and geographies. The private sector and academia are commonly considered to be at the cutting edge of technological development, but the full implications of this are not reflected in the current architecture of technology governance, which is still largely focused on, and designed by, states.

To be comprehensive and inclusive, technology governance should consider the wide range of actors involved in the different stages of technology development, production, transfer, deployment and use. This applies to both the development of new instruments of technology governance, and existing technology governance mechanisms. Multistakeholder engagement has become a buzzword that needs to be unpacked to be meaningful. In varying ways and degrees, it includes industry, government, parliaments, academia, research institutions and civil society more broadly, as well as regional and international organisations. At the industry level, this concerns not only larger, established companies but also smaller start-ups and new sectors that are not, or not yet, part of the security and defence industry as traditionally conceived. This aspect has recently been given more attention for an emerging group of actors in the space industry referred to as NewSpace, but also applies to other sectors (Brockmann & Héau, 2024). An important consideration is also that of which entity within a company is engaged since, to be meaningful, this engagement needs to go beyond government relations departments to encompass compliance units as well as those who actually develop and fully understand the technology, in other words, the engineers.

Within governments, stakeholder engagement requires coordination between different relevant ministries such as ministries of foreign affairs, of defence, of technology, research and innovation and of economy. It also concerns those based in the capital as well as those posted in relevant international hubs such as Geneva, New York and Vienna. Regional organisations such as the European Union (EU)

² While the author is not aware of a systematic mapping to this end, this finding has been confirmed in multiple conversations with natural and technical science students and professors from different disciplines and continents over a decade.

and international organisations such as the UN, and in particular its Office for Disarmament Affairs, have become important players in the international technology governance debate.

Parliaments have traditionally played an important role in science and technology assessments over many decades but are rarely mentioned in, nor part of, international technology governance and arms control discussions. The European Parliamentary Technology Assessment, a network with 24 parliamentary members in European countries, along with associates in other continents (<https://eptanetwork.org>), is just one example. The underestimated or underappreciated role of parliaments may be either because it has not necessarily or systematically been extended to security and military applications or due to a lack of awareness of relevant parliamentary activity in this field, or a combination of both.

Civil society actors have also played an important role with regard to specific technology areas, such as cybersurveillance, robotics and machine learning. The role of academia and other research institutions has only in recent years come into stronger focus, but an increasing number are setting up internal procedures and bodies and developing codes of conduct and compliance procedures, partly in response to a stronger recognition of responsibility, partly in response to increased legal obligations (Héau, 2025). Also, if one were to expand the concept of stakeholders in temporal terms, added weight would need to be given to education and involving the next generation.

Finally, while global access to technology is a crucial step forward in terms of sustainable development goals, diffusion poses not only opportunities but also challenges for technology governance, and needs to be at the forefront when designing and tailoring governance. Therefore, an additional aspect to keep in mind is that concern about specific components of technology governance and the way in which they are applied is at times voiced either by those worried about reduced or denied access to technologies. This has been the case not only for governments from Global South countries but also from the Global North, in relation to supply chain security and access to rare earths. In addition to access to technologies, which is important from the point of view of economic development, there is also the question of being given, or denied, a voice in governance approaches, thus perpetuating inequality, or resulting in simply being spoken for. Concerns have also been raised by researchers regarding overregulation, and impediments to international cooperation and freedom of academia. Some of those concerns can be accommodated if mechanisms are designed and implemented in a way that reflects and addresses these issues to the extent possible. An important consideration in this context is the right to peaceful uses of technology, which has been invoked by different states in UN discussions. A balanced discussion of these issues and careful consideration of ways to address the criticisms voiced, such as analysis of the role of different factors in impeding the sharing of nuclear technology for peaceful uses and inclusive discussion on how to facilitate such sharing, may go a long way in addressing emerging polarisation (Maletta et al., 2025).

3 Options for Technology Governance

3.1 Back to Basics: International Law and Diplomacy

Challenges arising from speed, complexity and diffusion are compounded by geopolitical tensions and competition that play out in increasing technonationalism (Capri, 2019). Additionally, multilateralism, international treaties and institutions, which have provided key and globally accepted frameworks of technology governance for decades, are increasingly contested, sidelined, or even actively deconstructed.

The norms underpinning much of technology governance have come under strain. While the norms against biological and chemical weapons remain strong, those around nuclear weapons have been facing substantial pressure, with mounting concerns about risk of use, salience and spread. The risk of horizontal nuclear proliferation, and thus of new nuclear weapon programmes, has increased, along with emerging fears of a resumption of nuclear testing. Vertical proliferation is ongoing, with all nuclear-weapon-possessing states modernising or increasing their arsenals (Kristensen & Korda, 2025).

Meanwhile, means and methods of warfare are evolving, but remain subject to the same principles of international humanitarian law (IHL) that have been in place for over a century. In other words: “While the use of certain technologies provides previously unconsidered scenarios, it does, however, not fundamentally change the applicable law” (Bauer, 2020). Thus, one way forward would be a back-to-basics approach. IHL has been the one consistent legal framework withstanding the test of time and technological developments, regardless of their nature and speed. Building on this, one could reaffirm those core principles through a paradigm for technology governance focusing on the protection of civilians.

Another way of reinforcing IHL norms is the reviews based on Article 36 of the 1977 Additional Protocol I of the 1949 Geneva Conventions requiring states to conduct legal reviews of new weapons, means and methods of warfare to determine whether their use is prohibited by international law—IHL reviews (Art. 36). Article 36 of Additional Protocol I states that:

In the study, development, acquisition or adoption of a new weapon, means or method of warfare, a High Contracting Party is under an obligation to determine whether its employment would, in some or all circumstances, be prohibited by this protocol or by any other rule of international law applicable to the High Contracting Party. (Protocol I, 1949)

This illustrates that there are key provisions in international law that are applicable to current and future technology developments (ICRC, 2015). The International Committee of the Red Cross (ICRC) in 2006 issued a *Guide to the Legal Review of New Weapons, Means and Methods of Warfare*, detailing measures to implement Article 36 (Boulanin & Verbruggen, 2017a; ICRC, 2006). However, practical application differs considerably from state to state (Boulanin & Verbruggen, 2017b; Cochrane, 2020; McClelland, 2003). These provisions need to

be discussed, or, at least, states could start with transparency and dialogue about how these are applied.

UN Security Council Resolution 1540 illustrates the potential of diplomacy and global institutions, and provides an example of a set of norms designed and decided in the UN framework that continue to shape technology governance. Moreover, it was adopted unanimously in a very specific context but has had considerable normative power going beyond its original narrow focus on preventing access by non-state actors to Weapons of Mass Destruction (WMD) (UN, 2004). The post-9/11 environment and revelations about the global activities of proliferation networks helped to forge a consensus across the permanent members of the UN Security Council, which secured unanimous agreement of the resolution, and Russia submitted the first draft. The UN Security Council acted as a legislator in the sense that it was creating general, binding and far-reaching obligations by declaring the proliferation of nuclear, biological and chemical weapons and their delivery systems to be a threat to peace and security. This provided the basis for adopting Resolution 1540 under Chapter VII of the UN Charter, which makes it binding on all states. It complements the non-proliferation conventions (BTWC, CWC and NPT), which focus on state obligations regarding state WMD programmes rather than the role of non-state actors. In addition to requiring domestic measures such as physical protection and accounting for WMD-related materials, the resolution for the first time creates an obligation on all states to establish dual-use trade controls. The way obligations were phrased and have been implemented resulted in a substantial increase in national trade control systems for dual-use items worldwide, governing both access to non-state actors and states. It provides that all states must:

Establish, develop, review and maintain appropriate effective national export and trans-shipment controls over such items [nuclear, chemical, or biological weapons and their means of delivery, and related materials, equipment and technology], including appropriate laws and regulations to control export, transit, trans-shipment and re-export and controls on providing funds and services related to such export and trans-shipment such as financing, and transporting that would contribute to proliferation, as well as establishing end-user controls; and establishing and enforcing appropriate criminal or civil penalties for violations of such export control laws and regulations.

Resolution 1540 thus became the legitimising factor of strategic trade controls/dual-use export, transit, transshipment and brokering controls, regardless of whether for non-state actors or state-run WMD programmes. Also, no international convention covers delivery systems, while Resolution 1540 explicitly applies to nuclear, biological and chemical weapon delivery systems, and also provides a definition:

Missiles, rockets and other unmanned systems capable of delivering nuclear, chemical, or biological weapons, that are specifically designed for such use.

Thus, a key factor leading to practical relevance is that operationalisation of the resolution was basically written into the text, highlighting the centrality of national

legislation as the obligation, whereas obligations in traditional WMD regimes are either more abstract, addressed to other entities such as a secretariat, or not verified. In spite of geopolitical challenges, Resolution 1540 remains an important reference point in the current contested environment, providing important lessons both for safeguarding existing frameworks and fora, and designing new ones, rather than further dismantling.

The maintenance of existing mechanisms and development of new frameworks will require another ‘back to basics’ component: to invest in a pool of technology-literate, forward-looking government representatives, to support multilateral engagement and foster common ground, to better understand escalation pathways and actively explore de-escalation, and to implement anticipatory governance. The international governance mechanisms that are in place would not exist without skilled diplomats, and will be difficult if not impossible to maintain and further develop without them.

3.2 Innovation: Upstream Approaches and Conceptual Re-Thinking

Much of traditional technology governance centres on downstream approaches, controlling technologies based on concrete and known applications. This can, however, only constitute one component of an overarching and forward-looking technology governance approach and needs to be complemented by upstream processes. Upstream processes necessarily involve those actors that are driving technological development, notably researchers in academic and private sector spaces, the latter often involving small start-ups in sectors that are not, or not yet, categorised as part of the security and defence sector. Not only state actors, but also a wide range of non-state actors are not only impacted by technology governance but are key to effectively designing and implementing it.

Such upstream approaches can include responsible research and innovation (RII), research ethics, strengthened inter- and cross-disciplinary education, and courses in natural science faculties across the globe on research ethics, societal responsibility, the peace and security implications of research, and responsible research and innovation. While these terms overlap and faculties may have preferences regarding the framework, it is crucial to include this thinking in education before scientists enter the professional world and perform advanced research in companies and at universities. Such courses could be made mandatory for graduation to highlight their importance for careers, but also for society more broadly. Somewhat connected is societal resilience in response to tech developments enabling unprecedented scale and ways of disinformation. Although only one tool, education is obviously key here. States could play a crucial role in strengthening these upstream approaches, and making them part of the technology governance toolkit, by drawing attention to success stories, disseminating resources and providing subsidies, in a manner that can encourage and incentivise further efforts.

Effort and resource will also need to be placed on exploring opportunities for new technologies to support arms control, non-proliferation and disarmament, for example through verification. The OPCW and its Scientific Advisory Board in 2024 pioneered such efforts by launching an AI research challenge in 2024, arguing that “AI could offer many benefits to the OPCW’s work, helping to achieve its mission of a world free of chemical weapons” (OPCW, n.d.).

Finally, while non-proliferation, arms control and disarmament as key features of technology governance have evolved considerably over the past decades, much of the language has stagnated. For example, in the export control community, the term dual-use is still the point of reference. However, many countries are considering end-uses in export control decisions that do not fit into the civilian-military dichotomy, such as uses by law enforcement authorities that impact human rights or security. The term ‘triple-use’ would thus be more correct than the dual-use, although the latter continues to be applied. Other examples of terms that continue to be utilised in spite of their limited technical accuracy or intellectual precision are terms such as strategic and tactical weapons, or strategic stability. While outdated language may facilitate regional and international agreement, it can also constrain innovative, outside-the-box thinking and perpetuate existing institutional, conceptual and legal stovepipes. Moreover, some of the concepts used need to be ‘translated’ or adapted to resonate in different communities. Finally, existing terminology is primarily dominated by either Cold War thinking, or by the West. However, it becomes abundantly clear, especially with the technological diffusion, that not revisiting this language can become counterproductive.

4 Conclusion

In the current fast-paced technology race, combined with a volatile global environment, traditional approaches are still very much needed but are insufficient. This chapter proposed a two-pronged approach reaffirming international legal norms and strengthening diplomatic capacities underpinning constructive international relations, but also new lines of effort, designing anticipatory governance approaches going beyond state actors and involving the private sector and research communities, rethinking conceptual boxes, and adjusting to the powerful interconnections between technologies. Moreover, both up- and downstream approaches designed along technological fields or established weapons categories are still necessary but will need to be complemented by a focus on nexus questions. While some of these questions can be considered in the framework of existing treaties and mechanisms, new instruments and formats may be necessary, and will need to go hand-in-hand with inclusive and multistakeholder engagement.

To conclude, responding to the big and urgent question of how to prevent or to govern warfare in cyberspace, outer space and mental space, how to apply existing and how to establish new norms governing technological developments, will

require creative and innovative problem-solving and the bridging of different communities, but also reinforcing the value of old-standing principles and treaties. The way forward will thus be a combination of ‘back to basics’ and innovation.

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Defence Technology Governance in the European Union

Daniel Fiott

Abstract

The 2025 EU White Paper on Defence underscores the need for a more risk-tolerant regulatory environment to foster European defence technologies and innovation. As the EU expands its role in defence, it faces complex governance challenges, particularly in managing emerging technologies such as AI, robotics, cyber and biotech. The Union's *sui generis* structure—comprising diverse state and non-state actors—complicates efforts to design cohesive regulatory frameworks, especially given the prevalence of dual-use technologies. Legal competencies outlined in EU Treaties further constrain centralised governance, as operational control remains with member states. Nonetheless, instruments like the EU Common Position on arms exports, the Dual-Use Regulation and the European Defence Fund introduce mechanisms for responsible oversight. This chapter examines both existing EU regulations and the political challenges of governing defence technologies amid rising global proliferation and reduced emphasis on governance by major powers like the US and China.

1 Introduction

The 2025 European Union (EU) White Paper on Defence boldly claims that “to accelerate the emergence of European defence tech players, the regulatory environment needs to be more conducive towards risk-taking” (European Commission,

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2025, p. 15). The idea of “risk” in the development and deployment of defence technologies is nothing new, but for the EU—which is rapidly expanding its presence in the defence innovation efforts—this creates a number of challenges. Indeed, as the EU steps up its financing of defence technologies, it will inevitably contend with a range of governance issues, including those related to responsible use and proliferation, to name a few. As the Union takes on more responsibility for defence, it must craft efficient and responsible governance structures to develop emerging technologies such as artificial intelligence (AI), robotics, cyber, biotechnologies and remotely piloted systems. Yet, designing such a governance framework for defence technologies is not simple in the EU, as there are diverse actors and diffuse competencies.

As one major work on the governance of security technologies stresses, the EU is a *sui generis* actor in the field of defence technologies. It combines a series of state and non-state actors, and there is a complex interplay of power, industrial interests and narratives framing how the Union should develop and utilise defence technologies (Calcara et al., 2022). What is more, the Union’s development of defence technology governance is complicated by the range of dual-use technologies that are integrated into defence systems. This implies that the political and legal framework for defence technology governance must include a wider range of civilian actors, including research institutes, specialised government agencies and civilian innovators from a range of non-defence sectors (Fiott, 2022). This means that any EU defence technology governance framework needs to be more broadly applicable to a range of actors that do not consider the defence market as their main sector of operations.

Governance of defence technologies in the EU is further complicated because of the legal competencies set down in the EU Treaties. We cannot overlook the fact that EU defence technology governance forms part of a larger Union approach to defence capability development. Overall, defence in the EU is governed in an inter-governmental fashion through the Common Security and Defence Policy (CSDP), and it is through this framework that the EU plans and prepares for a range of military and civilian missions and operations. Thus, if one consults the Union’s strategy documents, such as the Strategic Compass or the EU Defence White Paper, one will observe that a key objective of EU defence technology development is not only to support the innovation potential of the European defence sector, but also to ensure that defence technologies can be used effectively and within international legal parameters for military deployments. In the EU, institutions such as the European Commission have a large say in defence industrial policy terms, but not when it comes to the deployment of force or use of military equipment. The operational use of defence technologies and systems is still in the hands of EU member states.

Nevertheless, the EU has developed a range of legal and political instruments that are designed to govern the development and use of defence technologies. Two of the most noteworthy legal instruments in this regard are the EU Common Position on arms exports, which considers such questions as human rights, non-proliferation, risk and regional stability and the EU Dual-Use Regulation, which

governs technologies for both civilian and military purposes, particularly when it comes to cyber and surveillance technologies. What is more, the Union's development of financial instruments to support defence innovation, such as the European Defence Fund (EDF), comes with its own governance structures that cover issues such as responsible use and proliferation. For example, the EDF regulation states that the Commission is responsible for ethics screening and assessment as part of the selection and award procedures (EU, 2021).

This chapter considers the political challenges facing the EU when devising a defence technology governance framework. Essentially, one of the major issues facing the Union is how to effectively govern defence technologies in a global context where proliferation is on the rise, and where a range of major powers, such as the United States (US) and China, are investing heavily in emerging and disruptive technologies in the military sphere. In this sense, a question for us to investigate is how the Union can ensure effective and responsible development and proliferation of defence technologies in a context where most military actors outside of Europe are paying less attention to regulation and governance. To this end, this chapter is organised in two main parts: part one looks at existing EU defence technology regulations and measures; and part two analyses the challenges facing the Union in responsibly developing and using defence technologies.

2 The State of Defence Technology Governance in the EU

There are two core areas in which the EU has developed defence technology governance: (1) regulating and controlling the *development* of certain defence technologies at the EU level, mainly through its own defence funding mechanisms and adherence to international law; and (2) regulating the *proliferation* of defence technologies by legal tools such as the dual-use control and arms control legislation. These two governance areas have emerged on the back of successive historical events that have pushed the Union into collective action. We will look at the state of the art in each of these two main areas.

2.1 Defence Technology Development

Only recently did the EU venture into the realm of defence R&D. It did so after the creation of the EDF in 2021, where the Union, for the first time, dedicated funds from the EU budget to help stimulate defence innovation in the EU. The EDF, endowed with €8 billion until 2028, is the Union's tool for boosting collaborative defence research among industries, innovators and research institutes in the EU. It currently invests funds into the development of AI, cyber defence, space and remotely piloted systems, among other technologies. Yet in doing so, this has raised questions about defence technology governance for collaborative defence

innovation projects. So far, the European Commission has invested some €4 billion in 225 collaborative defence innovation projects (Fiott, 2025a). As part of this relatively new initiative, the Union has had to develop ethics controls for its investments, especially as they pertain to emerging and disruptive technologies such as AI or autonomous systems. To this end, recital 35 and articles 7 and 18 of the EDF Regulation make clear that ethical compliance is a precondition for participation in the Fund. Here, there is a dual system of self-assessment and assessment by an independent ethics review.

To this end, the Fund is supposed to reject any defence innovation projects that violate human rights or are banned under international conventions (e.g. cluster munitions, chemical and biological weapons, WMDs, etc.). However, beyond the need to ensure that EU-funded projects are developed in line with the Union's international commitments and obligations, the EDF Regulation also ensures the control of defence technologies through strict technology control provisions. Indeed, the EDF Regulation ensures that EU-funded defence technologies are not controlled by any third, non-EU states. This measure is designed to ensure that EU-developed defence technologies cannot be controlled by any states that cooperate with the Union (Klecza, 2024). Specifically, there is a fear that third states may seek to control the export and use of EU-funded defence technologies by extending their own extra-territorial regulations over EU rules (e.g. the United States' ITAR Regulation assumes that any defence system that integrates US technologies or software falls under the control of the US federal government) (Fiott, 2019a).

While the EDF is a relatively new EU funding instrument in defence, it has already raised questions within the academic literature about how it addresses ethical use. For example, one article looks at how visions of autonomy and technological competitiveness are embedded in the discourse surrounding the EDF. In this sense, these narratives raise questions about the ethical and political dimensions of defence technology, such as how the Union calibrates its risk appetite when nurturing dual-use technologies for EU security and defence (Csernatoní & Martins, 2019; Martins & Mawdsley, 2021). Another paper has discussed the need to develop defence technologies at the EU level, while also adhering to the Union's fundamental ethical principles (Gahler, 2016; Sabatino, 2022). And here, literature in security research has already uncovered some of the challenges the EU faces in relation to privacy, surveillance and proliferation when financing security research (Leese et al., 2018). What is more, scholarly work has also considered the political responsibility over defence technologies, with one article suggesting that EU-level defence technology investments complicate the traditional understanding of national sovereignty in defence (Hoeffler, 2023).

Overall, however, one of the major challenges facing defence technology governance under the EDF is the issue of scrutiny. We have seen how the EDF is presently governed by an ethics self-assessment process, plus an independent ethics committee that assesses prospective EDF projects against international law and conventions. However, such scrutiny may not always go far enough, and this raises the question of public scrutiny of EDF projects by the European Parliament. To date, the European Parliament is only partially involved in the scrutiny

of EDF projects, as under the Regulation, it plays a scrutiny role in evaluating the EDF (ending 31 December 2027 under the first cycle of the Fund) as a whole and not for individual projects financed under the Fund. Overall, the EU Treaties do not afford the European Parliament a major scrutiny role in security and defence policy, but under the EDF, the Parliament can play somewhat of a scrutiny role during the evaluation periods of the Fund, even though it cannot veto or scrutinise individual EDF projects, as they are agreed on an annual basis (Fiott, 2019b).

Aside from ethical dimensions, however, we should recognise that the EDF has introduced a novel form of defence technology governance in the EU. We should recognise that the Union has not traditionally supported defence innovation, so the EDF is by its very nature a new and relatively experimental form of defence technology governance. Even though the European Commission learned how to establish calls for tender and work programmes under the Fund from its broader civil research activities (e.g. the Horizon programme), the EDF is a novel form of governance given the sensitivities surrounding defence technology development. Indeed, the Fund is based around a programme committee comprised of the European Commission and the EU Member States. The committee essentially allows the EDF work programmes to synchronise better with national defence capability and technology priorities. This form of “comitology” allows the Commission to work directly with the member states and it gives national capitals a direct stake in the direction and prioritisation under the Fund (Simon & Marrone, 2021).

2.2 Proliferation

Aside from its diplomatic efforts to safeguard against proliferation in international fora, the EU has two specific legal instruments through which to maintain non-proliferation efforts: (1) the EU Common Position 2008/944/CFSP; and (2) the Dual-Use Regulation (EU) 2021/821. The Common Position 2008/944/CFSP was born of the Union’s desire to ensure that arms exports did not contribute to global instability, although this ran into the desire by certain member states to continue to export arms for defence industrial reasons. The post-Cold War period was one where the proliferation of weapons and technologies prolonged and intensified conflict in places such as Africa, the Balkans and the Middle East. An initial EU Code of Conduct on arms exports (1998) sought a politically—but not legally—binding agreement between member states, but it failed to impose a harmonised and binding framework on EU member states. However, EU enlargement raised fears of large stockpiles of weapons in post-Soviet states, and the range of counter-terror measures adopted globally after 2001 led the Union to reconsider its existing arms control regulations.

The EU Common Position 2008/944/CFSP has received sustained criticism from the academic community. For example, one study for the European Parliament argues that there are serious divergences among national arms export policies, especially during conflicts (Bauer et al., 2018). In particular, the study critiques reporting quality under the Common Position and the lack of any coherent

interpretation of criteria, not to mention the weak mechanisms to ensure compliance. Indeed, another paper reiterates the view that the Common Position is too vague and weak and that member states interpret the criteria in self-interested ways where commercial or industrial interests tend to dominate (Lüchow, 2015). Building on these criticisms, one study goes further in stating that licencing decisions under the Common Position are insufficiently transparent, information sharing between member states is inconsistent and the criteria—especially regarding human rights—are applied unevenly (Grünhage et al., 2013).

Equally, the Union's Dual-Use Regulation (2021) has its roots in the Cold War and the subsequent Wassenaar Arrangement (1996), which is the main international regime for regulating dual-use items. The Union's first dual-use legislation in 2009 sought to ensure security and non-proliferation (especially in nuclear, chemical and biological areas), as well as to break down barriers in the single market and harmonise export controls. In the 2010s, however, the rise of new and disruptive cyber, surveillance and biometric technologies, plus notable experiences during the 'Arab Spring', where European surveillance technologies were employed for internal repression, led the EU to revise its dual-use regulations. The EU's Dual-Use Regulation is a legally different instrument when compared to the EU Common Position 2008/944/CFSP. Whereas the Common Position is governed in an intergovernmental manner between EU member states through the Council of the EU, the Dual-Use Regulation is a community instrument linked to the Union's common commercial policy. This means that dual-use exports are governed in a supranational manner through the co-legislators.

Because of this different form of governance, the 2021 EU Dual-Use Regulation has greater power to ensure the harmonisation of dual-use standards in the EU. While it is true that EU member states can adopt stricter national dual-use controls in specific areas (e.g. cyber surveillance items), generally, member states cannot adopt national rules that duplicate or contradict the Dual-Use Regulation. Perhaps due to this supranational form of defence technology control, the academic literature has praised elements of the Regulation. For example, one study argues that the Regulation allows for more coordination between member states and enhances controls on emerging technologies, while also expanding transparency (Bromley & Brockmann, 2021). Other studies consider that the Regulation is useful in not only fostering defence innovation, but also civil-defence synergies, which can benefit SMEs, start-ups and innovators (Vella, 2025).

However, even the Dual-Use Regulation has been criticised in the scholarly literature. One study claims that while human rights norms are more explicitly part of the Regulation, there are tensions and contradictions—and some competing norms—in how they are implemented. Some EU member states resist stricter human rights criteria, especially when economic or strategic interests are at stake (Bromley & Brockmann, 2021). Relatedly, another study highlights how there is uneven application of the Regulation across the Union, with some member states having stronger capacities to enforce the Regulation compared to others (Bromley, 2023). Another work argues that the Regulation may impose burdens on research and knowledge exchange, especially in international collaborations, if

items or knowledge are caught by dual-use rules. This can create delays in projects, uncertainty and greater legal risk (Stalenhof et al., 2022).

Again, the Union's governance structure for dual-use technologies is also indicative of the flexible nature of governance in the EU, and its rather sophisticated regulatory architecture sits at the intersection of internal market integration and export control. Under the Dual-Use Regulation (EU) 2021/821, the Commission and Member States jointly manage a framework that seeks to reconcile security imperatives with the free movement of goods and technologies within the EU single market. Governance operates through a hybrid model: Member States retain control over export authorisations, but they do so within a common EU legal regime coordinated by the Dual-Use Coordination Group, chaired by the European Commission. This arrangement ensures a degree of policy convergence and transparency. The governance of dual-use exports has an undeniable supranational quality. The EU has effectively created a shared system for managing sensitive technologies, one that embeds security concerns within the legal and administrative machinery of the single market. In this sense, the dual-use regime exemplifies how the Union can pool competences, align national procedures and set ethical and strategic parameters for technologies that blur the civilian-military boundary.

In this first part of the chapter, we have considered two main areas in which the EU seeks to govern defence technologies: (1) development; and (2) proliferation. We have seen how the Union's move towards the financing of defence innovation has placed challenges on institutions such as the European Commission to ensure that defence technology projects adhere to the Union's commitment to international law and ethics. Here, we have shown that the EU has developed a governance framework that gives the Commission a large say over the assessment of defence innovation projects, but institutions such as the European Parliament retain an indirect scrutiny role. Under the EDF, ethical considerations are important, but the ethical assessment of projects is governed by a mixture of self-assessment (by industries, innovators and researchers) and an independent ethics committee. Such an approach raises the challenge of how best the European Commission can balance a military edge through its defence investments, compared to a need to control technologies in line with strategic and ethical norms.

Additionally, we have seen how the EU employs two primary legal instruments to advance non-proliferation objectives: the Common Position and the Dual-Use Regulation. The Common Position, developed to prevent arms exports from exacerbating global instability, has faced persistent criticism for its lack of binding enforcement, inconsistent interpretation of criteria and inadequate transparency, with member states often prioritising commercial interests over human rights concerns. Conversely, the Dual-Use Regulation, constantly updated in response to emerging technologies and geopolitical developments, operates under a supranational framework that better harmonises export controls across member states. This Regulation has been praised for enhancing coordination and transparency, fostering innovation and integrating human rights norms more explicitly. However, scholarly critiques highlight ongoing challenges, including uneven enforcement,

resistance from some member states on human rights criteria and potential impediments to international research collaboration due to regulatory burdens. Together, these instruments illustrate the EU's evolving but imperfect approach to balancing security, economic interests and ethical considerations for its non-proliferation efforts.

3 The Challenges Facing the EU in Defence Technology Governance

In the previous part of the chapter, we outlined the range of major EU instruments in the governance of defence technology. With this analytical basis, in this part, we turn our attention to some of the major challenges facing the Union in defence technology governance. As we will see, a critical dual challenge is the rapidly evolving nature of defence technologies and growing geopolitical rivalry between the major powers. In this respect, EU defence technology governance has the difficult task of simultaneously respecting international law while also ensuring that the Union can maintain and grow its military-technological edge. Indeed, in 2025, the EU inhabits a geostrategic context where military might and economic power have returned, and where a major hallmark of this geostrategic context is the ability to rapidly invest in and develop military technologies, including those principally being developed in the civilian domain. As we shall observe in this part of the chapter, the development of emerging and disruptive technologies creates a governance dilemma for the EU.

3.1 Emerging and Disruptive Technologies

The first major challenge facing the EU in the governance of defence technologies is the rapidly evolving nature of emerging and disruptive technologies. First, it is important to recognise that emerging and disruptive technologies encompass a wide array of fields, including AI, cyber-surveillance, autonomous systems, quantum technologies, advanced materials, biotechnologies, etc. These technologies are not static and are developing quickly, often unpredictably so, and with dual-use potential (i.e. civilian, security and military uses). As Csernatonì has noted, these technologies are increasingly central to the EU's strategy for technological sovereignty, economic competitiveness and defence policy (Csernatonì, 2021). However, the pace of innovation currently underway blurs civil and military boundaries, and this has led to global competition and even rivalry over the means of production of key technologies and resources. In particular, the United States and China are both presently seeking to compete over disruptive technologies in the hope that they may each gain a military advantage over the other.

The EU is not immune to this geopolitical competition and it has had to rapidly enhance its own strategic autonomy, partly because of the strategic uncertainty caused by the United States' commitment to European security. As the EU seeks

to enhance its strategic autonomy by investing in defence technologies, it runs into the challenge of how far to push technological innovation, especially if it raises concerns about existing legal and normative obligations. In this sense, the speed at which emerging and disruptive technologies are being developed strains existing regulatory frameworks and approaches. For example, the Union's Dual-Use Regulation and the international Wassenaar Arrangement have had to constantly amend regulations for cyber surveillance and intrusion software (Ruohonen & Kimppa, 2019). And here, the EU is faced with the dilemma of how to harness innovation in the commercial sector without engaging in unethical behaviour in the defence sphere. This is complicated by the fact that innovation is occurring in the health, agricultural, transport and life sciences sectors, but it may not always be ethical to engage such sectors through defence technologies (s. for example Calcara et al., 2022; or Rynchnovska, 2016). Not least because doing so raises questions about privacy, data use, discrimination, etc.

Rapidly emerging technologies imply that legal definitions often lag behind technological change, and this creates legal uncertainty and risk for governments, institutions, firms and citizens. This pace of innovation also raises the risk of a lack of accountability, as it is unclear how regulatory oversight should function in new technology domains such as AI and autonomous robotics, especially where there are questions about machine self-control. Case law and legal norms tend to appear in societies once any misuse of technologies has occurred, and by this time, misuse of technologies may be more widespread. In the EU, this becomes a doubly difficult challenge, as there is a tension between the high pace of technology development and national, territorially bounded legal systems and processes in the Union (Greenstein, 2021). On this basis, the academic literature notes that the EU's legal system has been unable to fully account for the safe and compliant deployment of AI across the Union (Novelli et al., 2024).

For the EU, a range of essential trade-offs emerge for defence technology governance. First, the Union must balance regulation of emerging and disruptive technologies with innovation: over-regulation can be viewed as a way to stifle technology development, even though ethical risks may emerge when opting for a light-touch approach to regulation. Second, it is not easy to develop or sustain Union-level regulatory approaches, as individual member states may have their own interests in strong or light regulation of emerging and disruptive technologies. Such an imbalance may result in uneven regulatory enforcement and create a competitive spiral between EU member states on disruptive technologies. Third, regulating "grey areas" in certain technology domains risks creating uncertainty and unintended consequences, so the temptation to be a first mover into certain defence technology domains needs to be offset against the risks of doing so. Finally, the EU's approach to the regulation of emerging and disruptive technologies is constantly being studied by adversaries and partners—EU regulation of defence technologies may or may not work to the (dis)advantage of third states.

3.2 The Erosion of International Norms and Conventions

The second major challenge facing the EU is an overarching erosion in international arms control and non-proliferation conventions, which threatens to risk greater global instability. In light of Russia's illegal invasion of Ukraine, we have seen increasing pressure placed on a range of international arms control and non-proliferation conventions. Once foundational agreements such as the Intermediate-Range Nuclear Forces (INF) Treaty, the Treaty on Open Skies, the Anti-Ballistic Missile (ABM) Treaty, the Strategic Arms Reduction Treaties (START) and even the Ottawa Convention banning anti-personnel landmines have all faced serious strain or outright abandonment. This deterioration reflects deeper structural changes in the international system, including resurgent great power competition, the emergence of novel military technologies and the weakening of verification mechanisms and normative commitments. Taken together, these developments represent not only a legal and institutional decline but also a dangerous shift in the strategic assumptions underpinning global security. This is particularly alarming for the EU, which has long adhered to and promoted such international conventions.

Multiple factors have driven this collective unravelling. First among them is the re-emergence of great power rivalry. Russia's actions in Ukraine and China's rapid military modernisation have all contributed to a strategic environment where arms control is increasingly viewed not as a shared interest, but as a potential constraint on national security (s. for example Anthony, 2017; Kulesa, 2018; Moeller, 2024; Risse, 2023). Treaties that were once perceived as mutual safeguards are now seen, particularly by revisionist or insecure states, as asymmetric liabilities. Moreover, many existing treaties were designed in an earlier technological era and do not adequately account for emerging threats such as hypersonic weapons, advanced missile defence systems or remotely piloted delivery vehicles. These new systems blur the lines between offensive and defensive capabilities, challenge traditional verification mechanisms and create incentives for states to modernise their arsenals outside existing treaty frameworks.

The erosion of arms control regimes carries profound implications, especially for the EU. It raises the likelihood of renewed arms races, lowers the threshold for weapons use in conflicts and increases the risk of accidental or miscalculated escalation. Smaller and medium-sized states, lacking the resources of great powers, are especially vulnerable to this strategic instability. Moreover, the loss of legally binding verification systems makes it harder for states to reliably assess one another's capabilities and intentions, fostering paranoia and worst-case scenario planning. Even more troubling is the broader normative impact: as treaty withdrawals become more common, the sense that certain weapons are illegitimate or unacceptable weakens, potentially reversing decades of progress in international humanitarian law and disarmament.

Of course, it might be argued that none of the above fears and developments should halt the Union's commitment to international conventions and norms. However, this is easier said than done, considering that individual EU member states

are rapidly developing weapons systems that provide additional defence in this post-normative age. Owing to the fact that actual and potential adversaries are not binding themselves to long-standing arms control conventions, the EU is faced with a difficult dilemma: adhere to international conventions, even if adversaries do not, and risk being placed at a military disadvantage; or develop defence technologies outside of now defunct international conventions and undermine the Union's diplomatic commitment to arms control and non-proliferation. In this sense, the EU's defence technology governance framework must simultaneously contend with military, political and normative considerations.

3.3 Military Power and Innovation

Finally, the Union faces the challenge of rapidly developing defence technologies for its own defence and autonomy, while also ensuring that it maintains its normative commitment and legal obligations. Traditionally, the EU has largely embraced a vision of security rooted in diplomacy, economic integration and multilateralism. Military power, while never irrelevant, was increasingly subordinated to the EU's emphasis on soft power, humanitarian engagement and crisis management. However, since Russia annexed Crimea in 2014 and its full-scale invasion of Ukraine in 2022, military power has returned to the centre of European strategic thinking. This shift has had a profound impact on the EU's approach to defence, not least in the domain of defence technology, where member states and EU institutions are seeking to develop greater strategic autonomy, enhance technological innovation and reduce dependence on external actors, especially the US (s. for example Csernaton, 2022; Fiott, 2025b).

The deterioration of the security environment in Europe has sharpened the EU's awareness of its own vulnerabilities. The war in Ukraine laid bare the extent of Europe's dependence on American military technology, from precision-guided munitions to satellite surveillance and logistics. This has accelerated discussions on the need for a robust European Defence Technological and Industrial Base (EDTIB), capable of producing, maintaining and innovating critical defence systems within Europe itself (s. for example Kollias & Tzeremes, 2024; Fiott, 2024). The aforementioned EDF is but one part of the Union's strategy of developing its defences. Indeed, in 2025, the EU combined its efforts through the EDF with novel initiatives such as a new defence loan-making facility, called the Security Action for Europe (SAFE), worth €150 billion, and regulatory changes to the Stability and Growth Pact to potentially unlock an additional €650 billion in national defence spending.

In October 2025, the Union also agreed to a new European Defence Industrial Programme (EDIP), designed to pull through defence innovation projects into major common development and procurement programmes, including common EU defence projects. The EDIP foresees a range of novel approaches to defence cooperation in the EU, including innovative institutional frameworks designed to bring together member states, EU institutions and industry. One such measure in EDIP

is the so-called Structures for European Armament Programme (SEAP), which will allow for joint cooperation and life-cycle management of defence equipment, including defence innovation. The EDIP gives due consideration to defence technologies and governance. It underlines the need to adhere to European ethical standards and fundamental rights in the development of defence technologies. The EDIP also stresses the need for a high level of information exchange with the European Parliament on these standards and rights as it develops defence technologies. (European Commission, 2024).

At the same time, the demands of high-intensity warfare, exemplified by the war in Ukraine, have underscored the urgent need for technological innovation. European defence firms and research institutions are now increasingly focused on dual-use technologies such as AI, cybersecurity, autonomous systems and space capabilities to ensure a military edge for Europe. The EU's efforts to coordinate these developments have led to new synergies between civilian innovation ecosystems and military requirements. Initiatives such as Permanent Structured Cooperation (PESCO) have helped foster cross-border cooperation on specific technological projects, including next-generation communication systems, battlefield drones and missile defence (Biscop, 2018). This reflects an evolution in the EU's strategic culture—away from crisis management and towards capabilities more aligned with deterrence and high-intensity conflict.

However, this transformation is not without tensions, as some member states and citizens caution against any militarisation of the Union. Nevertheless, the EU's pursuit of strategic autonomy in defence technology is increasingly seen as part of the Union's response to a resurgence of military power in the world (Fiott, 2023). While significant obstacles remain, the geopolitical shocks of recent years have pushed the EU to confront its defence deficiencies more directly than at any point in its history. The challenge now lies in converting this strategic awakening into sustained technological progress and a credible defence posture that reflects both European values and contemporary security imperatives.

Given these security challenges, however, questions remain about the effectiveness of EU defence technology governance. We have seen in this chapter that the Union has developed a flexible set of arrangements for the governance of defence technologies under the EDF and export controls under the dual-use regulation. In the initial stages of each of these institutional frameworks, there was a degree of experimental adaptation to balance member state interests, security/ethical concerns and economic competitiveness. For defence technology governance, however, there is a need to reflect on institutional overlap. The Commission has taken the lead in developing EU-level defence innovation via the EDF, and this brings the Commission into close political proximity to the member states—this is a novel development. However, other parts of the EU's defence policy architecture are involved in defence innovation efforts too. This includes the European Defence Agency (EDA), which is developing defence technology projects within its own mandate, not least via PESCO. Interestingly, over the course of the EDF's work programmes, the EDA has been entrusted with the management of individual projects under the Fund.

This again highlights the Union's ability to collectively adapt to new institutional mechanisms, but it raises questions about the driving ethos behind the EDF. For example, the European Commission is taking a largely industrial policy-driven approach to defence technologies, where it is trying to stimulate and support industrial programmes that correspond largely to commercial objectives in the defence sector. Other bodies, such as the EDA, have taken a different approach and have instead emphasised the importance of a capability-driven approach to defence technology development that corresponds mainly to the technology and capability needs identified by the EU's military representatives and defence ministries. This difference between industry- and capability-driven approaches to defence technology development is an issue that has still not really been addressed by the EU's governance structures, not least because each of these approaches is spread across the supranational and intergovernmental aspects of EU defence policy governance.

4 Conclusions

The governance of defence technologies in the EU sits at the intersection of ethics, innovation and geopolitics. As this chapter has shown, the EU's evolving governance architecture reflects both institutional innovation and structural limitations. On the one hand, instruments such as the European Defence Fund, the Common Position on Arms Exports and the Dual-Use Regulation provide a framework for regulating defence technology development and proliferation in line with international norms. On the other hand, the fragmentation of competences between intergovernmental and supranational domains continues to challenge coherent and transparent governance. The Union's growing engagement in defence R&D has necessitated new mechanisms for ethical oversight, but questions remain about the adequacy of scrutiny and democratic accountability, particularly in relation to the role of the European Parliament.

The EU's defence technology governance also faces mounting external pressures from the rapidly evolving landscape of emerging and disruptive technologies. AI, autonomy, quantum systems and biotechnology are advancing at a pace that outstrips current legal and regulatory frameworks. These technologies do not map neatly onto existing legal categories, nor do they align clearly with traditional distinctions between civilian and military technologies and domains. This ambiguity introduces legal uncertainty, regulatory lag and some degree of ethical risk. This creates a structural tension between innovation and control. The Union's challenge lies in fostering technological competitiveness and strategic autonomy without eroding its own ethical and legal standards. Failure to balance these imperatives risks either regulatory overreach that stifles innovation or, conversely, permissiveness that undermines human rights, privacy and international law. In this sense, the EU's approach to technology governance must be embedded in multilevel coordination between the European Commission, the European Parliament and national authorities.

In addition, the eroding framework of international arms control, the intensifying rivalries among great powers and the resurgence of military power as a central factor in global strategy place enormous strategic pressure on the EU. International treaties and conventions, once taken as almost foundational (e.g. the INF) are under strain. Verification regimes are weaker; some treaties have lapsed or have been abandoned; norms are less assumed and more contested. In such a context, the EU faces a stark choice between remaining committed to legal and normative frameworks that may increasingly be flouted by others, thereby risking strategic disadvantage, or adjusting its governance of defence technologies in recognition of the decline of norms, potentially undermining its own normative legitimacy. Rather than abandoning international conventions, the EU should redouble its efforts to modernise and reinterpret them in light of new technologies and geopolitical realities, thereby reinforcing its role as a standard-setter for responsible military innovation.

The return of military power to the forefront of European strategy has, moreover, reignited the question of how the EU reconciles its civilian ethos with the demands of deterrence and high-intensity warfare. The establishment of initiatives such as the EDF, SAFE and EDIP marks a significant deepening of EU-level defence industrial policy and cooperation. Yet these developments must be matched with stronger political guidance, parliamentary scrutiny and societal engagement if they are to command long-term legitimacy. The governance of defence technologies cannot be a purely technocratic exercise: it should be subject to public debate and rooted in shared European values. Achieving this will require sustained investment in regulatory expertise, ethics review capacity and transparency mechanisms within EU institutions.

In terms of future research, this chapter highlights several areas where knowledge remains incomplete. One is the evaluation of how effectively risk-taking, as advocated in the EU's 2025 White Paper on European Defence, is actually being calibrated in practice in defence innovation projects—who sets the risk thresholds, how are trade-offs handled and what kinds of failures are tolerated (and by whom). A second relates to comparative analysis: how do EU member states and EU institutions compare with counterparts in the US, China and Russia in balancing regulation and innovation in defence technologies, especially under conditions of strategic urgency. Third, empirical work is needed on the downstream effects of governance instruments: for example, on how regulatory burdens affect smaller firms and innovators, or how dual-use export rules affect international collaboration.

In addition to this future research agenda, there are several policy recommendations that could be considered. First, the EU should continue to strengthen parliamentary oversight of EU defence-related instruments, ensuring that ethical and legal assessments are complemented by democratic scrutiny. Second, the Union should invest in its technology foresight capacities, as past efforts (e.g., the Critical Technology Observatory) have proven underwhelming and largely ineffective. Without proper technology mapping or foresight, it will be difficult for the EU to properly assess the ethical dimensions of technology development. Third, the

EU should develop international initiatives on the responsible uptake of military innovation, working with partners to update global non-proliferation norms for the AI and autonomous era. Fourth, coordination between the EDF, PESCO, SAFE and the EDIP should be streamlined to avoid duplication and enhance coherence across civil-military innovation ecosystems. Finally, the EU and its member states should promote greater public dialogue, balancing the need for security with the imperative of accountability.

In sum, the return of military power in the international system has forced the EU to raise its strategic ambition in defence technology in ways that were not fully anticipated even a few years ago. The EU defence technology governance framework is expanding and being reformed. Yet none of these remove the fundamental tensions and trade-offs for the Union: between speed and oversight, innovation and norms and sovereignty and supranational coherence. How the EU navigates the present geopolitical period may well determine not only its own security and technological edge, but also whether it remains credible as a normative power in global affairs. And here, should international arms control measures continue to deteriorate, how will the EU decide to act in defence and with what technologies?

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